

Study Notes

Foundations of Real Analysis in Economics

Graduate course

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Notes based on the following bibliography:

Alabiso & Weiss, *A Primer on Hilbert Space Theory*

Moroşanu, *Functional Analysis for the Applied Sciences*

Pons, *Real Analysis for the Undergraduate*

Ok, *Real Analysis with Economic Applications*

Zorich, *Mathematical Analysis I–II*

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How to use these notes

These are study notes for *Foundations of Real Analysis in Economics*, written for a graduate course. They follow, week by week, the order in which the course is usually taught, and they are based on the bibliography listed below.

They are written for a specific reader: someone who comes to real analysis **from economics**, with mathematical training at a basic to intermediate level. This explains the two style choices that govern the whole text: **nothing is assumed** (every concept is built from scratch) and intuition always comes before formalism, without giving up rigor afterwards.

Remark 0.1 (Preliminary note). This is unofficial study material: it is not the course guide and it does not replace lectures, the syllabus or the textbooks, and it is not endorsed by any institution. The proofs follow the standard developments of the cited bibliography, with the wording and the order of presentation adapted to the goals of these notes. The priority marks explained below are a personal study criterion, not an official classification of contents. Any error belongs to these notes and not to the sources.

Structure of each chapter

- **Goals and place in the course:** which week of the schedule it covers and which book of the bibliography treats it.
- **Motivation:** why the concept exists and what problem it solves.
- **Definitions and theorems:** numbered, with proofs explained *step by step* (each step says what is done and why).
- Colored boxes:
 - **Intuition:** the idea in plain words, before the formalism.
 - **Careful (common mistake):** typical errors that cost points in exams.
 - **Exam strategy:** how the topic is usually asked and how to attack it.
 - **Link with economics:** where the concept appears in economic theory.
 - **What is this used for in Microeconomics?:** the concrete link with the topics of a graduate micro course.
 - **Chapter summary:** the minimum you have to know by heart.
- **Exam-style exercises with complete solutions.**

The priority marks and what they are for

A real analysis course covers much more than an economist will use later on a daily basis. Since these notes were also written as a *bridge* towards graduate microeconomics, each chapter and each section carries a **priority mark [P1]–[P5]** that answers a single question: how much is this material used in a typical **Microeconomics** course?

The calibration was made against the canonical syllabus of a graduate course based on MAS-COLELL, WHINSTON and GREEN (MWG), which is organized in five parts. The violet boxes throughout the text refer to them:

1. **Individual decision making** (MWG chs. 1–6): preference, choice and revealed preference; classical demand theory (duality, Slutsky equation, envelope theorem); aggregate demand; production and duality; choice under uncertainty (expected utility, risk aversion).
2. **Game theory** (MWG chs. 7–9): static games and Nash equilibrium in pure and mixed strategies; dynamic games; Bayesian games; repeated games and discounting.
3. **Market equilibrium and market failure** (MWG chs. 10–14): partial equilibrium and surplus; externalities and public goods; market power; adverse selection, signaling and screening; the principal-agent problem and moral hazard.
4. **General equilibrium** (MWG chs. 15–18): pure exchange and production economies, the Edgeworth box; Pareto optimum and the two welfare theorems; existence of Walrasian equilibrium; the core.
5. **Welfare and incentives** (MWG chs. 21–23): social choice and Arrow’s theorem; welfare economics; mechanism design, the revelation principle and implementation.

The marks mean the following:

- **[P1] Essential:** used directly and repeatedly in micro (existence of optima and equilibria, correspondences, first order conditions, separation of convex sets).
- **[P2] Very useful:** language and tools that appear often or that support the **[P1]** topics.
- **[P3] Useful as training:** it gives maturity and context; it is used indirectly.
- **[P4] Peripheral:** it appears little in micro (although it can be central in macroeconomics, econometrics or finance).
- **[P5] Marginal for micro:** it is almost never used in that course.

Careful (common mistake)

Two warnings about the marks. First: they are *relative to microeconomics*, not to the real analysis course. A **[P5]** topic is still examinable material of the course and can still appear in the midterm or in the final; the marks help you decide where to go deeper with micro in mind, not what to study for the analysis exams. Second: a specific program may weight things differently. A course with a lot of repeated game theory or with dynamic models will raise the value of series and discounting; one centered on general equilibrium and static contracts will lower it. It is a good idea to check the marks against the syllabus

you are given.

Remark 0.2 (What these notes do not cover and micro does use). To be honest with the reader, it is worth pointing out the gaps, which come from the syllabus of the analysis course and not from these notes. Three tools that modern microeconomics uses are missing here: **measure theory and probability**, needed to treat uncertainty, Bayesian games and general lottery spaces with rigor; **monotone comparative statics** (lattices, supermodularity, the theorems of Topkis and Milgrom–Shannon), now standard in micro courses; and the **implicit function theorem** together with optimization under equality and inequality constraints (Lagrange multipliers and Karush–Kuhn–Tucker conditions), which appear here only in the general form of the chapter on differentiation and optimality. If you aim at micro, you should cover these three fronts separately.

Bibliography and how it is used here

- ALABISO, C.; WEISS, I. *A Primer on Hilbert Space Theory* [1]. (Chs. 1–5). Main source for linear, topological, metric, normed and Hilbert spaces.
- MOROȘANU, G. *Functional Analysis for the Applied Sciences* [2]. (Chs. 1–4 and 6). Source for Banach spaces, linear operators and the Riesz theorem.
- PONS, M. *Real Analysis for the Undergraduate* [3]. The most “friendly” reference: useful when something here seems to move too fast.
- OK, E. *Real Analysis with Economic Applications* [4]. Source of the economic applications (fixed points, dynamic programming, Berge’s maximum theorem).
- ZORICH, V. *Mathematical Analysis I–II* [5, 6]. Complementary reference and source of differential calculus in normed spaces.

For the links with economic theory, the background reference is MAS-COLELL, A.; WHINSTON, M.; GREEN, J. *Microeconomic Theory* [7], cited as MWG; when a topic is easier to follow in a different presentation, JEHL, G.; RENY, P. *Advanced Microeconomic Theory* [8] covers the same syllabus with a somewhat slower treatment. The full references of all these works are in the bibliography at the end of these notes.

Suggested study plan

The course is usually graded with two exams of equal weight, a midterm around week 9 and a final around week 17, on the theory developed in class and on the exercise lists. An effective way to use these notes:

1. **Before each class:** read the motivation and the definitions of the corresponding chapter (30–45 min). Arriving with the definitions in mind lets you follow the proofs live, which is where you really learn.

2. **After each class:** go over the proofs and try the exercises *without looking* at the solution; compare afterwards.
3. **Memorize the definitions with surgical precision** (ε - δ , Cauchy, compactness, and so on). In this kind of course, stating a definition wrongly invalidates the whole problem.
4. **Before each exam:** use appendices A and B (midterm review and final review), which condense exact definitions, theorems with their hypotheses, the recurring proof techniques and a solved mock exam.

Careful (common mistake)

In analysis, a good share of exam mistakes are about *quantifier logic*: confusing “for every ε there exists N ” with “there exists N for every ε ”, or negating a definition wrongly. The first chapter (Preliminaries) attacks this head on; you should not skip it even if it looks elementary.

Chapter 1

Preliminaries: logic, sets and the real numbers [P1]

Reading before Week 1. Bibliography: Pons [3], Ch. 1; Zorich I [5], Chs. 1–2; Ok [4], Ch. A.

What is this used for in Microeconomics?

This chapter is the language in which the whole Microeconomics course is written. Choice theory starts with a *binary relation* of preference and its axioms (completeness and transitivity, MWG ch. 1), and the weak axiom of revealed preference is literally a statement with quantifiers about choice sets. The same language comes back later: a *Walrasian equilibrium* (Part IV) requires that *for every* consumer the chosen bundle maximizes utility and that market clearing *exists*; a mechanism is *incentive compatible* (Part V) if *for every* type and *for every* deviation telling the truth is optimal. Arrow’s impossibility theorem (MWG ch. 21) is in fact a purely logical argument about binary relations and quantifiers. Also, the maximum profit of the firm and the minimum expenditure of the consumer (Part I) are *suprema and infima*: the ε characterization of the supremum is used every time you approximate an optimum.

1.1 Motivation [P1]

Real analysis is, in essence, the art of handling statements with *quantifiers*: “for every $\varepsilon > 0$ there exists an N such that...”. All the central definitions of the course (convergence, continuity, Cauchy sequence, compactness) have this form. If you do not master the logical grammar of these sentences (what they mean, how they are negated, in which order they are read), it is impossible to write a correct proof, and in the exams of this course *everything* is proved.

Also, analysis lives on the real numbers $\mathbb{R}$, and the property that distinguishes $\mathbb{R}$ from the rationals $\mathbb{Q}$ is the *completeness axiom*, also called the supremum axiom. Almost all the important theorems of the course (Bolzano–Weierstrass, convergence of monotone sequences, existence of maxima of continuous functions on compact sets) come down from this single axiom. This chapter builds that foundation without assuming anything.

1.2 Logic of propositions and quantifiers [P1]

1.2.1 Propositions, implication and its variants

A **proposition** is a statement that is true or false, without ambiguity: “ $3 > 2$ ” is a true proposition; “ $x > 2$ ” is not a proposition until you say who x is.

Given two propositions P and Q , the **implication** $P \Rightarrow Q$ reads “if P then Q ”. It is essential to understand its truth table: $P \Rightarrow Q$ is false only when P is true and Q is false. In particular, if P is false, the implication is *automatically* true (we say it is “vacuously true”).

Intuition

Think of $P \Rightarrow Q$ as a contract: “if it rains (P), I lend you my umbrella (Q)”. The contract is broken only if it rains and I do not lend you the umbrella. If it does not rain, I have broken no promise, whatever I do: the contract is still valid. This is why an implication with a false hypothesis is true.

Attached to $P \Rightarrow Q$ there are three propositions that you must tell apart precisely:

- The **converse**: $Q \Rightarrow P$. It is *not* equivalent to the original one. Example: “if f is differentiable, then f is continuous” is true, but its converse is false ($f(x) = |x|$ is continuous and not differentiable at 0).
- The **contrapositive**: $\neg Q \Rightarrow \neg P$ (where $\neg$ means “not”). This one *is* logically equivalent to $P \Rightarrow Q$: proving one is proving the other.
- The **negation**: $\neg(P \Rightarrow Q)$ is equivalent to “ P and $\neg Q$ ”. Careful: the negation of an implication is *not* another implication.

We say that P and Q are **equivalent** ($P \Leftrightarrow Q$, “ P if and only if Q ”, abbreviated *iff*) when $P \Rightarrow Q$ and $Q \Rightarrow P$. To prove an equivalence you have to prove *both* directions, and in an exam you should separate them explicitly: $(\Rightarrow)$ and $(\Leftarrow)$.

1.2.2 Quantifiers and their order

The two quantifiers are:

- $\forall$: “for all”, the **universal** quantifier;
- $\exists$: “there exists”, the **existential** quantifier.

The most delicate point of the whole course: **the order of the quantifiers changes the meaning**. Compare:

$$\forall x \in \mathbb{R} \exists y \in \mathbb{R} : y > x \quad \text{versus} \quad \exists y \in \mathbb{R} \forall x \in \mathbb{R} : y > x.$$

The first one says: “for each real x there is some real y larger than it” (true: take $y = x + 1$; note that y may depend on x). The second one says: “there is a real y that is larger than all the reals” (false: such a y would have to be larger than itself). The general rule: *a variable*

quantified with $\exists$ may depend on all the variables quantified before it, but not on those that come after.

This is exactly what happens in the definition of limit: in “ $\forall \varepsilon > 0 \exists N \in \mathbb{N} : \dots$ ”, the N may (and almost always must) depend on ε .

1.2.3 Negating propositions with quantifiers

The mechanical rules are:

$$\neg(\forall x : P(x)) \equiv \exists x : \neg P(x), \quad \neg(\exists x : P(x)) \equiv \forall x : \neg P(x).$$

That is: when you negate, each $\forall$ becomes $\exists$, each $\exists$ becomes $\forall$, and the final property is negated. The quantifiers are handled *in order, from the outside in*.

Example 1.1. Let us negate the definition of convergence. “The sequence (x_k) converges to L ” means:

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall k \geq N : |x_k - L| < \varepsilon.$$

Applying the rules from the outside in, “ (x_k) does *not* converge to L ” means:

$$\exists \varepsilon > 0 \forall N \in \mathbb{N} \exists k \geq N : |x_k - L| \geq \varepsilon.$$

In words: there is a “bad” tolerance ε such that, no matter how far you go along the sequence, there are always terms at distance at least ε from L . Note the three changes: $\forall \rightarrow \exists$, $\exists \rightarrow \forall$, $\forall \rightarrow \exists$, and the final inequality was negated ($<$ became $\geq$).

Careful (common mistake)

Negating a definition with quantifiers in the wrong way is the most frequent and most expensive mistake in analysis exams. Worked example: “ f is continuous at a ” means

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x : |x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon.$$

Its correct negation (“ f is discontinuous at a ”) is:

$$\exists \varepsilon > 0 \forall \delta > 0 \exists x : |x - a| < \delta \text{ and } |f(x) - f(a)| \geq \varepsilon.$$

Typical mistakes: (i) writing “ $\forall \varepsilon$ ” in the negation (the quantifiers *must* change); (ii) negating the implication as another implication (the negation of $A \Rightarrow B$ is “ A and not B ”, never “ $A \Rightarrow \neg B$ ”); (iii) forgetting to change $<$ into $\geq$. Before the exam, practice negating: convergence, Cauchy, continuity and boundedness.

1.2.4 Proof techniques

In this course you will use four schemes again and again:

1. **Direct proof:** to prove $P \Rightarrow Q$, assume P and derive Q through a chain of justified steps. It is the default scheme.

2. **Proof by contrapositive:** to prove $P \Rightarrow Q$, prove $\neg Q \Rightarrow \neg P$. Useful when $\neg Q$ is easier to handle than P .
3. **Proof by contradiction** (*reductio ad absurdum*): assume that the conclusion is false and derive something absurd (for example, $0 > 0$). Classic case: the proof that $\sqrt{2}$ is irrational.
4. **Induction:** to prove that $P(n)$ holds for every $n \in \mathbb{N}$, prove the base case $P(1)$ and the inductive step $P(n) \Rightarrow P(n+1)$.

Example 1.2. Proof by induction of **Bernoulli's inequality**: if $x \geq -1$, then $(1+x)^n \geq 1+nx$ for every $n \in \mathbb{N}$.

Base case ($n = 1$): $(1+x)^1 = 1 + 1 \cdot x$, with equality. ✓

Inductive step: assume that $(1+x)^n \geq 1+nx$ (induction hypothesis). Since $x \geq -1$, we have $1+x \geq 0$, so we can multiply both sides of the hypothesis by $(1+x)$ *without reversing the inequality* (multiplying by nonnegative numbers preserves inequalities):

$$(1+x)^{n+1} \geq (1+nx)(1+x) = 1 + (n+1)x + nx^2 \geq 1 + (n+1)x,$$

where the last step uses $nx^2 \geq 0$. This is exactly $P(n+1)$, and the induction is complete. $\square$

This inequality will be used to study the convergence of r^n and of $(1+1/n)^n$.

1.3 Sets and functions [P1]

1.3.1 Set operations

A **set** is a collection of objects, called **elements**. We write $x \in A$ (“ x belongs to A ”) and $A \subseteq B$ (“ A is contained in B ”: every element of A is an element of B). Two sets are equal when $A \subseteq B$ and $B \subseteq A$; this “double inclusion” is the standard method to prove equalities of sets in exams.

Basic operations:

$$A \cup B = \{x : x \in A \text{ or } x \in B\}, \quad A \cap B = \{x : x \in A \text{ and } x \in B\},$$

$$A \setminus B = \{x \in A : x \notin B\}, \quad A^c = X \setminus A \text{ (complement with respect to a universe } X\text{)}.$$

De Morgan's laws, which we will use constantly in topology (Chapter 5), say that the complement swaps unions and intersections, even for arbitrary families of sets $\{A_i\}_{i \in I}$:

$$\left(\bigcup_{i \in I} A_i \right)^c = \bigcap_{i \in I} A_i^c, \quad \left(\bigcap_{i \in I} A_i \right)^c = \bigcup_{i \in I} A_i^c.$$

Proof of the first law. Double inclusion. ($\subseteq$) If $x \in \left(\bigcup_{i \in I} A_i \right)^c$, then $x \notin \bigcup_{i \in I} A_i$, that is, x is in *no* A_i (this is logical De Morgan: negating “there exists i with $x \in A_i$ ” gives “for every i , $x \notin A_i$ ”). Hence $x \in A_i^c$ for every i , that is, $x \in \bigcap_{i \in I} A_i^c$. ($\supseteq$) Every step above is reversible: all the statements used are equivalences, not just implications. $\square$

1.3.2 Functions

A **function** $f : A \rightarrow B$ assigns to each $x \in A$ exactly one element $f(x) \in B$. The set A is the **domain**, B is the **codomain**, and the **image** (or range) is $f(A) = \{f(x) : x \in A\} \subseteq B$.

Definition 1.3. Let $f : A \rightarrow B$.

1. f is **injective** (one-to-one) if distinct elements have distinct images: $x \neq y \Rightarrow f(x) \neq f(y)$.
Equivalently (contrapositive, more useful in practice): $f(x) = f(y) \Rightarrow x = y$.
2. f is **surjective** (onto) if $f(A) = B$: every element of the codomain is the image of someone.
3. f is **bijective** if it is injective and surjective; in that case the inverse function $f^{-1} : B \rightarrow A$ exists.

For $C \subseteq B$, the **preimage** (or inverse image) is

$$f^{-1}(C) = \{x \in A : f(x) \in C\}.$$

Careful with the notation! $f^{-1}(C)$ is *always* defined, even if f is not bijective and the inverse function does not exist. The preimage behaves better than the image: it respects unions, intersections and complements, that is, $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$, $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$ and $f^{-1}(C^c) = (f^{-1}(C))^c$. For the direct image, on the other hand, only $f(C \cap D) \subseteq f(C) \cap f(D)$ holds in general. This good behavior of the preimage is the reason why continuity is characterized with preimages of open sets (we will see this in Chapter 5).

1.3.3 Countability

A set A is **countable** if it is finite or if there is a bijection between $\mathbb{N}$ and A (that is, its elements can be “listed”: $a_1, a_2, a_3, \dots$). Facts you should know:

- $\mathbb{Z}$ and $\mathbb{Q}$ are countable (the rationals are listed by going through the table of fractions p/q along diagonals).
- A countable union of countable sets is countable.
- $\mathbb{R}$ is **not** countable (Cantor’s diagonal argument): there are “many more” reals than rationals.

The countability of $\mathbb{Q}$ together with its density in $\mathbb{R}$ (which we see below) will be the key point when we talk about *separability* of metric spaces.

1.4 The real numbers [P1]

1.4.1 Ordered field axioms

The set $\mathbb{R}$ with the operations $+$ and $\cdot$ and the order $\leq$ satisfies three groups of axioms:

1. **Field axioms:** $+$ and $\cdot$ are associative and commutative, there are neutral elements (0 and 1), every x has an additive inverse $-x$, every $x \neq 0$ has a multiplicative inverse x^{-1} , and distributivity $x(y + z) = xy + xz$ holds. (The same axioms that $\mathbb{Q}$ satisfies.)

2. **Order axioms:** $\leq$ is a total order compatible with the operations: if $x \leq y$ then $x + z \leq y + z$; and if in addition $z \geq 0$, then $xz \leq yz$. From these you derive all the rules for manipulating inequalities (for example: multiplying by a *negative* number reverses the inequality).
3. **Completeness axiom:** we state it in Section 1.4.3. This is the axiom that $\mathbb{Q}$ does *not* satisfy and the one that makes all of analysis work.

1.4.2 Absolute value and the triangle inequality

Definition 1.4. The **absolute value** of $x \in \mathbb{R}$ is

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

Intuition

$|x|$ is the distance from x to the origin, and $|x - y|$ is the distance between x and y . All of analysis in $\mathbb{R}$ is written in terms of $|x - y|$; when we move to metric spaces, we will simply replace $|x - y|$ by an abstract distance $d(x, y)$. So it is worth mastering the properties of the absolute value now: they are the prototype of all metrics.

Immediate properties: $|x| \geq 0$ with equality only if $x = 0$; $|xy| = |x| |y|$; and the very useful characterization

$$|x| \leq a \iff -a \leq x \leq a,$$

valid for $a \geq 0$. From it comes the most important property of the course:

Theorem 1.5 (Triangle inequality). *For all $x, y \in \mathbb{R}$:*

$$|x + y| \leq |x| + |y|.$$

The reverse triangle inequality also holds:

$$||x| - |y|| \leq |x - y|.$$

Proof. Step 1. By the definition of absolute value, we always have $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$ (if $x \geq 0$ the second inequality is an equality and the first one is obvious; if $x < 0$, the other way around).

Step 2. Adding the two chains of inequalities term by term (adding inequalities in the same direction is legitimate by the order axioms):

$$-(|x| + |y|) \leq x + y \leq |x| + |y|.$$

Step 3. By the characterization $|u| \leq a \iff -a \leq u \leq a$ with $u = x + y$ and $a = |x| + |y|$, we conclude $|x + y| \leq |x| + |y|$.

Step 4 (reverse). We write $x = (x - y) + y$ and apply the inequality just proved:

$$|x| = |(x - y) + y| \leq |x - y| + |y| \implies |x| - |y| \leq |x - y|.$$

Swapping the roles of x and y (the right hand side does not change because $|y - x| = |x - y|$): $|y| - |x| \leq |x - y|$. Since $|x| - |y|$ and its negative are both bounded by $|x - y|$, the characterization of Step 3 gives $||x| - |y|| \leq |x - y|$. $\square$

Exam strategy

The technique of Step 4 (“add and subtract” an intermediate term and apply the triangle inequality) is *the* standard move in analysis:

$$|a - c| = |(a - b) + (b - c)| \leq |a - b| + |b - c|.$$

It will appear in the uniqueness of the limit, in the algebra of limits, in uniform convergence and in dozens of problems. When in an exam you do not know how to start bounding $|a - c|$, ask yourself: which intermediate term b connects a with c ?

1.4.3 Supremum, infimum and the completeness axiom

Definition 1.6. Let $A \subseteq \mathbb{R}$ be nonempty.

1. $M \in \mathbb{R}$ is an **upper bound** of A if $a \leq M$ for every $a \in A$. If A has some upper bound, it is called **bounded above**.
2. $s \in \mathbb{R}$ is the **supremum** (least upper bound) of A , written $s = \sup A$, if: (i) s is an upper bound of A , and (ii) $s \leq M$ for every upper bound M of A (it is the *smallest* of the upper bounds).
3. In the same way we define **lower bound** and **infimum** (greatest lower bound), $\inf A$: the largest of the lower bounds.
4. If $\sup A \in A$, it is called the **maximum** of A ; if $\inf A \in A$, the **minimum**.

Intuition

The supremum is “the lowest possible ceiling”. The interval $A = (0, 1)$ has infinitely many upper bounds (1, 2, 17, ...), but only one is the best: $\sup A = 1$. Note that $1 \notin A$: *the supremum need not belong to the set* (in that case there is no maximum). This supremum versus maximum distinction is crucial in optimization: a problem may have a finite supremum value without any point attaining it.

Theorem 1.7 (Completeness axiom, supremum axiom). *Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.*

It is not proved: it is an *axiom*, the property that *defines* $\mathbb{R}$. To see what it says, note that $\mathbb{Q}$ does not satisfy it: the set $A = \{q \in \mathbb{Q} : q^2 < 2\}$ is nonempty and bounded above in $\mathbb{Q}$, but it has no *rational* supremum (the natural candidate, $\sqrt{2}$, is not rational). The reals “fill in the

holes” of the rationals; that is the intuitive idea of completeness, which in the chapter on Cauchy sequences and completeness will come back in metric form (every Cauchy sequence converges).

The following characterization is the everyday working tool:

Proposition 1.8 (ε characterization of the supremum). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let s be an upper bound of A . Then:*

$$s = \sup A \iff \forall \varepsilon > 0 \exists a \in A : a > s - \varepsilon.$$

Intuition

It says that s is the supremum exactly when no number strictly smaller than s (that is, no $s - \varepsilon$) is still an upper bound: as soon as you go a little below the supremum, there are already elements of the set above you. This is the “working” form of the supremum: it turns the definition (comparing with *all* upper bounds) into a statement about elements of the set, which is what you can manipulate.

Proof. ($\Rightarrow$) Let $s = \sup A$ and fix an arbitrary $\varepsilon > 0$. The number $s - \varepsilon$ is strictly smaller than s . Since s is the *smallest* upper bound, $s - \varepsilon$ cannot be an upper bound of A . Negating “ $s - \varepsilon$ is an upper bound” (that is, negating “ $\forall a \in A : a \leq s - \varepsilon$ ”) gives exactly: $\exists a \in A$ with $a > s - \varepsilon$. (Notice how we used the negation of quantifiers from Section 1.2!)

($\Leftarrow$) Assume that s is an upper bound and satisfies the ε condition. We want to show that it is the *smallest* upper bound. By contradiction: assume there is another upper bound M with $M < s$. Take $\varepsilon = s - M > 0$. By hypothesis there exists $a \in A$ with

$$a > s - \varepsilon = s - (s - M) = M,$$

which contradicts that M is an upper bound of A (there would be an element of A above M). Hence no such M exists, and $s = \sup A$. $\square$

Theorem 1.9 (Archimedean property). *For every $x \in \mathbb{R}$ there exists $n \in \mathbb{N}$ with $n > x$. Equivalently: for every $\varepsilon > 0$ there exists $n \in \mathbb{N}$ with $\frac{1}{n} < \varepsilon$.*

Proof. By contradiction. Assume there exists $x \in \mathbb{R}$ such that $n \leq x$ for every $n \in \mathbb{N}$. Then $\mathbb{N}$ is nonempty and bounded above (by x), so by the completeness axiom there exists $s = \sup \mathbb{N}$.

We apply the ε characterization of the supremum with $\varepsilon = 1$: there exists $n \in \mathbb{N}$ with $n > s - 1$. But then $n + 1 > s$, and $n + 1$ is also a natural number. We have found a natural number strictly larger than s , contradicting that s is an upper bound of $\mathbb{N}$.

For the second form: given $\varepsilon > 0$, apply the first form to $x = 1/\varepsilon$ to get $n > 1/\varepsilon$; since both sides are positive, taking reciprocals gives $1/n < \varepsilon$ (taking reciprocals of positive numbers reverses the inequality). $\square$

Remark 1.10. The Archimedean property looks obvious (“the naturals grow without a ceiling”), but its proof *needs* the completeness axiom. It is the result that justifies the final step of almost every convergence proof: “we choose N such that $1/N < \varepsilon$ ”.

Theorem 1.11 (Density of $\mathbb{Q}$ in $\mathbb{R}$). *If $x, y \in \mathbb{R}$ and $x < y$, then there exists $q \in \mathbb{Q}$ with $x < q < y$.*

Proof. Idea: if the step $1/n$ is smaller than the gap $y - x$, some multiple of $1/n$ must fall inside the interval (x, y) .

Step 1. Since $y - x > 0$, the Archimedean property gives $n \in \mathbb{N}$ with $\frac{1}{n} < y - x$, that is, $ny - nx > 1$.

Step 2. Since the interval (nx, ny) has length greater than 1, it contains an integer. Formally: let m be the smallest integer strictly greater than nx (it exists by the Archimedean property and the well-ordering principle, applied with care). Then $m - 1 \leq nx < m$, from which

$$m \leq nx + 1 < nx + (ny - nx) = ny.$$

So $nx < m < ny$.

Step 3. Dividing by $n > 0$ (which preserves the inequalities): $x < \frac{m}{n} < y$, and $q = m/n \in \mathbb{Q}$ is the rational we were looking for. $\square$

Link with economics

Why should an economist care about the completeness axiom? Because it *guarantees that optimal values exist*. The value of a maximization problem is defined as

$$v = \sup \{u(x) : x \in B(p, w)\},$$

where $B(p, w)$ is the budget set. Without the completeness axiom, this v might simply fail to exist as a real number, and the whole theory of the consumer (indirect utility function, value function in dynamic programming) would have no foundation. In the same way, in general equilibrium theory, the existence of prices that “clear the market” is proved with theorems (fixed point, intermediate value) that rest, in the end, on the completeness of $\mathbb{R}$. The supremum versus maximum distinction is also economically important: the Weierstrass theorem (chapter on Compactness) will give conditions (continuity of u plus compactness of $B(p, w)$) under which the supremum *is attained*, that is, under which an optimal demand exists.

1.5 Exam-style exercises [P1]

Exercise 1.12. Write the formal negation of the following proposition and decide which of the two is true:

$$\exists M > 0 \quad \forall n \in \mathbb{N} : \frac{n^2 + 1}{n} \leq M.$$

Solution. The proposition says that the sequence $a_n = \frac{n^2 + 1}{n}$ is bounded above. Its negation, changing $\exists \rightarrow \forall$, $\forall \rightarrow \exists$ and negating the inequality:

$$\forall M > 0 \quad \exists n \in \mathbb{N} : \frac{n^2 + 1}{n} > M.$$

The negation is the true one. Indeed, $\frac{n^2+1}{n} = n + \frac{1}{n} > n$, and by the Archimedean property, for every $M > 0$ there exists $n \in \mathbb{N}$ with $n > M$; for that n , $a_n > n > M$. Therefore the sequence is not bounded and the original proposition is false. $\square$

Exercise 1.13. Let $A = \left\{ \frac{n}{n+1} : n \in \mathbb{N} \right\} = \left\{ \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots \right\}$. Prove that $\sup A = 1$ and that A has no maximum.

Solution. *Step 1: 1 is an upper bound.* For every n , $\frac{n}{n+1} < 1$ because $n < n+1$ and dividing by the positive number $n+1$ preserves the inequality.

Step 2: it is the smallest upper bound. We use the ε characterization (Proposition 1.8). Let $\varepsilon > 0$; we must find an element of A greater than $1 - \varepsilon$. We set up

$$\frac{n}{n+1} > 1 - \varepsilon \iff 1 - \frac{1}{n+1} > 1 - \varepsilon \iff \frac{1}{n+1} < \varepsilon.$$

By the Archimedean property there exists $n \in \mathbb{N}$ with $\frac{1}{n+1} \leq \frac{1}{n} < \varepsilon$, and for that n the element $\frac{n}{n+1} \in A$ exceeds $1 - \varepsilon$. Since 1 is an upper bound and satisfies the ε condition, $\sup A = 1$.

Step 3: there is no maximum. If A had a maximum, it would be $\sup A = 1$ (the maximum always coincides with the supremum when it exists). But $1 \notin A$: if $\frac{n}{n+1} = 1$ then $n = n+1$, which is absurd. Hence there is no maximum. $\square$

Exercise 1.14. Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above, and define $A + B = \{a + b : a \in A, b \in B\}$. Prove that

$$\sup(A + B) = \sup A + \sup B.$$

Solution. Write $\alpha = \sup A$ and $\beta = \sup B$ (they exist by the completeness axiom). We prove the two inequalities.

Step 1: $\sup(A + B) \leq \alpha + \beta$. Every element of $A + B$ has the form $a + b$ with $a \leq \alpha$ and $b \leq \beta$; adding, $a + b \leq \alpha + \beta$. So $\alpha + \beta$ is an upper bound of $A + B$, and since the supremum is the *smallest* upper bound, $\sup(A + B) \leq \alpha + \beta$. (Along the way, this shows that $A + B$ is bounded above, so its supremum exists.)

Step 2: $\sup(A + B) \geq \alpha + \beta$. Let $\varepsilon > 0$. By the ε characterization of the supremum applied to A and to B separately (with $\varepsilon/2$ each): there exist $a \in A$ with $a > \alpha - \frac{\varepsilon}{2}$ and $b \in B$ with $b > \beta - \frac{\varepsilon}{2}$. Adding:

$$a + b > \alpha + \beta - \varepsilon.$$

Since $a + b \in A + B$ and $\sup(A + B) \geq a + b$, we get $\sup(A + B) > \alpha + \beta - \varepsilon$ for every $\varepsilon > 0$. A number that exceeds $\alpha + \beta - \varepsilon$ for every $\varepsilon > 0$ must be $\geq \alpha + \beta$ (if it were smaller, taking ε equal to half the difference gives a contradiction). Hence $\sup(A + B) \geq \alpha + \beta$.

From both inequalities, $\sup(A + B) = \alpha + \beta$. *Exam note:* the trick of “splitting ε ” in halves is completely standard; learn it here because it comes back in the algebra of limits. $\square$

Exercise 1.15. Prove by contradiction that $\sqrt{2} \notin \mathbb{Q}$.

Solution. Assume, for contradiction, that $\sqrt{2} = \frac{p}{q}$ with $p, q \in \mathbb{N}$ and the fraction in lowest

terms (no common factors; we can always reduce it until this holds). Squaring:

$$2 = \frac{p^2}{q^2} \implies p^2 = 2q^2.$$

Then p^2 is even. We claim that p is even: if p were odd, $p = 2k + 1$, then $p^2 = 4k^2 + 4k + 1$ would be odd (contrapositive). Write $p = 2m$. Substituting: $4m^2 = 2q^2$, that is, $q^2 = 2m^2$. By the same argument, q is even. But then p and q share the factor 2, contradicting that the fraction was in lowest terms. The contradiction shows that $\sqrt{2}$ is not rational.

Moral: this is the canonical example that $\mathbb{Q}$ has “holes” and of a well structured proof by contradiction: (1) state the assumption, (2) derive valid consequences, (3) exhibit the contradiction, (4) conclude. $\square$

Exercise 1.16. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $A, B \subseteq \mathbb{R}$. Prove that $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$, and give an example where $f(A \cap B) \neq f(A) \cap f(B)$.

Solution. *Equality of preimages* (double inclusion, although here all the steps are equivalences):

$$x \in f^{-1}(A \cap B) \iff f(x) \in A \cap B \iff f(x) \in A \text{ and } f(x) \in B \iff x \in f^{-1}(A) \text{ and } x \in f^{-1}(B),$$

and this last statement is $x \in f^{-1}(A) \cap f^{-1}(B)$. Each step uses only the definition of preimage and of intersection.

Counterexample for the direct image: let $f(x) = x^2$, $A = \{-1\}$, $B = \{1\}$. Then $A \cap B = \emptyset$, so $f(A \cap B) = \emptyset$. But $f(A) = \{1\} = f(B)$, so $f(A) \cap f(B) = \{1\} \neq \emptyset$. The failure happens because f is not injective: two distinct points collapse to the same image. (In fact, if f is injective the equality does hold; a good extra exercise.) $\square$

Chapter summary

- Negation of quantifiers: $\forall \leftrightarrow \exists$ and the final predicate is negated; the negation of $A \Rightarrow B$ is “ A and $\neg B$ ”.
- In “ $\forall \varepsilon \exists N$ ”, N may depend on ε . The order of quantifiers matters.
- Four proof techniques: direct, contrapositive, contradiction, induction.
- Triangle inequality: $|x + y| \leq |x| + |y|$; reverse: $||x| - |y|| \leq |x - y|$; the add and subtract trick.
- $\sup A$ = smallest upper bound; ε characterization: $\forall \varepsilon > 0 \exists a \in A : a > \sup A - \varepsilon$.
- Completeness axiom $\Rightarrow$ Archimedean property $\Rightarrow$ density of $\mathbb{Q}$ in $\mathbb{R}$. It is the property that separates $\mathbb{R}$ from $\mathbb{Q}$ and the engine of the whole course.

Chapter 2

The vector space $\mathbb{R}^n$ and its norms

[P2]

Week 1 of the course. Bibliography: Alabiso and Weiss [1], Ch. 2; Pons [3], Ch. 8; Zorich I [5], Ch. 7; Zorich II [6], Ch. 9.

What is this used for in Microeconomics?

In Microeconomics, $\mathbb{R}^n$ is the commodity space: a consumption bundle is a vector $x \in \mathbb{R}_+^n$, a price vector is $p \in \mathbb{R}_{++}^n$, and the budget constraint is the dot product $p \cdot x \leq w$ (Part I: demand theory, MWG chs. 2–3). The Edgeworth box, the indifference curves, the budget set and the production set of the firm (Part I, MWG ch. 5) are geometric objects in $\mathbb{R}^n$, and the dot product comes back as revenue $p \cdot y$ and as cost $w \cdot z$. Two more spaces of the theory are subsets of $\mathbb{R}^n$: the *simplex* of lotteries over finitely many prizes, the ground of von Neumann–Morgenstern expected utility (MWG ch. 6), and the simplex of *mixed strategies* of a finite game (Part II), which is where Nash equilibrium lives. The section on vector spaces and the dot product are essential; norms other than the Euclidean one and the equivalence of norms are mathematical training (they matter more in the analysis course than in micro).

2.1 Motivation [P2]

In economics almost nothing lives on the real line: a consumption bundle with n goods is a point of $\mathbb{R}^n$, a price vector is a point of $\mathbb{R}^n$, an allocation in an economy with m agents lives in $\mathbb{R}^{nm}$. To do analysis (to speak of convergence, continuity, optimization) in these spaces we need two structures:

1. an *algebraic* structure that lets you add vectors and rescale them: the **vector space** structure;
2. a notion of *size* or *length* of a vector, which in turn induces a notion of *distance*: the **norm**.

This chapter builds both in the concrete case of $\mathbb{R}^n$. Everything we do here is the model to imitate when, later on, we work with abstract spaces (metric, normed, Banach, Hilbert): $\mathbb{R}^n$ is the laboratory where the ideas can be seen by hand.

A central lesson of the chapter: in $\mathbb{R}^n$ there are *many* reasonable norms, but (we announce the result now) **all of them are equivalent**: they define the same notion of convergence. This convenience belongs to finite dimension only and will be lost in function spaces.

2.2 Vector spaces [P1]

Definition 2.1 (Vector space). A **real vector space** is a set V (whose elements we call *vectors*) with two operations,

$$+ : V \times V \rightarrow V \quad (\text{vector addition}), \quad \cdot : \mathbb{R} \times V \rightarrow V \quad (\text{scalar multiplication}),$$

that satisfy the following eight axioms, for all $u, v, w \in V$ and $\alpha, \beta \in \mathbb{R}$:

1. **Associativity of addition:** $(u + v) + w = u + (v + w)$.
2. **Commutativity of addition:** $u + v = v + u$.
3. **Zero vector:** there exists $0 \in V$ such that $v + 0 = v$ for every v . (Do not confuse the vector 0 with the scalar 0 ; the context decides.)
4. **Additive inverse:** for each v there exists $-v \in V$ with $v + (-v) = 0$.
5. **Compatibility:** $\alpha(\beta v) = (\alpha\beta)v$.
6. **Scalar identity:** $1 \cdot v = v$.
7. **Distributivity over vector addition:** $\alpha(u + v) = \alpha u + \alpha v$.
8. **Distributivity over scalar addition:** $(\alpha + \beta)v = \alpha v + \beta v$.

Intuition

The eight axioms say, as a block, a single thing: in V you can form *linear combinations* $\alpha u + \beta v$ and these are handled with the usual rules of school algebra. Axioms 1 to 4 say that addition behaves well (commutative group); axioms 5 to 8 say that rescaling interacts well with adding. In practice, when you check that something is a vector space, the real work is almost always to check that the operations *do not leave the set* (closure), because the eight properties are usually inherited from a larger space.

Example 2.2. The leading example:

$$\mathbb{R}^n = \{x = (x_1, x_2, \dots, x_n) : x_i \in \mathbb{R} \text{ for } i = 1, \dots, n\},$$

with the operations defined *coordinate by coordinate*:

$$x + y = (x_1 + y_1, \dots, x_n + y_n), \quad \alpha x = (\alpha x_1, \dots, \alpha x_n).$$

The eight axioms are checked coordinate by coordinate using that $\mathbb{R}$ is a field: for example, commutativity $x + y = y + x$ holds because in each coordinate $x_i + y_i = y_i + x_i$ (commutativity in $\mathbb{R}$). The zero vector is $0 = (0, \dots, 0)$ and the additive inverse of x is $-x = (-x_1, \dots, -x_n)$.

Example 2.3. Other vector spaces that will come back in the course (check closure mentally in each case):

1. $\mathcal{P}_k$, the polynomials of degree $\leq k$: the sum of two polynomials of degree $\leq k$ has degree $\leq k$, and rescaling does not raise the degree.
2. $C[a, b]$, the continuous functions $f : [a, b] \rightarrow \mathbb{R}$: the sum of continuous functions is continuous and a multiple of a continuous function is continuous (we prove this in the chapter on sequences and continuity). This space is *infinite dimensional* and will be the star example of the second half of the course.
3. The set of bounded real sequences, denoted ℓ^∞ .

On the other hand, $\{x \in \mathbb{R}^2 : x_1 \geq 0\}$ is **not** a vector space: it is not closed under multiplication by the scalar -1 .

2.2.1 The dot product in $\mathbb{R}^n$

Definition 2.4 (Dot product). For $x, y \in \mathbb{R}^n$, the **dot product** is

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n.$$

Immediate properties (checked directly from the definition, using commutativity and distributivity in $\mathbb{R}$):

1. *Symmetry*: $\langle x, y \rangle = \langle y, x \rangle$.
2. *Bilinearity*: $\langle \alpha x + \beta z, y \rangle = \alpha \langle x, y \rangle + \beta \langle z, y \rangle$ (linear in each argument).
3. *Positivity*: $\langle x, x \rangle = \sum_i x_i^2 \geq 0$, with equality if and only if $x = 0$ (a sum of squares is zero only if each term is zero).

Link with economics

The dot product is everywhere in economics: if $p \in \mathbb{R}^n$ is a price vector and $x \in \mathbb{R}^n$ a bundle, expenditure is exactly $\langle p, x \rangle = \sum_i p_i x_i$. The budget constraint $\langle p, x \rangle \leq w$ defines a half space; its boundary $\langle p, x \rangle = w$ is a hyperplane whose “normal” is p . All the geometry of consumer theory (separating hyperplanes, duality) is the geometry of the dot product.

2.3 Norms in $\mathbb{R}^n$ [P3]

Definition 2.5 (Norm). A **norm** on a vector space V is a function $\|\cdot\| : V \rightarrow \mathbb{R}$ that satisfies, for all $x, y \in V$ and $\alpha \in \mathbb{R}$:

- (N1) **Positive definiteness**: $\|x\| \geq 0$, and $\|x\| = 0$ if and only if $x = 0$.
- (N2) **Absolute homogeneity**: $\|\alpha x\| = |\alpha| \|x\|$.
- (N3) **Triangle inequality**: $\|x + y\| \leq \|x\| + \|y\|$.

The pair $(V, \|\cdot\|)$ is called a **normed space**.

Intuition

A norm is a way to measure the *length* of a vector, and each axiom captures something that any reasonable notion of length must satisfy: (N1) lengths are not negative and only the zero vector measures zero; (N2) stretching a vector to twice its size doubles its length, and reversing its direction does not change it (this is why $|\alpha|$ appears and not α); (N3) going “straight” from 0 to $x + y$ cannot be longer than going first to x and then translating by y : the straight line is the shortest path. Note that (N1) to (N3) are exactly the properties of the absolute value in $\mathbb{R}$: the norm is the generalization of the absolute value to dimension n .

The three norms you must master:

Definition 2.6. For $x = (x_1, \dots, x_n) \in \mathbb{R}^n$:

$$\begin{aligned} \|x\|_1 &= \sum_{i=1}^n |x_i| && \textbf{1-norm} \text{ (taxicab or Manhattan norm)}, \\ \|x\|_2 &= \left(\sum_{i=1}^n x_i^2 \right)^{1/2} = \sqrt{\langle x, x \rangle} && \textbf{Euclidean norm}, \\ \|x\|_\infty &= \max_{1 \leq i \leq n} |x_i| && \textbf{infinity norm} \text{ (sup norm or max norm)}. \end{aligned}$$

More generally, for $1 \leq p < \infty$, the **p -norm** is $\|x\|_p = (\sum_i |x_i|^p)^{1/p}$.

2.3.1 Checking that they are norms

For $\|\cdot\|_1$ and $\|\cdot\|_\infty$ the check is direct; we do it first, and then we attack the Euclidean norm, which needs Cauchy–Schwarz.

Proposition 2.7. $\|\cdot\|_1$ and $\|\cdot\|_\infty$ are norms on $\mathbb{R}^n$.

Proof. The 1-norm. (N1): a sum of absolute values is ≥ 0 ; it is 0 if and only if each $|x_i| = 0$ (a sum of nonnegative numbers is zero only if all are zero), that is, if and only if $x = 0$. (N2): $\|\alpha x\|_1 = \sum_i |\alpha x_i| = \sum_i |\alpha| |x_i| = |\alpha| \sum_i |x_i| = |\alpha| \|x\|_1$, using $|ab| = |a| |b|$ in $\mathbb{R}$. (N3): by the triangle inequality in $\mathbb{R}$ applied in each coordinate, $|x_i + y_i| \leq |x_i| + |y_i|$; summing over i :

$$\|x + y\|_1 = \sum_i |x_i + y_i| \leq \sum_i (|x_i| + |y_i|) = \|x\|_1 + \|y\|_1.$$

The infinity norm. (N1): a maximum of numbers ≥ 0 is ≥ 0 , and it is 0 if and only if all the $|x_i|$ are 0. (N2): $\max_i |\alpha x_i| = \max_i |\alpha| |x_i| = |\alpha| \max_i |x_i|$ (pulling a nonnegative constant out of a maximum is legitimate). (N3): for each fixed index j ,

$$|x_j + y_j| \leq |x_j| + |y_j| \leq \max_i |x_i| + \max_i |y_i| = \|x\|_\infty + \|y\|_\infty,$$

where the second step bounds each term by the corresponding maximum. Since this holds for *every* j , in particular it holds for the j that attains the maximum of $|x_j + y_j|$, and therefore

$$\|x + y\|_\infty \leq \|x\|_\infty + \|y\|_\infty. \quad \square$$

For the Euclidean norm, the hard property is (N3), and the key is the most famous inequality in analysis:

Theorem 2.8 (Cauchy–Schwarz inequality). *For all $x, y \in \mathbb{R}^n$:*

$$|\langle x, y \rangle| \leq \|x\|_2 \|y\|_2,$$

with equality if and only if x and y are linearly dependent (one is a multiple of the other).

Proof. The proof uses the famous “discriminant trick”; understand it well because it is an exam classic.

Step 1: setup. If $y = 0$, both sides are 0 and there is nothing to prove. Assume $y \neq 0$. Consider the real function

$$q(t) = \|x - ty\|_2^2, \quad t \in \mathbb{R}.$$

Being a squared norm, $q(t) \geq 0$ for every t : this single observation is the heart of the proof.

Step 2: expand. Using bilinearity and symmetry of the dot product (exactly as you expand a binomial):

$$\begin{aligned} q(t) &= \langle x - ty, x - ty \rangle = \langle x, x \rangle - t \langle x, y \rangle - t \langle y, x \rangle + t^2 \langle y, y \rangle \\ &= \|y\|_2^2 t^2 - 2 \langle x, y \rangle t + \|x\|_2^2. \end{aligned}$$

This is a quadratic polynomial in t with leading coefficient $\|y\|_2^2 > 0$ (a parabola that opens upwards).

Step 3: discriminant. A parabola that opens upwards and never goes below zero cannot have two distinct real roots; by the quadratic formula, this means that its discriminant is ≤ 0 :

$$\Delta = (2 \langle x, y \rangle)^2 - 4 \|y\|_2^2 \|x\|_2^2 \leq 0.$$

Rearranging: $\langle x, y \rangle^2 \leq \|x\|_2^2 \|y\|_2^2$, and taking the square root (increasing on the nonnegative numbers): $|\langle x, y \rangle| \leq \|x\|_2 \|y\|_2$.

Step 4: equality case. There is equality if and only if $\Delta = 0$, that is, if the parabola has one (double) root t^* : then $q(t^*) = \|x - t^*y\|_2^2 = 0$, and by (N1), $x = t^*y$: the vectors are linearly dependent. Conversely, if $x = \lambda y$, both sides equal $|\lambda| \|y\|_2^2$. $\square$

Theorem 2.9 (Minkowski inequality; $\|\cdot\|_2$ is a norm). *For all $x, y \in \mathbb{R}^n$: $\|x + y\|_2 \leq \|x\|_2 + \|y\|_2$. Consequently, $\|\cdot\|_2$ is a norm on $\mathbb{R}^n$.*

Proof. (N1) and (N2) for $\|\cdot\|_2$ are direct: $\sqrt{\sum x_i^2} \geq 0$ with equality only at $x = 0$, and $\|\alpha x\|_2 = \sqrt{\alpha^2 \sum x_i^2} = |\alpha| \|x\|_2$ (note: $\sqrt{\alpha^2} = |\alpha|$, not α). For (N3), we expand the square and use

Cauchy–Schwarz on the cross term:

$$\begin{aligned}
 \|x + y\|_2^2 &= \langle x + y, x + y \rangle = \|x\|_2^2 + 2\langle x, y \rangle + \|y\|_2^2 \\
 &\leq \|x\|_2^2 + 2|\langle x, y \rangle| + \|y\|_2^2 \\
 &\leq \|x\|_2^2 + 2\|x\|_2\|y\|_2 + \|y\|_2^2 \quad (\text{Cauchy–Schwarz}) \\
 &= (\|x\|_2 + \|y\|_2)^2.
 \end{aligned}$$

Taking square roots at both ends (legitimate: the root is increasing and both sides are ≥ 0) gives the Minkowski inequality. $\square$

Exam strategy

Classic exam pattern: “prove that such a function is a norm”. Routine: check (N1), (N2), (N3) *in that order and separately*, stating at each step which property of $\mathbb{R}$ (or which known inequality) you use. The property that almost always concentrates the difficulty is (N3); if the expression involves squares or products, suspect that you will need Cauchy–Schwarz, as in Minkowski. Another classic: asking for the proof of Cauchy–Schwarz itself; the discriminant trick must come out of memory.

2.3.2 Balls and their geometry

Definition 2.10 (Open ball). Given a norm $\|\cdot\|$ on $\mathbb{R}^n$, the **open ball** with center a and radius $r > 0$ is

$$B(a, r) = \{x \in \mathbb{R}^n : \|x - a\| < r\}.$$

The **unit ball** is $B(0, 1)$.

The *shape* of the unit ball depends on the norm. In $\mathbb{R}^2$:

- With $\|\cdot\|_2$: the usual round disk $\{x_1^2 + x_2^2 < 1\}$.
- With $\|\cdot\|_1$: the *diamond* (a square rotated by 45 degrees) with vertices $(\pm 1, 0)$ and $(0, \pm 1)$, since $|x_1| + |x_2| < 1$.
- With $\|\cdot\|_\infty$: the *square* of side 2 aligned with the axes, $\max\{|x_1|, |x_2|\} < 1$, that is, $|x_1| < 1$ and $|x_2| < 1$.

Picture the three balls nested: the diamond inside the disk, the disk inside the square. That geometric nesting *is* the equivalence of norms that we prove next.

2.4 Equivalence of norms [P4]

Definition 2.11 (Equivalent norms). Two norms $\|\cdot\|_a$ and $\|\cdot\|_b$ on a vector space V are **equivalent** if there exist constants $c, C > 0$ such that

$$c\|x\|_a \leq \|x\|_b \leq C\|x\|_a \quad \text{for every } x \in V.$$

Intuition

That two norms are equivalent means that they measure lengths in a “comparable” way: neither can become arbitrarily large while the other stays small. The practical consequence (proved in the next chapter) is that *they define the same convergent sequences, the same open sets and the same continuous functions*: for all purposes of analysis, they are interchangeable. This is why in $\mathbb{R}^n$ you can choose in each proof the most convenient norm (almost always $\|\cdot\|_\infty$, because it works coordinate by coordinate).

Remark 2.12. The relation “being equivalent” is indeed an equivalence relation: reflexive ($c = C = 1$), symmetric (if $c\|x\|_a \leq \|x\|_b \leq C\|x\|_a$, then $\frac{1}{C}\|x\|_b \leq \|x\|_a \leq \frac{1}{c}\|x\|_b$) and transitive (by chaining constants).

Theorem 2.13. *For every $x \in \mathbb{R}^n$ the following hold:*

$$\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1 \leq n \|x\|_\infty, \quad \|x\|_2 \leq \sqrt{n} \|x\|_\infty, \quad \|x\|_1 \leq \sqrt{n} \|x\|_2.$$

In particular, the three norms are equivalent to one another.

Proof. We prove each inequality separately, with all the steps.

(a) $\|x\|_\infty \leq \|x\|_2$. Let j be the index where the maximum is attained: $\|x\|_\infty = |x_j|$. Then

$$\|x\|_\infty^2 = x_j^2 \leq x_1^2 + x_2^2 + \cdots + x_n^2 = \|x\|_2^2,$$

because adding the other squares (all ≥ 0) can only increase the sum. Taking square roots: $\|x\|_\infty \leq \|x\|_2$.

(b) $\|x\|_2 \leq \|x\|_1$. We compare squares. Expanding the square of the sum:

$$\|x\|_1^2 = \left(\sum_i |x_i|\right)^2 = \sum_i |x_i|^2 + \sum_{i \neq j} |x_i| |x_j| \geq \sum_i x_i^2 = \|x\|_2^2,$$

since the cross terms $|x_i| |x_j|$ are all ≥ 0 . Take the square root and you are done.

(c) $\|x\|_1 \leq n \|x\|_\infty$. Each term is bounded by the maximum:

$$\|x\|_1 = \sum_{i=1}^n |x_i| \leq \sum_{i=1}^n \|x\|_\infty = n \|x\|_\infty.$$

(d) $\|x\|_2 \leq \sqrt{n} \|x\|_\infty$. Same as (c) but with squares:

$$\|x\|_2^2 = \sum_{i=1}^n x_i^2 \leq \sum_{i=1}^n \|x\|_\infty^2 = n \|x\|_\infty^2,$$

and taking the root: $\|x\|_2 \leq \sqrt{n} \|x\|_\infty$.

(e) $\|x\|_1 \leq \sqrt{n} \|x\|_2$. This one is sharper and uses Cauchy–Schwarz with a clever auxiliary vector: let $u = (1, 1, \dots, 1) \in \mathbb{R}^n$ and $v = (|x_1|, \dots, |x_n|)$. Then

$$\|x\|_1 = \sum_i 1 \cdot |x_i| = \langle u, v \rangle \leq \|u\|_2 \|v\|_2 = \sqrt{n} \cdot \|x\|_2,$$

where we used $\|u\|_2 = \sqrt{1 + \cdots + 1} = \sqrt{n}$ and $\|v\|_2 = \|x\|_2$ (the squares remove the absolute values).

Putting things together: each pair of norms is trapped between constant multiples of the other; for example, from (a) and (d), $\|x\|_\infty \leq \|x\|_2 \leq \sqrt{n} \|x\|_\infty$, with constants $c = 1$, $C = \sqrt{n}$ that *do not depend on x* (only on the dimension n , which is fixed). This uniformity of the constants is exactly what Definition 2.11 requires. $\square$

Careful (common mistake)

Two frequent mistakes: (i) writing the inequalities with the constants “the wrong way round” (always check with a concrete vector, for example $x = (1, 1, \dots, 1)$, for which $\|x\|_\infty = 1$, $\|x\|_2 = \sqrt{n}$, $\|x\|_1 = n$: this lets you remember which norm is the big one and which constants are the sharp ones); (ii) believing that equivalence allows constants that depend on x ; the definition requires *uniform* constants, valid for all vectors at once. An “inequality” with a constant that depends on x is empty: any pair of norms satisfies it.

The next result is one of the structural theorems of finite dimension. We state it now, but its proof needs compactness and the Weierstrass theorem, so we give it in the chapter on compactness.

Theorem 2.14 (Equivalence of all norms in finite dimension). *On $\mathbb{R}^n$ (and on any finite dimensional vector space) all norms are equivalent to one another.*

Remark 2.15. The proof is postponed to the chapter on compactness (the strategy: compare any norm with $\|\cdot\|_\infty$ using that the unit sphere is compact and that every norm is continuous). What you should keep from now on: (i) the result *is false in infinite dimension*: in $C[0, 1]$, the norms $\|f\|_\infty = \max_t |f(t)|$ and $\|f\|_1 = \int_0^1 |f|$ are *not* equivalent, and that failure of equivalence is the source of almost all the subtlety of functional analysis; (ii) thanks to this theorem, sentences like “ $x_k \rightarrow x$ in $\mathbb{R}^n$ ” do not need to specify the norm.

2.5 The distance induced by a norm [P3]

Definition 2.16. Every norm $\|\cdot\|$ on V induces a **distance** (or metric)

$$d(x, y) = \|x - y\|.$$

Proposition 2.17. *The induced distance satisfies, for all $x, y, z \in V$:*

1. $d(x, y) \geq 0$, and $d(x, y) = 0 \iff x = y$;
2. $d(x, y) = d(y, x)$ (symmetry);
3. $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality).

Proof. (1) This is (N1) applied to the vector $x - y$: the norm is ≥ 0 and vanishes only if $x - y = 0$, that is, $x = y$. (2) $d(y, x) = \|y - x\| = \|(-1)(x - y)\| = |-1| \|x - y\| = d(x, y)$, using (N2) with $\alpha = -1$. (3) We add and subtract y and apply (N3):

$$d(x, z) = \|x - z\| = \|(x - y) + (y - z)\| \leq \|x - y\| + \|y - z\| = d(x, y) + d(y, z). \quad \square$$

These three properties will be, in the next block of the course, the *definition* of metric space: we will abstract the distance by forgetting that it came from a norm. It is worth noting the hierarchy of structures that this chapter leaves installed:

$$\text{inner product} \implies \text{norm} \implies \text{distance},$$

where each arrow means “induces”: the dot product induces $\|x\|_2 = \sqrt{\langle x, x \rangle}$, and every norm induces $d(x, y) = \|x - y\|$. The arrows do *not* reverse in general: there are norms that do not come from an inner product ($\|\cdot\|_1$ and $\|\cdot\|_\infty$, as we will see with the parallelogram law in the chapter on Hilbert spaces) and distances that do not come from a norm (the discrete metric, next block).

Link with economics

Norms are the natural way to measure the “size” of multidimensional economic objects, and choosing the norm means choosing what you want to emphasize. Examples: (i) if $x - y$ is the difference between two consumption bundles, $\|x - y\|_1$ adds up the discrepancies good by good (relevant if goods are comparable unit by unit), while $\|x - y\|_\infty$ measures the worst discrepancy in a single good (relevant for uniform guarantees); (ii) an aggregate shock $\varepsilon \in \mathbb{R}^n$ on n sectors can be measured in the 2-norm (total energy of the shock) or in the infinity norm (the sector hit hardest); (iii) in econometrics, ordinary least squares minimizes a 2-norm (hence “least squares”), LAD regression minimizes a 1-norm, and LASSO regularization penalizes with $\|\beta\|_1$ while ridge uses $\|\beta\|_2^2$: the geometry of the balls (a diamond with corners versus a round disk) explains why LASSO produces coefficients that are exactly zero and ridge does not. The equivalence of norms in $\mathbb{R}^n$ guarantees that *qualitative* statements (“the estimator converges”, “demand is continuous”) do not depend on the chosen norm.

2.6 Exam-style exercises [P2]

Exercise 2.18. Prove directly from the vector space axioms that in any vector space V : (a) the zero vector is unique; (b) $0 \cdot v = 0$ for every $v \in V$ (the zero scalar times a vector gives the zero vector); (c) $(-1)v = -v$.

Solution. (a) Assume that 0 and $0'$ are both zero vectors (each one satisfies axiom 3). Then

$$0 = 0 + 0' = 0',$$

where the first equality uses that $0'$ is neutral (adding it to 0 leaves it unchanged) and the second uses that 0 is neutral together with commutativity. Hence the neutral element is unique.

(b) Standard trick: $0 + 0 = 0$ in $\mathbb{R}$, so by axiom 8 (distributivity over scalars):

$$0 \cdot v = (0 + 0)v = 0 \cdot v + 0 \cdot v.$$

Adding to both sides the additive inverse $-(0 \cdot v)$ (axiom 4) and using associativity:

$$0 = 0 \cdot v + (-(0 \cdot v)) = (0 \cdot v + 0 \cdot v) + (-(0 \cdot v)) = 0 \cdot v + 0 = 0 \cdot v.$$

(c) We must show that $(-1)v$ works as the additive inverse of v ; since that inverse is unique (same argument as in (a)), this gives $(-1)v = -v$. We compute:

$$v + (-1)v = 1 \cdot v + (-1)v = (1 + (-1))v = 0 \cdot v = 0,$$

using axiom 6, axiom 8 and part (b). Done. $\square$

Exercise 2.19. Decide whether $f(x_1, x_2) = |x_1| + 3|x_2|$ and $g(x_1, x_2) = |x_1 + x_2|$ are norms on $\mathbb{R}^2$. Justify each axiom or give a counterexample.

Solution. *f is a norm.* (N1): $f(x) \geq 0$ is clear; $f(x) = 0$ forces $|x_1| = 0$ and $3|x_2| = 0$ (a sum of nonnegative numbers), that is, $x = 0$. (N2): $f(\alpha x) = |\alpha x_1| + 3|\alpha x_2| = |\alpha|(|x_1| + 3|x_2|) = |\alpha|f(x)$. (N3): by the triangle inequality in $\mathbb{R}$ in each term,

$$f(x + y) = |x_1 + y_1| + 3|x_2 + y_2| \leq |x_1| + |y_1| + 3|x_2| + 3|y_2| = f(x) + f(y).$$

(It is a “weighted 1-norm”; any choice of positive weights gives a norm.)

g is not a norm. It fails (N1): the vector $x = (1, -1) \neq 0$ satisfies $g(x) = |1 - 1| = 0$. This counterexample is enough; g is only a *seminorm*: it satisfies (N2) and (N3) but vanishes away from the origin. $\square$

Exercise 2.20. Prove that for all $x, y \in \mathbb{R}^n$ the *parallelogram law* holds for the Euclidean norm:

$$\|x + y\|_2^2 + \|x - y\|_2^2 = 2\|x\|_2^2 + 2\|y\|_2^2,$$

and check with $x = (1, 0)$, $y = (0, 1)$ that the norm $\|\cdot\|_1$ on $\mathbb{R}^2$ does *not* satisfy it.

Solution. We expand both squares with the bilinearity of the dot product:

$$\begin{aligned}\|x + y\|_2^2 &= \langle x + y, x + y \rangle = \|x\|_2^2 + 2\langle x, y \rangle + \|y\|_2^2, \\ \|x - y\|_2^2 &= \langle x - y, x - y \rangle = \|x\|_2^2 - 2\langle x, y \rangle + \|y\|_2^2.\end{aligned}$$

Adding them, the cross terms $\pm 2\langle x, y \rangle$ cancel and we are left with $2\|x\|_2^2 + 2\|y\|_2^2$, as wanted.

For $\|\cdot\|_1$ with $x = (1, 0)$, $y = (0, 1)$: $\|x + y\|_1 = \|(1, 1)\|_1 = 2$ and $\|x - y\|_1 = \|(1, -1)\|_1 = 2$, so the left hand side is $4 + 4 = 8$; but the right hand side is $2 \cdot 1 + 2 \cdot 1 = 4 \neq 8$. The identity fails.

Moral (we come back to this in the chapter on Hilbert spaces): the parallelogram law characterizes the norms that come from an inner product; $\|\cdot\|_1$ comes from none. $\square$

Exercise 2.21. Using Cauchy–Schwarz, prove that for every $x \in \mathbb{R}^n$:

$$\left(\frac{1}{n} \sum_{i=1}^n x_i\right)^2 \leq \frac{1}{n} \sum_{i=1}^n x_i^2,$$

that is, the square of the average is less than or equal to the average of the squares. When is there equality? Interpret this statistically.

Solution. We apply Cauchy–Schwarz to the vectors $u = (1, \dots, 1)$ and $x = (x_1, \dots, x_n)$:

$$|\langle u, x \rangle| = \left| \sum_{i=1}^n x_i \right| \leq \|u\|_2 \|x\|_2 = \sqrt{n} \left(\sum_i x_i^2 \right)^{1/2}.$$

Squaring (both sides are ≥ 0): $(\sum_i x_i)^2 \leq n \sum_i x_i^2$. Dividing by n^2 :

$$\left(\frac{1}{n} \sum_i x_i \right)^2 \leq \frac{1}{n} \sum_i x_i^2.$$

Equality in Cauchy–Schwarz happens exactly when x is a multiple of u , that is, when all the coordinates x_i are equal.

Interpretation: if the x_i are data, the inequality says $\bar{x}^2 \leq \overline{x^2}$, which is exactly $\text{Var}(x) = \overline{x^2} - \bar{x}^2 \geq 0$: the sample variance is nonnegative, and it is zero only when all the data coincide. In this sense, Cauchy–Schwarz is the mathematical reason why variances cannot be negative (and why correlations live in $[-1, 1]$). $\square$

Exercise 2.22. Let $\|\cdot\|$ be any norm on $\mathbb{R}^n$. Prove the reverse triangle inequality:

$$|\|x\| - \|y\|| \leq \|x - y\| \quad \text{for all } x, y \in \mathbb{R}^n,$$

and explain why this implies that the function $x \mapsto \|x\|$ is continuous (a preview of the next chapter: it will be enough to see that it is “Lipschitz”).

Solution. We add and subtract: $x = (x - y) + y$, and apply (N3):

$$\|x\| \leq \|x - y\| + \|y\| \implies \|x\| - \|y\| \leq \|x - y\|.$$

Swapping the roles of x and y , and using that $\|y - x\| = \|(-1)(x - y)\| = \|x - y\|$ by (N2):

$$\|y\| - \|x\| \leq \|x - y\|.$$

Since $\|x\| - \|y\|$ and its negative are both bounded by $\|x - y\|$, we conclude $|\|x\| - \|y\|| \leq \|x - y\|$.

Continuity: the inequality says that the function $f(x) = \|x\|$ changes at most as much as its argument changes ($|f(x) - f(y)| \leq 1 \cdot \|x - y\|$: it is Lipschitz with constant 1). In ε - δ terms: given $\varepsilon > 0$, it is enough to take $\delta = \varepsilon$; if $\|x - y\| < \delta$ then $|f(x) - f(y)| \leq \|x - y\| < \varepsilon$. This exact argument will be formalized in the next chapter and is an exam favorite. $\square$

Chapter summary

- Vector space: a set closed under linear combinations, with 8 axioms; $\mathbb{R}^n$, polynomials and $C[a, b]$ are the key examples.
- Norm: (N1) positive definite, (N2) absolutely homogeneous, (N3) triangle inequality. Every norm induces the distance $d(x, y) = \|x - y\|$.

- Three norms on $\mathbb{R}^n$: $\|x\|_1 = \sum |x_i|$ (diamond), $\|x\|_2 = \sqrt{\sum x_i^2}$ (disk), $\|x\|_\infty = \max |x_i|$ (square).
- Cauchy–Schwarz: $|\langle x, y \rangle| \leq \|x\|_2 \|y\|_2$ (discriminant trick); from it comes Minkowski, that is, (N3) for the Euclidean norm.
- Key inequalities: $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1 \leq n \|x\|_\infty$ and $\|x\|_2 \leq \sqrt{n} \|x\|_\infty$, with constants uniform in x : the three norms are equivalent.
- Theorem (proof postponed to compactness): in finite dimension *all* norms are equivalent. In infinite dimension this is false.
- Hierarchy: inner product $\Rightarrow$ norm $\Rightarrow$ distance; the arrows do not reverse.

Chapter 3

Sequences, convergence and continuity in $\mathbb{R}^n$ [P1]

Week 2 of the course. Bibliography: Pons [3], Chs. 2 and 8; Zorich I [5], Chs. 3 and 7.

What is this used for in Microeconomics?

This is a top priority chapter for micro. Continuity of preferences is the assumption that lets you represent them by a continuous utility function and that guarantees that consumer demand exists (Part I, MWG Chs. 3–4). The sequential criterion is the tool that Mas-Colell, Whinston and Green use to prove continuity of the indirect utility function, the expenditure function, and the Marshallian and Hicksian demands, and also continuity of the profit and cost functions in production theory (MWG Ch. 5): you take a sequence of prices $p_k \rightarrow p$, extract convergent subsequences and pass to the limit. That same pattern (sequences plus passing to the limit) supports the existence proofs in general equilibrium (Part IV) and the limit arguments in game theory (Part II). Master this chapter and the MWG proofs will stop looking like magic.

3.1 Motivation [P1]

The basic question of analysis is: what does it mean to *get close*? A sequence of approximations $x_1, x_2, x_3, \dots$ (iterations of an algorithm, the capital path of an economy, estimators with growing samples) “gets close” to a limit L ; a function is “continuous” if nearby inputs produce nearby outputs. This chapter makes both ideas precise in $\mathbb{R}^n$ with the ε – N definition (for sequences) and the ε – δ definition (for functions), and it proves the bridge between them: the **sequential criterion for continuity**, which in practice is the most used tool to prove (and to refute!) continuity.

Mastering this chapter pays off: the same definitions, word for word, will be reused in general metric spaces by replacing $\|x - y\|$ with $d(x, y)$. If you understand convergence in $\mathbb{R}^n$, you understand 80% of convergence in metric spaces.

3.2 Sequences and convergence [P1]

Definition 3.1 (Sequence). A **sequence** in $\mathbb{R}^n$ is a function $x : \mathbb{N} \rightarrow \mathbb{R}^n$; we write x_k instead of $x(k)$ and denote the sequence by $(x_k)_{k \in \mathbb{N}}$ or simply (x_k) . Each $x_k \in \mathbb{R}^n$ is a **term** of the sequence.

Definition 3.2 (Convergence). The sequence (x_k) in $\mathbb{R}^n$ **converges** to the point $L \in \mathbb{R}^n$ (written $x_k \rightarrow L$ or $\lim_{k \rightarrow \infty} x_k = L$) if:

$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N} \quad \forall k \geq N : \quad \|x_k - L\| < \varepsilon.$$

If no such L exists, the sequence **diverges**. (By the equivalence of norms from the previous chapter, it does not matter which norm we use: if the condition holds for one, it holds for all of them, only with a different N ; we make this explicit below.)

Intuition

Read it as a challenge between two players. The *skeptic* proposes a tolerance $\varepsilon > 0$, as small as they want (“I do not believe the sequence gets closer to L than one millionth”). You answer with a time N (which may depend on ε : tighter tolerances require waiting longer) after which *all* terms are at distance less than ε from L . Convergence means that you win the challenge *always*, for every ε .

A picture in words: around L draw the ball $B(L, \varepsilon)$. The sequence converges to L if, for each radius ε , the ball traps *the whole tail* of the sequence: at most finitely many terms $(x_1, \dots, x_{N-1})$ stay outside the ball. As ε shrinks, the ball shrinks, and the trapped tail starts later, but it always ends up trapped.

Example 3.3. Let us prove in full detail that $x_k = \frac{1}{k} \rightarrow 0$ in $\mathbb{R}$.

Scratch work: we want $\left| \frac{1}{k} - 0 \right| = \frac{1}{k} < \varepsilon$, which is the same as $k > \frac{1}{\varepsilon}$. This tells us which N to choose.

Clean proof: let $\varepsilon > 0$ be arbitrary. By the Archimedean property there is $N \in \mathbb{N}$ with $N > \frac{1}{\varepsilon}$, that is, $\frac{1}{N} < \varepsilon$. Then, for every $k \geq N$:

$$|x_k - 0| = \frac{1}{k} \leq \frac{1}{N} < \varepsilon,$$

where the first inequality uses that $k \geq N$ implies $\frac{1}{k} \leq \frac{1}{N}$ (inverting positive numbers reverses the order). Since ε was arbitrary, $x_k \rightarrow 0$. $\square$

Note the two step format: first the *scratch work*, where you solve for k in terms of ε ; then the *write up*, which starts with “let $\varepsilon > 0$ ” and presents N as if it fell from the sky. In the exam you hand in only the write up, but always do the scratch work first.

Careful (common mistake)

The order of the quantifiers is sacred. “ $\exists N \forall \varepsilon$ ” (one single N that works for all ε) would mean that $x_k = L$ for all $k \geq N$: much stronger than converging. The N *depends* on ε , and you are allowed (it is standard) to write $N(\varepsilon)$. Second common mistake: checking the

condition only for *some* ε (for instance $\varepsilon = 0.1$); it must hold for *all* of them. Third: the negation of convergence is NOT “the sequence moves away”, it is the one in Example 1.1 of the preliminaries chapter: $\exists \varepsilon > 0$ such that infinitely many terms stay outside $B(L, \varepsilon)$.

3.2.1 First properties

Theorem 3.4 (Uniqueness of the limit). *If $x_k \rightarrow L$ and $x_k \rightarrow M$, then $L = M$.*

Proof. By contradiction: suppose $L \neq M$, so that $\|L - M\| > 0$. Take $\varepsilon = \frac{\|L - M\|}{2} > 0$: half the distance between the two candidate limits (balls of this radius around L and M do not touch each other; that is the geometric idea).

By convergence to L with this ε : there is N_1 such that $\|x_k - L\| < \varepsilon$ for every $k \geq N_1$. By convergence to M : there is N_2 such that $\|x_k - M\| < \varepsilon$ for every $k \geq N_2$. Let $k = \max\{N_1, N_2\}$: for this k both inequalities hold at the same time.

Now the add and subtract trick plus the triangle inequality:

$$\|L - M\| = \|(L - x_k) + (x_k - M)\| \leq \|x_k - L\| + \|x_k - M\| < \varepsilon + \varepsilon = \|L - M\|.$$

We obtained $\|L - M\| < \|L - M\|$, which is absurd. Therefore $L = M$. □

Definition 3.5 (Bounded sequence). (x_k) is **bounded** if there is $M > 0$ such that $\|x_k\| \leq M$ for every $k \in \mathbb{N}$ (all terms fit inside a ball of radius M centered at the origin).

Theorem 3.6. *Every convergent sequence is bounded.*

Proof. Let $x_k \rightarrow L$. We apply the definition of convergence with the specific value $\varepsilon = 1$ (we may: the definition holds for *every* ε , in particular for 1): there is N such that $\|x_k - L\| < 1$ for every $k \geq N$. For those k , the triangle inequality gives

$$\|x_k\| = \|(x_k - L) + L\| \leq \|x_k - L\| + \|L\| < 1 + \|L\|.$$

Only the *finitely many* initial terms $x_1, \dots, x_{N-1}$ are out of control. We take

$$M = \max\{\|x_1\|, \dots, \|x_{N-1}\|, 1 + \|L\|\},$$

a maximum of finitely many numbers (which is why it exists). Then $\|x_k\| \leq M$ for every k : the first $N - 1$ by construction of the maximum, and the rest because they are bounded by $1 + \|L\| \leq M$. □

Exam strategy

Two moves in this proof are standard and reusable: (i) *choosing a specific ε* (here $\varepsilon = 1$) when you only need “some” bound, not an arbitrary one; (ii) *splitting tail from head*: the tail is controlled by convergence, the head is finite and is controlled with a maximum. The converse of the theorem is **false**: $x_k = (-1)^k$ is bounded and does not converge (the subsequences of even and odd indices go to different limits, see Theorem 3.11). Memorize

this counterexample.

3.2.2 Limit laws

Theorem 3.7 (Limit laws). *Let $x_k \rightarrow L$ and $y_k \rightarrow M$ in $\mathbb{R}^n$, and let $\alpha \in \mathbb{R}$. Then:*

1. $x_k + y_k \rightarrow L + M$;
2. $\alpha x_k \rightarrow \alpha L$;
3. (in $\mathbb{R}$, $n = 1$) $x_k y_k \rightarrow LM$;
4. (in $\mathbb{R}$) if $M \neq 0$ and $y_k \neq 0$ for every k , then $\frac{x_k}{y_k} \rightarrow \frac{L}{M}$.

Proof. We prove (1) and (3) in detail, since they contain all the ideas; (2) is similar to (1), and (4) reduces to (3) by first proving $\frac{1}{y_k} \rightarrow \frac{1}{M}$.

(1) *Sum.* Let $\varepsilon > 0$. The goal is to make $\|(x_k + y_k) - (L + M)\|$ small. We regroup and use the triangle inequality:

$$\|(x_k + y_k) - (L + M)\| = \|(x_k - L) + (y_k - M)\| \leq \|x_k - L\| + \|y_k - M\|.$$

We want the sum to be $< \varepsilon$, so we ask each summand to be $< \frac{\varepsilon}{2}$ (the split into halves from the previous chapter). Since $x_k \rightarrow L$ there is N_1 such that $\|x_k - L\| < \frac{\varepsilon}{2}$ if $k \geq N_1$; since $y_k \rightarrow M$ there is N_2 such that $\|y_k - M\| < \frac{\varepsilon}{2}$ if $k \geq N_2$. With $N = \max\{N_1, N_2\}$, for every $k \geq N$ both bounds hold and

$$\|(x_k + y_k) - (L + M)\| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

(3) *Product (in $\mathbb{R}$).* The trick of adding and subtracting the mixed term $x_k M$:

$$|x_k y_k - LM| = |x_k y_k - x_k M + x_k M - LM| \leq |x_k| |y_k - M| + |M| |x_k - L|.$$

Let us look at each term. For the first one: (x_k) is convergent, hence bounded (Theorem 3.6): there is $C > 0$ with $|x_k| \leq C$ for every k . So the first term is $\leq C |y_k - M|$, which we can make small. For the second one: $|M|$ is a fixed constant, and $|x_k - L|$ becomes small.

Formally: given $\varepsilon > 0$, choose N_1 such that $|y_k - M| < \frac{\varepsilon}{2C}$ for $k \geq N_1$, and N_2 such that $|x_k - L| < \frac{\varepsilon}{2(|M|+1)}$ for $k \geq N_2$ (we use $|M| + 1$ in the denominator so that we never divide by zero when $M = 0$; this is a small trick worth copying). For $k \geq N = \max\{N_1, N_2\}$:

$$|x_k y_k - LM| \leq C \cdot \frac{\varepsilon}{2C} + |M| \cdot \frac{\varepsilon}{2(|M|+1)} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \quad \square$$

3.2.3 Coordinatewise convergence

Theorem 3.8. *Let $x_k = (x_k^{(1)}, \dots, x_k^{(n)}) \in \mathbb{R}^n$ and $L = (L^{(1)}, \dots, L^{(n)})$. Then*

$$x_k \rightarrow L \text{ in } \mathbb{R}^n \iff x_k^{(i)} \rightarrow L^{(i)} \text{ in } \mathbb{R} \text{ for each } i = 1, \dots, n.$$

Proof. The key is the following inequalities (previous chapter):

$$\left| x_k^{(i)} - L^{(i)} \right| \leq \|x_k - L\|_\infty \leq \|x_k - L\|_2 \leq \sqrt{n} \|x_k - L\|_\infty.$$

($\Rightarrow$) Suppose $x_k \rightarrow L$ (say in the Euclidean norm; by equivalence of norms it makes no difference). Given $\varepsilon > 0$, there is N with $\|x_k - L\|_2 < \varepsilon$ for $k \geq N$. By the first chain of inequalities, for each coordinate i : $\left| x_k^{(i)} - L^{(i)} \right| \leq \|x_k - L\|_2 < \varepsilon$. So each coordinate converges (with the same N !).

($\Leftarrow$) Suppose each coordinate converges. Given $\varepsilon > 0$, for each i there is N_i such that $\left| x_k^{(i)} - L^{(i)} \right| < \frac{\varepsilon}{\sqrt{n}}$ if $k \geq N_i$. Let $N = \max \{N_1, \dots, N_n\}$ (a maximum of *finitely many* indices: here it is essential that the dimension n is finite). For $k \geq N$, all coordinates are controlled at the same time, and

$$\|x_k - L\|_2 \leq \sqrt{n} \|x_k - L\|_\infty = \sqrt{n} \max_i \left| x_k^{(i)} - L^{(i)} \right| < \sqrt{n} \cdot \frac{\varepsilon}{\sqrt{n}} = \varepsilon. \quad \square$$

Remark 3.9. This theorem turns convergence in $\mathbb{R}^n$ into n convergence problems in $\mathbb{R}$: for example, $(\frac{1}{k}, 2 + \frac{(-1)^k}{k^2}, e^{-k}) \rightarrow (0, 2, 0)$ with no more work than looking at each coordinate. In infinite dimensional spaces the argument with the maximum of the N_i breaks down, and coordinatewise (pointwise) convergence stops agreeing with convergence in norm: that gap is the topic of the chapter on pointwise and uniform convergence.

3.2.4 Subsequences and monotone sequences

Definition 3.10 (Subsequence). A **subsequence** of (x_k) is a sequence of the form $(x_{k_j})_{j \in \mathbb{N}}$ where $k_1 < k_2 < k_3 < \dots$ is a strictly increasing sequence of indices. (You pick infinitely many terms of the original sequence, keeping the order.) A useful fact, proved by induction: $k_j \geq j$ for every j .

Theorem 3.11. If $x_k \rightarrow L$, then every subsequence of (x_k) converges to L . As a consequence, if a sequence has two subsequences with different limits, the sequence does not converge.

Proof. Let (x_{k_j}) be a subsequence and let $\varepsilon > 0$. By convergence there is N such that $\|x_k - L\| < \varepsilon$ for every $k \geq N$. If $j \geq N$, then $k_j \geq j \geq N$ (using the fact in the definition), so $\|x_{k_j} - L\| < \varepsilon$. Hence $x_{k_j} \rightarrow L$ with the same N .

The consequence is the contrapositive combined with uniqueness of the limit: if the sequence converged to some L , both subsequences would have to converge to that same L . $\square$

Example 3.12. $x_k = (-1)^k$ does not converge: the subsequence of even indices is constant $1 \rightarrow 1$ and the one of odd indices is constant $-1 \rightarrow -1$; two different limits. This is *the* practical method to refute convergence.

Definition 3.13. A sequence (x_k) in $\mathbb{R}$ is **increasing** if $x_k \leq x_{k+1}$ for every k , and **decreasing** if $x_k \geq x_{k+1}$. In both cases we call it **monotone**.

Theorem 3.14 (Monotone Convergence Theorem). Every increasing sequence in $\mathbb{R}$ that is bounded above converges, and its limit is $\sup \{x_k : k \in \mathbb{N}\}$. (Similarly, a decreasing sequence bounded below converges to the infimum.)

Proof. Let $A = \{x_k : k \in \mathbb{N}\}$ be the set of values of the sequence. It is nonempty and bounded above (by assumption), so by the *supremum axiom* there is $s = \sup A$. We claim that $x_k \rightarrow s$.

Let $\varepsilon > 0$. By the ε characterization of the supremum, there is an element of A greater than $s - \varepsilon$: that is, there is an index N with $x_N > s - \varepsilon$. Now monotonicity enters: for every $k \geq N$,

$$s - \varepsilon < x_N \leq x_k \leq s < s + \varepsilon,$$

where $x_N \leq x_k$ is monotonicity (increasing) and $x_k \leq s$ holds because s is an upper bound of all the values. The two ends of the chain say exactly that $|x_k - s| < \varepsilon$. Hence $x_k \rightarrow s$. $\square$

Intuition

An increasing sequence has only two possible fates: either it escapes to $+\infty$, or it hits a ceiling; and if there is a ceiling, the supremum axiom builds the *best* ceiling, against which the sequence gets squeezed. This theorem is the first big payoff of the supremum axiom, and it is the number one tool to prove that sequences *defined by recursion* converge (see Exercise 3.23): first you prove monotonicity and boundedness (usually by induction), you conclude that the limit L exists, and *only then* you pass to the limit in the recursion to identify L .

3.3 Functions from $\mathbb{R}^n$ to $\mathbb{R}$: limit and continuity [P1]

Definition 3.15 (Limit of a function). Let $f : A \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ and let a be an accumulation point of A (every neighborhood of a contains points of A other than a ; we do not need $a \in A$). We say that $\lim_{x \rightarrow a} f(x) = \ell$ if:

$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \forall x \in A : \quad 0 < \|x - a\| < \delta \implies |f(x) - \ell| < \varepsilon.$$

Definition 3.16 (Continuity). $f : A \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is **continuous at** $a \in A$ if:

$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \forall x \in A : \quad \|x - a\| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

f is **continuous** (without further qualification) if it is continuous at every point of A .

Intuition

Again the two player game, now with the function: the skeptic demands precision ε in the *output* (“I want $f(x)$ to be less than ε away from $f(a)$ ”); you answer with a tolerance δ in the *input* (“it is enough that x is less than δ away from a ”). Continuity at a means you can always answer. As a picture: however small the output window $(f(a) - \varepsilon, f(a) + \varepsilon)$ is, there is a ball $B(a, \delta)$ whose image under f falls inside that window.

Differences with the limit: for continuity we require $a \in A$, we compare with $f(a)$ (not with an external ℓ), and we do not exclude $x = a$ (the condition $0 < \|x - a\|$ disappears, since at $x = a$ the conclusion is trivial).

3.3.1 The sequential criterion

Theorem 3.17 (Sequential criterion for continuity). *Let $f : A \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ and $a \in A$. The following are equivalent:*

1. f is continuous at a ;
2. for every sequence (x_k) in A with $x_k \rightarrow a$, we have $f(x_k) \rightarrow f(a)$.

Proof. (1) $\Rightarrow$ (2), *direct.* Suppose f is continuous at a and take any sequence $x_k \rightarrow a$ in A . We want: $f(x_k) \rightarrow f(a)$. Let $\varepsilon > 0$. By continuity, there is $\delta > 0$ such that

$$x \in A, \|x - a\| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

By convergence of the sequence, *using δ in the role of the tolerance* (this chaining is the heart of the argument): there is N such that $\|x_k - a\| < \delta$ for every $k \geq N$. Combining both: for $k \geq N$, $|f(x_k) - f(a)| < \varepsilon$. That is exactly $f(x_k) \rightarrow f(a)$.

(2) $\Rightarrow$ (1), *by contradiction.* We prove the contrapositive: if f is *not* continuous at a , we build a sequence $x_k \rightarrow a$ whose images do not converge to $f(a)$ (negating (2)).

We negate continuity carefully (quantifier rules!): there is $\varepsilon_0 > 0$ such that *for every* $\delta > 0$ there is a point $x \in A$ with $\|x - a\| < \delta$ but $|f(x) - f(a)| \geq \varepsilon_0$.

Since this holds for every δ , we can use it with a family of deltas that goes to zero: for each $k \in \mathbb{N}$, take $\delta = \frac{1}{k}$ and obtain a point $x_k \in A$ with

$$\|x_k - a\| < \frac{1}{k} \quad \text{and} \quad |f(x_k) - f(a)| \geq \varepsilon_0.$$

(This is the creative step: turning “for every δ ” into a *sequence* by choosing $\delta = 1/k$. Remember it: this technique comes back throughout the course.)

The first condition forces $x_k \rightarrow a$: given $\varepsilon > 0$, by the Archimedean property there is N with $\frac{1}{N} < \varepsilon$, and for $k \geq N$, $\|x_k - a\| < \frac{1}{k} \leq \frac{1}{N} < \varepsilon$. The second condition blocks $f(x_k) \rightarrow f(a)$: the values $f(x_k)$ stay at distance $\geq \varepsilon_0$ from $f(a)$ for *every* k , so no N can work for the tolerance ε_0 .

We have exhibited a sequence converging to a whose images do not converge to $f(a)$: (2) is false. This completes the contrapositive. $\square$

Exam strategy

The sequential criterion has two exam uses, in opposite directions:

1. To *refute* continuity: exhibit ONE sequence $x_k \rightarrow a$ with $f(x_k) \not\rightarrow f(a)$ (or two sequences going to a whose images tend to different values). A single bad sequence is enough.
2. To *prove* properties of continuous functions by inheriting the limit laws for sequences (next proposition): without touching a single ε .

To *prove* continuity, on the other hand, the criterion does not help directly (you would have to check *all* sequences): there you use ε - δ or the algebra of continuous functions.

Proposition 3.18 (Algebra of continuous functions). *If $f, g : A \rightarrow \mathbb{R}$ are continuous at $a \in A$ and $\alpha \in \mathbb{R}$, then $f + g$, αf and fg are continuous at a ; and if $g(a) \neq 0$, f/g is continuous at a . Also, the composition of continuous functions is continuous.*

Proof. Everything follows for free from the sequential criterion plus the limit laws for sequences. For example, for the product: let $x_k \rightarrow a$ be any sequence in A . By continuity of f and g (through the criterion, direction (1) $\Rightarrow$ (2)): $f(x_k) \rightarrow f(a)$ and $g(x_k) \rightarrow g(a)$. By the limit laws (Theorem 3.7(3)): $f(x_k)g(x_k) \rightarrow f(a)g(a)$. Since the sequence was arbitrary, the criterion (direction (2) $\Rightarrow$ (1)) gives continuity of fg at a . The same scheme works for the other operations and for composition. $\square$

It follows that polynomials in several variables are continuous on all of $\mathbb{R}^n$ (the projections $x \mapsto x_i$ are continuous, as shown in Exercise 3.24, and polynomials are built with sums and products of projections and constants), and that quotients of polynomials are continuous wherever the denominator does not vanish.

3.3.2 Classical examples in two variables

Example 3.19 (A function that is discontinuous at the origin). Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be given by

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Let us see that f is **not** continuous at $(0, 0)$, using the sequential criterion with two sequences that reach the origin along different paths.

Path 1 (along the x axis): $u_k = (\frac{1}{k}, 0) \rightarrow (0, 0)$, and

$$f(u_k) = \frac{\frac{1}{k} \cdot 0}{\frac{1}{k^2} + 0} = 0 \longrightarrow 0.$$

Path 2 (along the diagonal): $v_k = (\frac{1}{k}, \frac{1}{k}) \rightarrow (0, 0)$, and

$$f(v_k) = \frac{\frac{1}{k^2}}{\frac{1}{k^2} + \frac{1}{k^2}} = \frac{1}{2} \longrightarrow \frac{1}{2}.$$

Two sequences converging to $(0, 0)$ whose images tend to different values ($0 \neq \frac{1}{2}$): if f were continuous at $(0, 0)$, both would have to tend to $f(0, 0)$. Hence f is discontinuous at the origin, and in fact $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ does not exist.

One more remark: the *iterated limits* $\lim_{x \rightarrow 0}(\lim_{y \rightarrow 0} f(x, y)) = \lim_{x \rightarrow 0} 0 = 0$ and $\lim_{y \rightarrow 0}(\lim_{x \rightarrow 0} f(x, y)) = 0$ exist and agree, and still the joint limit does not exist. Iterated limits do not detect the problem along the diagonal.

Careful (common mistake)

In several variables, checking the limit “along lines” (or along any particular family of paths) **never proves** that the limit exists: it can only *refute* it. A devastating example: $g(x, y) = \frac{x^2 y}{x^4 + y^2}$ has limit 0 at $(0, 0)$ along *all* lines $y = mx$, but along the parabola $y = x^2$ it is constantly $\frac{1}{2}$: the limit does not exist. To *prove* existence you must bound: find $h(x, y) \rightarrow 0$ with $|f(x, y) - \ell| \leq h(x, y)$ (for instance using $|xy| \leq \frac{x^2 + y^2}{2}$, or polar coordinates with a bound uniform in the angle).

Example 3.20 (The norm is continuous). The function $f(x) = \|x\|$ (any norm on $\mathbb{R}^n$) is continuous at every point. Full ε - δ proof: let $a \in \mathbb{R}^n$ and $\varepsilon > 0$. Choose $\delta = \varepsilon$. If $\|x - a\| < \delta$, the reverse triangle inequality (last exercise of the previous chapter) gives

$$|f(x) - f(a)| = |\|x\| - \|a\|| \leq \|x - a\| < \delta = \varepsilon.$$

Done. Functions that, like this one, satisfy $|f(x) - f(y)| \leq K \|x - y\|$ for a constant K are called **Lipschitz**, and for them $\delta = \varepsilon/K$ always works: they are the “easy” case of continuity and they will come back as main characters in the Banach fixed point theorem.

Link with economics

Continuity is a structural assumption in microeconomics, and the sequential criterion is the way it is actually used: (i) Debreu’s representation theorem says that rational and *continuous* preferences admit a continuous utility function; continuity of preferences is defined sequentially ($x_k \succeq y_k$, $x_k \rightarrow x$, $y_k \rightarrow y \Rightarrow x \succeq y$); (ii) to prove that demand $x(p, w)$ is continuous, the standard argument (through Berge’s maximum theorem, final chapter) consists of taking sequences $(p_k, w_k) \rightarrow (p, w)$ and checking that the optima converge; (iii) in dynamic programming, continuity of the value function is inherited through limits of sequences of iterations. Without the sequential criterion, none of these arguments gets started. And the counterexample $xy/(x^2 + y^2)$ teaches an applied lesson: with several variables, “continuity in each variable separately” does NOT imply joint continuity, an assumption that is sometimes overlooked in modeling.

3.4 Exam-style exercises [P1]

Exercise 3.21. Prove with the ε - N definition that $x_k = \frac{3k+1}{k+2} \rightarrow 3$.

Solution. *Scratch work:* we compute the distance to the limit and simplify it:

$$|x_k - 3| = \left| \frac{3k+1 - 3(k+2)}{k+2} \right| = \left| \frac{-5}{k+2} \right| = \frac{5}{k+2}.$$

We want $\frac{5}{k+2} < \varepsilon$, that is, $k > \frac{5}{\varepsilon} - 2$.

Write up: let $\varepsilon > 0$. By the Archimedean property, there is $N \in \mathbb{N}$ with $N > \frac{5}{\varepsilon}$. Then, for every $k \geq N$:

$$|x_k - 3| = \frac{5}{k+2} < \frac{5}{k} \leq \frac{5}{N} < \varepsilon,$$

where we used $k+2 > k$ (which enlarges the denominator and shrinks the fraction) and $k \geq N$. Since ε was arbitrary, $x_k \rightarrow 3$. *Style note:* simplifying the bound (using $\frac{5}{k}$ instead of $\frac{5}{k+2}$) is good practice: do not look for the optimal N , look for a *convenient* one that works. $\square$

Exercise 3.22. Let (x_k) be a sequence in $\mathbb{R}$ with $x_k \rightarrow L > 0$. Prove that there is N such that $x_k > \frac{L}{2}$ for every $k \geq N$. (“Sequences with a positive limit end up being positive, and bounded away from zero”.)

Solution. The idea: choose ε so small that the ball around L stays entirely to the right of $\frac{L}{2}$. Take the specific value $\varepsilon = \frac{L}{2} > 0$ (this is where $L > 0$ is used). By convergence, there is N such that $|x_k - L| < \frac{L}{2}$ for every $k \geq N$. Unfolding the absolute value:

$$-\frac{L}{2} < x_k - L < \frac{L}{2} \implies x_k > L - \frac{L}{2} = \frac{L}{2},$$

for every $k \geq N$, as required. *Comment:* this small lemma is used all the time (for instance, to prove the quotient rule in the limit laws: it guarantees that $|y_k| \geq \frac{|M|}{2}$ and that the denominators do not blow up). Choosing a specific ε adapted to the goal is the skill the exam wants to see. $\square$

Exercise 3.23. Define $x_1 = 1$ and $x_{k+1} = \sqrt{2 + x_k}$. Prove that (x_k) converges and compute its limit.

Solution. Strategy in three acts: (I) boundedness by induction, (II) monotonicity by induction, (III) monotone convergence plus identification of the limit.

(I) $x_k < 2$ for every k . Base: $x_1 = 1 < 2$. Step: if $x_k < 2$, then $x_{k+1} = \sqrt{2 + x_k} < \sqrt{2 + 2} = 2$ (the square root is increasing). So, by induction, the sequence is bounded above by 2.

(II) (x_k) is increasing. Base: $x_2 = \sqrt{3} \approx 1.732 > 1 = x_1$. Step: if $x_{k+1} \geq x_k$, then $x_{k+2} = \sqrt{2 + x_{k+1}} \geq \sqrt{2 + x_k} = x_{k+1}$ (again, monotonicity of the square root). By induction, increasing.

(III) By the Monotone Convergence Theorem (Theorem 3.14), (x_k) converges to some $L \in \mathbb{R}$ (with $L \leq 2$). To identify L , we pass to the limit in the recursion: since $x_k \rightarrow L$, also $x_{k+1} \rightarrow L$ (it is a shifted subsequence), and since the function $t \mapsto \sqrt{2 + t}$ is continuous, the sequential criterion gives $\sqrt{2 + x_k} \rightarrow \sqrt{2 + L}$. By uniqueness of the limit:

$$L = \sqrt{2 + L} \implies L^2 = 2 + L \implies L^2 - L - 2 = 0 \implies (L - 2)(L + 1) = 0,$$

so $L = 2$ or $L = -1$. Since all terms are positive, $L \geq 0$ and therefore $L = 2$.

Built in warning: the step “ $L = \sqrt{2 + L}$ ” is legitimate only *after* you know that the limit exists. Solving the equation first and declaring “the limit is 2” without (I) and (II) is a classical mistake that loses most of the points: a divergent sequence would satisfy the fixed point equation as well. $\square$

Exercise 3.24. Prove with ε - δ that the projection $\pi_i : \mathbb{R}^n \rightarrow \mathbb{R}$, $\pi_i(x) = x_i$, is continuous at every point, and use this (with the algebra of continuous functions) to justify that $f(x, y) = \frac{x^2 y - 3xy}{1 + x^2 + y^2}$ is continuous on all of $\mathbb{R}^2$.

Solution. *Projection.* Let $a \in \mathbb{R}^n$ and $\varepsilon > 0$; take $\delta = \varepsilon$. If $\|x - a\|_2 < \delta$, then

$$|\pi_i(x) - \pi_i(a)| = |x_i - a_i| \leq \|x - a\|_\infty \leq \|x - a\|_2 < \delta = \varepsilon,$$

using the chain $|x_i - a_i| \leq \max_j |x_j - a_j| \leq \|x - a\|_2$ from the previous chapter. So π_i is continuous (in fact Lipschitz with constant 1).

The given function. The numerator $x^2y - 3xy$ is built from the projections $\pi_1(x, y) = x$ and $\pi_2(x, y) = y$ through products, scalar multiples and sums: by Proposition 3.18 it is continuous on all of $\mathbb{R}^2$. The denominator $1 + x^2 + y^2$ is continuous for the same reason, and it *never vanishes* (it is $\geq 1 > 0$ at every point). By the quotient rule for continuous functions, f is continuous on all of $\mathbb{R}^2$. *Exam format:* this “structural” reasoning (2 or 3 lines citing the algebra) is exactly what is expected; nobody wants to see a handmade ε - δ for a rational function whose denominator has no zeros. $\square$

Exercise 3.25. Study continuity at $(0, 0)$ of

$$g(x, y) = \begin{cases} \frac{x^2y}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Solution. Unlike the example $xy/(x^2 + y^2)$, here the numerator has degree 3 and the denominator degree 2: there is one degree to spare, and that will save continuity. The correct strategy is to *bound* (not to test paths, which only serve to refute).

We use the elementary inequality $x^2 \leq x^2 + y^2$, which implies (for $(x, y) \neq (0, 0)$, dividing by the positive number $x^2 + y^2$):

$$|g(x, y) - 0| = \frac{x^2}{x^2 + y^2} \cdot |y| \leq 1 \cdot |y| = |y| \leq \|(x, y)\|_2.$$

ε - δ proof: given $\varepsilon > 0$, take $\delta = \varepsilon$. If $0 < \|(x, y) - (0, 0)\|_2 < \delta$, then $|g(x, y) - 0| \leq \|(x, y)\|_2 < \delta = \varepsilon$. Hence $\lim_{(x, y) \rightarrow (0, 0)} g(x, y) = 0 = g(0, 0)$ and g is *continuous* at the origin.

Comparative comment (a frequent theory question): in $xy/(x^2 + y^2)$ numerator and denominator have the same degree (2 and 2) and the quotient “remembers the direction” of approach; here (3 over 2) the extra degree in the numerator crushes any directional dependence. A practical rule to guess (you still have to prove it by bounding!): if the degree of the numerator is strictly larger than the degree of the denominator, the limit at the origin usually exists and equals 0; if they tie, suspect the diagonal. $\square$

Chapter summary

- $x_k \rightarrow L$: $\forall \varepsilon > 0 \exists N \forall k \geq N : \|x_k - L\| < \varepsilon$. The N depends on ε . Method: scratch work (solve for k) plus write up.
- The limit is unique; convergent $\Rightarrow$ bounded (converse false: $(-1)^k$); every subsequence inherits the limit (two subsequences with different limits $\Rightarrow$ no convergence).
- Limit laws: sum ($\varepsilon/2$ plus triangle inequality), product (add and subtract the mixed

term plus a bound), quotient (keep the denominator away from zero).

- In $\mathbb{R}^n$: convergence $\iff$ coordinatewise convergence (through $|x_k^{(i)} - L^{(i)}| \leq \|x_k - L\| \leq \sqrt{n} \|x_k - L\|_\infty$).
- Monotone plus bounded $\Rightarrow$ convergent (to the sup or inf; uses the supremum axiom).
Recipe for recursions: induction (bound, monotonicity) $\rightarrow L$ exists $\rightarrow$ fixed point equation.
- Continuity at a : $\forall \varepsilon \exists \delta$ such that $\|x - a\| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon$. Lipschitz $\Rightarrow$ continuous with $\delta = \varepsilon/K$.
- Sequential criterion: continuous at $a \iff$ every sequence $x_k \rightarrow a$ satisfies $f(x_k) \rightarrow f(a)$. To refute: one bad sequence (or two paths with different limits). To prove existence of limits in $\mathbb{R}^2$: *bound*, never “test paths”.

Chapter 4

Metric spaces [P3]

Week 3 of the course. Bibliography: Alabiso and Weiss [1], Ch. 4; Pons [3], Ch. 8; Ok [4], Ch. C.

What is this used for in Microeconomics?

In Microeconomics almost everything happens in $\mathbb{R}^n$, so this chapter is more training than a daily tool: it gives the general framework in which $\mathbb{R}^n$ is a particular case. Its value for micro lies in two points. First, there are central objects of the theory that are not subsets of $\mathbb{R}^n$ but are metric spaces: the spaces of lotteries over continuous prizes (Part I, choice under uncertainty), the strategy spaces in infinite games and the spaces of histories in repeated games (Part II), and the spaces of contracts, menus and incentive schemes in the principal agent problem and in mechanism design (Parts III and V). Second, Lipschitz functions and contractions come back in the fixed point arguments that support the existence of equilibrium. This is a chapter about mathematical maturity: it lets you read literature where the space is not $\mathbb{R}^n$ without getting lost.

4.1 Motivation [P3]

In the previous chapters we worked in $\mathbb{R}^n$, where the distance between two points is computed with a norm: $d(x, y) = \|x - y\|$. But many of the ideas we used (convergence, continuity, balls) depended only on *three properties* of that distance, not on the particular structure of $\mathbb{R}^n$.

The natural question is: what happens if we take *any* set X (points, functions, sequences, strategies of a game...) and put on it a notion of distance satisfying those three properties? The answer is the theory of **metric spaces**: one of the most profitable abstractions in all of mathematics. With a single definition we get, at once, a theory of convergence and continuity valid in $\mathbb{R}$, in $\mathbb{R}^n$, in function spaces (where the value function of a dynamic programming problem lives), in sequence spaces, and so on.

Intuition

A metric space is simply “a set where we know how to measure how far apart two elements are”. We do not need to add elements, or multiply them by scalars, or anything else: only

distance. All the topology of this course (open sets, closed sets, convergence, completeness, compactness) is built only from that distance.

4.2 Definition of metric and first examples [P3]

Definition 4.1 (Metric and metric space). Let X be a nonempty set. A **metric** or *distance function* on X is a function

$$d : X \times X \rightarrow \mathbb{R}$$

such that, for all $x, y, z \in X$:

(M1) **Positivity and identity of indiscernibles:** $d(x, y) \geq 0$, and moreover $d(x, y) = 0$ if and only if $x = y$.

(M2) **Symmetry:** $d(x, y) = d(y, x)$.

(M3) **Triangle inequality:**

$$d(x, z) \leq d(x, y) + d(y, z).$$

The pair (X, d) is called a **metric space**.

Let us look at what each axiom says and why it captures the intuitive idea of “distance”:

- (M1) says two things. First, that distances are never negative: it makes no sense to be “−3 meters away” from something. Second, that distance zero characterizes equality: if two points are at distance zero, they *are the same point*, and every point is at distance zero from itself. This second part is what makes the limit in “ $x_k \rightarrow x$ ” unique, as we will see.
- (M2) says that the distance one way is the same as the distance back. (In real life this can fail, think of one way streets, but classical theory requires it.)
- (M3) is the most important axiom and the one most used in proofs. It says that “taking a detour through an intermediate point never shortens the trip”: going directly from x to z costs at most what it costs to go from x to y and then from y to z . Almost every estimate in analysis consists of inserting a clever intermediate point and applying (M3).

Example 4.2 (The usual metric on $\mathbb{R}$). Let $X = \mathbb{R}$ and $d(x, y) = |x - y|$. Let us check the three axioms in full detail.

(M1): $|x - y| \geq 0$ always, because the absolute value of any real number is nonnegative. Moreover $|x - y| = 0 \iff x - y = 0 \iff x = y$, since the only real number with absolute value zero is zero.

(M2): $|x - y| = |-(y - x)| = |y - x|$, using that $|-a| = |a|$ for every $a \in \mathbb{R}$.

(M3): we write $x - z = (x - y) + (y - z)$ (we add and subtract y : the trick of “inserting the intermediate point”) and apply the triangle inequality for the absolute value, $|a + b| \leq |a| + |b|$:

$$d(x, z) = |x - z| = |(x - y) + (y - z)| \leq |x - y| + |y - z| = d(x, y) + d(y, z).$$

Therefore $(\mathbb{R}, |\cdot - \cdot|)$ is a metric space. It is *the* example you should always keep in mind.

Example 4.3 (Metrics induced by norms on $\mathbb{R}^n$). If $\|\cdot\|$ is any norm on $\mathbb{R}^n$ (seen in the chapter on $\mathbb{R}^n$), then

$$d(x, y) := \|x - y\|$$

is a metric on $\mathbb{R}^n$. Let us check it using only the properties of a norm (nonnegativity and vanishing only at 0; homogeneity $\|\lambda v\| = |\lambda| \|v\|$; triangle inequality $\|u + v\| \leq \|u\| + \|v\|$):

(M1): $d(x, y) = \|x - y\| \geq 0$, and $\|x - y\| = 0 \iff x - y = 0 \iff x = y$, by the vanishing property of the norm.

(M2): $d(y, x) = \|y - x\| = \|(-1)(x - y)\| = |-1| \|x - y\| = \|x - y\| = d(x, y)$, using homogeneity with $\lambda = -1$.

(M3): again we insert the intermediate point:

$$d(x, z) = \|x - z\| = \|(x - y) + (y - z)\| \leq \|x - y\| + \|y - z\| = d(x, y) + d(y, z).$$

In particular, on $\mathbb{R}^n$ we have the metrics

$$d_2(x, y) = \left(\sum_{i=1}^n (x_i - y_i)^2 \right)^{1/2}, \quad d_1(x, y) = \sum_{i=1}^n |x_i - y_i|, \quad d_\infty(x, y) = \max_{1 \leq i \leq n} |x_i - y_i|,$$

called the Euclidean metric, the taxicab metric and the max (or Chebyshev) metric, respectively.

Example 4.4 (The discrete metric). On *any* nonempty set X we can define the **discrete metric**:

$$d(x, y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } x \neq y. \end{cases}$$

(M1): the only possible values are 0 and 1, both ≥ 0 ; and by construction $d(x, y) = 0$ exactly when $x = y$.

(M2): the condition “ $x = y$ ” is symmetric in x and y , so $d(x, y) = d(y, x)$.

(M3): we must prove $d(x, z) \leq d(x, y) + d(y, z)$. We argue by cases:

- If $x = z$, the left side is 0 and the right side is ≥ 0 : it holds.
- If $x \neq z$, the left side is 1. We claim the right side is at least 1: if it were 0, we would have $d(x, y) = 0$ and $d(y, z) = 0$, that is, $x = y$ and $y = z$, hence $x = z$, contradicting $x \neq z$. Therefore $d(x, y) + d(y, z) \geq 1 = d(x, z)$.

This example looks artificial, but it is a *factory of counterexamples*: it shows that a metric space may look nothing like $\mathbb{R}^n$.

Example 4.5 (The sup metric on $C[a, b]$). Let $C[a, b]$ be the set of all continuous functions $f : [a, b] \rightarrow \mathbb{R}$. We define the **sup metric** (also called the uniform metric):

$$d_\infty(f, g) := \sup_{t \in [a, b]} |f(t) - g(t)|.$$

Step 0 (well defined). Before checking the axioms we must verify that $d_\infty(f, g)$ is a real number (not $+\infty$). Since f and g are continuous on the closed bounded interval $[a, b]$, the

function $t \mapsto |f(t) - g(t)|$ is continuous on $[a, b]$, and by the Weierstrass theorem (a continuous function on a compact set attains its maximum, which we will prove in the chapter on compactness) the supremum is finite and is in fact a maximum: there is $t^* \in [a, b]$ with $d_\infty(f, g) = |f(t^*) - g(t^*)| < \infty$.

(M1): each $|f(t) - g(t)| \geq 0$, so the supremum (which is an upper bound of those values) is ≥ 0 . If $d_\infty(f, g) = 0$, then $|f(t) - g(t)| \leq 0$ for every t (the supremum bounds each term), so $f(t) = g(t)$ for every $t \in [a, b]$, that is, $f = g$ as functions. Conversely, if $f = g$ the supremum of zeros is zero.

(M2): $|f(t) - g(t)| = |g(t) - f(t)|$ for each t , so the suprema agree.

(M3): fix $f, g, h \in C[a, b]$ and an arbitrary $t \in [a, b]$. By the triangle inequality in $\mathbb{R}$:

$$|f(t) - h(t)| \leq |f(t) - g(t)| + |g(t) - h(t)| \leq d_\infty(f, g) + d_\infty(g, h),$$

where in the last step we used that each term is bounded by its supremum. The inequality holds for *every* t , that is, the number $d_\infty(f, g) + d_\infty(g, h)$ is an upper bound of the set $\{|f(t) - h(t)| : t \in [a, b]\}$. Since the supremum is the *least* upper bound, we conclude

$$d_\infty(f, h) = \sup_t |f(t) - h(t)| \leq d_\infty(f, g) + d_\infty(g, h).$$

This reasoning (“I bound for each t , and then I pass to the supremum”) is a pattern that will repeat many times in the course: learn it as a template.

Geometric reading: $d_\infty(f, g) \leq \varepsilon$ means that the graph of g lies inside a “band” or “tube” of radius ε around the graph of f . Two functions are close in d_∞ when they are close *at all points at the same time* (uniformly).

Example 4.6 (Bounded sequences: the space ℓ^∞). Let ℓ^∞ be the set of all bounded real sequences $x = (x_1, x_2, x_3, \dots)$, that is, such that there is M with $|x_k| \leq M$ for every k . We define

$$d_\infty(x, y) := \sup_{k \in \mathbb{N}} |x_k - y_k|.$$

The supremum is finite: if $|x_k| \leq M_1$ and $|y_k| \leq M_2$ for every k , then $|x_k - y_k| \leq |x_k| + |y_k| \leq M_1 + M_2$, so $M_1 + M_2$ is an upper bound of the set of differences and the supremum exists (supremum axiom in $\mathbb{R}$). The check of (M1) to (M3) is identical, word for word, to the one in the previous example, replacing “ $t \in [a, b]$ ” by “ $k \in \mathbb{N}$ ”. That is exactly the point of abstraction: one argument works for very different spaces.

Example 4.7 (Metric subspaces). If (X, d) is a metric space and $A \subseteq X$ is any nonempty subset, then the restriction of d to $A \times A$ is a metric on A (the axioms are inherited automatically, since they hold for all points of X , in particular for those of A). The pair $(A, d|_{A \times A})$ is called a **metric subspace**. For example, $(0, 1]$ and $\mathbb{Q}$ are metric spaces with the distance inherited from $\mathbb{R}$. This example matters: we will see that properties such as completeness are *not* inherited by subspaces.

Careful (common mistake)

A very common mistake: believing that a “metric space” is always something similar to $\mathbb{R}^n$. It is not: the *points* of a metric space can be functions (Example 4.5), sequences (Example 4.6), or anything at all (Example 4.4). When an exam says “let (X, d) be a metric space”, you may only use (M1), (M2) and (M3): no coordinates, no sums of points, no multiplication by scalars, unless the problem says so.

4.3 Immediate consequences of the axioms [P3]

Proposition 4.8 (Reverse triangle inequality). *Let (X, d) be a metric space. For all $x, y, z \in X$:*

$$|d(x, z) - d(y, z)| \leq d(x, y).$$

Proof. Recall that $|a| \leq b$ is equivalent to the two inequalities $a \leq b$ and $-a \leq b$. We prove both.

Step 1. By the triangle inequality (M3) applied to the path $x \rightarrow y \rightarrow z$:

$$d(x, z) \leq d(x, y) + d(y, z) \implies d(x, z) - d(y, z) \leq d(x, y).$$

(We just moved $d(y, z)$ to the other side by subtracting.)

Step 2. Now we apply (M3) to the path $y \rightarrow x \rightarrow z$:

$$d(y, z) \leq d(y, x) + d(x, z) \implies d(y, z) - d(x, z) \leq d(y, x) = d(x, y),$$

where in the last equality we used symmetry (M2). This is exactly $-(d(x, z) - d(y, z)) \leq d(x, y)$.

Conclusion. Since $d(x, z) - d(y, z) \leq d(x, y)$ and $-(d(x, z) - d(y, z)) \leq d(x, y)$, we conclude $|d(x, z) - d(y, z)| \leq d(x, y)$. $\square$

Remark 4.9. Proposition 4.8 says that the function “distance to a fixed point z ”, that is $x \mapsto d(x, z)$, changes little when x changes little: it is a Lipschitz function with constant 1 (a concept we define below). In particular, distance is a continuous function. This is used all the time: for example, to pass to the limit inside a distance.

4.4 Balls, bounded sets and diameter [P3]

Definition 4.10 (Open and closed balls). Let (X, d) be a metric space, $c \in X$ and $r > 0$. We define:

- the **open ball** with center c and radius r :

$$B(c, r) := \{x \in X : d(x, c) < r\},$$

- the **closed ball**:

$$\overline{B}(c, r) := \{x \in X : d(x, c) \leq r\}.$$

The only difference is the strict inequality ($<$) versus the non strict one ($\leq$): the open ball does *not* include the “shell”.

Example 4.11 (What balls look like in each space). It is worth visualizing balls in our examples:

1. In $(\mathbb{R}, |\cdot|)$: $B(c, r) = (c - r, c + r)$, an open interval centered at c . Indeed, $|x - c| < r \iff -r < x - c < r \iff c - r < x < c + r$.
2. In $(\mathbb{R}^2, d_2)$: $B(c, r)$ is the circular disk with center c and radius r *without* the boundary circle.
3. In $(\mathbb{R}^2, d_1)$: the ball is a square “rotated 45 degrees” (a diamond) with vertices at distance r from the center along the axes: the condition is $|x_1 - c_1| + |x_2 - c_2| < r$.
4. In $(\mathbb{R}^2, d_\infty)$: the ball is a square of side $2r$ with sides parallel to the axes: the condition $\max\{|x_1 - c_1|, |x_2 - c_2|\} < r$ says that *both* coordinates differ by less than r .
5. In the discrete metric: if $r \leq 1$, $B(c, r) = \{c\}$ (the only point at distance $< r \leq 1$ from c is c itself); if $r > 1$, $B(c, r) = X$, the whole space. Balls are either one point or everything!
6. In $(C[a, b], d_\infty)$: $B(f, r)$ is the set of continuous functions whose graph stays strictly inside the “tube” of radius r around the graph of f .

Lesson: the “shape” of a ball depends entirely on the metric. The word “ball” should not always make you picture a round sphere.

Definition 4.12 (Bounded set and diameter). Let (X, d) be a metric space and let $A \subseteq X$ be nonempty.

- A is **bounded** if it is contained in some ball: there are $c \in X$ and $r > 0$ such that $A \subseteq B(c, r)$.
- The **diameter** of A is

$$\text{diam}(A) := \sup \{d(x, y) : x, y \in A\},$$

which may be $+\infty$ if the set of distances is unbounded.

Proposition 4.13. *A is bounded if and only if $\text{diam}(A) < \infty$.*

Proof. ($\Rightarrow$) Suppose $A \subseteq B(c, r)$. Let $x, y \in A$ be arbitrary. Inserting the center c as intermediate point (M3):

$$d(x, y) \leq d(x, c) + d(c, y) < r + r = 2r,$$

where both distances to c are $< r$ because $x, y \in B(c, r)$. Then $2r$ is an upper bound of the set of distances, so $\text{diam}(A) \leq 2r < \infty$.

($\Leftarrow$) Suppose $\text{diam}(A) = D < \infty$. Fix any point $c \in A$ (it exists because $A \neq \emptyset$). For every $x \in A$ we have $d(x, c) \leq D < D + 1$, so $A \subseteq B(c, D + 1)$: A is bounded. $\square$

4.5 Equivalent metrics [P4]

In $\mathbb{R}^n$ we saw that all norms are equivalent. The analogous notion for metrics has *two* versions that must be told apart carefully.

Definition 4.14 (Strong and topological equivalence). Let d and d' be two metrics on the same set X .

1. d and d' are **strongly equivalent** (also called Lipschitz equivalent) if there are constants $\alpha, \beta > 0$ such that

$$\alpha d(x, y) \leq d'(x, y) \leq \beta d(x, y) \quad \text{for all } x, y \in X.$$

2. d and d' are **topologically equivalent** if they generate the same open sets; equivalently (and more manageable): for every $x \in X$ and every $r > 0$ there are $r_1, r_2 > 0$ such that

$$B_{d'}(x, r_1) \subseteq B_d(x, r) \quad \text{and} \quad B_d(x, r_2) \subseteq B_{d'}(x, r).$$

(“Inside each ball of one metric there fits a ball of the other one with the same center”).

Intuition

Strong equivalence: the two distances are comparable “at a global scale”, they differ at most by a fixed multiplicative factor. *Topological* equivalence: the two metrics have the same notion of “close” point by point, and therefore the same convergent sequences, the same continuous functions and the same open sets, even if the numerical sizes of the distances get distorted. Strong $\Rightarrow$ topological, but not the other way round, as the next classical example shows.

Proposition 4.15. *If d and d' are strongly equivalent, then they are topologically equivalent.*

Proof. Suppose $\alpha d \leq d' \leq \beta d$ with $\alpha, \beta > 0$. Fix $x \in X$ and $r > 0$.

Inclusion 1: we claim that $B_{d'}(x, \alpha r) \subseteq B_d(x, r)$. Indeed, if $d'(x, y) < \alpha r$, then using $\alpha d(x, y) \leq d'(x, y)$ we get $\alpha d(x, y) < \alpha r$, and dividing by $\alpha > 0$ (which preserves the inequality): $d(x, y) < r$. That is, $y \in B_d(x, r)$.

Inclusion 2: we claim that $B_d(x, r/\beta) \subseteq B_{d'}(x, r)$. If $d(x, y) < r/\beta$, then $d'(x, y) \leq \beta d(x, y) < \beta \cdot r/\beta = r$, that is, $y \in B_{d'}(x, r)$.

Both inclusions give exactly the condition of topological equivalence (with $r_1 = \alpha r$ and $r_2 = r/\beta$). $\square$

Example 4.16 (Topologically equivalent but not strongly equivalent). On $X = \mathbb{R}$ consider the usual metric $d(x, y) = |x - y|$ and

$$d'(x, y) := \min \{1, |x - y|\}.$$

First, d' is a metric: (M1) and (M2) follow at once from those of d (taking the minimum with 1 does not change when the distance is zero, nor symmetry). For (M3), let $x, y, z \in \mathbb{R}$ and let us prove $d'(x, z) \leq d'(x, y) + d'(y, z)$. Case 1: if $d'(x, y) + d'(y, z) \geq 1$, we are done, because $d'(x, z) \leq 1$ always (it is a minimum with 1). Case 2: if $d'(x, y) + d'(y, z) < 1$, then in particular $d'(x, y) < 1$ and $d'(y, z) < 1$, which forces $d'(x, y) = |x - y|$ and $d'(y, z) = |y - z|$ (the minimum

is attained at the second argument). Then

$$d'(x, z) \leq |x - z| \leq |x - y| + |y - z| = d'(x, y) + d'(y, z),$$

where the first inequality holds because d' is a minimum, and the second is the usual triangle inequality.

They are topologically equivalent. The key: for small distances the two metrics *agree*. Fix $x \in \mathbb{R}$ and $r > 0$, and let $\rho := \min\{r, 1/2\} > 0$. If $d'(x, y) < \rho \leq 1/2 < 1$, then necessarily $d'(x, y) = |x - y|$ (if the minimum were 1, we would have $d'(x, y) = 1 \geq \rho$, a contradiction), so $|x - y| < \rho \leq r$: this proves $B_{d'}(x, \rho) \subseteq B_d(x, r)$. Similarly, if $|x - y| < \rho$ then $d'(x, y) \leq |x - y| < \rho \leq r$, which proves $B_d(x, \rho) \subseteq B_{d'}(x, r)$. The two inclusions give topological equivalence.

They are not strongly equivalent. Suppose, looking for a contradiction, that there were $\alpha > 0$ with $\alpha |x - y| \leq d'(x, y)$ for all x, y . Take $x = 0$ and $y = n \in \mathbb{N}$. Then

$$\alpha \cdot n = \alpha |0 - n| \leq d'(0, n) = \min\{1, n\} \leq 1 \quad \text{for every } n \in \mathbb{N},$$

that is, $\alpha \leq 1/n$ for every n , which forces $\alpha \leq 0$: a contradiction with $\alpha > 0$. Intuitively: d' is bounded by 1, while d is unbounded, and no constant can compress arbitrarily large distances inside a bounded metric.

Lesson: d' changes the “sizes” radically (all of $\mathbb{R}$ has d' -diameter ≤ 1 , so $\mathbb{R}$ is d' -bounded!) but it does not change the notion of convergence nor the open sets. Boundedness is *not* a topological property.

Careful (common mistake)

Exams often ask you to “prove that these metrics are equivalent”. The first thing to ask is *which* equivalence: if you can find constants α, β , prove the strong one (which also implies the topological one, Proposition 4.15). If you suspect there are no constants (for example, a bounded metric versus an unbounded one), prove the topological one through inclusions of balls and look for the counterexample to the strong one with “far apart” points, as in Example 4.16.

4.6 Functions between metric spaces [P3]

Definition 4.17 (Continuity in metric spaces). Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f : X \rightarrow Y$ is **continuous at the point** $x_0 \in X$ if:

$$\forall \varepsilon > 0 \exists \delta > 0 \text{ such that } d_X(x, x_0) < \delta \implies d_Y(f(x), f(x_0)) < \varepsilon.$$

f is **continuous** (without further qualification) if it is continuous at every point of X .

Intuition

It is exactly the ε - δ definition from $\mathbb{R}$, replacing each absolute value by the corresponding metric. Reading: “I can guarantee that $f(x)$ is as close to $f(x_0)$ as required (ε), provided I take x close enough to x_0 (δ)”. Note that δ may depend on ε and on the point x_0 .

Definition 4.18 (Isometry, Lipschitz function, contraction). Let (X, d_X) , (Y, d_Y) be metric spaces and $f : X \rightarrow Y$.

1. f is an **isometry** if it preserves distances: $d_Y(f(x), f(y)) = d_X(x, y)$ for all $x, y \in X$.
2. f is **Lipschitz** with constant $L \geq 0$ if

$$d_Y(f(x), f(y)) \leq L d_X(x, y) \quad \text{for all } x, y \in X.$$

3. f is a **contraction** if it is Lipschitz with some constant $L < 1$: it brings points closer together in a proportion strictly smaller than 1.

Theorem 4.19 (Lipschitz $\Rightarrow$ continuous). *Every Lipschitz function is continuous. In particular, every isometry and every contraction is continuous.*

Proof. Let f be Lipschitz with constant L . If $L = 0$, then $d_Y(f(x), f(y)) \leq 0$, hence $= 0$, for all pairs: f is constant, and a constant function is trivially continuous (any δ works). Suppose $L > 0$. Fix $x_0 \in X$ and $\varepsilon > 0$; we must exhibit $\delta > 0$. We propose

$$\delta := \frac{\varepsilon}{L}.$$

Let us check that it works: if $d_X(x, x_0) < \delta$, then by the Lipschitz condition

$$d_Y(f(x), f(x_0)) \leq L d_X(x, x_0) < L \cdot \frac{\varepsilon}{L} = \varepsilon.$$

This is exactly the definition of continuity at x_0 . Since x_0 was arbitrary, f is continuous on all of X . Note something important: the δ we found *does not depend on the point* x_0 ; this uniformity is called *uniform continuity*, a property stronger than continuity, which will come back in the chapter on compactness.

For isometries, the definition gives Lipschitz with $L = 1$; for contractions, with $L < 1$. In both cases the argument above applies. $\square$

Example 4.20. The function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{1}{2} \cos x$, is a contraction with the usual metric. Indeed, by the mean value theorem (reminder: if g is differentiable, $g(x) - g(y) = g'(\xi)(x - y)$ for some ξ between x and y), applied to $g = f$:

$$|f(x) - f(y)| = |f'(\xi)| |x - y| = \frac{1}{2} |\sin \xi| |x - y| \leq \frac{1}{2} |x - y|,$$

since $|\sin \xi| \leq 1$. So $L = 1/2 < 1$. In the chapter on the fixed point theorem we will see that this guarantees that the equation $x = \frac{1}{2} \cos x$ has *exactly one* solution, and that it can be approximated by iterating f from any starting point. Contractions are the central tool of that

chapter (Banach fixed point theorem) and of its applications to dynamic programming; here we only introduce them.

Link with economics

“Exotic” metric spaces are the natural habitat of modern economic analysis:

- In **dynamic programming** (neoclassical growth model, job search, and so on), the unknown is not a number but a *function*: the value function V . The relevant space is a space of bounded or continuous functions with the sup metric $d_\infty(V, W) = \sup_x |V(x) - W(x)|$: two value functions are close if they give similar values *at all states at the same time*. The Bellman operator turns out to be a contraction in this space, and from there (fixed point chapter) we get existence and uniqueness of V .
- In **game theory**, a profile of mixed strategies lives in a subset of $\mathbb{R}^n$; the distance between profiles measures how much behaviors differ, and continuity of payoffs with respect to that metric is a key assumption in the existence theorems for equilibrium.
- The sup distance between **policies** (decision rules) g, h , given by $\sup_x |g(x) - h(x)|$, formalizes the idea that “two economic policy rules are similar if they prescribe similar actions in all scenarios”.

Exam strategy

Typical questions on this topic and how to attack them:

1. “Prove that d is a metric”: check (M1), (M2), (M3) in that order, without skipping the part “ $d(x, y) = 0 \Rightarrow x = y$ ” (the one most often forgotten). For (M3), the almost universal trick is to insert the intermediate point and use the triangle inequality of $\mathbb{R}$ or of a norm, and if there are suprema, bound for each index and then pass to the supremum (template of Example 4.5).
2. “Is $d'(x, y) = \min\{1, d(x, y)\}$ (or $\frac{d}{1+d}$) a metric? Is it equivalent to d ?”: yes, it is a metric (argument by cases as in Example 4.16); it is topologically equivalent but in general not strongly equivalent (counterexample with far apart points).
3. “Prove that $x \mapsto d(x, z)$ is continuous”: use the reverse triangle inequality (Proposition 4.8) and take $\delta = \varepsilon$.
4. “Prove that this f is Lipschitz or a contraction”: if f is differentiable on an interval, bound $|f'|$ and use the mean value theorem as in Example 4.20.

Chapter summary

The minimum you should know by heart from this chapter:

- Axioms (M1) positivity and $d(x, y) = 0 \iff x = y$; (M2) symmetry; (M3) triangle inequality $d(x, z) \leq d(x, y) + d(y, z)$.
- Key examples: $|x - y|$ on $\mathbb{R}$; d_1, d_2, d_∞ on $\mathbb{R}^n$; the discrete metric; $d_\infty(f, g) =$

$\sup_t |f(t) - g(t)|$ on $C[a, b]$; subspaces.

- Reverse triangle inequality: $|d(x, z) - d(y, z)| \leq d(x, y)$.
- Balls: $B(c, r) = \{x : d(x, c) < r\}$; their shape depends on the metric. Bounded $\iff$ finite diameter.
- Strong equivalence ($\alpha d \leq d' \leq \beta d$) $\Rightarrow$ topological equivalence (same open sets); the converse is false ($\min\{1, d\}$).
- ε - δ continuity with metrics; isometry (preserves d), Lipschitz ($d(f(x), f(y)) \leq L d(x, y)$), contraction ($L < 1$); Lipschitz $\Rightarrow$ continuous with $\delta = \varepsilon/L$.

4.7 Exam-style exercises [P3]

Exercise 4.21. Let $X = \mathbb{R}$ and $d(x, y) = \sqrt{|x - y|}$. Prove that (X, d) is a metric space.

Solution. (M1): the square root of a nonnegative number is nonnegative, so $d(x, y) = \sqrt{|x - y|} \geq 0$. Moreover $\sqrt{|x - y|} = 0 \iff |x - y| = 0 \iff x = y$ (the square root vanishes only at zero, and the absolute value only when its argument is zero).

(M2): $|x - y| = |y - x|$ implies $\sqrt{|x - y|} = \sqrt{|y - x|}$.

(M3): we must prove $\sqrt{|x - z|} \leq \sqrt{|x - y|} + \sqrt{|y - z|}$. We will use the following auxiliary fact: for $a, b \geq 0$ we have $\sqrt{a + b} \leq \sqrt{a} + \sqrt{b}$. Proof of the fact: since both sides are nonnegative, it is enough to compare their squares (squaring preserves the order between nonnegative numbers):

$$(\sqrt{a + b})^2 = a + b \quad \text{and} \quad (\sqrt{a} + \sqrt{b})^2 = a + 2\sqrt{a}\sqrt{b} + b \geq a + b,$$

since $2\sqrt{a}\sqrt{b} \geq 0$. This proves the fact.

Now, with $a = |x - y|$ and $b = |y - z|$: by the usual triangle inequality, $|x - z| \leq |x - y| + |y - z| = a + b$. Since the square root is increasing:

$$\sqrt{|x - z|} \leq \sqrt{a + b} \leq \sqrt{a} + \sqrt{b} = \sqrt{|x - y|} + \sqrt{|y - z|},$$

using the auxiliary fact in the second inequality. This is (M3).

Conclusion: (X, d) is a metric space. (Comment: in general, if d is a metric and $\varphi : [0, \infty) \rightarrow [0, \infty)$ is increasing, subadditive and vanishes only at 0, then $\varphi \circ d$ is a metric; here $\varphi(t) = \sqrt{t}$.) $\square$

Exercise 4.22. Let (X, d) be a metric space. Prove that

$$d'(x, y) := \frac{d(x, y)}{1 + d(x, y)}$$

is a metric on X and that $d'(x, y) < 1$ for all x, y .

Solution. First the bound: since $d(x, y) \geq 0$, we have $d'(x, y) = \frac{d(x, y)}{1 + d(x, y)} < \frac{1 + d(x, y)}{1 + d(x, y)} = 1$ (the numerator is strictly smaller than the denominator because they differ by 1).

(M1): numerator ≥ 0 and denominator > 0 , so $d' \geq 0$. Moreover $d'(x, y) = 0 \iff d(x, y) = 0 \iff x = y$ (a fraction is zero only if its numerator is; and d is a metric).

(M2): immediate, since $d(x, y) = d(y, x)$ and the formula depends only on $d(x, y)$.

(M3): we use that the function $\varphi(t) = \frac{t}{1+t}$ is *increasing* on $[0, \infty)$: indeed, if $0 \leq s \leq t$,

$$\varphi(t) - \varphi(s) = \frac{t}{1+t} - \frac{s}{1+s} = \frac{t(1+s) - s(1+t)}{(1+t)(1+s)} = \frac{t-s}{(1+t)(1+s)} \geq 0,$$

since the numerator $t-s \geq 0$ and the denominator is positive.

Write $a = d(x, y)$, $b = d(y, z)$, $c = d(x, z)$. We know $c \leq a + b$ (triangle inequality for d). Then, by monotonicity of φ :

$$\begin{aligned} d'(x, z) = \varphi(c) &\leq \varphi(a+b) = \frac{a+b}{1+a+b} = \frac{a}{1+a+b} + \frac{b}{1+a+b} \\ &\leq \frac{a}{1+a} + \frac{b}{1+b} = d'(x, y) + d'(y, z). \end{aligned}$$

In the last step we used that enlarging the denominator shrinks the fraction: since $1+a+b \geq 1+a$ and $1+a+b \geq 1+b$ (because $a, b \geq 0$), we have $\frac{a}{1+a+b} \leq \frac{a}{1+a}$ and $\frac{b}{1+a+b} \leq \frac{b}{1+b}$. This completes (M3), and (X, d') is a *bounded* metric space (all distances are < 1), topologically equivalent to (X, d) by the same kind of argument as in the example of the chapter. $\square$

Exercise 4.23. In $C[0, 1]$ with the sup metric d_∞ , compute $d_\infty(f, g)$ for $f(t) = t$ and $g(t) = t^2$, and describe the ball $B(0, 1)$, where 0 denotes the identically zero function.

Solution. Computing the distance. By definition,

$$d_\infty(f, g) = \sup_{t \in [0, 1]} |t - t^2|.$$

For $t \in [0, 1]$ we have $t \geq t^2$ (since $t^2 \leq t \iff t(t-1) \leq 0$, which is true when $0 \leq t \leq 1$), so $|t - t^2| = t - t^2$. We maximize the function $h(t) = t - t^2$ on $[0, 1]$ with elementary calculus: $h'(t) = 1 - 2t = 0$ gives the critical point $t = 1/2$; since $h''(t) = -2 < 0$ it is a maximum. We compare with the endpoints of the interval: $h(0) = 0$, $h(1) = 0$, $h(1/2) = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$. The maximum is $1/4$, attained at $t = 1/2$. Therefore

$$d_\infty(f, g) = \frac{1}{4}.$$

Geometrically: the largest vertical gap between the line $y = t$ and the parabola $y = t^2$ over $[0, 1]$ occurs at $t = 1/2$ and equals $1/4$.

Describing the ball. By definition,

$$B(0, 1) = \{h \in C[0, 1] : d_\infty(h, 0) < 1\} = \left\{h \in C[0, 1] : \sup_{t \in [0, 1]} |h(t)| < 1\right\}.$$

It is the set of continuous functions whose graph stays strictly inside the horizontal band between the lines $y = -1$ and $y = 1$; more precisely, those for which there is $c < 1$ with $|h(t)| \leq c$ on all of $[0, 1]$ (the supremum, which is a maximum by continuity on the compact set $[0, 1]$, is smaller than 1). For example, $h(t) = 0.9 \sin(10t) \in B(0, 1)$, but $h(t) = t$ does not belong to the ball, since $\sup_t |t| = 1 \not< 1$. $\square$

Exercise 4.24. Let (X, d) be a metric space and fix $z \in X$. Prove, with ε - δ , that the function $\varphi : X \rightarrow \mathbb{R}$ given by $\varphi(x) = d(x, z)$ is continuous (on $\mathbb{R}$ we use the usual metric).

Solution. Fix arbitrary $x_0 \in X$ and $\varepsilon > 0$. We must find $\delta > 0$ such that

$$d(x, x_0) < \delta \implies |\varphi(x) - \varphi(x_0)| < \varepsilon.$$

We propose $\delta := \varepsilon$. Check: suppose $d(x, x_0) < \delta = \varepsilon$. By the reverse triangle inequality (Proposition 4.8, with the roles $x \rightarrow x$, $y \rightarrow x_0$, $z \rightarrow z$):

$$|\varphi(x) - \varphi(x_0)| = |d(x, z) - d(x_0, z)| \leq d(x, x_0) < \varepsilon.$$

That is exactly what we had to prove. Since x_0 and ε were arbitrary, φ is continuous on all of X . Note that $\delta = \varepsilon$ does not depend on the point: φ is even Lipschitz with constant 1 and uniformly continuous. $\square$

Exercise 4.25. Consider $f : [1, \infty) \rightarrow [1, \infty)$ given by $f(x) = \frac{x}{2} + \frac{1}{x}$.

1. Check that f really takes values in $[1, \infty)$.
2. Prove that f is a contraction on $[1, \infty)$ with the usual metric, finding an explicit Lipschitz constant $L < 1$.

Solution. (a) For $x \geq 1$, by the inequality between the arithmetic and geometric means (AM-GM: $\frac{a+b}{2} \geq \sqrt{ab}$ for $a, b \geq 0$) applied to $a = x/2$ and $b = 1/x$:

$$f(x) = \frac{x}{2} + \frac{1}{x} \geq 2\sqrt{\frac{x}{2} \cdot \frac{1}{x}} = 2\sqrt{\frac{1}{2}} = \sqrt{2} \geq 1.$$

So $f(x) \in [\sqrt{2}, \infty) \subseteq [1, \infty)$: the function is well defined as a map from $[1, \infty)$ into itself.

(b) Mean value method. The derivative is

$$f'(x) = \frac{1}{2} - \frac{1}{x^2}.$$

For $x \geq 1$ we have $0 < \frac{1}{x^2} \leq 1$, hence

$$-\frac{1}{2} = \frac{1}{2} - 1 \leq f'(x) < \frac{1}{2}.$$

Therefore $|f'(x)| \leq \frac{1}{2}$ for every $x \geq 1$. Now, given $x, y \in [1, \infty)$ with $x \neq y$, the mean value theorem (applicable because f is continuous on the closed interval with endpoints x, y and differentiable in its interior) gives a ξ strictly between x and y (in particular $\xi \geq 1$) such that

$$f(x) - f(y) = f'(\xi)(x - y).$$

Taking absolute values:

$$|f(x) - f(y)| = |f'(\xi)| |x - y| \leq \frac{1}{2} |x - y|.$$

(If $x = y$ the inequality is trivial, $0 \leq 0$.) So f is Lipschitz with constant $L = \frac{1}{2} < 1$: it is a contraction. (Preview: by the Banach fixed point theorem, which we will see later, f will have a unique fixed point in $[1, \infty)$; solving $x = \frac{x}{2} + \frac{1}{x}$ gives $x = \sqrt{2}$. This is exactly the Babylonian iteration for approximating $\sqrt{2}$.) $\square$

Chapter 5

Basic topology in metric spaces: open and closed sets [P2]

Week 4 of the course. Bibliography: Alabiso and Weiss [1], Chs. 3–4; Pons [3], Ch. 8; Ok [4], Ch. C.

What is this used for in Microeconomics?

Here lives the official definition of *continuous preferences* in MWG (Ch. 3): the upper contour sets $\{y : y \succeq x\}$ and the lower contour sets are *closed*. The sequential characterization of closed sets is exactly what you use to work with that assumption throughout Part I (choice and demand theory). The budget set $\{x : p \cdot x \leq w\}$ is closed (and bounded if $p \gg 0$), which sets up the existence of demand through compactness; the production set Y is assumed closed for the same reason (MWG Ch. 5). The interior versus boundary distinction is the distinction between *interior solutions*, where the first order conditions hold with equality, and *corner solutions* in the consumer and firm problems. It is a good idea to give priority to the section on closed sets.

5.1 Motivation [P2]

We already know how to measure distances in a metric space (X, d) . The next step is to classify the subsets of X according to how they relate to the notion of “closeness”. Two families of sets organize the whole theory:

- **Open sets:** sets where every point has “room to move”, that is, you can move a little in any direction without leaving the set.
- **Closed sets:** sets that “trap” their limits: if a sequence of points of the set converges, the limit stays inside.

These notions are the language in which the important theorems of the course are stated: completeness uses closed sets, compactness uses open and closed sets, continuity is characterized with open sets. Mastering this chapter is a necessary condition for everything that follows.

5.2 Open sets [P2]

Recall the open ball: $B(c, r) = \{x \in X : d(x, c) < r\}$.

Definition 5.1 (Open set). Let (X, d) be a metric space. A set $U \subseteq X$ is **open** if for every point $x \in U$ there is a radius $r > 0$ such that $B(x, r) \subseteq U$.

Intuition

Think of U as a territory and of each $x \in U$ as a person standing inside it. That U is open means that *nobody is stuck to the border*: every person has a complete bubble around them (perhaps a very small one) that stays inside the territory. Every point has its “room to move”: it can be perturbed in any direction by a distance smaller than r and stay in U . The radius r may depend on the point: points “near the edge” will have smaller bubbles. By contrast, in $[0, 1]$ the point 0 has no bubble: any move to the left, however small, takes it out of the set; that is why $[0, 1]$ is not open.

Example 5.2. In $\mathbb{R}$ with the usual metric: $(0, 1)$ is open (we prove this in general in a moment), $[0, 1]$ is not open (the point $0 \in [0, 1]$ admits no ball $B(0, r) = (-r, r) \subseteq [0, 1]$, since $-r/2 \in B(0, r)$ but $-r/2 \notin [0, 1]$). The sets $\emptyset$ and X are *always* open in any metric space: X because every ball is contained in X by definition; $\emptyset$ because the condition “for every $x \in \emptyset \dots$ ” holds vacuously (there are no points to check).

Proposition 5.3 (Every open ball is an open set). *In any metric space (X, d) , the ball $B(c, r)$ is an open set.*

Proof. We must check Definition 5.1: take an arbitrary point of the ball and build its bubble.

Step 1: choose the radius of the bubble. Let $x \in B(c, r)$ be arbitrary; by definition $d(x, c) < r$. The quantity

$$\rho := r - d(x, c) > 0$$

is positive precisely because $d(x, c) < r$. Geometrically, ρ is “what x still needs to reach the edge of the big ball”: if x is at distance $d(x, c)$ from the center and the ball has radius r , there is a margin of $r - d(x, c)$ up to the shell. That margin will be the radius of the bubble.

Step 2: prove $B(x, \rho) \subseteq B(c, r)$. Let $y \in B(x, \rho)$, that is, $d(y, x) < \rho$. We want to see that $y \in B(c, r)$, that is, $d(y, c) < r$. We insert the intermediate point x with the triangle inequality:

$$d(y, c) \leq d(y, x) + d(x, c) < \rho + d(x, c) = (r - d(x, c)) + d(x, c) = r.$$

In the strict inequality we used $d(y, x) < \rho$. We conclude $d(y, c) < r$, that is, $y \in B(c, r)$.

Since $x \in B(c, r)$ was arbitrary and we found $\rho > 0$ with $B(x, \rho) \subseteq B(c, r)$, the ball $B(c, r)$ is open. $\square$

Theorem 5.4 (Properties of open sets). *In a metric space (X, d) :*

1. $\emptyset$ and X are open.
2. The union of any family (finite or infinite, even uncountable) of open sets is open.

3. The intersection of a finite family of open sets is open.

Proof. (a) Already seen in the previous example.

(b) Let $\{U_i\}_{i \in I}$ be a family of open sets (the index set I is arbitrary) and let $U = \bigcup_{i \in I} U_i$. Take $x \in U$. By definition of union, there is *some* index $i_0 \in I$ with $x \in U_{i_0}$. Since U_{i_0} is open, there is $r > 0$ with $B(x, r) \subseteq U_{i_0}$. But $U_{i_0} \subseteq U$ (every member of the family is contained in the union), so $B(x, r) \subseteq U$. This proves that U is open. Note how cheap the argument is: it is enough that the point lies in *one* of the open sets.

(c) Let $U_1, \dots, U_m$ be open and let $V = U_1 \cap \dots \cap U_m$. If $V = \emptyset$, it is open by (a). If not, let $x \in V$. Then x belongs to *each* U_j , and since each U_j is open, for each $j = 1, \dots, m$ there is $r_j > 0$ with $B(x, r_j) \subseteq U_j$. We define

$$r := \min \{r_1, \dots, r_m\} > 0,$$

which is positive because it is the minimum of a *finite* number of positive numbers. Since $r \leq r_j$, we have $B(x, r) \subseteq B(x, r_j) \subseteq U_j$ for each j , hence $B(x, r) \subseteq \bigcap_{j=1}^m U_j = V$. Therefore V is open. $\square$

Example 5.5 (Finiteness in (c) is essential). In $\mathbb{R}$, consider the open sets $U_n = \left(-\frac{1}{n}, \frac{1}{n}\right)$ for $n \in \mathbb{N}$. We claim that

$$\bigcap_{n=1}^{\infty} U_n = \{0\},$$

which is *not* open (no ball $(-r, r)$ fits inside $\{0\}$).

Let us prove the equality of sets by double inclusion. ($\supseteq$) The point 0 is in every U_n since $-\frac{1}{n} < 0 < \frac{1}{n}$. ($\subseteq$) Let $x \in \bigcap_n U_n$, and suppose $x \neq 0$. Then $|x| > 0$, and by the Archimedean property of $\mathbb{R}$ (for every positive real there is a natural number whose reciprocal is smaller) there is $n_0 \in \mathbb{N}$ with $\frac{1}{n_0} < |x|$. But then $x \notin U_{n_0}$, a contradiction. Hence $x = 0$.

Where does the argument in the proof of (c) fail? In that $r = \inf \{r_1, r_2, r_3, \dots\}$ of infinitely many positive radii can be 0: the room to move shrinks to zero and no common bubble is left.

5.3 Closed sets [P1]

Definition 5.6 (Closed set). A set $F \subseteq X$ is **closed** if its complement $X \setminus F$ is open.

This definition is convenient but not very intuitive. The real “personality” of closed sets is given by the following characterization through sequences, which in practice is the one most used.

Theorem 5.7 (Sequential characterization of closed sets). *Let (X, d) be a metric space and $F \subseteq X$. The following are equivalent:*

1. F is closed.
2. For every sequence (x_k) with $x_k \in F$ for all k , if $x_k \rightarrow x$ in X , then $x \in F$.

That is: F is closed if and only if it contains the limits of all its convergent sequences.

Proof. (a) $\Rightarrow$ (b). Suppose F is closed, that is, $X \setminus F$ is open. Let (x_k) be a sequence of points of F with $x_k \rightarrow x$; we want to prove $x \in F$. We argue by contradiction: suppose $x \notin F$, that is, $x \in X \setminus F$. Since $X \setminus F$ is open, there is $r > 0$ such that

$$B(x, r) \subseteq X \setminus F.$$

On the other hand, $x_k \rightarrow x$ means: for every $\varepsilon > 0$ there is K such that $d(x_k, x) < \varepsilon$ for every $k \geq K$. Applying this with $\varepsilon = r$, there is K such that $x_k \in B(x, r)$ for every $k \geq K$. But then $x_K \in B(x, r) \subseteq X \setminus F$, that is, $x_K \notin F$. This contradicts that *all* terms of the sequence are in F . The contradiction proves $x \in F$.

(b) $\Rightarrow$ (a). Suppose F contains the limits of all its convergent sequences; we will prove that $X \setminus F$ is open. Again by contradiction: suppose $X \setminus F$ is *not* open. Then there is a point $x \in X \setminus F$ with no room to move: for *no* $r > 0$ does $B(x, r) \subseteq X \setminus F$ hold. In other words, *every* ball centered at x touches F :

$$\text{for every } r > 0 : \quad B(x, r) \cap F \neq \emptyset.$$

We use this with the radii $r = \frac{1}{k}$, $k = 1, 2, 3, \dots$: for each k we can choose a point

$$x_k \in B(x, \frac{1}{k}) \cap F.$$

This gives a sequence (x_k) of points of F with $d(x_k, x) < \frac{1}{k} \rightarrow 0$, so that $x_k \rightarrow x$ (given $\varepsilon > 0$, taking $K > 1/\varepsilon$ we get $d(x_k, x) < 1/k \leq 1/K < \varepsilon$ for $k \geq K$). By assumption (b), the limit x must be in F . But we had taken $x \in X \setminus F$: a contradiction. Hence $X \setminus F$ is open and F is closed. $\square$

Intuition

A closed set is a set “with no leaks”: approaching through points of the set you never escape from it in the limit. The interval $(0, 1]$ is *not* closed in $\mathbb{R}$ because the sequence $x_k = \frac{1}{k}$ lives in the set but its limit 0 leaks out. The technique in the proof of (b) $\Rightarrow$ (a), choosing a point in each ball of radius $1/k$ to build a convergent sequence, is one of the most reused tricks in analysis: memorize it as a tool.

Example 5.8 (The full catalogue of possibilities). In $\mathbb{R}$ with the usual metric:

- $[0, 1]$ is closed and not open. (Closed: if $x_k \in [0, 1]$ and $x_k \rightarrow x$, the inequalities $0 \leq x_k \leq 1$ are preserved in the limit, so $0 \leq x \leq 1$.)
- $(0, 1)$ is open and not closed ($\frac{1}{k} \rightarrow 0 \notin (0, 1)$).
- $[0, 1]$ is **neither open nor closed**: it is not open because the point 0 has no bubble inside; it is not closed because the sequence $x_k = 1 - \frac{1}{k} \in [0, 1]$ converges to $1 \notin [0, 1]$. This example destroys the false dichotomy “either open or closed”.
- $\emptyset$ and $\mathbb{R}$ are **open and closed at the same time** (clopen): each one is the complement of the other and both are open.

- In a space with the **discrete metric**, *every* subset is open (for any $x \in A$, the ball $B(x, \frac{1}{2}) = \{x\} \subseteq A$) and therefore every subset is also closed (its complement, being a subset, is open). Everything is clopen.

Careful (common mistake)

Two classical exam mistakes:

1. **“It is not closed, so it is open”**: FALSE. Open and closed are not logical opposites; they are independent properties. There are sets with neither property ($[0, 1)$ in $\mathbb{R}$) and with both ($\emptyset, X$). The correct statement is: F closed $\iff X \setminus F$ open.
2. **Forgetting the ambient space**: being open or closed is relative to the whole space (X, d) . For example, $(0, 1]$ is not closed in $\mathbb{R}$, but *it is closed in the metric space* $X = (0, \infty)$ (with the inherited metric): the sequence $1/k$ converges to 0, but 0 is not even a point of X , so it does not count as a “leak”. Always ask: closed *in which space*?

By De Morgan’s laws (the complement of a union is the intersection of the complements and vice versa), Theorem 5.4 translates immediately to closed sets:

Corollary 5.9 (Properties of closed sets). 1. $\emptyset$ and X are closed.

2. The intersection of any family of closed sets is closed.

3. The finite union of closed sets is closed.

Proof. (a) $X \setminus \emptyset = X$ and $X \setminus X = \emptyset$ are open. (b) If each F_i is closed, then $X \setminus \bigcap_i F_i = \bigcup_i (X \setminus F_i)$ (De Morgan) is an arbitrary union of open sets, hence open; so $\bigcap_i F_i$ is closed. (c) $X \setminus \bigcup_{j=1}^m F_j = \bigcap_{j=1}^m (X \setminus F_j)$ is a finite intersection of open sets, hence open. $\square$

5.4 Interior, closure, boundary and accumulation points [P3]

Definition 5.10. Let (X, d) be a metric space and $A \subseteq X$.

1. The **interior** of A is

$$\text{int } A := \{x \in A : \exists r > 0 \text{ with } B(x, r) \subseteq A\},$$

the set of points of A that have room to move (“interior points”).

2. The **closure** of A is

$$\text{cl } A := \{x \in X : \forall r > 0, B(x, r) \cap A \neq \emptyset\},$$

the set of points “adherent” to A : every ball around them, however small, touches A . (It is also denoted $\overline{A}$.)

3. The **boundary** of A is

$$\partial A := \text{cl } A \setminus \text{int } A.$$

Equivalently: $x \in \partial A$ if every ball $B(x, r)$ contains points of A and points of $X \setminus A$.

4. A point $x \in X$ is an **accumulation point** (also called a limit point) of A if every ball $B(x, r)$ contains some point of A different from x : $B(x, r) \cap (A \setminus \{x\}) \neq \emptyset$ for every $r > 0$. The set of accumulation points is denoted A' .
5. A point $a \in A$ is an **isolated point** of A if it is not an accumulation point: there is $r > 0$ with $B(a, r) \cap A = \{a\}$ (a ball that touches A only at a).

Intuition

Four “zones” associated with a set A :

- $\text{int } A$: the deep part of A , well inside.
- $\text{cl } A$: A together with all the points stuck to it (its limits).
- ∂A : the “shell”: points that see both A and its complement at any distance.
- A' : points that have companions from A arbitrarily close (the point itself does not count). An isolated point of A belongs to A but *not* to A' .

The inclusions $\text{int } A \subseteq A \subseteq \text{cl } A$ always hold, and $\text{cl } A = A \cup A'$.

Remark 5.11 (Closure and sequences). $x \in \text{cl } A$ if and only if there is a sequence (a_k) of points of A with $a_k \rightarrow x$. The proof is the same trick with balls of radius $1/k$: if $x \in \text{cl } A$, choose $a_k \in B(x, 1/k) \cap A$ and you get $a_k \rightarrow x$; conversely, if $a_k \rightarrow x$ with $a_k \in A$, every ball $B(x, r)$ contains terms a_k (those with large index), so it meets A .

Theorem 5.12 (Structural properties). *Let $A \subseteq X$. Then:*

1. $\text{int } A$ is open, and it is the largest open set contained in A : if U is open and $U \subseteq A$, then $U \subseteq \text{int } A$.
2. $\text{cl } A$ is closed, and it is the smallest closed set containing A : if F is closed and $A \subseteq F$, then $\text{cl } A \subseteq F$.
3. A is open $\iff A = \text{int } A$; A is closed $\iff A = \text{cl } A$.

Proof. (a) $\text{int } A$ is open: let $x \in \text{int } A$; there is $r > 0$ with $B(x, r) \subseteq A$. We claim something stronger: $B(x, r) \subseteq \text{int } A$. Indeed, let $y \in B(x, r)$; since $B(x, r)$ is open (Proposition 5.3) there is $\rho > 0$ with $B(y, \rho) \subseteq B(x, r) \subseteq A$, which says exactly that y is an interior point of A , that is, $y \in \text{int } A$. So every point of $\text{int } A$ has a ball contained in $\text{int } A$: it is open.

Maximality: let U be open with $U \subseteq A$, and let $x \in U$. Since U is open there is $r > 0$ with $B(x, r) \subseteq U \subseteq A$; then x satisfies the definition of interior point of A , that is, $x \in \text{int } A$. Hence $U \subseteq \text{int } A$.

(b) $\text{cl } A$ is closed: we will prove that $X \setminus \text{cl } A$ is open. Let $x \notin \text{cl } A$. Negating the definition of closure: there is $r > 0$ such that $B(x, r) \cap A = \emptyset$. We claim that even $B(x, r/2) \subseteq X \setminus \text{cl } A$ (this is the subtle part). Let $y \in B(x, r/2)$; to see that $y \notin \text{cl } A$ it is enough to exhibit a ball around y that does not touch A . We take $B(y, r/2)$: if $z \in B(y, r/2)$, then

$$d(z, x) \leq d(z, y) + d(y, x) < \frac{r}{2} + \frac{r}{2} = r,$$

so $z \in B(x, r)$, which is disjoint from A ; hence $B(y, r/2) \cap A = \emptyset$ and $y \notin \text{cl } A$. This proves that $X \setminus \text{cl } A$ is open.

Minimality: let F be closed with $A \subseteq F$, and let $x \in \text{cl } A$. By Remark 5.11, there is a sequence (a_k) in A (hence in F) with $a_k \rightarrow x$. Since F is closed, by the sequential characterization (Theorem 5.7) the limit x belongs to F . So $\text{cl } A \subseteq F$.

(c) If $A = \text{int } A$, then A is open by (a). If A is open, then A is an open set contained in A , so $A \subseteq \text{int } A$ by maximality; the other inclusion $\text{int } A \subseteq A$ holds by definition. The argument for closed sets is symmetric using (b): if A is closed, it is a closed set containing A , so $\text{cl } A \subseteq A$ by minimality, and always $A \subseteq \text{cl } A$. $\square$

Example 5.13 (Full computation: $\mathbb{Q}$ inside $\mathbb{R}$). We work in $X = \mathbb{R}$ with the usual metric and $A = \mathbb{Q}$.

- $\text{int } \mathbb{Q} = \emptyset$: let $q \in \mathbb{Q}$ and let $r > 0$ be arbitrary. The ball $B(q, r) = (q - r, q + r)$ contains irrational numbers (between two different reals there is always an irrational, by density of the irrationals), so $B(q, r) \not\subseteq \mathbb{Q}$. No rational is an interior point.
- $\text{cl } \mathbb{Q} = \mathbb{R}$: let $x \in \mathbb{R}$ and $r > 0$. The interval $(x - r, x + r)$ contains rationals (density of $\mathbb{Q}$ in $\mathbb{R}$), so $B(x, r) \cap \mathbb{Q} \neq \emptyset$ for every r : every real is in the closure.
- $\partial \mathbb{Q} = \text{cl } \mathbb{Q} \setminus \text{int } \mathbb{Q} = \mathbb{R} \setminus \emptyset = \mathbb{R}$: the boundary of $\mathbb{Q}$ is the whole line! A set can have a huge boundary.
- $\mathbb{Q}' = \mathbb{R}$ (every real is an accumulation point of $\mathbb{Q}$: the ball $(x - r, x + r)$ contains infinitely many rationals, in particular one different from x) and $\mathbb{Q}$ has no isolated points.

Example 5.14 (Full computation: $(0, 1]$ in $\mathbb{R}$). With $A = (0, 1] \subseteq \mathbb{R}$:

- $\text{int } A = (0, 1)$: each $x \in (0, 1)$ has the ball $B(x, r)$ with $r = \min\{x, 1 - x\} > 0$ contained in $(0, 1) \subseteq A$. The point $1 \in A$ is not interior: every ball $(1 - r, 1 + r)$ contains points larger than 1, which are not in A .
- $\text{cl } A = [0, 1]$: the points of $[0, 1]$ are limits of sequences from A ($0 = \lim 1/k$ with $1/k \in A$; each $x \in (0, 1]$ is the limit of the constant sequence x). No point outside $[0, 1]$ is adherent: if $x > 1$, the ball of radius $r = x - 1$ does not touch A ; if $x < 0$, neither does the one of radius $|x|$.
- $\partial A = [0, 1] \setminus (0, 1) = \{0, 1\}$: the two endpoints.
- $A' = [0, 1]$ and A has no isolated points.

Example 5.15 (Open ball in $\mathbb{R}^2$). Let $A = B(0, 1) = \{x \in \mathbb{R}^2 : \|x\|_2 < 1\}$ be the open unit disk in $(\mathbb{R}^2, d_2)$. Then:

$$\text{int } A = A \quad (A \text{ is open}), \quad \text{cl } A = \overline{B}(0, 1) = \{x : \|x\|_2 \leq 1\}, \quad \partial A = \{x : \|x\|_2 = 1\},$$

the unit circle. Let us check $\text{cl } A$: if $\|x\|_2 = 1$, the points $x_k = (1 - \frac{1}{k})x$ satisfy $\|x_k\|_2 = 1 - \frac{1}{k} < 1$ (they are in A) and $x_k \rightarrow x$ since $\|x_k - x\|_2 = \frac{1}{k} \|x\|_2 = \frac{1}{k} \rightarrow 0$; so the sphere is adherent to the disk. If $\|x\|_2 > 1$, the ball of radius $\|x\|_2 - 1$ around x does not meet A (by the reverse triangle inequality). **A subtle warning:** in a general metric space, $\text{cl } B(c, r)$ can be *strictly smaller*

than the closed ball $\overline{B}(c, r)$. Example: in the discrete metric, $B(c, 1) = \{c\}$, which is already closed, so $\text{cl } B(c, 1) = \{c\}$; but $\overline{B}(c, 1) = X$, the whole space. In $\mathbb{R}^n$ with norms they do agree.

5.5 Continuity through open sets [P2]

Continuity, which we defined with ε - δ , has a purely topological characterization: it mentions no numbers, only open sets.

Theorem 5.16 (Topological characterization of continuity). *Let (X, d_X) , (Y, d_Y) be metric spaces and $f : X \rightarrow Y$. The following are equivalent:*

1. *f is continuous at every point of X (ε - δ definition).*
2. *For every open set $V \subseteq Y$, the preimage $f^{-1}(V) := \{x \in X : f(x) \in V\}$ is open in X .*

Proof. (a) $\Rightarrow$ (b). Let $V \subseteq Y$ be open; we will prove that $f^{-1}(V)$ is open. If $f^{-1}(V) = \emptyset$, it is open and we are done. If not, let $x_0 \in f^{-1}(V)$, that is, $f(x_0) \in V$. Since V is open, there is $\varepsilon > 0$ such that

$$B_Y(f(x_0), \varepsilon) \subseteq V.$$

By continuity of f at x_0 applied to *this* ε , there is $\delta > 0$ such that

$$d_X(x, x_0) < \delta \implies d_Y(f(x), f(x_0)) < \varepsilon.$$

In the language of sets, this says: $x \in B_X(x_0, \delta) \implies f(x) \in B_Y(f(x_0), \varepsilon) \subseteq V$, that is, $x \in f^{-1}(V)$. Therefore $B_X(x_0, \delta) \subseteq f^{-1}(V)$: the point x_0 has its bubble inside the preimage. Since x_0 was arbitrary, $f^{-1}(V)$ is open.

(b) $\Rightarrow$ (a). Let $x_0 \in X$ and $\varepsilon > 0$; we must produce δ . Consider the set

$$V := B_Y(f(x_0), \varepsilon),$$

which is open in Y (balls are open). By assumption, $f^{-1}(V)$ is open in X . Moreover $x_0 \in f^{-1}(V)$, because $f(x_0) \in V$ (it is at distance $0 < \varepsilon$ from itself). By the definition of open set applied to the point $x_0 \in f^{-1}(V)$, there is $\delta > 0$ with $B_X(x_0, \delta) \subseteq f^{-1}(V)$. Translation: if $d_X(x, x_0) < \delta$, then $x \in f^{-1}(V)$, that is, $f(x) \in V$, that is, $d_Y(f(x), f(x_0)) < \varepsilon$. That is exactly continuity at x_0 . $\square$

Remark 5.17. Taking complements (and using $f^{-1}(Y \setminus B) = X \setminus f^{-1}(B)$), condition (b) is equivalent to: *the preimage of every closed set is closed*. Careful with the direction! It is the *preimage*, not the image. The direct image of an open set under a continuous function may fail to be open (example: $f(x) = x^2$ sends the open set $(-1, 1)$ to the non open set $[0, 1)$), and the image of a closed set may fail to be closed (example: $f(x) = \arctan x$ sends the closed set $\mathbb{R}$ to the non closed set $(-\pi/2, \pi/2)$).

Exam strategy

Star application of the topological characterization: *deciding quickly whether a set is open or closed without using ε - δ* . If $g : X \rightarrow \mathbb{R}$ is continuous, then:

$$\{x : g(x) > c\} = g^{-1}((c, \infty)) \text{ is open,} \quad \{x : g(x) \geq c\} = g^{-1}([c, \infty)) \text{ is closed,}$$

$$\{x : g(x) = c\} = g^{-1}(\{c\}) \text{ is closed.}$$

For example, in $\mathbb{R}^2$, the set $\{(x, y) : x^2 + y^2 \leq 1, x + y \geq 0\}$ is closed because it is the intersection of two preimages of closed sets under continuous functions. In an exam this technique saves pages: cite the theorem, exhibit the continuous function and the model closed or open set, and conclude.

Link with economics

- **Budget sets:** the set $B(p, w) = \{x \in \mathbb{R}_+^n : \langle p, x \rangle \leq w\}$ (bundles affordable at prices p with wealth w) is *closed*, because it is the intersection of the preimages of the closed sets $\{x : p \cdot x \leq w\}$ and $\{x : x_i \geq 0\}$ under continuous (linear) functions. Being closed (and bounded if $p \gg 0$, hence compact) is exactly what will guarantee that the consumer problem has a solution (Weierstrass theorem, compactness chapter).
- **Strict inequalities and robustness:** sets defined by strict inequalities ($u(x) > u(y)$: “ x is strictly preferred to y ”) are open when u is continuous. Economic reading: a strict preference is *robust to small perturbations* of the bundle; if x is strictly preferred, so is any bundle close enough to x . This “stability of strict inequalities” is the economic translation of the notion of open set.

Chapter summary

- U open: every $x \in U$ has $B(x, r) \subseteq U$. Every open ball is open (radius $\rho = r - d(x, c)$).
- Open sets: arbitrary unions and *finite* intersections (infinite counterexample: $\bigcap(-\frac{1}{n}, \frac{1}{n}) = \{0\}$).
- F closed $\iff X \setminus F$ open $\iff F$ contains the limits of its convergent sequences (sequential characterization: the one most used in practice).
- There are sets that are neither open nor closed ($[0, 1)$) and clopen sets ($\emptyset, X$; every set in the discrete metric).
- $\text{int } A = \text{largest open set } \subseteq A$; $\text{cl } A = \text{smallest closed set } \supseteq A$; $\partial A = \text{cl } A \setminus \text{int } A$; $\text{cl } A = A \cup A'$; $x \in \text{cl } A \iff$ some sequence from A converges to x .
- Model computations: $\text{int } \mathbb{Q} = \emptyset$, $\text{cl } \mathbb{Q} = \mathbb{R}$, $\partial \mathbb{Q} = \mathbb{R}$.
- f continuous $\iff$ the preimage of every open set is open $\iff$ the preimage of every closed set is closed. With g continuous: $\{g > c\}$ open, $\{g \geq c\}$ and $\{g = c\}$

closed.

5.6 Exam-style exercises [P2]

Exercise 5.18. Prove directly (with the definition) that the set $U = \{(x, y) \in \mathbb{R}^2 : y > 0\}$ (upper half plane) is open in $(\mathbb{R}^2, d_2)$, and that $F = \{(x, y) \in \mathbb{R}^2 : y \geq 0\}$ is closed.

Solution. U is open. Let $p = (x_0, y_0) \in U$, that is, $y_0 > 0$. We propose the radius $r := y_0 > 0$ and prove $B(p, r) \subseteq U$. Let $q = (x, y) \in B(p, r)$; then

$$|y - y_0| \leq \sqrt{(x - x_0)^2 + (y - y_0)^2} = d_2(q, p) < r = y_0,$$

where the first inequality says that one coordinate never exceeds the Euclidean norm of the vector. From $|y - y_0| < y_0$ we get $-y_0 < y - y_0 < y_0$, and from the left inequality: $y > 0$. Hence $q \in U$, and $B(p, y_0) \subseteq U$: U is open. (Intuition: the room to move of a point is exactly its height above the horizontal axis.)

F is closed. We use the sequential characterization. Let $(p_k) = (x_k, y_k)$ be a sequence in F (that is, $y_k \geq 0$ for all k) with $p_k \rightarrow p = (x, y)$ in $\mathbb{R}^2$. Convergence in $\mathbb{R}^2$ implies coordinatewise convergence: $y_k \rightarrow y$ (since $|y_k - y| \leq d_2(p_k, p) \rightarrow 0$). Non strict inequalities are preserved in the limit: if $y_k \geq 0$ for every k and $y_k \rightarrow y$, then $y \geq 0$ (if we had $y < 0$, taking $\varepsilon = |y|/2$ there would be k with $y_k < y + \varepsilon = y/2 < 0$, a contradiction). Hence $p \in F$, and F contains the limits of its sequences: it is closed. Express alternative: $F = g^{-1}([0, \infty))$ with $g(x, y) = y$ continuous, the preimage of a closed set. $\square$

Exercise 5.19. Let (X, d) be a metric space and $A \subseteq X$. Prove the identities:

$$X \setminus \text{cl } A = \text{int}(X \setminus A) \quad \text{and} \quad X \setminus \text{int } A = \text{cl}(X \setminus A).$$

Solution. First identity. We prove the equivalence of memberships point by point. For $x \in X$:

$$\begin{aligned} x \in X \setminus \text{cl } A &\iff x \notin \text{cl } A \\ &\iff \text{not every ball } B(x, r) \text{ meets } A \\ &\iff \exists r > 0 \text{ with } B(x, r) \cap A = \emptyset \\ &\iff \exists r > 0 \text{ with } B(x, r) \subseteq X \setminus A \\ &\iff x \in \text{int}(X \setminus A). \end{aligned}$$

Justifications: the second equivalence is the negation of the definition of closure (negating “for every r ” produces “there is r ” with the negated property); the fourth uses that “not touching A ” is the same as “being contained in the complement of A ”; the last one is the definition of interior applied to the set $X \setminus A$.

Second identity. It follows from the first one applied to the set $B := X \setminus A$ (whose complement is A):

$$X \setminus \text{cl}(X \setminus A) = \text{int}(X \setminus (X \setminus A)) = \text{int } A.$$

Taking complements on both sides (if $X \setminus P = Q$ then $P = X \setminus Q$):

$$\text{cl}(X \setminus A) = X \setminus \text{int } A,$$

which is what we wanted. Mnemonic: “the complement swaps interior and closure”. $\square$

Exercise 5.20. Compute, justifying each step, the interior, the closure, the boundary and the set of accumulation points of

$$A = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \dots \right\} \subseteq \mathbb{R}.$$

Does A have isolated points? Is A open, closed, neither?

Solution. Interior. $\text{int } A = \emptyset$: any ball $(\frac{1}{n} - r, \frac{1}{n} + r)$ around a point of A is an interval, which contains infinitely many reals, in particular irrationals and rationals that are not of the form $1/m$; so no ball fits inside the countable set A .

Accumulation points. We claim $A' = \{0\}$. *0 is an accumulation point:* given any ball $(-r, r)$, by the Archimedean property there is n with $\frac{1}{n} < r$, and $\frac{1}{n} \in A$, $\frac{1}{n} \neq 0$: the ball contains points of A different from 0. *No other point is:* (i) if $x = \frac{1}{n} \in A$, the distance from x to its nearest neighbor in A is $\min \left\{ \frac{1}{n} - \frac{1}{n+1}, \frac{1}{n-1} - \frac{1}{n} \right\}$ (only the first term if $n = 1$), which is a positive number ρ_n ; the ball $B(x, \rho_n)$ contains no points of A other than x , so x is *isolated*, not an accumulation point. (ii) If $x \notin A \cup \{0\}$, we argue directly: if $x < 0$, the ball of radius $|x|$ does not touch A ; if $x > 1$, neither does the one of radius $x - 1$; if $0 < x \leq 1$ and $x \notin A$, then x falls between two consecutive terms, $\frac{1}{n+1} < x < \frac{1}{n}$ for some n , and the ball of radius $\min \left\{ x - \frac{1}{n+1}, \frac{1}{n} - x \right\} > 0$ does not meet A .

Closure and boundary. $\text{cl } A = A \cup A' = A \cup \{0\}$. Since $\text{int } A = \emptyset$: $\partial A = \text{cl } A \setminus \text{int } A = A \cup \{0\}$.

Conclusions. All points of A are isolated. A is not open ($\text{int } A = \emptyset \neq A$). A is not closed: the sequence $\frac{1}{k} \in A$ converges to $0 \notin A$ (a leak). On the other hand, $A \cup \{0\}$ is closed (it is its own closure). $\square$

Exercise 5.21. Let $f : X \rightarrow \mathbb{R}$ be continuous (X a metric space). Prove using Theorem 5.16 that the set $Z = \{x \in X : f(x) = 0\}$ (the zero set of f) is closed. Deduce that if two continuous functions $f, g : X \rightarrow \mathbb{R}$ agree on a dense set D (that is, $\text{cl } D = X$), then $f = g$ on all of X .

Solution. Z is closed. $\{0\}$ is closed in $\mathbb{R}$ (its complement $(-\infty, 0) \cup (0, \infty)$ is a union of two open sets, hence open). Since f is continuous, the preimage of every closed set is closed (complementary form of Theorem 5.16); therefore $Z = f^{-1}(\{0\})$ is closed in X .

Agreement on a dense set. Define $h := f - g$, which is continuous (difference of continuous functions; proved with the limit laws). Its zero set

$$Z_h = \{x : f(x) = g(x)\}$$

is closed by the first part. By assumption $D \subseteq Z_h$ (they agree on D). Applying minimality of

the closure (the smallest closed set containing D , Theorem 5.12(b)):

$$X = \text{cl } D \subseteq Z_h \subseteq X,$$

so $Z_h = X$: the functions agree at every point. Reading: a continuous function is completely determined by its values on a dense set (for example, on $\mathbb{Q}$ inside $\mathbb{R}$). This is used all the time: to check an identity between continuous functions it is enough to check it on the rationals. $\square$

Exercise 5.22. Decide, with full justification, whether each set is open, closed, both or neither, in the indicated space:

1. $A = \{(x, y) \in \mathbb{R}^2 : xy > 1\}$ in $(\mathbb{R}^2, d_2)$.
2. $B = \{f \in C[0, 1] : f(0) = 0\}$ in $(C[0, 1], d_\infty)$.
3. $C = [0, 1)$ in the metric space $X = [0, \infty)$ with the metric inherited from $\mathbb{R}$.

Solution. (a) The function $g(x, y) = xy$ is continuous on $\mathbb{R}^2$ (a product of projections, which are continuous). Then $A = g^{-1}((1, \infty))$ is the preimage of an open set under a continuous function: *open*. It is not closed: take $p_k = (1 + \frac{1}{k}, 1) \in A$ (since $(1 + \frac{1}{k}) \cdot 1 = 1 + \frac{1}{k} > 1$); then $p_k \rightarrow (1, 1)$ and $g(1, 1) = 1 \not> 1$, so the limit $(1, 1) \notin A$: there is a leak, A is not closed. Conclusion: open and not closed.

(b) The evaluation map $E : C[0, 1] \rightarrow \mathbb{R}$, $E(f) = f(0)$, is continuous with respect to d_∞ : it is even Lipschitz with constant 1, since

$$|E(f) - E(g)| = |f(0) - g(0)| \leq \sup_{t \in [0, 1]} |f(t) - g(t)| = d_\infty(f, g).$$

Then $B = E^{-1}(\{0\})$ is the preimage of the closed set $\{0\}$ under a continuous function: *closed*. It is not open: given any $f \in B$ and any $r > 0$, the function $g = f + \frac{r}{2}$ (adding the constant $r/2$) satisfies $d_\infty(f, g) = \frac{r}{2} < r$, but $g(0) = \frac{r}{2} \neq 0$, that is, $g \in B(f, r) \setminus B$: no ball around f stays inside B . Conclusion: closed and not open.

(c) Watch out for the ambient space $X = [0, \infty)$! C is open in X : for $x \in [0, 1)$ take $r = 1 - x > 0$. The ball in X is $B_X(x, r) = \{y \geq 0 : |y - x| < r\} = [0, \infty) \cap (x - r, x + r)$. Every y in it satisfies $y < x + r = 1$ and $y \geq 0$, hence $y \in [0, 1) = C$. (The point $x = 0$ causes no trouble: the ball in X is $[0, r)$, contained in C ; the “neighbors on the left” of 0 do not exist in X .) C is not closed in X : the sequence $x_k = 1 - \frac{1}{k} \in C$ converges to $1 \in X$, but $1 \notin C$: a leak inside the ambient space. Conclusion: open in X and not closed in X . (Compare: in $\mathbb{R}$, $[0, 1)$ is neither open nor closed. The lesson of the exercise is that both properties depend on the ambient space.) $\square$

Chapter 6

Cauchy sequences, completeness and density [P4]

Week 5 of the course. Bibliography: Alabiso and Weiss [1], Ch. 4; Moroşanu [2], Ch. 1; Pons [3], Chs. 2 and 8.

What is this used for in Microeconomics?

To be honest: completeness almost never shows up explicitly in the static part of Microeconomics, which is most of the course. It is, however, the invisible foundation of everything else: the fact that $\mathbb{R}$ is complete is what makes Bolzano–Weierstrass work, and with it the Weierstrass theorem on existence of optima, which are [P1] for micro. Where it does appear directly is in *repeated games* with discounting (Part II, MWG ch. 9): the payoff of a strategy is the sum of a geometric series of discounted payoffs, and that series converges because its partial sums form a Cauchy sequence and $\mathbb{R}$ is complete. Beyond this course, the chapter becomes central in dynamic macroeconomics and in recursive models, where value function iteration converges precisely because the space of functions is complete.

6.1 Motivation [P4]

So far, to know whether a sequence converges we need to *know its limit*: the definition of $x_k \rightarrow x$ mentions x explicitly. In practice the situation is the opposite: we build a sequence of approximations (iterations of an algorithm, partial sums of a series, iterations of the Bellman operator) and we want to guarantee that it converges *without knowing in advance what the limit is*.

The tool for this is the notion of a Cauchy sequence: an *internal* convergence condition that only involves the terms of the sequence among themselves. The central question of the chapter is: in which spaces does “being Cauchy” guarantee convergence? Those spaces are called *complete*, and they are the setting where fixed point theorems work, and with them half of dynamic economic theory.

6.2 Convergence in metric spaces: quick review [P4]

Definition 6.1 (Convergence). Let (X, d) be a metric space. A sequence (x_k) in X **converges** to the point $x \in X$ (written $x_k \rightarrow x$ or $\lim_k x_k = x$) if

$$\forall \varepsilon > 0 \exists K \in \mathbb{N} \text{ such that } k \geq K \implies d(x_k, x) < \varepsilon.$$

Equivalently: the sequence of real numbers $d(x_k, x)$ converges to 0.

Proposition 6.2 (Uniqueness of the limit). *In a metric space, the limit of a convergent sequence is unique.*

Proof. Suppose $x_k \rightarrow x$ and $x_k \rightarrow y$. For each k , the triangle inequality with intermediate point x_k gives

$$0 \leq d(x, y) \leq d(x, x_k) + d(x_k, y).$$

The right hand side tends to $0 + 0 = 0$ as $k \rightarrow \infty$. A fixed number $d(x, y)$ trapped between 0 and quantities that tend to 0 must be 0 (if we had $d(x, y) = c > 0$, then for large k we would get $c \leq d(x, x_k) + d(x_k, y) < c/2 + c/2 = c$, which is absurd). By (M1), $d(x, y) = 0 \implies x = y$. $\square$

6.3 Cauchy sequences [P4]

Definition 6.3 (Cauchy sequence). A sequence (x_k) in a metric space (X, d) is a **Cauchy sequence** if

$$\forall \varepsilon > 0 \exists K \in \mathbb{N} \text{ such that } m, k \geq K \implies d(x_m, x_k) < \varepsilon.$$

Intuition

Compare the two definitions carefully:

- *Convergence*: after some point, all terms are close *to the limit* x .
- *Cauchy*: after some point, all terms are close *to each other* (they bunch together).

The Cauchy condition does not mention any limit: you can check it by looking at the sequence alone. It is the formal version of “the sequence is settling down”. The big question is whether that bunching guarantees that there is a point of the space they converge to. The answer depends on the space: in $\mathbb{R}$ yes (completeness of $\mathbb{R}$), in $\mathbb{Q}$ or in $(0, 1]$ no, because the point they bunch around may be a hole that the space does not contain.

Theorem 6.4 (Convergent $\implies$ Cauchy). *Every convergent sequence is Cauchy.*

Proof. Let $x_k \rightarrow x$ and let $\varepsilon > 0$ be given. We apply the definition of convergence with the number $\varepsilon/2$ (the reason for this choice is clear at the end): there is K such that

$$k \geq K \implies d(x_k, x) < \frac{\varepsilon}{2}.$$

Now take any two indices $m, k \geq K$. Inserting the limit x as an intermediate point (triangle

inequality):

$$d(x_m, x_k) \leq d(x_m, x) + d(x, x_k) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

(We used symmetry (M2) to write $d(x, x_k) = d(x_k, x)$.) This is exactly the Cauchy condition. The “ $\varepsilon/2$ ” trick is standard: since the argument was going to add two small distances, we asked each of them to be less than half of the target. $\square$

Proposition 6.5 (Cauchy $\Rightarrow$ bounded). *Every Cauchy sequence is bounded (its set of terms is a bounded set).*

Proof. Let (x_k) be Cauchy. We apply the definition with $\varepsilon = 1$: there is K such that $d(x_m, x_k) < 1$ for all $m, k \geq K$. In particular, fixing $k = K$:

$$d(x_m, x_K) < 1 \quad \text{for all } m \geq K.$$

That is, *the tail* of the sequence (from index K on) lives in the ball $B(x_K, 1)$. Only the finitely many initial terms $x_1, \dots, x_{K-1}$ are left uncontrolled. We define

$$R := 1 + \max \{d(x_1, x_K), d(x_2, x_K), \dots, d(x_{K-1}, x_K)\},$$

which is a well defined real number because it is the maximum of a *finite* collection. Then *all* terms satisfy $d(x_m, x_K) < R$: those in the tail because $d(x_m, x_K) < 1 \leq R$, and the initial ones because their distance to x_K is among the numbers in the maximum, which is $< R$. Hence $\{x_k : k \in \mathbb{N}\} \subseteq B(x_K, R)$: bounded. $\square$

Theorem 6.6 (Cauchy + convergent subsequence $\Rightarrow$ convergent). *Let (x_k) be a Cauchy sequence in (X, d) . If some subsequence (x_{k_j}) converges to $x \in X$, then the whole sequence converges to x .*

Proof. This result is *key* (we will use it to prove completeness of $\mathbb{R}$ and of other spaces) and its proof combines the two hypotheses with the $\varepsilon/2$ trick.

Let $\varepsilon > 0$. We must find K such that $d(x_k, x) < \varepsilon$ for all $k \geq K$.

Step 1 (use Cauchy). There is K_1 such that

$$m, k \geq K_1 \implies d(x_m, x_k) < \frac{\varepsilon}{2}.$$

Step 2 (use the subsequence). Since $x_{k_j} \rightarrow x$, there is J such that

$$j \geq J \implies d(x_{k_j}, x) < \frac{\varepsilon}{2}.$$

Step 3 (combine). Recall that the indices of a subsequence increase strictly, and by induction $k_j \geq j$ for every j . We pick a specific index of the subsequence that satisfies *both* conditions: take $j^* := \max \{J, K_1\}$; then $j^* \geq J$ and $k_{j^*} \geq j^* \geq K_1$.

Now let $k \geq K_1$ be arbitrary. We insert the term $x_{k_{j^*}}$ as an intermediate point:

$$d(x_k, x) \leq \underbrace{d(x_k, x_{k_{j^*}})}_{< \varepsilon/2} + \underbrace{d(x_{k_{j^*}}, x)}_{< \varepsilon/2} < \varepsilon.$$

The first brace uses the Cauchy condition of Step 1 (both indices, k and k_{j^*} , are $\geq K_1$); the second uses Step 2 (since $j^* \geq J$). Taking $K = K_1$ proves the convergence $x_k \rightarrow x$.

Intuitive reading: the Cauchy condition says that the whole sequence travels bunched together; if one part of it (the subsequence) reaches x , the rest, which travels attached to it, reaches x too. $\square$

Example 6.7 (Cauchy without a limit: the holes of the space). **(a)** In $X = (0, 1]$ with the usual metric inherited from $\mathbb{R}$, the sequence $x_k = \frac{1}{k}$ is Cauchy: given $m, k \geq K$,

$$\left| \frac{1}{m} - \frac{1}{k} \right| \leq \frac{1}{m} + \frac{1}{k} \leq \frac{2}{K},$$

which is $< \varepsilon$ if $K > 2/\varepsilon$. However, (x_k) does not converge in X : its only possible limit would be 0 (in $\mathbb{R}$ it converges to 0 and the limit is unique), but $0 \notin X$. The space has a hole exactly where the sequence wanted to land.

(b) In $X = \mathbb{Q}$ with the usual metric, define the sequence of decimal truncations of $\sqrt{2}$:

$$x_1 = 1.4, \quad x_2 = 1.41, \quad x_3 = 1.414, \quad x_4 = 1.4142, \dots$$

Each x_k is rational (a finite decimal). It is Cauchy: if $m \geq k$, the first k decimals of x_m and x_k agree, so $|x_m - x_k| \leq 10^{-k}$, which becomes arbitrarily small. But it does not converge in $\mathbb{Q}$: in $\mathbb{R}$ its limit is $\sqrt{2}$, which is irrational (classical proof: if $\sqrt{2} = p/q$ in lowest terms, then $p^2 = 2q^2$ forces p to be even, $p = 2s$, and then $q^2 = 2s^2$ forces q to be even, which contradicts minimality). By uniqueness of the limit in $\mathbb{R}$, no rational number can be its limit. $\mathbb{Q}$ is full of holes, and that is why analysis is done in $\mathbb{R}$.

Careful (common mistake)

The converse of Theorem 6.4 is **false in general**: Cauchy does not imply convergent, as the examples just showed. Writing “the sequence is Cauchy, therefore it converges” *without first justifying that the space is complete* is a serious exam mistake. The implication Cauchy $\Rightarrow$ convergent is exactly the *definition* of completeness of the space, not a general theorem.

6.4 Complete metric spaces [P4]

Definition 6.8 (Completeness). A metric space (X, d) is **complete** if every Cauchy sequence in X converges to a point of X .

Intuition

A complete space is a space *without holes*: every approximation process that settles down internally does find a landing point inside the space. The previous examples show that $(0, 1]$ and $\mathbb{Q}$ are not complete. Completeness is the property that makes a space reliable for analysis: it lets you assert that limits exist without exhibiting them.

6.4.1 $\mathbb{R}$ is complete

The completeness of $\mathbb{R}$ rests on the *supremum axiom* (every nonempty family of reals that is bounded above has a supremum). The classical route goes through the Bolzano–Weierstrass theorem in $\mathbb{R}$.

Lemma 6.9 (Bolzano–Weierstrass in $\mathbb{R}$). *Every bounded sequence of real numbers has a convergent subsequence.*

Proof. We use the **bisection method**. Let (x_k) be a sequence with all its terms in an interval $[a, b]$.

Inductive construction of nested intervals. We define $I_0 = [a_0, b_0] := [a, b]$, which contains infinitely many terms of the sequence (in fact all of them; here “infinitely many terms” means: infinitely many indices k with $x_k \in I_0$). Suppose we have built $I_j = [a_j, b_j]$ containing x_k for infinitely many indices k . We split I_j at its midpoint $c_j = \frac{a_j + b_j}{2}$ into two halves $[a_j, c_j]$ and $[c_j, b_j]$. The infinitely many indices with $x_k \in I_j$ are distributed between the two halves, so *at least one* half contains terms for infinitely many indices (if both contained only finitely many, their union would contain finitely many, which is absurd). We choose that half as I_{j+1} . By construction:

$$I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots, \quad \text{length}(I_j) = \frac{b-a}{2^j} \rightarrow 0.$$

The limit point. The left endpoints a_j form an increasing sequence that is bounded above (by b); by the supremum axiom there is $x := \sup_j a_j$. We claim that $x \in I_j$ for every j : indeed, $a_j \leq x$ (the supremum is an upper bound) and $x \leq b_j$ (each b_j is an upper bound of *all* the a_i , because the intervals are nested, and the supremum is the least upper bound).

Extraction of the subsequence. We choose indices $k_1 < k_2 < k_3 < \cdots$ with $x_{k_j} \in I_j$: this is possible because each I_j contains terms of the sequence for *infinitely many* indices, so there is always one larger than the previous index k_{j-1} . Then, since x_{k_j} and x both lie in I_j :

$$|x_{k_j} - x| \leq \text{length}(I_j) = \frac{b-a}{2^j} \xrightarrow{j \rightarrow \infty} 0,$$

so $x_{k_j} \rightarrow x$. □

Theorem 6.10 ($\mathbb{R}$ is complete). $(\mathbb{R}, |\cdot - \cdot|)$ is a complete metric space.

Proof. Let (x_k) be a Cauchy sequence in $\mathbb{R}$. By Proposition 6.5, it is bounded. By Bolzano–Weierstrass (Lemma 6.9), it has a subsequence converging to some $x \in \mathbb{R}$. By Theorem 6.6 (Cauchy plus convergent subsequence), the whole sequence converges to $x \in \mathbb{R}$. Done: three pieces already proved, assembled in three lines. This is how mathematics is built. □

Theorem 6.11 ($\mathbb{R}^n$ is complete). $\mathbb{R}^n$ with any of the metrics d_1, d_2, d_∞ is complete.

Proof. We work with d_∞ (the three metrics are strongly equivalent, and strong equivalence preserves both Cauchy sequences and convergent ones, since it compresses and stretches distances by fixed constants; that is why one case is enough).

Let $(x^{(k)})$ be Cauchy in $(\mathbb{R}^n, d_\infty)$, where we write $x^{(k)} = (x_1^{(k)}, \dots, x_n^{(k)})$. **Step 1: each coordinate is Cauchy in $\mathbb{R}$.** For each $i \in \{1, \dots, n\}$:

$$|x_i^{(m)} - x_i^{(k)}| \leq \max_{1 \leq j \leq n} |x_j^{(m)} - x_j^{(k)}| = d_\infty(x^{(m)}, x^{(k)}),$$

so given $\varepsilon > 0$, the K that works for the vector sequence works for each coordinate. **Step 2: limits coordinate by coordinate.** By the completeness of $\mathbb{R}$ (Theorem 6.10), each numerical sequence $(x_i^{(k)})_k$ converges to some $x_i \in \mathbb{R}$. We define $x := (x_1, \dots, x_n)$. **Step 3: vector convergence.** Given $\varepsilon > 0$, for each i there is K_i with $|x_i^{(k)} - x_i| < \varepsilon$ if $k \geq K_i$. Taking $K = \max \{K_1, \dots, K_n\}$ (a finite maximum), for $k \geq K$:

$$d_\infty(x^{(k)}, x) = \max_i |x_i^{(k)} - x_i| < \varepsilon.$$

Hence $x^{(k)} \rightarrow x \in \mathbb{R}^n$. □

6.4.2 $C[a, b]$ with the supremum metric is complete

This is the most important example of a complete space of *infinite dimension* in the course: the home of dynamic programming.

Theorem 6.12 (Completeness of $(C[a, b], d_\infty)$). *The space $C[a, b]$ of continuous functions $f : [a, b] \rightarrow \mathbb{R}$, with $d_\infty(f, g) = \sup_t |f(t) - g(t)|$, is complete.*

Proof. Let (f_k) be a Cauchy sequence in $(C[a, b], d_\infty)$. The proof follows the standard three act scheme: (1) build a candidate limit, (2) prove that the convergence to it is in the metric of the space (uniform), (3) prove that the candidate belongs to the space (it is continuous).

Act 1: pointwise candidate. Fix a point $t \in [a, b]$. For all m, k :

$$|f_m(t) - f_k(t)| \leq \sup_{s \in [a, b]} |f_m(s) - f_k(s)| = d_\infty(f_m, f_k).$$

Since (f_k) is Cauchy in d_∞ , this inequality shows that the sequence of numbers $(f_k(t))_k$ is Cauchy in $\mathbb{R}$. By the completeness of $\mathbb{R}$, it converges; we call $f(t)$ its limit. Doing this for each t defines a function $f : [a, b] \rightarrow \mathbb{R}$, $f(t) := \lim_k f_k(t)$ (the pointwise limit).

Act 2: the convergence is uniform. We must prove $d_\infty(f_k, f) \rightarrow 0$, which is stronger than pointwise convergence. Let $\varepsilon > 0$. By Cauchy, there is K such that

$$m, k \geq K \implies d_\infty(f_m, f_k) < \frac{\varepsilon}{2}, \quad \text{in particular} \quad |f_m(t) - f_k(t)| < \frac{\varepsilon}{2} \quad \forall t \in [a, b].$$

Now we fix $k \geq K$ and $t \in [a, b]$, and let $m \rightarrow \infty$ in the inequality $|f_m(t) - f_k(t)| < \varepsilon/2$. Since $f_m(t) \rightarrow f(t)$ (Act 1) and the absolute value is continuous, the limit preserves the *non strict* inequality:

$$|f(t) - f_k(t)| \leq \frac{\varepsilon}{2}.$$

This step (“freeze k and t , let m run”) is the heart of the argument: it turns information between terms into information against the limit. Since the bound holds for *every* t , taking the

supremum:

$$d_\infty(f, f_k) = \sup_t |f(t) - f_k(t)| \leq \frac{\varepsilon}{2} < \varepsilon \quad \text{for all } k \geq K.$$

That is, $f_k \rightarrow f$ uniformly.

Act 3: f is continuous. This is exactly the theorem “the uniform limit of continuous functions is continuous”, proved in full detail in the next chapter (pointwise and uniform convergence) with the $\varepsilon/3$ argument; we use it here and announce it. With it, $f \in C[a, b]$.

Conclusion: the arbitrary Cauchy sequence (f_k) converges in d_∞ to an element f of the space. $(C[a, b], d_\infty)$ is complete. $\square$

6.4.3 Complete subsets

Theorem 6.13 (Completeness and closed sets). *Let (X, d) be a metric space and $A \subseteq X$ with the inherited metric.*

1. *If X is complete and A is closed in X , then A is complete.*
2. *If A is complete (whether X is complete or not), then A is closed in X .*

Slogan: inside a complete space, complete = closed.

Proof. (a) Let (a_k) be a Cauchy sequence in A . In particular it is Cauchy in X (the metric is the same), and since X is complete, it converges to some $x \in X$. But A is closed and $a_k \in A$: by the sequential characterization of closed sets, the limit x belongs to A . So (a_k) converges in A , and A is complete.

(b) We use the sequential characterization. Let (a_k) be a sequence in A with $a_k \rightarrow x \in X$; we must prove $x \in A$. Since (a_k) converges in X , it is Cauchy (Theorem 6.4); and being Cauchy depends only on the distances between terms, so it is Cauchy *as a sequence of A* . By the completeness of A , it converges to some $a \in A$. But limits in X are unique (Proposition 6.2) and $a_k \rightarrow x$, so $x = a \in A$. Therefore A is closed. $\square$

Example 6.14. $[0, 1]$ is complete (closed inside the complete space $\mathbb{R}$). $(0, 1]$ is not complete (it is not closed in $\mathbb{R}$; or directly by Example 6.7). The set $B = \{f \in C[0, 1] : f(0) = 0\}$ is complete: it is closed in the complete space $(C[0, 1], d_\infty)$ (we proved this in the previous chapter). This type of reasoning (“it is closed in a known complete space, hence complete”) is the standard and fast way to check completeness in exams.

6.5 Density [P4]

Definition 6.15 (Dense set). Let (X, d) be a metric space. A subset $D \subseteq X$ is **dense** in X if $\text{cl } D = X$: every point of the space is adherent to D .

Theorem 6.16 (Characterizations of density). *For $D \subseteq X$, the following are equivalent:*

1. $\text{cl } D = X$ (D is dense).
2. Every open ball of X contains points of D : for all $x \in X$, $r > 0$: $B(x, r) \cap D \neq \emptyset$.
3. Every point of X is the limit of a sequence of points of D .

Proof. (a) $\Leftrightarrow$ (b). By definition, $\text{cl } D = \{x : \forall r > 0, B(x, r) \cap D \neq \emptyset\}$. Saying $\text{cl } D = X$ is saying that *every* $x \in X$ satisfies that condition, which is literally (b). (The inclusion $\text{cl } D \subseteq X$ is automatic; the real content is $X \subseteq \text{cl } D$.)

(b) $\Rightarrow$ (c). Let $x \in X$. We apply (b) to the balls $B(x, \frac{1}{k})$, $k = 1, 2, \dots$: for each k we choose $d_k \in B(x, \frac{1}{k}) \cap D$. Then $d(d_k, x) < \frac{1}{k} \rightarrow 0$, that is, $d_k \rightarrow x$ with $d_k \in D$. (The trick of radii $1/k$ again.)

(c) $\Rightarrow$ (b). Let $x \in X$ and $r > 0$. By (c) there is a sequence (d_k) in D with $d_k \rightarrow x$; applying the definition of convergence with $\varepsilon = r$, there is K with $d(d_K, x) < r$, that is, $d_K \in B(x, r) \cap D \neq \emptyset$. $\square$

Example 6.17 ($\mathbb{Q}$ is dense in $\mathbb{R}$). We check condition (b): every interval $(x - r, x + r)$ contains a rational number. Proof with the Archimedean property: choose $n \in \mathbb{N}$ with $\frac{1}{n} < r$ (it exists by Archimedes: the reciprocals $1/n$ go below any positive number). Consider the integer $m := \lfloor nx \rfloor$ (the largest integer $\leq nx$), so that $m \leq nx < m + 1$. Dividing by n : $\frac{m}{n} \leq x < \frac{m+1}{n}$. The rational $q := \frac{m+1}{n}$ satisfies

$$x < q = \frac{m}{n} + \frac{1}{n} \leq x + \frac{1}{n} < x + r,$$

so $q \in (x, x + r) \subseteq (x - r, x + r)$. Every ball of $\mathbb{R}$ therefore contains rationals: $\text{cl } \mathbb{Q} = \mathbb{R}$. (The irrationals are dense in $\mathbb{R}$ too: add $\sqrt{2}$ to a suitable rational.)

A practical consequence we already saw: a continuous function on $\mathbb{R}$ is determined by its values on $\mathbb{Q}$.

Remark 6.18 (Separability and completion). Two related notions worth knowing by name:

- A metric space is **separable** if it has a *countable* dense subset. Examples: $\mathbb{R}$ (with $\mathbb{Q}$), $\mathbb{R}^n$ (with $\mathbb{Q}^n$), and also $C[a, b]$ (polynomials with rational coefficients are dense: Weierstrass approximation theorem). Separability makes infinite dimensional spaces manageable and will be useful in the theory of Hilbert spaces.
- Every metric space (X, d) admits a **completion**: a complete metric space $(\hat{X}, \hat{d})$ that contains an isometric copy of X as a dense subset, unique up to isometry. The idea of the construction: the points of $\hat{X}$ are (equivalence classes of) Cauchy sequences of X ; each hole is filled with the sequence that was pointing at it. Standard example: the completion of $\mathbb{Q}$ is $\mathbb{R}$. We will not need the technical details, only the idea.

Link with economics

For an economist, completeness is a *guarantee of existence*:

- **Value function iteration**: in dynamic programming one builds the sequence $V_{k+1} = TV_k$, where T is the Bellman operator acting on a space of functions with the supremum metric. One proves that (V_k) is Cauchy (because T is a contraction; see the coming chapters). The step “and therefore V_k converges to a function V of the space” uses *exactly* the completeness of a space of continuous and bounded functions with the supremum metric, of the same type as the one proved in this chapter (Theorem 6.12). Without completeness, the algorithm would settle towards a hole: there would be no value function.

- **Why $\mathbb{R}$ and not $\mathbb{Q}$:** economic models require equilibrium prices and quantities to *exist*. Equilibria are obtained as limits (of Walrasian adjustment processes, of maximizing sequences). In $\mathbb{Q}$ those limits may fail to exist (Example 6.7(b)); in $\mathbb{R}$ they do exist. The completeness of $\mathbb{R}$ is the silent background assumption of all economic analysis.

Exam strategy

Strategies for the typical questions:

1. “*Prove that this sequence is Cauchy*”: bound $d(x_m, x_k)$ by something that depends only on $\min\{m, k\}$ and tends to 0 (like $2/K$ or 10^{-K} in the examples). Do not try to guess the limit: the point of Cauchy is that you do not need it.
2. “*Prove that this space is complete*”: if it is a subset of a known complete space ($\mathbb{R}^n$, $C[a, b]$, ℓ^∞), prove that it is *closed* and cite Theorem 6.13(a). If it is a new space of functions, reproduce the three act script: pointwise candidate, uniformity (freeze k , let $m \rightarrow \infty$, take the supremum), membership in the space.
3. “*Prove that this space is NOT complete*”: exhibit a concrete Cauchy sequence whose natural limit is outside the space, and argue by uniqueness of the limit (template of Example 6.7).
4. “*Prove that D is dense*”: choose the most convenient characterization in Theorem 6.16; often (b) (every ball touches D) is the most direct one.

Chapter summary

- Cauchy: $\forall \varepsilon \exists K: m, k \geq K \Rightarrow d(x_m, x_k) < \varepsilon$. An internal condition, with no mention of a limit.
- Chain of implications: $\text{convergent} \Rightarrow \text{Cauchy} \Rightarrow \text{bounded}$; no arrow can be reversed in general.
- Key tool: $\text{Cauchy} + \text{convergent subsequence} \Rightarrow \text{the whole sequence converges (to the same limit)}$.
- Complete: every Cauchy sequence converges inside the space. Complete: $\mathbb{R}$ (via Bolzano–Weierstrass), $\mathbb{R}^n$ (coordinates), $(C[a, b], d_\infty)$ (three act script). Not complete: $\mathbb{Q}$, $(0, 1]$.
- In a complete space: $A \text{ complete} \iff A \text{ closed}$. (And $\text{complete} \Rightarrow \text{closed}$ always.)
- D dense: $\text{cl } D = X \iff \text{every ball meets } D \iff \text{every point is a limit of points of } D$. $\mathbb{Q}$ is dense in $\mathbb{R}$ (Archimedes plus integer part).
- Ideas: separable = countable dense subset; every space has a completion (this is how $\mathbb{R}$ is built from $\mathbb{Q}$).

6.6 Exam-style exercises [P4]

Exercise 6.19. Let (x_k) be a sequence in a metric space (X, d) such that

$$d(x_{k+1}, x_k) \leq \frac{1}{2^k} \quad \text{for all } k \in \mathbb{N}.$$

Prove that (x_k) is Cauchy. Is it true that every sequence with $d(x_{k+1}, x_k) \rightarrow 0$ is Cauchy? Justify your answer.

Solution. Part 1. Let $m > k$. We insert *all* the intermediate points $x_{k+1}, \dots, x_{m-1}$ by applying the triangle inequality repeatedly:

$$d(x_m, x_k) \leq \sum_{j=k}^{m-1} d(x_{j+1}, x_j) \leq \sum_{j=k}^{m-1} \frac{1}{2^j} < \sum_{j=k}^{\infty} \frac{1}{2^j} = \frac{1/2^k}{1 - 1/2} = \frac{1}{2^{k-1}},$$

where we used the geometric series formula $\sum_{j \geq k} r^j = \frac{r^k}{1-r}$ with $r = \frac{1}{2}$. Given $\varepsilon > 0$, we choose K with $2^{1-K} < \varepsilon$ (possible because $2^{1-K} \rightarrow 0$); then for $m > k \geq K$: $d(x_m, x_k) < 2^{1-k} \leq 2^{1-K} < \varepsilon$ (and the case $m = k$ gives distance 0). Hence (x_k) is Cauchy. Note the mechanism: it is not enough that the individual steps get small; what controls everything is that the *sum of the tails* of the steps tends to zero (a convergent series).

Part 2. *False.* Standard counterexample: $x_k = \sqrt{k}$ in $\mathbb{R}$. The steps tend to zero:

$$x_{k+1} - x_k = \sqrt{k+1} - \sqrt{k} = \frac{(\sqrt{k+1} - \sqrt{k})(\sqrt{k+1} + \sqrt{k})}{\sqrt{k+1} + \sqrt{k}} = \frac{1}{\sqrt{k+1} + \sqrt{k}} \rightarrow 0$$

(we rationalized by multiplying and dividing by the conjugate sum). But the sequence is not bounded ($\sqrt{k} \rightarrow \infty$), and every Cauchy sequence is bounded (Proposition of this chapter): hence it is not Cauchy. Exam lesson: “small consecutive steps” is NOT the same as Cauchy; the definition of Cauchy requires control of $d(x_m, x_k)$ for *all* large pairs, not only consecutive ones. $\square$

Exercise 6.20. Prove that the space $X = (0, \infty)$ with the usual metric of $\mathbb{R}$ is not complete, but that the same set with the metric

$$d'(x, y) := |\ln x - \ln y|$$

is complete. (Hint: relate the Cauchy sequences of (X, d') to those of $\mathbb{R}$ through the logarithm.)

Solution. Failure of completeness with the usual metric. The sequence $x_k = \frac{1}{k}$ lives in X , is Cauchy (seen in this chapter), and its only candidate limit is $0 \notin X$. If it converged to some $x \in X \subseteq \mathbb{R}$, then by uniqueness of the limit in $\mathbb{R}$ we would have $x = 0$, a contradiction. So there is a Cauchy sequence with no limit in X : not complete.

Completeness with d' . First, d' is a metric: it is the image of the usual metric of $\mathbb{R}$ under the bijection $\ln : (0, \infty) \rightarrow \mathbb{R}$; formally, $d'(x, y) = |L(x) - L(y)|$ with $L = \ln$ injective, and the axioms follow immediately (M1: $d' = 0 \iff \ln x = \ln y \iff x = y$ by injectivity; M2 and M3 are inherited from the absolute value).

Structural remark: L is an *isometry* from (X, d') onto $(\mathbb{R}, |\cdot|)$, with inverse $L^{-1}(t) = e^t$. Bijective isometries carry Cauchy sequences to Cauchy sequences and convergent ones to convergent ones in both directions, because the relevant distances are *equal* number by number.

Let (x_k) be Cauchy in (X, d') . We define $t_k := \ln x_k \in \mathbb{R}$. Then

$$|t_m - t_k| = |\ln x_m - \ln x_k| = d'(x_m, x_k),$$

so (t_k) is Cauchy in $\mathbb{R}$. By the completeness of $\mathbb{R}$, $t_k \rightarrow t^* \in \mathbb{R}$. We define $x^* := e^{t^*} \in (0, \infty) = X$ (the exponential is always positive, so the candidate stays inside the space). Finally:

$$d'(x_k, x^*) = |\ln x_k - \ln x^*| = |t_k - t^*| \rightarrow 0,$$

so $x_k \rightarrow x^*$ in (X, d') . Every Cauchy sequence converges: (X, d') is complete.

Lesson: completeness is NOT a topological property of the set: it depends on the metric. The same set $(0, \infty)$ is incomplete with one metric and complete with another (which is moreover topologically equivalent to the usual one, since $\ln$ and $\exp$ are continuous). The metric d' pushes the hole at 0 off to infinity: the sequences that were falling towards 0 stop being Cauchy because $\ln(1/k) = -\ln k \rightarrow -\infty$. $\square$

Exercise 6.21. Let c_0 be the set of real sequences that converge to 0, seen as a subset of ℓ^∞ (bounded sequences) with the supremum metric $d_\infty(x, y) = \sup_k |x_k - y_k|$. Prove that c_0 is closed in ℓ^∞ and conclude that (c_0, d_∞) is complete. (You may use without proof that ℓ^∞ is complete, since its proof is analogous to the one for $C[a, b]$.)

Solution. c_0 is closed. We use the sequential characterization. Let $(x^{(j)})_j$ be a sequence of elements of c_0 (each $x^{(j)} = (x_1^{(j)}, x_2^{(j)}, \dots)$ is a sequence tending to 0) such that $x^{(j)} \rightarrow x$ in d_∞ , with $x = (x_1, x_2, \dots) \in \ell^\infty$. We must prove that $x \in c_0$, that is, $x_k \rightarrow 0$ as $k \rightarrow \infty$.

Let $\varepsilon > 0$. This is a two level $\varepsilon/2$ argument:

Level 1 (approximate the limit sequence by one of the family). Since $x^{(j)} \rightarrow x$ in d_∞ , there is J such that

$$d_\infty(x^{(J)}, x) = \sup_k |x_k^{(J)} - x_k| < \frac{\varepsilon}{2}.$$

Level 2 (use that the approximating element is in c_0). Since $x^{(J)} \in c_0$, we have $x_k^{(J)} \rightarrow 0$ in k : there is K such that

$$k \geq K \implies |x_k^{(J)}| < \frac{\varepsilon}{2}.$$

Combination. For every $k \geq K$:

$$|x_k| \leq |x_k - x_k^{(J)}| + |x_k^{(J)}| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

where the first distance is dominated by the supremum of Level 1. Hence $x_k \rightarrow 0$, that is, $x \in c_0$: the set contains the limits of its convergent sequences and is closed.

Completeness. ℓ^∞ is complete (admitted) and c_0 is closed in it; by the theorem “a closed subset of a complete space is complete”, (c_0, d_∞) is complete. $\square$

Exercise 6.22. Let (X, d) be a complete metric space and let $F_1 \supseteq F_2 \supseteq F_3 \supseteq \cdots$ be a decreasing sequence of nonempty closed sets with $\text{diam}(F_k) \rightarrow 0$. Prove the *Cantor intersection theorem*: there is exactly one point x^* with $x^* \in \bigcap_{k=1}^{\infty} F_k$.

Solution. Existence. For each k , we choose a point $x_k \in F_k$ (possible because each $F_k \neq \emptyset$).

(x_k) is Cauchy: let $m \geq k$. Since the family is decreasing, $F_m \subseteq F_k$, so both points $x_m \in F_m \subseteq F_k$ and $x_k \in F_k$ belong to F_k . By the definition of diameter:

$$d(x_m, x_k) \leq \text{diam}(F_k).$$

Given $\varepsilon > 0$, since $\text{diam}(F_k) \rightarrow 0$ there is K with $\text{diam}(F_K) < \varepsilon$; for $m, k \geq K$, both points lie in $F_{\min\{m, k\}} \subseteq F_K$, so $d(x_m, x_k) \leq \text{diam}(F_K) < \varepsilon$. Cauchy is proved.

Limit in the intersection: by completeness of X , $x_k \rightarrow x^*$ for some $x^* \in X$. Fix an arbitrary index k_0 . The tail $(x_k)_{k \geq k_0}$ lives in F_{k_0} (by the decreasing property) and converges to x^* ; since F_{k_0} is closed, the sequential characterization gives $x^* \in F_{k_0}$. As k_0 is arbitrary, $x^* \in \bigcap_k F_k$.

Uniqueness. If $x^*, y^* \in \bigcap_k F_k$, then both lie in F_k for every k , so

$$0 \leq d(x^*, y^*) \leq \text{diam}(F_k) \xrightarrow{k \rightarrow \infty} 0,$$

which forces $d(x^*, y^*) = 0$ and, by (M1), $x^* = y^*$.

Comment: the three hypotheses are essential. Without closed sets, take $F_k = (0, \frac{1}{k})$ in $\mathbb{R}$ (empty intersection); without $\text{diam} \rightarrow 0$, take $F_k = [k, \infty)$ (nested closed sets, empty intersection); without completeness, take $F_k = (0, \frac{1}{k}]$ inside the space $X = (0, 1]$. This theorem is the abstract version of nested intervals and will reappear in proofs by bisection. $\square$

Exercise 6.23. Let $P \subseteq C[0, 1]$ be the set of polynomial functions restricted to $[0, 1]$.

1. Assuming the Weierstrass approximation theorem (every $f \in C[0, 1]$ is the uniform limit of polynomials), prove that P is dense in $(C[0, 1], d_\infty)$.
2. Prove that $P \neq C[0, 1]$ and conclude that P is *not* closed in $C[0, 1]$ and that (P, d_∞) is *not* complete.

Solution. (a) The Weierstrass theorem says: for every $f \in C[0, 1]$ and every $\varepsilon > 0$ there is a polynomial p with $\sup_{t \in [0, 1]} |f(t) - p(t)| < \varepsilon$, that is, $d_\infty(f, p) < \varepsilon$, that is, $p \in B(f, \varepsilon) \cap P$. Since f and ε are arbitrary, *every ball of $C[0, 1]$ contains polynomials*: this is exactly characterization (b) of density in Theorem 6.16. Hence $\text{cl } P = C[0, 1]$: P is dense.

(b) P is a proper subset. The function $f(t) = \left|t - \frac{1}{2}\right|$ is continuous but is not a polynomial: polynomials are differentiable at every point, while f is not differentiable at $t = \frac{1}{2}$ (the one sided difference quotients equal -1 and $+1$). Another option: $f(t) = e^t$ is not a polynomial (a polynomial of degree n has zero $(n+1)$ -th derivative; e^t never does). Hence $P \subsetneq C[0, 1]$.

P is not closed. If P were closed, then $P = \text{cl } P = C[0, 1]$ (by part (a)), contradicting $P \subsetneq C[0, 1]$. Therefore P is not closed. Concretely: by density there are polynomials p_k with $d_\infty(p_k, f) \rightarrow 0$ for the non polynomial f above; that sequence of points of P converges outside P .

P is not complete. The sequence (p_k) above converges in $C[0, 1]$, hence it is Cauchy (convergent $\Rightarrow$ Cauchy), and all its terms are in P ; but it does not converge in P : if it did, to some

$q \in P$, uniqueness of the limit in $C[0, 1]$ would give $q = f \notin P$, a contradiction. A Cauchy sequence in P with no limit in P : not complete. (You can also cite: “complete $\Rightarrow$ closed” (Theorem 6.13(b)) and P is not closed.)

Interpretation: polynomials are the rationals of the world of continuous functions: dense but full of holes; their completion inside d_∞ is all of $C[0, 1]$. $\square$

Chapter 7

Pointwise and uniform convergence of sequences of functions [P4]

Week 6 of the course. Bibliography: Pons [3], Ch. 5; Zorich II [6], Ch. 16.

What is this used for in Microeconomics?

This chapter is peripheral for the canonical syllabus of microeconomics: demand theory, game theory and general equilibrium (Mas-Colell, Whinston and Green, chs. 1 to 18) are essentially static, and there you do not need to take limits of sequences of functions. Where this machinery does shine is in econometrics (convergence of estimators and of likelihood functions) and in dynamic and recursive models, where value function iteration converges *uniformly* to the true value function. For micro it is enough to keep the central idea: uniform means that the maximum error over the whole domain tends to zero, and that uniformity is what preserves continuity in the limit.

7.1 Motivation [P4]

So far we have studied sequences of *points*: real numbers, vectors of $\mathbb{R}^n$, or elements of an abstract metric space. In this chapter we make a basic conceptual jump: we study sequences whose terms are *functions*. That is, for each $n \in \mathbb{N}$ we have a function $f_n: A \rightarrow \mathbb{R}$, and we ask what it means for the sequence (f_n) to converge to a limit function f .

Why does this matter? Because in mathematics (and in economics) the functions we care about are rarely known exactly: they are *approximated*. Some examples:

- Taylor series approximate complicated functions by polynomials: the function e^x is the limit of the polynomials $1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!}$.
- In dynamic programming, the value function is obtained as the limit of a sequence of iterated approximations.
- In econometrics, nonparametric estimators of a density or of a regression function are sequences of functions that converge to the true object.

When a sequence of functions converges, some crucial questions appear at once about *which*

properties are inherited in the limit:

1. If each f_n is continuous and $f_n \rightarrow f$, is f continuous?
2. If each f_n is integrable, does $\lim_n \int f_n = \int \lim_n f_n$ hold? That is, can we exchange the limit and the integral?
3. If each f_n is differentiable, is the limit differentiable, and does $(\lim f_n)' = \lim f_n'$ hold?

The answer to all three questions is, in general, **no** if we use the most naive notion of convergence (pointwise convergence). The lesson of the chapter is that we need a stronger notion, *uniform convergence*, for the good properties (continuity, integrability) to pass to the limit. This contrast between pointwise and uniform convergence is one of the favorite topics in analysis exams.

7.2 Pointwise convergence [P4]

Definition 7.1 (Pointwise convergence). Let $A \subseteq \mathbb{R}^n$ (or, more generally, a metric space) and let $f_n: A \rightarrow \mathbb{R}$ be a function for each $n \in \mathbb{N}$. We say that the sequence (f_n) *converges pointwise* to the function $f: A \rightarrow \mathbb{R}$ if for each fixed point $x \in A$, the sequence of real numbers $(f_n(x))_{n \in \mathbb{N}}$ converges to $f(x)$. In symbols:

$$\forall x \in A, \forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } n \geq N \implies |f_n(x) - f(x)| < \varepsilon.$$

We write $f_n \rightarrow f$ pointwise on A .

Intuition

We fix a point x of the domain, like sticking a pin on the horizontal axis, and we look at the column of values $f_1(x), f_2(x), f_3(x), \dots$. This is an ordinary sequence of real numbers, and we ask it to converge to $f(x)$. Pointwise convergence is, literally, convergence point by point: in each vertical column the sequence of heights converges, but each column goes its own way, with no coordination with the others.

The key to the definition is the *order of the quantifiers*: N is chosen *after* knowing x and ε . So N may depend on both: $N = N(\varepsilon, x)$. At one point the convergence may be very fast, and at another point hopelessly slow.

Example 7.2 (The canonical example: $f_n(x) = x^n$ on $[0, 1]$). Let $f_n: [0, 1] \rightarrow \mathbb{R}$ be given by $f_n(x) = x^n$. We compute the pointwise limit, that is, we fix x and compute $\lim_{n \rightarrow \infty} x^n$.

- If $0 \leq x < 1$: we know from geometric sequences that $x^n \rightarrow 0$ (a quantity smaller than 1 raised to larger and larger powers becomes arbitrarily small). Formally, if $0 < x < 1$ we write $x = \frac{1}{1+h}$ with $h > 0$; by Bernoulli's inequality, $(1+h)^n \geq 1 + nh \rightarrow \infty$, so $x^n = \frac{1}{(1+h)^n} \rightarrow 0$.
- If $x = 1$: the sequence is constant, $f_n(1) = 1^n = 1$ for every n , so $f_n(1) \rightarrow 1$.

Therefore the pointwise limit is the function

$$f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1, \\ 1 & \text{if } x = 1. \end{cases}$$

Note the central phenomenon of the chapter. Each $f_n(x) = x^n$ is a polynomial, that is, a continuous and even infinitely differentiable function. But the pointwise limit f is **discontinuous** at $x = 1$: the function equals 0 at every point to the left of 1 and jumps abruptly to 1 at the endpoint. Continuity does *not* survive pointwise convergence.

Example 7.3 ($f_n(x) = x/n$: pointwise convergence to zero). Let $f_n: \mathbb{R} \rightarrow \mathbb{R}$, $f_n(x) = \frac{x}{n}$. For any fixed $x \in \mathbb{R}$, we have $|f_n(x) - 0| = \frac{|x|}{n} \rightarrow 0$ as $n \rightarrow \infty$. So $f_n \rightarrow 0$ (the identically zero function) pointwise on $\mathbb{R}$.

Let us look at how fast this is. Given $\varepsilon > 0$, we need $\frac{|x|}{n} < \varepsilon$, that is, $n > \frac{|x|}{\varepsilon}$. The smallest N that works is approximately $N(\varepsilon, x) = \frac{|x|}{\varepsilon}$: it *depends on* x , and it grows without bound as $|x| \rightarrow \infty$. For $\varepsilon = 0.01$ and $x = 1$ it is enough to take $n > 100$; but for $x = 10^6$ you need $n > 10^8$. There is no single N that works for all $x \in \mathbb{R}$ at once. (We will see that this means exactly that the convergence is not uniform on $\mathbb{R}$, although it is uniform on any bounded interval.)

Example 7.4 ($f_n(x) = nx(1 - x^2)^n$ on $[0, 1]$: the limit of the integrals is not the integral of the limit). Let $f_n: [0, 1] \rightarrow \mathbb{R}$, $f_n(x) = nx(1 - x^2)^n$. First the pointwise limit:

- If $x = 0$: $f_n(0) = 0$ for every n , so $f_n(0) \rightarrow 0$.
- If $0 < x \leq 1$: let $r = 1 - x^2$, which satisfies $0 \leq r < 1$. Then $f_n(x) = nxr^n$. Since $r < 1$, the geometric factor r^n decays *exponentially* and crushes the linear factor n : it is known that $nr^n \rightarrow 0$ for $0 \leq r < 1$ (for example, by the ratio test: $\frac{(n+1)r^{n+1}}{nr^n} = r \frac{n+1}{n} \rightarrow r < 1$). So $f_n(x) \rightarrow 0$.

Hence $f_n \rightarrow 0$ pointwise on $[0, 1]$. Now we integrate each f_n with the substitution $u = 1 - x^2$, $du = -2x dx$:

$$\int_0^1 nx(1 - x^2)^n dx = \frac{n}{2} \int_0^1 u^n du = \frac{n}{2} \cdot \frac{1}{n+1} = \frac{n}{2(n+1)} \rightarrow \frac{1}{2}.$$

But the integral of the pointwise limit is $\int_0^1 0 dx = 0$. Conclusion:

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx = \frac{1}{2} \neq 0 = \int_0^1 \lim_{n \rightarrow \infty} f_n(x) dx.$$

The exchange of limit and integral **fails** under merely pointwise convergence. What happens geometrically? The graph of f_n has a bump near 0 whose height grows with n (the maximum of f_n is of order $\sqrt{n}$) while its base becomes narrower; the area under the bump settles at $1/2$, but the bump escapes towards the vertical axis and at each fixed point the function ends up being almost zero.

Example 7.5 (The moving bump). Define $f_n: \mathbb{R} \rightarrow \mathbb{R}$ as a triangular tent of height 1 supported

on the interval $[n, n+1]$:

$$f_n(x) = \begin{cases} 2(x-n) & \text{if } n \leq x \leq n + \frac{1}{2}, \\ 2(n+1-x) & \text{if } n + \frac{1}{2} < x \leq n+1, \\ 0 & \text{otherwise.} \end{cases}$$

Fix $x \in \mathbb{R}$. Since the bump is supported on $[n, n+1]$ and this interval slides towards $+\infty$, for every $n > x$ we have $f_n(x) = 0$. That is, the sequence $(f_n(x))$ is eventually zero, so $f_n(x) \rightarrow 0$. Therefore $f_n \rightarrow 0$ pointwise on $\mathbb{R}$.

The convergence is bad, however: at every stage there is a complete bump of height 1 somewhere on the line, so $\sup_{x \in \mathbb{R}} |f_n(x) - 0| = 1$ for every n ; the bump never flattens, it only moves. Moreover $\int_{\mathbb{R}} f_n = \frac{1}{2}$ for every n , while $\int_{\mathbb{R}} 0 = 0$: again the exchange of limit and integral fails.

Careful (common mistake)

When you compute a pointwise limit, **the point x is fixed and what moves is n** . A very frequent mistake is to treat x and n as if they moved together. For example, in $f_n(x) = x^n$ it is tempting to say “near $x = 1$ the function is almost 1, so the limit cannot be 0”. False: for each *fixed* $x < 1$, however close to 1 it is, $x^n \rightarrow 0$. What is true is that the *speed* of convergence degrades as $x \rightarrow 1^-$, and that is exactly what uniform convergence detects (and punishes).

7.3 Uniform convergence [P4]

In the previous examples the problem was always the same: the N of the definition depended on the point x , and there was no N that worked for all points at once. Uniform convergence demands exactly that: a universal N .

Definition 7.6 (Uniform convergence). Let $f_n, f: A \rightarrow \mathbb{R}$. We say that (f_n) *converges uniformly* to f on A if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } n \geq N \implies |f_n(x) - f(x)| < \varepsilon \text{ for every } x \in A.$$

We write $f_n \rightrightarrows f$ on A , or $f_n \rightarrow f$ uniformly on A .

Intuition

Compare the quantifiers carefully:

Pointwise: $\forall x, \forall \varepsilon, \exists N(\varepsilon, x), \dots$

Uniform: $\forall \varepsilon, \exists N(\varepsilon), \forall x, \dots$

In uniform convergence, the “ $\forall x$ ” appears *after* the “ $\exists N$ ”: the same N must work simultaneously for all points of the domain. The word uniform means exactly that: the speed of convergence is uniform in x , nobody lags behind.

The geometric picture is the **tube of radius ε** : draw the graph of f and around it a band or tube made of all points at vertical distance less than ε from the graph (the curves $f - \varepsilon$ and $f + \varepsilon$ are the walls of the tube). Uniform convergence says: for each tube, however narrow, from some N on *the whole graph* of f_n lies inside the tube. In pointwise convergence, by contrast, each point of the graph eventually enters the tube, but at each stage n there may be pieces of graph outside it (like the bump of Example 7.4 or the moving bump of Example 7.5).

Remark 7.7. Uniform $\implies$ pointwise, always: if a single N works for all x , in particular it works for each fixed x . The converse implication is false, as Examples 7.2, 7.4 and 7.5 show. An important practical consequence: **the only candidate for a uniform limit is the pointwise limit**. This is why the standard procedure is: (1) compute the pointwise limit f ; (2) decide whether the convergence to that f is also uniform.

7.3.1 The supremum criterion

The number one practical tool to decide uniform convergence is to restate it in terms of the supremum of the error.

Theorem 7.8 (Supremum criterion). *Let $f_n, f: A \rightarrow \mathbb{R}$ and define the maximum error*

$$M_n := \sup_{x \in A} |f_n(x) - f(x)|.$$

Then $f_n \rightarrow f$ uniformly on A if and only if $M_n \rightarrow 0$ as $n \rightarrow \infty$.

Proof. We prove the two implications separately.

($\implies$) Suppose $f_n \rightrightarrows f$. Let $\varepsilon > 0$. By the definition of uniform convergence applied to $\varepsilon/2$, there is N such that for every $n \geq N$ and every $x \in A$:

$$|f_n(x) - f(x)| < \frac{\varepsilon}{2}.$$

Fix $n \geq N$. The number $\varepsilon/2$ is then an *upper bound* of the set $\{|f_n(x) - f(x)| : x \in A\}$. Since the supremum is the *least* upper bound, we conclude

$$M_n = \sup_{x \in A} |f_n(x) - f(x)| \leq \frac{\varepsilon}{2} < \varepsilon.$$

(Note why we used $\varepsilon/2$: the strict inequality $< \varepsilon$ at each point only guarantees $\leq \varepsilon$ for the supremum; with the $\varepsilon/2$ trick we recover a strict inequality.) We have proved that for every $\varepsilon > 0$ there is N such that $n \geq N \implies 0 \leq M_n < \varepsilon$; this is exactly $M_n \rightarrow 0$.

($\impliedby$) Suppose $M_n \rightarrow 0$. Let $\varepsilon > 0$. There is N such that $n \geq N \implies M_n < \varepsilon$. But for *any* $x \in A$, by the definition of supremum,

$$|f_n(x) - f(x)| \leq \sup_{y \in A} |f_n(y) - f(y)| = M_n < \varepsilon.$$

So the same N (which depends only on ε) works for all $x \in A$ simultaneously: that is uniform convergence. $\square$

Exam strategy

The supremum criterion turns a problem with quantifiers into a problem of **one variable calculus**: for each n , maximize the function $g_n(x) = |f_n(x) - f(x)|$ over A (differentiate, set equal to zero, compare with the endpoints of the interval) and then study whether the maximum value tends to 0. Exam recipe:

1. Compute the pointwise limit f (fix x , move n).
2. Form $g_n(x) = |f_n(x) - f(x)|$ and compute $M_n = \sup_A g_n$ (often with $g'_n(x) = 0$).
3. If $M_n \rightarrow 0$: uniform convergence. If $M_n \not\rightarrow 0$ (or $M_n = \infty$): only pointwise.

Shortcut to refute: it is enough to find *one* sequence of points $x_n \in A$ with $|f_n(x_n) - f(x_n)| \not\rightarrow 0$, because $M_n \geq |f_n(x_n) - f(x_n)|$. Choosing x_n by chasing the bump (for example x_n = the point where f_n attains its maximum) almost always works.

Example 7.9 (x^n does not converge uniformly on $[0, 1]$, but it does on $[0, a]$ with $a < 1$). Back to Example 7.2: $f_n(x) = x^n$ on $[0, 1]$, with pointwise limit f (equal to 0 on $[0, 1)$ and to 1 at $x = 1$).

On $[0, 1]$ the convergence is not uniform. We compute

$$M_n = \sup_{x \in [0, 1]} |x^n - f(x)| = \sup_{x \in [0, 1]} x^n = 1,$$

because at $x = 1$ the error is $|1 - 1| = 0$, while on $[0, 1)$ the function x^n is increasing and gets as close to 1 as we want when $x \rightarrow 1^-$ (for example, $x = (1/2)^{1/n}$ gives $x^n = 1/2$; and values of x even closer to 1 give x^n as close to 1 as desired). Since $M_n = 1 \not\rightarrow 0$, there is no uniform convergence. Alternatively, with the shortcut: taking $x_n = (1/2)^{1/n} \in [0, 1)$ we get $|f_n(x_n) - f(x_n)| = |1/2 - 0| = \frac{1}{2} \not\rightarrow 0$.

There is also a structural argument, which we will see in Theorem 7.11: each x^n is continuous, but the limit f is discontinuous; if the convergence were uniform, the limit would have to be continuous. Contradiction.

On $[0, a]$ with $0 < a < 1$ the convergence is uniform. Here the pointwise limit is $f \equiv 0$, and since x^n is increasing on $[0, a]$:

$$M_n = \sup_{x \in [0, a]} |x^n - 0| = a^n \rightarrow 0$$

because $0 < a < 1$. By the supremum criterion, $x^n \rightarrow 0$ on $[0, a]$. Lesson: uniformity depends crucially on the domain; by cutting the domain away from the problematic point ($x = 1$), the convergence becomes uniform.

Example 7.10 (Complete application of the criterion with calculus). We study the convergence of $f_n(x) = \frac{x}{1 + nx^2}$ on $\mathbb{R}$.

Step 1: pointwise limit. For $x = 0$, $f_n(0) = 0$. For fixed $x \neq 0$, the denominator $1 + nx^2 \rightarrow \infty$, so $f_n(x) \rightarrow 0$. Hence $f \equiv 0$ pointwise.

Step 2: supremum of the error. We must maximize $g_n(x) = |f_n(x)|$ on $\mathbb{R}$. Since f_n is odd, $x \geq 0$ is enough. We differentiate:

$$f'_n(x) = \frac{(1 + nx^2) - x(2nx)}{(1 + nx^2)^2} = \frac{1 - nx^2}{(1 + nx^2)^2}.$$

The derivative vanishes when $1 - nx^2 = 0$, that is at $x_n = \frac{1}{\sqrt{n}}$, which is a maximum (the derivative goes from positive to negative). The maximum value is

$$M_n = f_n\left(\frac{1}{\sqrt{n}}\right) = \frac{1/\sqrt{n}}{1 + n \cdot \frac{1}{n}} = \frac{1}{2\sqrt{n}}.$$

Step 3: conclusion. $M_n = \frac{1}{2\sqrt{n}} \rightarrow 0$, so $f_n \rightrightarrows 0$ **on all of** $\mathbb{R}$. This example shows that the supremum criterion can give uniformity even on unbounded domains.

7.4 Uniform convergence and continuity [P4]

We now reach the star theorem of the chapter: uniform convergence *carries* continuity to the limit. This result explains why uniform convergence is the right notion of convergence.

Theorem 7.11 (The uniform limit of continuous functions is continuous). *Let (X, d) be a metric space, $A \subseteq X$, and let $f_n: A \rightarrow \mathbb{R}$ be continuous functions such that $f_n \rightrightarrows f$ on A . Then f is continuous on A .*

Proof. This is the famous $\varepsilon/3$ argument. Fix a point $x_0 \in A$; we want to prove that f is continuous at x_0 , that is: given $\varepsilon > 0$, find $\delta > 0$ such that $d(x, x_0) < \delta$ (with $x \in A$) implies $|f(x) - f(x_0)| < \varepsilon$.

The idea: we know nothing direct about f , but we do know that f is uniformly close to some f_N (by uniform convergence) and that this f_N is continuous. So we travel from $f(x)$ to $f(x_0)$ with three steps: from $f(x)$ to $f_N(x)$, from $f_N(x)$ to $f_N(x_0)$, and from $f_N(x_0)$ to $f(x_0)$. Formally, we add and subtract and apply the triangle inequality:

$$|f(x) - f(x_0)| \leq \underbrace{|f(x) - f_N(x)|}_{(I)} + \underbrace{|f_N(x) - f_N(x_0)|}_{(II)} + \underbrace{|f_N(x_0) - f(x_0)|}_{(III)}.$$

Now we control each term with $\varepsilon/3$:

Terms (I) and (III), controlled by uniform convergence. Applying the definition of uniform convergence with $\varepsilon/3$, there is $N \in \mathbb{N}$ such that

$$|f_N(y) - f(y)| < \frac{\varepsilon}{3} \quad \text{for every } y \in A.$$

It is crucial that the inequality holds for all y at once, because we will use it at two different points: at $y = x$ (which will vary) and at $y = x_0$. **This is where pointwise convergence would not be enough:** with pointwise convergence we would have an N that is good for x_0 ,

but when x moves the required N could change, and the argument collapses. With this fixed N , (I) $< \varepsilon/3$ and (III) $< \varepsilon/3$.

Term (II), controlled by the continuity of f_N . The function f_N (a single function, with N already fixed) is continuous at x_0 . So there is $\delta > 0$ such that

$$x \in A, d(x, x_0) < \delta \implies |f_N(x) - f_N(x_0)| < \frac{\varepsilon}{3}.$$

Conclusion. For $x \in A$ with $d(x, x_0) < \delta$:

$$|f(x) - f(x_0)| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

Since $x_0 \in A$ was arbitrary, f is continuous on A . □

Remark 7.12. The theorem gives a very useful **negative criterion**: if the f_n are continuous but the pointwise limit f is discontinuous, then the convergence *cannot* be uniform. This settles Example 7.9 in one line. Careful with the logic: the converse of the theorem is false; the limit can be continuous without the convergence being uniform (Examples 7.4 and 7.5: the limit 0 is continuous, the convergence is not uniform).

7.5 The supremum metric: uniform convergence as convergence in a metric space [P4]

Uniform convergence is not an isolated curiosity: it is exactly the convergence of sequences in a metric space of functions, which connects this chapter with the whole theory of metric spaces in the course.

Definition 7.13 (The space $C[a, b]$ with the supremum metric). Let $C[a, b]$ be the set of all continuous functions $f: [a, b] \rightarrow \mathbb{R}$. We define the *supremum metric* (also called the uniform metric or the metric of uniform convergence):

$$d_\infty(f, g) := \sup_{x \in [a, b]} |f(x) - g(x)|.$$

(The supremum is finite: $f - g$ is continuous on the compact set $[a, b]$, hence bounded; in fact the supremum is a maximum, by the Weierstrass theorem.)

Proposition 7.14. $f_n \rightarrow f$ uniformly on $[a, b] \iff d_\infty(f_n, f) \rightarrow 0$, that is, $\iff f_n \rightarrow f$ as a sequence of points of the metric space $(C[a, b], d_\infty)$.

Proof. This is a direct restatement of the supremum criterion (Theorem 7.8): $d_\infty(f_n, f)$ is exactly the number M_n of that theorem, and $M_n \rightarrow 0$ is equivalent to uniform convergence. □

Intuition

A basic change of perspective: each continuous function f becomes a *single point* of a large space $C[a, b]$, and the distance between two functions is the largest vertical gap between their graphs. With these glasses, everything learned about metric spaces (balls,

Cauchy sequences, completeness, compactness and so on) applies to spaces of functions. This is the entrance to functional analysis, the second half of the course: Banach and Hilbert spaces are spaces of functions with structures of this type.

7.5.1 Uniform Cauchy criterion and completeness

How can we know that (f_n) converges uniformly if we do not know the limit? With a Cauchy criterion, just as in $\mathbb{R}$.

Theorem 7.15 (Uniform Cauchy criterion). *Let $f_n: A \rightarrow \mathbb{R}$. The sequence (f_n) converges uniformly on A (to some function f) if and only if*

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} : m, n \geq N \implies |f_n(x) - f_m(x)| < \varepsilon \quad \forall x \in A.$$

Proof. $(\implies)$ If $f_n \rightrightarrows f$, given $\varepsilon > 0$ we take N such that $n \geq N$ implies $|f_n(x) - f(x)| < \varepsilon/2$ for every x . For $m, n \geq N$ and every $x \in A$, the triangle inequality gives

$$|f_n(x) - f_m(x)| \leq |f_n(x) - f(x)| + |f(x) - f_m(x)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

$(\impliedby)$ Suppose the uniform Cauchy condition holds. The argument has three steps.

Step 1: build the candidate f . Fix $x \in A$. The condition implies in particular that the sequence of real numbers $(f_n(x))_n$ is Cauchy in $\mathbb{R}$. Since $\mathbb{R}$ is complete, it converges; call $f(x) := \lim_n f_n(x)$. This defines a function $f: A \rightarrow \mathbb{R}$ and we already know that $f_n \rightarrow f$ pointwise.

Step 2: upgrade from pointwise to uniform. Let $\varepsilon > 0$ and take N from the Cauchy condition with $\varepsilon/2$: for $m, n \geq N$ and every $x \in A$,

$$|f_n(x) - f_m(x)| < \frac{\varepsilon}{2}.$$

Fix $n \geq N$ and $x \in A$, and let $m \rightarrow \infty$ in the inequality above. Since $f_m(x) \rightarrow f(x)$ and the absolute value is continuous, taking the limit preserves the non strict inequality:

$$|f_n(x) - f(x)| = \lim_{m \rightarrow \infty} |f_n(x) - f_m(x)| \leq \frac{\varepsilon}{2} < \varepsilon.$$

(Note the typical manoeuvre: the strict inequality $< \varepsilon/2$ may weaken to $\leq \varepsilon/2$ when taking the limit, which is why we start with $\varepsilon/2$ and finish comfortably below ε .)

Step 3: conclusion. The N we found depends only on ε , and the final bound holds for every $x \in A$: this is exactly $f_n \rightrightarrows f$. $\square$

Remark 7.16. Combining Theorem 7.15 with Theorem 7.11 we get a central result of the course: $(C[a, b], d_\infty)$ is a **complete metric space**. Indeed, if (f_n) is Cauchy in d_∞ , it is uniformly Cauchy, hence it converges uniformly to some f ; and by Theorem 7.11, that limit f is continuous, that is, $f \in C[a, b]$: the limit does not leave the space. This fact is the basis of the Banach fixed point theorem applied to spaces of functions (dynamic programming) and it will reappear when we study Banach spaces: $C[a, b]$ with the supremum norm is the canonical example.

7.6 Exchanging limit and integral [P4]

Theorem 7.17 (Passing to the limit under the integral). *Let $f_n: [a, b] \rightarrow \mathbb{R}$ be continuous with $f_n \Rightarrow f$ on $[a, b]$. Then (f is continuous, hence integrable, and)*

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b f(x) dx.$$

Proof. By Theorem 7.11, f is continuous on $[a, b]$ and therefore integrable: the statement makes sense.

We estimate the difference of integrals. We use two basic properties of the integral: linearity and the inequality $|\int f| \leq \int |f|$ (the integral of a function is dominated by the integral of its absolute value). Then:

$$\left| \int_a^b f_n - \int_a^b f \right| = \left| \int_a^b (f_n - f) \right| \leq \int_a^b |f_n(x) - f(x)| dx.$$

Now uniform convergence enters: the integrand is dominated by a *constant* (the same one for all x):

$$|f_n(x) - f(x)| \leq \sup_{y \in [a, b]} |f_n(y) - f(y)| = d_\infty(f_n, f) \quad \forall x \in [a, b],$$

and the integral of a constant over $[a, b]$ is the constant times the length of the interval:

$$\left| \int_a^b f_n - \int_a^b f \right| \leq d_\infty(f_n, f) (b - a).$$

Since $f_n \Rightarrow f$ means $d_\infty(f_n, f) \rightarrow 0$, the right hand side tends to 0, and by the squeeze theorem the difference of integrals tends to 0. This is exactly what we had to prove. $\square$

Remark 7.18. The proof shows why uniformity is the right hypothesis: the bound $d_\infty(f_n, f)(b-a)$ requires an error bound *valid on the whole interval at once* and an interval of *finite length*. Example 7.4 shows that without uniformity the theorem fails; Example 7.5 shows that on domains of infinite length extra care is needed. For derivatives the situation is more delicate: uniform convergence of f_n is *not* enough for $f'_n \rightarrow f'$ (classical counterexample: $f_n(x) = \frac{\sin(nx)}{\sqrt{n}} \Rightarrow 0$, but $f'_n(x) = \sqrt{n} \cos(nx)$ does not converge); the correct theorem asks for uniform convergence of the *derivatives*.

7.7 Series of functions and the Weierstrass M-test [P5]

A *series of functions* $\sum_{k=1}^{\infty} g_k$ converges (pointwise or uniformly) if the sequence of partial sums $S_n = g_1 + \dots + g_n$ does. The most used criterion for uniformity is:

Theorem 7.19 (Weierstrass M-test). *Let $g_k: A \rightarrow \mathbb{R}$. Suppose there are constants $M_k \geq 0$ such that*

$$|g_k(x)| \leq M_k \quad \forall x \in A, \quad \text{and} \quad \sum_{k=1}^{\infty} M_k < \infty.$$

Then $\sum_{k=1}^{\infty} g_k$ converges uniformly (and absolutely) on A .

Proof. We check the uniform Cauchy criterion (Theorem 7.15) for the partial sums S_n . For $m > n$ and every $x \in A$, the triangle inequality gives

$$|S_m(x) - S_n(x)| = \left| \sum_{k=n+1}^m g_k(x) \right| \leq \sum_{k=n+1}^m |g_k(x)| \leq \sum_{k=n+1}^m M_k.$$

Since the numerical series $\sum M_k$ converges, its tails are small: given $\varepsilon > 0$ there is N such that $m > n \geq N$ implies $\sum_{k=n+1}^m M_k < \varepsilon$ (Cauchy criterion for series of real numbers). The bound we obtained does not depend on x , so (S_n) is uniformly Cauchy and by Theorem 7.15 it converges uniformly on A . $\square$

Example 7.20. The series $\sum_{k=1}^{\infty} \frac{\cos(kx)}{k^2}$ converges uniformly on all of $\mathbb{R}$: taking $M_k = \frac{1}{k^2}$ we have $\left| \frac{\cos(kx)}{k^2} \right| \leq M_k$ for every x , and $\sum \frac{1}{k^2} < \infty$ (a p series with $p = 2 > 1$). By the M-test and Theorem 7.11, the sum is a continuous function on $\mathbb{R}$, even though we have no closed formula for it. This is how functions defined by series are built and studied (power series and Fourier series, for example).

Careful (common mistake)

Classical exam mistakes with convergence of functions:

- **Believing that pointwise implies uniform.** Never. Always check with the supremum. And remember the three pocket counterexamples: x^n on $[0, 1]$ (discontinuous limit), $nx(1 - x^2)^n$ (growing bump: the integral fails), the moving bump (the mass escapes to infinity).
- **Computing the supremum wrongly:** the supremum is taken over the given domain, not over all of $\mathbb{R}$. The same sequence can converge uniformly on $[0, a]$ and not on $[0, 1]$ (Example 7.9); if the exam changes the domain, the answer can change.
- **Evaluating the supremum at a fixed point:** M_n must be the supremum over x , which in general is attained at a point x_n that depends on n . If you only evaluate at fixed points, you may wrongly conclude that $M_n \rightarrow 0$.
- **Using the theorem backwards:** “the limit is continuous, therefore the convergence is uniform” is FALSE (the moving bump has a continuous limit without uniformity). The theorem only works in the direction: discontinuous limit $\implies$ convergence is not uniform.

Link with economics

Dynamic programming and the supremum norm. In macroeconomics and dynamic decision theory, the value function V of an agent’s problem (consumption and saving, optimal growth) is characterized as the solution of the Bellman equation $V = T(V)$, where T is the *Bellman operator*. The value function iteration algorithm generates the sequence of functions $V_{n+1} = T(V_n)$, and the Banach fixed point theorem (which we will

see in the chapter on contractions) guarantees that $V_n \rightarrow V$ *uniformly*, precisely because T is a contraction on the space of continuous and bounded functions with the supremum metric d_∞ , which is complete thanks to Theorems 7.15 and 7.11. Uniformity matters: it guarantees that the limit V is continuous (inherited from the V_n) and that the numerical error $\sup_x |V_n(x) - V(x)|$ is controlled globally, not only state by state. Similarly, in nonparametric econometrics, uniform convergence of estimators ($\sup_x |\hat{f}_n(x) - f(x)| \rightarrow 0$) is what lets you transfer properties of the estimator to the true function.

Chapter summary

- **Pointwise:** $\forall x \forall \varepsilon \exists N(\varepsilon, x)$: each column converges on its own. **Uniform:** $\forall \varepsilon \exists N(\varepsilon) \forall x$: a single N for the whole domain (the graph enters the tube of radius ε).
- Uniform $\implies$ pointwise; the converse is false (x^n , bumps, moving bumps). The only candidate for a uniform limit is the pointwise limit.
- **Supremum criterion:** $f_n \rightrightarrows f \iff M_n = \sup_A |f_n - f| \rightarrow 0$. In practice: maximize $|f_n - f|$ with calculus.
- The uniform limit of continuous functions is continuous ($\varepsilon/3$ argument). Useful contrapositive: discontinuous pointwise limit $\implies$ no uniformity.
- Uniform convergence = convergence in $(C[a, b], d_\infty)$, and this space is **complete** (uniform Cauchy plus continuity of the limit).
- Under uniform convergence on $[a, b]$: $\int_a^b f_n \rightarrow \int_a^b f$, with the bound $|\int f_n - \int f| \leq d_\infty(f_n, f)(b - a)$.
- **Weierstrass M-test:** $|g_k(x)| \leq M_k$ with $\sum M_k < \infty \implies \sum g_k$ converges uniformly.

7.8 Exam-style exercises [P4]

Exercise 7.21. For $f_n(x) = \frac{nx}{1+nx}$ on $A = [0, 1]$: (a) compute the pointwise limit; (b) determine whether the convergence is uniform on $[0, 1]$; (c) determine whether it is uniform on $[c, 1]$ for $0 < c < 1$.

Solution. (a) If $x = 0$: $f_n(0) = 0 \rightarrow 0$. If $0 < x \leq 1$ is fixed: dividing numerator and denominator by n , $f_n(x) = \frac{x}{\frac{1}{n}+x} \rightarrow \frac{x}{x} = 1$. The pointwise limit is

$$f(x) = \begin{cases} 0 & x = 0, \\ 1 & 0 < x \leq 1, \end{cases}$$

which is discontinuous at 0.

(b) Each f_n is continuous on $[0, 1]$ (a quotient of continuous functions with denominator $1 + nx \geq 1 > 0$), but the pointwise limit f is discontinuous at $x = 0$. By the uniform limit

theorem (Theorem 7.11), if the convergence were uniform, f would be continuous. Contradiction: the convergence **is not uniform** on $[0, 1]$. (Direct check: at $x = 0$ the error is 0, and for $x \in (0, 1]$ we have $|f_n(x) - 1| = \frac{1}{1+nx}$. Since $\frac{1}{1+nx} \rightarrow 1$ as $x \rightarrow 0^+$, the supremum over $(0, 1]$ is 1, even though it is not attained at any point. So $M_n = 1 \not\rightarrow 0$.)

(c) On $[c, 1]$ the pointwise limit is $f \equiv 1$. The error is $|f_n(x) - 1| = \frac{1}{1+nx}$, decreasing in x , so its supremum on $[c, 1]$ is attained at $x = c$:

$$M_n = \frac{1}{1+nc} \rightarrow 0.$$

By the supremum criterion, the convergence **is uniform** on $[c, 1]$. $\square$

Exercise 7.22. Let $f_n(x) = nxe^{-nx^2}$ on $[0, 1]$. Prove that $f_n \rightarrow 0$ pointwise, compute $M_n = \sup_{[0,1]} |f_n|$ and decide whether the convergence is uniform. Compute also $\lim_n \int_0^1 f_n$ and compare it with $\int_0^1 \lim_n f_n$.

Solution. *Pointwise limit.* $f_n(0) = 0$. For fixed $0 < x \leq 1$: $f_n(x) = \frac{nx}{e^{nx^2}}$; the denominator grows exponentially in n and the numerator only linearly, so $f_n(x) \rightarrow 0$ (for example, $e^{nx^2} \geq \frac{(nx^2)^2}{2}$, so $0 \leq f_n(x) \leq \frac{2nx}{n^2x^4} = \frac{2}{nx^3} \rightarrow 0$). Pointwise limit: $f \equiv 0$.

Supremum. We maximize f_n on $[0, 1]$ (it is ≥ 0):

$$f'_n(x) = ne^{-nx^2} + nx(-2nx)e^{-nx^2} = ne^{-nx^2}(1 - 2nx^2) = 0 \iff x_n = \frac{1}{\sqrt{2n}}.$$

For $n \geq 1$, $x_n \in (0, 1]$, and there is a maximum there (the derivative changes from $+$ to $-$). Then

$$M_n = f_n(x_n) = n \cdot \frac{1}{\sqrt{2n}} \cdot e^{-1/2} = \sqrt{\frac{n}{2}} e^{-1/2} \rightarrow \infty.$$

Since $M_n \not\rightarrow 0$ (it blows up), the convergence **is not uniform** on $[0, 1]$: there is a bump of increasing height $\sim \sqrt{n}$ attached to the origin.

Integrals. With the substitution $u = nx^2$, $du = 2nx \, dx$:

$$\int_0^1 nxe^{-nx^2} \, dx = \frac{1}{2} \int_0^n e^{-u} \, du = \frac{1 - e^{-n}}{2} \rightarrow \frac{1}{2},$$

while $\int_0^1 0 \, dx = 0$. So $\lim_n \int_0^1 f_n = \frac{1}{2} \neq 0 = \int_0^1 \lim_n f_n$: the exchange fails, consistent with the lack of uniformity (and confirming that the uniformity in Theorem 7.17 is not a decorative hypothesis). $\square$

Exercise 7.23. Let $f_n(x) = \sqrt{x^2 + \frac{1}{n}}$ on $\mathbb{R}$. (a) Find the pointwise limit f . (b) Prove that $f_n \rightrightarrows f$ on $\mathbb{R}$ by estimating $|f_n(x) - f(x)|$ without computing the supremum by differentiation. (c) Comment: what interesting property does this example have regarding differentiability?

Solution. (a) For fixed x , $x^2 + \frac{1}{n} \rightarrow x^2$ and the square root is continuous, so $f_n(x) \rightarrow \sqrt{x^2} = |x|$. The pointwise limit is $f(x) = |x|$.

(b) We use the rationalization trick (multiply and divide by the conjugate sum), valid because

$f_n(x) + |x| > 0$:

$$0 \leq f_n(x) - |x| = \frac{\left(x^2 + \frac{1}{n}\right) - x^2}{\sqrt{x^2 + \frac{1}{n} + |x|}} = \frac{1/n}{\sqrt{x^2 + \frac{1}{n} + |x|}} \leq \frac{1/n}{\sqrt{1/n}} = \frac{1}{\sqrt{n}},$$

where in the last inequality we dropped the term $|x| \geq 0$ from the denominator and used $\sqrt{x^2 + \frac{1}{n}} \geq \sqrt{\frac{1}{n}}$. The bound $\frac{1}{\sqrt{n}}$ does not depend on x , so

$$M_n = \sup_{x \in \mathbb{R}} |f_n(x) - f(x)| \leq \frac{1}{\sqrt{n}} \rightarrow 0,$$

and by the supremum criterion $f_n \Rightarrow |x|$ on all of $\mathbb{R}$.

(c) Each f_n is differentiable on all of $\mathbb{R}$ (it is the square root of a strictly positive polynomial: $x^2 + \frac{1}{n} \geq \frac{1}{n} > 0$), but the uniform limit $|x|$ **is not differentiable at 0** (it has a corner). Lesson: uniform convergence preserves continuity but *not* differentiability. To differentiate the limit you need uniform convergence of the derivatives, a hypothesis that fails here ($f'_n(x) = \frac{x}{\sqrt{x^2 + 1/n}} \rightarrow \text{sgn}(x)$, discontinuous at 0). $\square$

Exercise 7.24. Prove that the series $\sum_{k=1}^{\infty} \frac{x^k}{k^2}$ converges uniformly on $[-1, 1]$, and conclude that its sum is a continuous function on $[-1, 1]$.

Solution. We apply the Weierstrass M-test (Theorem 7.19). For every $x \in [-1, 1]$ and every $k \geq 1$:

$$\left| \frac{x^k}{k^2} \right| = \frac{|x|^k}{k^2} \leq \frac{1}{k^2} =: M_k,$$

because $|x| \leq 1$ implies $|x|^k \leq 1$. The majorant series $\sum_{k \geq 1} \frac{1}{k^2}$ is a p series with $p = 2 > 1$, hence it converges. By the M-test, $\sum_k \frac{x^k}{k^2}$ converges uniformly (and absolutely) on $[-1, 1]$.

Each partial sum $S_n(x) = \sum_{k=1}^n \frac{x^k}{k^2}$ is a polynomial, in particular continuous on $[-1, 1]$. Since S_n converges uniformly to the sum S , Theorem 7.11 guarantees that S is continuous on $[-1, 1]$. (Comment: in the interior $(-1, 1)$ this is a power series and the general theory already gave continuity; the extra value of the M-test here is uniformity *up to the endpoints* $x = \pm 1$ included.) $\square$

Exercise 7.25. Let $f_n, g_n: A \rightarrow \mathbb{R}$ with $f_n \Rightarrow f$ and $g_n \Rightarrow g$ on A . (a) Prove that $f_n + g_n \Rightarrow f + g$ on A . (b) Give a counterexample showing that $f_n g_n$ may *fail* to converge uniformly to $f g$. (c) Prove that if in addition f and g are bounded, then $f_n g_n \Rightarrow f g$ does hold.

Solution. (a) Let $\varepsilon > 0$. There are N_1, N_2 such that for every $x \in A$: $n \geq N_1 \Rightarrow |f_n(x) - f(x)| < \varepsilon/2$ and $n \geq N_2 \Rightarrow |g_n(x) - g(x)| < \varepsilon/2$. For $n \geq N := \max\{N_1, N_2\}$ and every $x \in A$, the triangle inequality gives

$$|(f_n + g_n)(x) - (f + g)(x)| \leq |f_n(x) - f(x)| + |g_n(x) - g(x)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

The N does not depend on x : uniform convergence.

(b) Counterexample on $A = \mathbb{R}$: take $f_n(x) = x$ (a sequence that is constant in n ; it converges uniformly to $f(x) = x$, with error identically 0) and $g_n(x) = \frac{1}{n}$ (constant functions; $\sup_x |g_n - 0| = \frac{1}{n} \rightarrow 0$, so $g_n \Rightarrow g \equiv 0$). The product is $f_n(x)g_n(x) = \frac{x}{n}$, whose pointwise limit is $fg \equiv 0$; but

$$\sup_{x \in \mathbb{R}} \left| \frac{x}{n} - 0 \right| = \infty \quad \text{for every } n,$$

so the convergence of the product is not uniform on $\mathbb{R}$ (this is Example 7.3). The problem: $f(x) = x$ is not bounded.

(c) Suppose $|f| \leq K$ and $|g| \leq K$ on A for some constant $K > 0$. First, the g_n are *uniformly bounded for large n* : there is N_0 such that $n \geq N_0$ implies $|g_n(x)| \leq |g(x)| + \sup_y |g_n(y) - g(y)| \leq K + 1$ for every x . Now we add and subtract fg_n (the standard trick for products):

$$|f_n g_n - fg| = |(f_n - f)g_n + f(g_n - g)| \leq |f_n - f| |g_n| + |f| |g_n - g|.$$

Taking suprema over $x \in A$, for $n \geq N_0$:

$$\sup_A |f_n g_n - fg| \leq (K + 1) \sup_A |f_n - f| + K \sup_A |g_n - g| \rightarrow 0,$$

since both suprema tend to 0 by hypothesis. By the supremum criterion, $f_n g_n \Rightarrow fg$ on A . $\square$

Chapter 8

Compactness and the Bolzano–Weierstrass theorem [P1]

Week 6 of the course. Bibliography: Alabiso and Weiss [1], Chs. 3–4; Pons [3], Ch. 8; Ok [4], Ch. C.

What is this used for in Microeconomics?

If there is one analysis chapter that supports a whole microeconomics course, this is it. The Weierstrass theorem (a continuous function on a compact set attains its maximum) is what guarantees that the optimization problems of the syllabus *have a solution*: consumer demand exists because the budget set is compact (closed and bounded, by Heine–Borel) and utility is continuous (demand theory, Mas-Colell, Whinston and Green, chs. 2 and 3), and the same argument gives the producer’s optimum (ch. 5). In game theory, the simplex of mixed strategies is compact and convex, which is exactly the hypothesis of the equilibrium existence theorems (ch. 8). And the existence of Walrasian equilibrium (ch. 17), of the optimal contract in principal agent problems (ch. 14) and of the optimal mechanism (ch. 23) all rest on the same idea. Give priority to sequential compactness, Bolzano–Weierstrass, Heine–Borel and Weierstrass.

8.1 Motivation [P1]

Compactness is, without exaggeration, **the most important concept of the course for an economist**. The reason is one word: *existence*. The basic question of economic theory (does an optimum exist? does an equilibrium exist?) is almost always answered with a compactness argument.

Consider the problem of maximizing a function f over a set A . Three things can go wrong:

1. f is not bounded above on A : for example, $f(x) = x$ on $A = \mathbb{R}$. There is no maximum because you can always do better.
2. f is bounded but the supremum escapes through a hole in the boundary: $f(x) = x$ on $A = (0, 1)$. The supremum is 1, but it is not attained: the point where it would be attained, $x = 1$, does not belong to A .

3. f is discontinuous and the supremum escapes through a jump: on $A = [0, 1]$, the function $f(x) = x$ for $x < 1$ with $f(1) = 0$ has supremum 1, not attained.

The Weierstrass theorem, which we prove here, says that if A is *compact* (which in $\mathbb{R}^n$ means closed and bounded: no escape to infinity and no holes in the boundary) and f is *continuous* (no jumps), then the maximum and the minimum **exist and are attained**. This is the theorem behind the existence of consumer demand, of firm supply, and (combined with fixed point theorems) of general equilibrium.

Along the way we prove the Bolzano–Weierstrass theorem (every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence), the Heine–Borel theorem (in $\mathbb{R}^n$, compact = closed + bounded), and we settle a pending debt: the equivalence of all norms on $\mathbb{R}^n$.

8.2 Compactness by open covers [P3]

We start with the official definition of compactness, the one that works in any topological space. It is the most abstract of the course, so we go slowly.

Definition 8.1 (Open cover). Let (X, d) be a metric space and $K \subseteq X$. An *open cover* of K is a family of open sets $\{U_i\}_{i \in I}$ (the index set I may be finite, countable or even uncountable) whose union contains K :

$$K \subseteq \bigcup_{i \in I} U_i.$$

A *finite subcover* is a finite subfamily $U_{i_1}, \dots, U_{i_m}$ that still covers: $K \subseteq U_{i_1} \cup \dots \cup U_{i_m}$.

Intuition

Imagine you want to cover the set K with open patches (balls, open intervals and so on), possibly using infinitely many patches. The question of compactness is: was it really necessary to use infinitely many patches, or would finitely many of them already cover everything? A set is compact if *any* way of covering it with open sets can be reduced to a finite number of patches. Compactness is a form of topological finiteness: the set may be infinite as a set of points, but it behaves as if it were finite when faced with open covers.

Definition 8.2 (Compact set). $K \subseteq X$ is *compact* if **every** open cover of K admits a finite subcover.

Careful (common mistake)

The quantifier is “**every** cover admits a finite subcover”, not “there is a finite cover”. Every set admits some trivial finite cover (for example $\{X\}$, a single open set), and that says nothing. To prove that K is compact you must start with an *arbitrary* open cover and extract a finite subcover from it. To prove that K is *not* compact it is enough to exhibit *one* concrete open cover with no finite subcover.

Example 8.3 ($(0, 1)$ is not compact). Consider the open sets $U_n = \left(\frac{1}{n}, 1\right)$ for $n = 2, 3, 4, \dots$. We claim that they cover $(0, 1)$: if $x \in (0, 1)$, then $x > 0$, and by the Archimedean property there is n with $\frac{1}{n} < x$, so $x \in U_n$. This is an open cover of $(0, 1)$.

However, no finite subfamily covers. Indeed, if we take $U_{n_1}, \dots, U_{n_m}$ and set $N = \max\{n_1, \dots, n_m\}$, then since the U_n are nested ($U_2 \subseteq U_3 \subseteq \dots$), the union of the chosen ones is $U_N = (\frac{1}{N}, 1)$, which does *not* contain the points of $(0, \frac{1}{N}]$, for example the point $\frac{1}{2N} \in (0, 1)$. Conclusion: $(0, 1)$ is not compact. The problem is the hole at the endpoint 0: points close to 0 need finer and finer patches and no finite set of patches covers them all.

Example 8.4 ($\mathbb{R}$ is not compact). Take $U_n = (-n, n)$ for $n \in \mathbb{N}$. They cover $\mathbb{R}$ (every real is in some $(-n, n)$, again by the Archimedean property). But a finite subfamily has union $(-N, N)$ with N the largest index, and leaves out, for example, the point N . Here the problem is not a hole but *unboundedness*: the set escapes to infinity.

These two examples show the two enemies of compactness: holes in the boundary (not closed) and escape to infinity (not bounded). The Heine–Borel theorem will say that, in $\mathbb{R}^n$, those are the *only* enemies.

8.3 Sequential compactness [P1]

In metric spaces there is a much more manageable characterization, in terms of sequences. Recall that a *subsequence* of (x_n) is a sequence $(x_{n_k})_k$ obtained by choosing strictly increasing indices $n_1 < n_2 < n_3 < \dots$.

Definition 8.5 (Sequential compactness). $K \subseteq X$ is *sequentially compact* if every sequence $(x_n) \subseteq K$ has a subsequence (x_{n_k}) that converges to a point $x \in K$. (The limit must stay in K .)

Intuition

A sequentially compact set is a trap for sequences: no matter how hard a sequence tries to escape (by oscillating, by spreading out), some part of it (a subsequence) is forced to accumulate at a point of the set itself. In $(0, 1)$ the sequence $x_n = \frac{1}{n}$ accumulates at 0, but $0 \notin (0, 1)$: the trap has a hole. In $\mathbb{R}$ the sequence $x_n = n$ does not accumulate anywhere: the trap is infinitely large and the prey escapes.

Theorem 8.6 (Equivalence in metric spaces). *Let (X, d) be a metric space and $K \subseteq X$. Then K is compact (by covers) if and only if K is sequentially compact.*

Proof (of the direct implication; the converse is sketched). **Compact $\implies$ sequentially compact.** We argue by contradiction. Suppose K is compact but there is a sequence $(x_n) \subseteq K$ without any subsequence converging to a point of K .

Preliminary claim: then no point $y \in K$ is an accumulation point of the sequence, that is, for each $y \in K$ there is a radius $r_y > 0$ such that the ball $B(y, r_y)$ contains x_n only for *finitely many* indices n . Indeed, if for some y every ball $B(y, \frac{1}{k})$ contained infinitely many terms of the sequence, we could choose increasing indices $n_1 < n_2 < \dots$ with $x_{n_k} \in B(y, \frac{1}{k})$, and that subsequence would converge to $y \in K$, contradicting our assumption.

Now we use compactness: the family $\{B(y, r_y)\}_{y \in K}$ is an open cover of K (each $y \in K$ lies in its own ball). By compactness there is a finite subcover:

$$K \subseteq B(y_1, r_{y_1}) \cup \cdots \cup B(y_m, r_{y_m}).$$

But each of these m balls contains only finitely many terms of the sequence, so their union contains only finitely many terms. This is absurd: the union contains K , and therefore *all* the terms x_n (which are infinitely many, counted with index). Contradiction.

Sequentially compact $\implies$ compact (idea). This direction is more technical and is usually omitted in the course; we give the scheme for anyone who wants to reconstruct it. One first proves that a sequentially compact metric space is *totally bounded* (for each $\varepsilon > 0$ it is covered by finitely many balls of radius ε ; otherwise one builds a sequence of points mutually separated by ε , with no convergent subsequence) and that every open cover admits a *Lebesgue number* $\delta > 0$ (every subset of diameter $< \delta$ fits inside some open set of the cover; otherwise there would be bad sets of diameter $\rightarrow 0$ and a subsequence argument gives a contradiction). Combining: cover K with finitely many balls of radius $\delta/3$, and each one fits inside an open set of the cover; those finitely many open sets are the subcover we wanted. The details are in Pons [3], Ch. 8, and Ok [4], Ch. C. $\square$

Remark 8.7. Thanks to this theorem, in metric spaces (which is where the whole course lives) we can freely use the sequential characterization, which is much more convenient for proving theorems: “take a sequence, extract a convergent subsequence, and work with the limit”. This will be our standard tool from now on.

8.4 The Bolzano–Weierstrass theorem [P1]

The heart of the whole theory of compactness in $\mathbb{R}^n$ is the following result about bounded sequences. Recall that $(x_n) \subseteq \mathbb{R}$ is *bounded* if there is $M > 0$ with $|x_n| \leq M$ for every n .

Theorem 8.8 (Bolzano–Weierstrass in $\mathbb{R}$). *Every bounded sequence of real numbers has a convergent subsequence.*

Proof. We use the *bisection method*, a hunt for the accumulation point that is worth mastering because it reappears in many proofs.

Step 0: setup. Let (x_n) be bounded: all its terms live in some interval $I_0 = [a_0, b_0]$. Key observation that drives everything: I_0 contains infinitely many terms of the sequence (it contains all of them), where “infinitely many terms” means that the set of indices $\{n : x_n \in I_0\}$ is infinite.

Step 1: bisect and keep the populated half. We split I_0 at its midpoint $m_0 = \frac{a_0+b_0}{2}$ into two halves $[a_0, m_0]$ and $[m_0, b_0]$. The infinitely many indices of I_0 are distributed between the two halves, so **at least one half contains infinitely many terms** (if both contained finitely many, their union would contain finitely many, which is absurd). We call $I_1 = [a_1, b_1]$ a half with infinitely many terms. Its length is half as large: $b_1 - a_1 = \frac{b_0 - a_0}{2}$.

Step 2: repeat indefinitely. Iterating the argument we obtain nested intervals

$$I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots, \quad \text{length}(I_k) = \frac{b_0 - a_0}{2^k} \longrightarrow 0,$$

each with infinitely many terms of the sequence.

Step 3: the limit point. The left endpoints a_k form an increasing sequence bounded above (by b_0); by the supremum axiom, $a_k \rightarrow x^* := \sup_k a_k$. Similarly b_k decreases to its infimum, and since $b_k - a_k \rightarrow 0$, both endpoints converge to the same x^* , which belongs to every I_k (it is the point trapped by the nesting; this is the *nested interval property*, a consequence of the completeness axiom of $\mathbb{R}$).

Step 4: build the subsequence. We choose indices as follows: n_1 is any index with $x_{n_1} \in I_1$; then, since I_2 contains *infinitely many* terms, there is $n_2 > n_1$ with $x_{n_2} \in I_2$ (here it is essential that there are infinitely many: there are always indices larger than the ones already used); in general, $n_k > n_{k-1}$ with $x_{n_k} \in I_k$.

Step 5: the subsequence converges to x^ .* Both x_{n_k} and x^* are in I_k , so

$$|x_{n_k} - x^*| \leq \text{length}(I_k) = \frac{b_0 - a_0}{2^k} \rightarrow 0.$$

Therefore $x_{n_k} \rightarrow x^*$, and we have found the convergent subsequence. $\square$

Theorem 8.9 (Bolzano–Weierstrass in $\mathbb{R}^n$). *Every bounded sequence in $\mathbb{R}^n$ (with any norm; say the Euclidean one) has a convergent subsequence.*

Proof. The idea: apply the real case *coordinate by coordinate*, extracting nested subsequences. We write it for $n = 2$; the general case is identical with n stages.

Let $(x_k) = (x_k^{(1)}, x_k^{(2)}) \subseteq \mathbb{R}^2$ be bounded: $\|x_k\| \leq M$. Since $|x_k^{(1)}| \leq \|x_k\| \leq M$, the first coordinate is a bounded real sequence; by Theorem 8.8 there is a subsequence of indices (k_j) such that $x_{k_j}^{(1)} \rightarrow a_1 \in \mathbb{R}$.

Now we look at the *second* coordinate, but only along those indices: $(x_{k_j}^{(2)})_j$ is also bounded by M , so it has in turn a convergent subsequence: there is a further subsequence of indices $(k_{j_l})_l$ with $x_{k_{j_l}}^{(2)} \rightarrow a_2$. Crucial point: when passing to a subsequence of a convergent sequence, convergence is preserved, so *also* $x_{k_{j_l}}^{(1)} \rightarrow a_1$ (the first coordinate is not spoiled).

Along the final indices k_{j_l} , both coordinates converge, and since in $\mathbb{R}^n$ convergence in norm is equivalent to coordinatewise convergence, we conclude $x_{k_{j_l}} \rightarrow (a_1, a_2)$.

For general $\mathbb{R}^n$ the refinement is repeated n times (extract, inside the previous subsequence, a new subsequence that fixes the next coordinate); after n stages, all coordinates converge simultaneously. $\square$

Exam strategy

The pattern “extract nested subsequences, one coordinate at a time” is an absolute exam classic, both in the proof of B–W in $\mathbb{R}^n$ and in variants (sequences in products of compact sets, diagonal arguments). Practice writing the subscripts carefully: the typical mistake is to extract the subsequence for the second coordinate from the *original* sequence instead of from the subsequence already obtained, so that the two convergences do not happen along the same indices and the proof does not close.

8.5 Compact implies closed and bounded; Heine–Borel [P1]

Definition 8.10. $A \subseteq X$ is *bounded* if it is contained in some ball: there are $x_0 \in X$ and $R > 0$ with $A \subseteq B(x_0, R)$. Equivalently, if its diameter $\text{diam}(A) = \sup\{d(x, y) : x, y \in A\}$ is finite.

Theorem 8.11 (Compact $\implies$ closed and bounded, in any metric space). *Let (X, d) be a metric space and $K \subseteq X$ compact. Then K is closed and bounded.*

Proof. We use the sequential characterization (Theorem 8.6).

K is closed. It is enough to show that K contains the limits of its convergent sequences (sequential characterization of closed sets). Let $(x_n) \subseteq K$ with $x_n \rightarrow x \in X$; we must prove $x \in K$. By sequential compactness, some subsequence $x_{n_k} \rightarrow y \in K$. But every subsequence of a convergent sequence converges to the *same* limit: $x_{n_k} \rightarrow x$. By uniqueness of the limit in metric spaces, $x = y \in K$. Done.

K is bounded. By contradiction: if K were not bounded, fixing any point $x_0 \in X$, for each $n \in \mathbb{N}$ there would be $x_n \in K$ with $d(x_n, x_0) > n$ (if there were none, $K \subseteq B(x_0, n)$ and it would be bounded). The sequence (x_n) lives in K ; by compactness it has a convergent subsequence $x_{n_k} \rightarrow y \in K$. But convergent sequences are bounded, while $d(x_{n_k}, x_0) > n_k \rightarrow \infty$: a contradiction (in detail: $d(x_{n_k}, x_0) \leq d(x_{n_k}, y) + d(y, x_0)$, and the right hand side converges to $d(y, x_0) < \infty$, hence it is bounded). $\square$

The converse is **false in general** (we will see it in Example 8.14), but **true in $\mathbb{R}^n$** : that is the famous Heine–Borel theorem.

Theorem 8.12 (Heine–Borel in $\mathbb{R}^n$). *$K \subseteq \mathbb{R}^n$ is compact if and only if K is closed and bounded.*

Proof. $(\implies)$ This is Theorem 8.11, valid in any metric space.

$(\impliedby)$ Let $K \subseteq \mathbb{R}^n$ be closed and bounded; we prove sequential compactness (which is equivalent to compactness, by Theorem 8.6). Let $(x_k) \subseteq K$ be an arbitrary sequence. Since K is bounded, the sequence is bounded; by Bolzano–Weierstrass in $\mathbb{R}^n$ (Theorem 8.9) there is a convergent subsequence $x_{k_j} \rightarrow x \in \mathbb{R}^n$. The delicate part remains: does the limit belong to K ? Here closedness enters: x is the limit of a sequence of points of K , and K is closed, so $x \in K$. Therefore every sequence in K has a convergent subsequence with limit in K : K is (sequentially) compact. $\square$

Remark 8.13. Anatomy of the theorem: **boundedness** provides the convergent subsequence (via B–W) and **closedness** guarantees that the limit does not escape from the set. If either one fails, there is a counterexample: $(0, 1]$ is bounded but not closed (the sequence $\frac{1}{n}$ has all its subsequences converging to $0 \notin (0, 1]$); $[0, \infty)$ is closed but not bounded (the sequence n has no convergent subsequence).

Example 8.14 (The closed unit ball of $C[0, 1]$ is not compact). This is *the* counterexample that separates finite dimension from infinite dimension, and an exam favorite. Consider the metric space $(C[0, 1], d_\infty)$ of continuous functions on $[0, 1]$ with the supremum metric, and its closed unit ball

$$\bar{B} = \left\{ f \in C[0, 1] : d_\infty(f, 0) = \sup_{x \in [0, 1]} |f(x)| \leq 1 \right\}.$$

$\bar{B}$ is closed and bounded in $(C[0, 1], d_\infty)$ (it is a closed ball). We claim that it is **not compact**.

Take the sequence $f_n(x) = x^n$, which satisfies $f_n \in \bar{B}$ ($\sup |x^n| = 1$). Suppose, seeking a contradiction, that some subsequence (f_{n_k}) converged in d_∞ to some $f \in \bar{B}$. Convergence in d_∞ is *uniform* convergence, which implies *pointwise* convergence; but we already know that the pointwise limit of x^n (and of any subsequence x^{n_k} , by the same computation) is the function equal to 0 on $[0, 1)$ and to 1 at $x = 1$, which is **discontinuous**. On the other hand, the uniform limit of continuous functions must be continuous. The two conclusions are incompatible: there is no such convergent subsequence in $C[0, 1]$.

Conclusion: in $(C[0, 1], d_\infty)$ there are closed and bounded sets that are not compact; **Heine–Borel is a theorem about $\mathbb{R}^n$, not a universal truth**. (In fact, a theorem of Riesz says that the closed unit ball of a normed space is compact *if and only if* the space is finite dimensional. In spaces of functions, compactness requires extra equicontinuity hypotheses: the Arzelà–Ascoli theorem, which these notes do not develop.)

8.6 Compactness and continuity [P1]

Theorem 8.15 (The continuous image of a compact set is compact). *Let (X, d_X) , (Y, d_Y) be metric spaces, $K \subseteq X$ compact and $f: X \rightarrow Y$ continuous. Then $f(K)$ is compact in Y .*

Proof. We use sequential compactness on both sides. Let (y_n) be an arbitrary sequence in $f(K)$; we must find a convergent subsequence with limit in $f(K)$.

Step 1: lift the sequence to K . By the definition of image, for each n there is $x_n \in K$ with $f(x_n) = y_n$.

Step 2: use the compactness of K . The sequence (x_n) lives in the compact set K , so it has a subsequence $x_{n_k} \rightarrow x^* \in K$.

Step 3: come back down with continuity. By the sequential criterion for continuity, $f(x_{n_k}) \rightarrow f(x^*)$, that is, $y_{n_k} \rightarrow f(x^*)$. And $f(x^*) \in f(K)$ because $x^* \in K$.

We have found a subsequence of (y_n) converging to a point of $f(K)$: $f(K)$ is sequentially compact, hence compact. $\square$

Theorem 8.16 (Weierstrass extreme value theorem). *Let K be a nonempty compact metric space and $f: K \rightarrow \mathbb{R}$ continuous. Then f is bounded and **attains** its maximum and its minimum: there are $x_{\max}, x_{\min} \in K$ such that*

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}) \quad \text{for every } x \in K.$$

Proof. We prove it in full detail, because it is *the* theorem of the course for an economist.

Step 1: $f(K)$ is compact. By Theorem 8.15, $f(K) \subseteq \mathbb{R}$ is compact; by Theorem 8.11, $f(K)$ is closed and bounded in $\mathbb{R}$. In particular f is a bounded function.

Step 2: the supremum exists. $f(K)$ is a nonempty subset of $\mathbb{R}$ (since $K \neq \emptyset$) and bounded above; by the supremum axiom there is $s := \sup f(K) \in \mathbb{R}$.

Step 3: the supremum belongs to $f(K)$. By the characterization of the supremum, for each

$n \in \mathbb{N}$ the number $s - \frac{1}{n}$ is not an upper bound, so there is $y_n \in f(K)$ with

$$s - \frac{1}{n} < y_n \leq s.$$

By the squeeze theorem, $y_n \rightarrow s$: the supremum is a limit of points of $f(K)$. Since $f(K)$ is *closed*, it contains that limit: $s \in f(K)$.

Step 4: translate. $s \in f(K)$ means that there is $x_{\max} \in K$ with $f(x_{\max}) = s = \sup f(K)$, that is, $f(x_{\max}) \geq f(x)$ for every $x \in K$: the maximum **is attained**. The argument for the minimum is symmetric (use the infimum, or apply the above to $-f$, which is also continuous). $\square$

Link with economics

Existence of consumer demand. The consumer's problem is

$$\max_{x \in B(p, w)} u(x), \quad B(p, w) = \{x \in \mathbb{R}_+^n : p \cdot x \leq w\},$$

where u is the utility function, $p \gg 0$ the price vector and $w > 0$ wealth. The budget set $B(p, w)$ is **closed** (it is defined by non strict inequalities with continuous functions: $x_i \geq 0$ and $p \cdot x \leq w$) and **bounded** (each coordinate satisfies $0 \leq x_i \leq w/p_i$). By Heine–Borel (Theorem 8.12), $B(p, w)$ is compact; it is also nonempty ($0 \in B$). If u is continuous, the Weierstrass theorem (Theorem 8.16) guarantees that at least one optimal bundle $x^*(p, w)$ **exists**: demand is well defined. Without compactness existence fails: if some price were 0, the budget set would be unbounded and an insatiable consumer would have no optimal demand. The same template (compact constraint plus continuous objective) gives the existence of firm supply with compact production sets, and it is the first link in the proofs of existence of general equilibrium (where it is combined with fixed point theorems and with Berge's maximum theorem, which we will see at the end of the course).

8.7 Uniform continuity and Heine's theorem [P3]

Definition 8.17 (Uniform continuity). $f: (X, d_X) \rightarrow (Y, d_Y)$ is *uniformly continuous* on X if

$$\forall \varepsilon > 0, \exists \delta > 0 : \quad d_X(x, x') < \delta \implies d_Y(f(x), f(x')) < \varepsilon \quad \forall x, x' \in X.$$

Intuition

Compare with ordinary continuity: there, once a point x_0 is fixed, the δ may depend on x_0 (and on ε). In uniform continuity, a single $\delta = \delta(\varepsilon)$ must work at *all* points of the domain at once. It is the same improvement of quantifiers that took us from pointwise to uniform convergence. Example of failure: $f(x) = x^2$ on $\mathbb{R}$ is continuous but not uniformly continuous: to move f by less than ε near a large point x_0 , you need $\delta \approx \frac{\varepsilon}{2x_0}$, which shrinks as x_0 grows; no fixed δ works on the whole line.

Theorem 8.18 (Heine–Cantor). *If K is a compact metric space and $f: K \rightarrow (Y, d_Y)$ is continuous, then f is uniformly continuous.*

Proof. By contradiction, with the subsequence argument. Suppose f is continuous but *not* uniformly continuous. Negating uniform continuity (careful with the negation of quantifiers) gives: there is $\varepsilon_0 > 0$ such that for *every* $\delta > 0$ there are points at distance $< \delta$ whose images are at distance $\geq \varepsilon_0$. Taking $\delta = \frac{1}{n}$ for each $n \in \mathbb{N}$, we obtain sequences $(x_n), (z_n) \subseteq K$ with

$$d(x_n, z_n) < \frac{1}{n} \quad \text{and} \quad d_Y(f(x_n), f(z_n)) \geq \varepsilon_0.$$

By compactness of K , (x_n) has a subsequence $x_{n_k} \rightarrow x^* \in K$. We claim that $z_{n_k} \rightarrow x^*$ as well: by the triangle inequality,

$$d(z_{n_k}, x^*) \leq d(z_{n_k}, x_{n_k}) + d(x_{n_k}, x^*) < \frac{1}{n_k} + d(x_{n_k}, x^*) \rightarrow 0.$$

Now the continuity of f at x^* (sequential criterion) gives $f(x_{n_k}) \rightarrow f(x^*)$ and $f(z_{n_k}) \rightarrow f(x^*)$. Then

$$d_Y(f(x_{n_k}), f(z_{n_k})) \leq d_Y(f(x_{n_k}), f(x^*)) + d_Y(f(x^*), f(z_{n_k})) \rightarrow 0,$$

which contradicts $d_Y(f(x_{n_k}), f(z_{n_k})) \geq \varepsilon_0 > 0$ for every k . The contradiction proves the theorem. $\square$

8.8 Star application: equivalence of norms on $\mathbb{R}^n$ [P4]

In the chapter on norms we announced that on $\mathbb{R}^n$ *all* norms are equivalent. We now have the tools (compactness plus Weierstrass) to prove it.

Theorem 8.19 (Equivalence of all norms on $\mathbb{R}^n$). *Let $\|\cdot\|$ be any norm on $\mathbb{R}^n$ and let $\|x\|_1 = \sum_{i=1}^n |x_i|$. There are constants $0 < c \leq C$ such that*

$$c \|x\|_1 \leq \|x\| \leq C \|x\|_1 \quad \forall x \in \mathbb{R}^n.$$

Consequently, any two norms on $\mathbb{R}^n$ are equivalent to each other (both are comparable with $\|\cdot\|_1$).

Proof. Step 1: upper bound (the easy one). Let $\{e_1, \dots, e_n\}$ be the canonical basis. For $x = \sum_i x_i e_i$, the triangle inequality and homogeneity give

$$\|x\| = \left\| \sum_{i=1}^n x_i e_i \right\| \leq \sum_{i=1}^n |x_i| \|e_i\| \leq \left(\max_{1 \leq i \leq n} \|e_i\| \right) \sum_{i=1}^n |x_i| = C \|x\|_1,$$

with $C := \max_i \|e_i\| > 0$.

Step 2: $\|\cdot\|$ is continuous with respect to the 1 norm. By the reverse triangle inequality and Step 1:

$$| \|x\| - \|y\| | \leq \|x - y\| \leq C \|x - y\|_1,$$

so the function $x \mapsto \|x\|$ is (Lipschitz) continuous on $(\mathbb{R}^n, \|\cdot\|_1)$.

Step 3: compactness of the unit sphere. Let $S = \{x \in \mathbb{R}^n : \|x\|_1 = 1\}$. S is closed (the preimage of the closed set $\{1\}$ under the continuous function $\|\cdot\|_1$) and bounded (its points have norm 1), so, by Heine–Borel (Theorem 8.12), S is compact. And it is nonempty ($e_1 \in S$).

Step 4: Weierstrass on the sphere. The continuous function $x \mapsto \|x\|$ attains its minimum on the compact set S (Theorem 8.16): there is $x^* \in S$ with

$$c := \|x^*\| = \min_{x \in S} \|x\|.$$

Crucial: $c > 0$. Indeed, $x^* \in S$ implies $\|x^*\|_1 = 1 \neq 0$, so $x^* \neq 0$, and a norm vanishes only at the zero vector; by the definition of norm, $\|x^*\| > 0$.

Step 5: scale to all of $\mathbb{R}^n$. Let $x \neq 0$ be arbitrary. The normalized vector $\frac{x}{\|x\|_1}$ belongs to S , so $\left\| \frac{x}{\|x\|_1} \right\| \geq c$. By homogeneity of the norm,

$$\frac{\|x\|}{\|x\|_1} \geq c \implies \|x\| \geq c \|x\|_1.$$

For $x = 0$ the inequality is trivial ($0 \geq 0$). Together with Step 1, $c\|x\|_1 \leq \|x\| \leq C\|x\|_1$ for every x . $\square$

Remark 8.20. Consequences: in $\mathbb{R}^n$, the notions of convergence, Cauchy sequence, open, closed, bounded and compact set, and continuity **do not depend on which norm is used**. This is why in finite dimension one picks the most convenient norm for each computation. In infinite dimension this is radically false (the norms $\|\cdot\|_1$ and $\|\cdot\|_\infty$ on $C[0, 1]$ are not equivalent), and that is why in functional analysis you must always specify the norm. Note where the proof used finite dimension: in the compactness of the sphere (Heine–Borel), exactly what fails in infinite dimension (Example 8.14).

Careful (common mistake)

Common mistakes with compactness:

- **“Closed and bounded = compact” outside $\mathbb{R}^n$.** False: it holds in $\mathbb{R}^n$ (Heine–Borel) but not in spaces of functions (Example 8.14). The implication compact $\implies$ closed and bounded is universal.
- **Forgetting that the limit must stay in K :** in the sequential definition it is not enough that a subsequence converges to something; the limit must belong to K . $(0, 1)$ fails exactly for this reason.
- **Weierstrass without checking hypotheses:** the theorem requires compact *and* continuous *and* nonempty. In an exam, check the three things explicitly (closed + bounded + Heine–Borel is the standard route in $\mathbb{R}^n$).
- **Confusing maximum with supremum:** Weierstrass does not only say that f is bounded (finite supremum), it says that the supremum *is attained* (it is a maximum). That difference is the whole point of the theorem.

Chapter summary

- **Compact (covers):** every open cover admits a finite subcover. **Compact (sequential):** every sequence has a convergent subsequence with limit inside the set. In metric spaces they are equivalent.
- **Bolzano–Weierstrass:** every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence (bisection in $\mathbb{R}$; coordinate by coordinate with nested subsequences in $\mathbb{R}^n$).
- $\text{Compact} \implies \text{closed and bounded}$ (always). **Heine–Borel:** in $\mathbb{R}^n$ the converse holds. In infinite dimension it does not: the closed ball of $C[0, 1]$ is not compact.
- The continuous image of a compact set is compact. **Weierstrass:** a continuous f on a nonempty compact set attains its maximum and minimum. This is the existence theorem for optima.
- **Heine–Cantor:** continuous on a compact set $\implies$ uniformly continuous.
- With compactness plus Weierstrass one proves that **all norms on $\mathbb{R}^n$ are equivalent** (positive minimum of the norm on the compact unit sphere).

8.9 Exam-style exercises [P1]

Exercise 8.21. Decide, justifying with the theorems of this chapter, which of the following subsets of $\mathbb{R}^2$ (with the Euclidean norm) are compact:

$$A = \{(x, y) : x^2 + y^2 \leq 4\}, \quad B = \{(x, y) : x^2 + y^2 < 4\},$$

$$C = \{(x, y) : x \geq 0, y \geq 0\}, \quad D = \{(x, y) : x \in [0, 1], y = \sin(1/x)\} \cup \{(0, 0)\}.$$

(In D , read $x \in (0, 1]$ for the part with $\sin(1/x)$.)

Solution. We use Heine–Borel: in $\mathbb{R}^2$, compact $\iff$ closed and bounded.

A (*closed disc*): it is bounded (contained in the ball of radius 4) and closed (the preimage of the closed set $(-\infty, 4]$ under the continuous function $(x, y) \mapsto x^2 + y^2$). **Compact.**

B (*open disc*): bounded but *not closed*: the sequence $(2 - \frac{1}{n}, 0) \in B$ converges to $(2, 0) \notin B$. **Not compact.**

C (*first quadrant*): closed (intersection of the closed preimages $\{x \geq 0\}$ and $\{y \geq 0\}$) but *not bounded*: it contains $(n, 0)$ for every n . **Not compact.** (The sequence $(n, 0)$ has no convergent subsequence.)

D : it is bounded: $x \in [0, 1]$ and $|y| \leq 1$. Is it closed? The limit points of the graph $\{(x, \sin(1/x)) : x \in (0, 1]\}$ that are not on it are those of the segment $\{0\} \times [-1, 1]$: when $x_n \rightarrow 0^+$, $\sin(1/x_n)$ oscillates and can converge to *any* value of $[-1, 1]$ with a suitable choice of x_n . But D only adds the point $(0, 0)$, not the whole segment. For example, taking x_n such that $1/x_n = \frac{\pi}{2} + 2\pi n$, we get $(x_n, 1) \in D$ and $(x_n, 1) \rightarrow (0, 1) \notin D$. So D is not closed: **not compact.** (If the whole segment $\{0\} \times [-1, 1]$ were added, the resulting set would be closed and bounded, hence compact.) $\square$

Exercise 8.22. Let (X, d) be a metric space, $K \subseteq X$ compact and $F \subseteq K$ closed (in X). Prove that F is compact. (“A closed subset of a compact set is compact.”)

Solution. Sequential route. Let $(x_n) \subseteq F$ be an arbitrary sequence. Since $F \subseteq K$ and K is compact, there is a subsequence $x_{n_k} \rightarrow x^* \in K$. Now we use that F is closed: x^* is the limit of a sequence of points of F (the subsequence), and closed sets contain the limits of their convergent sequences; so $x^* \in F$. Every sequence of F then has a convergent subsequence with limit in F : F is sequentially compact and, by the equivalence in metric spaces, compact.

(Cover route, for comparison: if $\{U_i\}$ is an open cover of F , then $\{U_i\} \cup \{X \setminus F\}$ is an open cover of K , since $X \setminus F$ is open and covers whatever the U_i do not cover of K ; we extract a finite subcover of K , and discarding $X \setminus F$, which contributes no points of F , we are left with a finite subcover of F .) $\square$

Exercise 8.23. Let $K \subseteq \mathbb{R}^n$ be nonempty and compact and let $x_0 \in \mathbb{R}^n$. Prove that there is a point $k^* \in K$ *closest* to x_0 , that is,

$$\|x_0 - k^*\| = \inf_{k \in K} \|x_0 - k\| = \text{dist}(x_0, K).$$

Is this still true if K is only closed (not bounded)? And if it is only bounded (not closed)?

Solution. Define $f: K \rightarrow \mathbb{R}$ by $f(k) = \|x_0 - k\|$. The function f is continuous: by the reverse triangle inequality, $|f(k) - f(k')| = ||x_0 - k| - |x_0 - k'||| \leq \|k' - k\|$, so f is 1-Lipschitz. Since K is compact and nonempty, the Weierstrass theorem (applied to the *minimum*) gives $k^* \in K$ with $f(k^*) = \min_{k \in K} f(k) = \text{dist}(x_0, K)$. That is what was asked.

K closed and unbounded: existence **still holds**, with a local compactification trick: fix any $k_0 \in K$ and let $R = \|x_0 - k_0\|$. The points of K at distance greater than R from x_0 are irrelevant for the infimum, so it is enough to minimize over $K \cap \bar{B}(x_0, R)$, which is closed (an intersection of closed sets) and bounded, hence compact and nonempty (k_0 is there); Weierstrass concludes.

K bounded and not closed: it **fails**. Example in $\mathbb{R}$: $K = (0, 1)$ and $x_0 = 2$. Then $\text{dist}(x_0, K) = \inf_{k \in (0, 1)} (2 - k) = 1$, but no $k \in (0, 1)$ satisfies $2 - k = 1$ (it would require $k = 1 \notin K$): the infimum is not attained. $\square$

Exercise 8.24. Let $f: K \rightarrow \mathbb{R}$ be continuous on a compact set K , with $f(x) > 0$ for every $x \in K$. Prove that there is $\alpha > 0$ such that $f(x) \geq \alpha$ for every $x \in K$ (“ f is uniformly bounded away from zero”). Give a counterexample showing that this fails without compactness.

Solution. By Weierstrass (Theorem 8.16), f attains its minimum: there is $x_{\min} \in K$ with $f(x_{\min}) \leq f(x)$ for every $x \in K$. Define $\alpha := f(x_{\min})$. By hypothesis $f(x_{\min}) > 0$ (positivity holds *at each point*, in particular at $x_{\min}$), so $\alpha > 0$ and $f(x) \geq \alpha$ for every x : exactly what was asked. The subtle point is that without compactness the infimum could be 0 without being attained.

Counterexample without compactness: $f(x) = x$ on $K = (0, 1)$ (not closed): $f > 0$ at every point but $\inf f = 0$; there is no $\alpha > 0$ below f . Another one: $f(x) = e^{-x}$ on $K = [0, \infty)$ (not bounded): $f > 0$ always, $\inf f = 0$. This result (“positive on a compact set implies uniformly positive”) is used constantly, for example to bound denominators in proofs. $\square$

Exercise 8.25. Let $K_1 \supseteq K_2 \supseteq K_3 \supseteq \cdots$ be nonempty compact sets in a metric space X (a nested sequence). Prove that $\bigcap_{n=1}^{\infty} K_n \neq \emptyset$ (Cantor intersection property). Show that the result fails if the K_n are only nonempty closed sets.

Solution. For each n , choose a point $x_n \in K_n$ (possible: $K_n \neq \emptyset$). All the terms live in K_1 : indeed, $x_n \in K_n \subseteq K_1$ by nesting. By compactness of K_1 , there is a subsequence $x_{n_k} \rightarrow x^* \in K_1$.

We claim that $x^* \in K_m$ for *every* m . Fix m . For $n_k \geq m$ we have $x_{n_k} \in K_{n_k} \subseteq K_m$ (nesting again). So the tail of the subsequence from the point where $n_k \geq m$ is a sequence of points of K_m converging to x^* ; since K_m is compact, it is in particular *closed* (Theorem 8.11), so it contains the limit: $x^* \in K_m$. As m was arbitrary, $x^* \in \bigcap_m K_m$, which is therefore nonempty.

With closed but not compact sets it fails: in $\mathbb{R}$, take $K_n = [n, \infty)$. They are closed, nonempty and nested ($[n+1, \infty) \subseteq [n, \infty)$), but their intersection is empty: no real number is greater than or equal to all the naturals (Archimedean property). The ingredient that saves the compact version is exactly the sequential trap: the compactness of K_1 prevents the witnesses x_n from escaping to infinity. $\square$

Chapter 9

Connectedness and the intermediate value theorem [P3]

Week 7 of the course. Bibliography: Alabiso and Weiss [1], Ch. 3; Pons [3]; Ok [4], Ch. D.

What is this used for in Microeconomics?

The part of this chapter that shows up again in microeconomics is the intermediate value theorem (IVT): it is the simplest existence argument for equilibrium that there is. In an exchange economy with two goods (the Edgeworth box, Mas-Colell, Whinston and Green, ch. 15) you normalize $p_2 = 1$ and study the excess demand for good 1 as a function of the price p_1 : if it is continuous, positive at low prices and negative at high prices, the IVT guarantees a price where excess demand is exactly zero, and that is the Walrasian equilibrium. The rest of the chapter (connectedness in general) is background: it explains *why* the IVT works and it prepares the fixed point theorems, which are the same argument in higher dimension (ch. 17).

9.1 Motivation [P3]

Some sets are intuitively “one single piece” (an interval, a disk, a spiral) and others are “broken into pieces”, such as the union of two separated intervals or the set $\mathbb{Z}$ of the integers. *Connectedness* is the topological way of making this idea of “a single piece” precise.

Why does this matter to an analyst or to an economist? Because of the **intermediate value theorem**: a continuous function defined on a set that is one single piece cannot “skip” values. If the excess demand of a market is positive at low prices and negative at high prices, and it varies continuously with the price, then *at some intermediate price it is exactly zero*: there you have an equilibrium. This type of argument (continuity + connected domain $\implies$ no values are skipped) is one of the most elementary and most powerful existence tools in economic theory, and it is the one dimensional ancestor of the fixed point theorems.

The plan of the chapter: define connectedness, prove that the connected subsets of $\mathbb{R}$ are exactly the intervals, show that continuity preserves connectedness, deduce the intermediate value theorem, and present the more geometric variant of path-connectedness.

9.2 Connected spaces and sets [P3]

Definition 9.1 (Separation and connected space). Let (X, d) be a metric space. A *separation* of X is a pair of sets $U, V \subseteq X$ such that:

1. U and V are open in X ;
2. $U \neq \emptyset$ and $V \neq \emptyset$;
3. $U \cap V = \emptyset$;
4. $U \cup V = X$.

The space X is *connected* if it admits **no** separation. A subset $A \subseteq X$ is connected if it is connected as a metric subspace (with the inherited metric); concretely, A is *disconnected* if there are open sets U, V of X with

$$A \subseteq U \cup V, \quad U \cap A \neq \emptyset, \quad V \cap A \neq \emptyset, \quad U \cap V \cap A = \emptyset,$$

and connected otherwise.

Intuition

A separation is a “clean cut” of the space into two nonempty pieces that do not touch each other. The requirement that both pieces be *open* is what makes the cut clean: each piece is well detached from the other, with no shared boundary points. Note that in a separation each piece is also closed (it is the complement of the other open set): the pieces are open and closed at the same time (“clopen”). In fact, a useful restatement is: X is connected if and only if its only subsets that are both open and closed are $\emptyset$ and X .

Why is it not enough to ask for any two disjoint pieces? Because every set with two or more points splits trivially into disjoint pieces (one point on one side, the rest on the other). It is the topology (openness) that captures that the pieces are *really separated* and not just different.

Example 9.2 (A disconnected set). Let $A = [0, 1] \cup [2, 3] \subseteq \mathbb{R}$. Take the open sets of $\mathbb{R}$

$$U = \left(-\frac{1}{2}, \frac{3}{2}\right), \quad V = \left(\frac{3}{2}, \frac{7}{2}\right).$$

Then $A \subseteq U \cup V$; $U \cap A = [0, 1] \neq \emptyset$; $V \cap A = [2, 3] \neq \emptyset$; and $U \cap V = \emptyset$ (so in particular $U \cap V \cap A = \emptyset$). This is a separation of A : the set is **disconnected**. The two “blocks” $[0, 1]$ and $[2, 3]$ each sit inside a different open set, and the gap $(1, 2)$ is what lets you build the clean cut.

Other useful disconnected sets: $\mathbb{Q}$ is disconnected (cut it with $U = (-\infty, \sqrt{2})$, $V = (\sqrt{2}, \infty)$: both are open, they cover $\mathbb{Q}$ because $\sqrt{2} \notin \mathbb{Q}$, and both contain rationals); $\mathbb{Z}$ is disconnected; and in general any set with a gap that you can cut through.

Careful (common mistake)

To prove that a set is *disconnected* it is enough to **exhibit** one concrete separation (two open sets with the four properties). To prove that it is *connected* you must show that

no separation exists, which is an argument by contradiction: “suppose U, V separate...”. Mixing up the direction of the quantifier is the number one mistake in this topic. And when you check a separation, do not forget to verify *all four* conditions: points are usually lost by not checking that both pieces are nonempty or that the sets are open.

9.3 The connected subsets of $\mathbb{R}$ are the intervals [P3]

Recall that $I \subseteq \mathbb{R}$ is an *interval* if together with each pair of points it contains everything between them:

$$x, y \in I, x < z < y \implies z \in I.$$

(The intervals are: $\emptyset$, single points, and the sets of the form (a, b) , $[a, b]$, $[a, b)$, $(a, b]$, $(-\infty, b)$, $(-\infty, b]$, (a, ∞) , $[a, \infty)$, $\mathbb{R}$.)

Theorem 9.3 (Characterization of the connected subsets of $\mathbb{R}$). *A subset $A \subseteq \mathbb{R}$ is connected if and only if A is an interval.*

Proof. (**Connected $\implies$ interval**). We prove the contrapositive: if A is not an interval, then A is disconnected. That A is not an interval means that there are $x, y \in A$ and $z \notin A$ with $x < z < y$: there is a gap z between two points of the set. That gap gives you the cut:

$$U = (-\infty, z), \quad V = (z, \infty).$$

Both are open in $\mathbb{R}$; $A \subseteq U \cup V$ because the only real number missing is z , which is not in A ; $x \in U \cap A$ and $y \in V \cap A$, so both intersections are nonempty; and $U \cap V = \emptyset$. This is a separation: A is disconnected.

(**Interval $\implies$ connected**). This is the deep direction: it uses the supremum axiom (the completeness of $\mathbb{R}$). Let I be an interval and suppose, for contradiction, that a separation exists: open sets U, V of $\mathbb{R}$ with $I \subseteq U \cup V$, $U \cap I \neq \emptyset$, $V \cap I \neq \emptyset$, $U \cap V \cap I = \emptyset$.

Choose $a \in U \cap I$ and $b \in V \cap I$; without loss of generality $a < b$ (if $a > b$, swap the roles of U and V). Since I is an interval, the whole segment $[a, b]$ is contained in I .

The idea: you walk from a towards b staying in U as long as possible, and you look at the “last reachable point”. Define

$$s := \sup S, \quad \text{where } S = \{x \in [a, b] : [a, x] \subseteq U\}.$$

The set S is nonempty ($a \in S$, since $[a, a] = \{a\} \subseteq U$) and is bounded above by b ; by the supremum axiom, s exists and $a \leq s \leq b$. In particular $s \in [a, b] \subseteq I$, so s is in U or in V (and not in both). We rule out both possibilities:

Case 1: $s \in U$. Since U is open, there is $\varepsilon > 0$ with $(s - \varepsilon, s + \varepsilon) \subseteq U$. First, this implies $[a, s] \subseteq U$: the points of $[a, s - \varepsilon]$ are in U because, by the definition of supremum, there is $x \in S$ with $x > s - \varepsilon$, and then $[a, x] \subseteq U$ covers $[a, s - \varepsilon]$; and the points of $(s - \varepsilon, s]$ are in the neighborhood $(s - \varepsilon, s + \varepsilon) \subseteq U$. Now, if $s < b$, you can move a bit further: taking $\delta > 0$ small with $s + \delta < b$ and $s + \delta < s + \varepsilon$, you get $[a, s + \delta] \subseteq U$, so $s + \delta \in S$, which contradicts that s

is an upper bound of S . Therefore it would have to be $s = b$; but then $b = s \in U$, and $b \in V$, that is, $b \in U \cap V \cap I = \emptyset$: contradiction.

Case 2: $s \in V$. Since V is open, there is $\varepsilon > 0$ with $(s - \varepsilon, s + \varepsilon) \subseteq V$. Note that $s > a$ (since $a \in U$ and $U \cap V \cap I = \emptyset$ prevent $a \in V$, and if $s = a$ then $s \in U$, which is Case 1). By the definition of supremum, there is $x \in S$ with $x > s - \varepsilon$. That same x produces the contradiction. On one hand, $x \in S$ means $[a, x] \subseteq U$, so $x \in U$. On the other hand, $x \leq s$ (because s is an upper bound of S) and $x > s - \varepsilon$, so that $x \in (s - \varepsilon, s + \varepsilon) \subseteq V$. And also $x \in [a, b] \subseteq I$. Then $x \in U \cap V \cap I = \emptyset$: contradiction.

Both cases are impossible: no separation exists, and I is connected. $\square$

Remark 9.4. This proof is the model case of the “supremum argument”, or last reachable point method, a direct relative of the classical proofs of the intermediate value theorem. Notice where the completeness of $\mathbb{R}$ enters: in the mere existence of $s = \sup S$. Over $\mathbb{Q}$ the theorem is false ($\mathbb{Q}$ is an “interval with holes” and is totally disconnected), precisely because suprema may fail to exist in $\mathbb{Q}$.

9.4 Continuity and connectedness [P2]

Theorem 9.5 (The continuous image of a connected set is connected). *Let (X, d_X) , (Y, d_Y) be metric spaces, $f: X \rightarrow Y$ continuous and $A \subseteq X$ connected. Then $f(A) \subseteq Y$ is connected.*

Proof. By contradiction. Suppose $f(A)$ is disconnected: there are open sets U, V of Y such that

$$f(A) \subseteq U \cup V, \quad U \cap f(A) \neq \emptyset, \quad V \cap f(A) \neq \emptyset, \quad U \cap V \cap f(A) = \emptyset.$$

The strategy: *pull the separation back with f and contradict the connectedness of A .* Define

$$U' := f^{-1}(U) = \{x \in X : f(x) \in U\}, \quad V' := f^{-1}(V).$$

We check that U', V' separate A :

1. *Open:* by the topological characterization of continuity, the preimage of an open set under a continuous function is open. So U' and V' are open in X .
2. *They cover A :* if $x \in A$, then $f(x) \in f(A) \subseteq U \cup V$, so $f(x) \in U$ or $f(x) \in V$, that is, $x \in U'$ or $x \in V'$.
3. *They meet A :* since $U \cap f(A) \neq \emptyset$, there is a point $f(x) \in U$ with $x \in A$; that x lies in $U' \cap A$. In the same way $V' \cap A \neq \emptyset$.
4. *Disjoint over A :* if there were $x \in U' \cap V' \cap A$, then $f(x) \in U \cap V \cap f(A) = \emptyset$, which is absurd.

So U', V' form a separation of A , contradicting that A is connected. Therefore $f(A)$ is connected. $\square$

Theorem 9.6 (Intermediate value theorem, general version). *Let X be a connected metric space and $f: X \rightarrow \mathbb{R}$ continuous. If $a, b \in X$ and $c \in \mathbb{R}$ satisfies $f(a) < c < f(b)$, then there is $x \in X$ with $f(x) = c$: the function attains every intermediate value between two values that it takes.*

Proof. By Theorem 9.5, $f(X) \subseteq \mathbb{R}$ is connected. By Theorem 9.3, $f(X)$ is an interval. The numbers $f(a)$ and $f(b)$ belong to $f(X)$ and $f(a) < c < f(b)$; by the definition of interval (it contains every point between two of its points), $c \in f(X)$. That is, there is $x \in X$ with $f(x) = c$. $\square$

Corollary 9.7 (Classical intermediate value theorem (Bolzano)). *Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous with $f(a) < c < f(b)$ (or $f(b) < c < f(a)$). Then there is $x \in (a, b)$ with $f(x) = c$. In particular (the case $c = 0$, Bolzano's theorem): if $f(a)$ and $f(b)$ have opposite signs, f has a root in (a, b) .*

Proof. $[a, b]$ is an interval, hence connected (Theorem 9.3); apply Theorem 9.6. The point x with $f(x) = c$ cannot be a or b because $f(a) \neq c \neq f(b)$, so $x \in (a, b)$. $\square$

Example 9.8 (Existence of roots). Every polynomial of odd degree $p(x) = x^{2k+1} + a_{2k}x^{2k} + \dots + a_0$ has at least one real root. Indeed, for large x the leading term dominates: $p(x) \rightarrow +\infty$ as $x \rightarrow +\infty$ and $p(x) \rightarrow -\infty$ as $x \rightarrow -\infty$; so there are $a < b$ with $p(a) < 0 < p(b)$. Since p is continuous on $[a, b]$, Bolzano (Corollary 9.7) gives a root in (a, b) . The same method shows, for example, that $x = \cos x$ has a solution: the function $g(x) = \cos x - x$ satisfies $g(0) = 1 > 0$ and $g(\frac{\pi}{2}) = -\frac{\pi}{2} < 0$.

Proposition 9.9 (Fixed point on $[0, 1]$: a preview of the fixed point chapter). *Every continuous function $f: [0, 1] \rightarrow [0, 1]$ has at least one fixed point: there is $x^* \in [0, 1]$ with $f(x^*) = x^*$.*

Proof. Define $g: [0, 1] \rightarrow \mathbb{R}$ by $g(x) = f(x) - x$, which is continuous (a difference of continuous functions). Evaluate at the endpoints, using that f takes values in $[0, 1]$:

$$g(0) = f(0) - 0 = f(0) \geq 0, \quad g(1) = f(1) - 1 \leq 0.$$

If $g(0) = 0$, then $x^* = 0$ is a fixed point; if $g(1) = 0$, then $x^* = 1$ is one. In the remaining case, $g(0) > 0 > g(1)$, and Bolzano's theorem (Corollary 9.7) applied to g on $[0, 1]$ gives $x^* \in (0, 1)$ with $g(x^*) = 0$, that is, $f(x^*) = x^*$.

Geometrically: the graph of f enters the square $[0, 1]^2$ on the left (at height $f(0) \geq 0$, on or above the diagonal) and leaves on the right (at height $f(1) \leq 1$, on or below the diagonal); being continuous, at some moment it must cross the diagonal $y = x$. This result is the one dimensional case of **Brouwer's fixed point theorem**, a central piece in the existence proofs of general equilibrium (Arrow–Debreu); in the next chapter you will see the Banach fixed point theorem, which also gives uniqueness and an algorithm under stronger assumptions. $\square$

Link with economics

Existence of an equilibrium price. Consider a market for one good with excess demand function $z: (0, \infty) \rightarrow \mathbb{R}$, $z(p) = D(p) - S(p)$, continuous in the price p . Suppose that at very low prices there is excess demand ($z(p_1) > 0$ for some p_1 : a good that is almost free is demanded more than it is supplied) and at very high prices there is excess supply ($z(p_2) < 0$ for some $p_2 > p_1$). The interval $[p_1, p_2]$ is connected and z is continuous, so by the intermediate value theorem there is $p^* \in (p_1, p_2)$ with $z(p^*) = 0$: **the market clears at p^*** ; that is the equilibrium price. Note the anatomy of the argument:

the *connectedness* of the price domain prevents excess demand from “jumping” from a positive sign to a negative one without passing through zero. Similar arguments prove other “middle value” results: the existence of a tax rate that collects a given amount (with a continuous Laffer curve), of an internal rate of return (net present value is continuous and changes sign), or of an exchange rate that balances the trade account. In higher dimension (many markets at once) the intermediate value theorem is no longer enough and you need Brouwer or Kakutani, but the logic (continuity plus a topological property of the domain give existence) is the same.

9.5 Path-connectedness [P4]

There is a more intuitive and more geometric notion of “one single piece”: being able to travel continuously between any two points without leaving the set.

Definition 9.10 (Path and path-connectedness). Let (X, d) be a metric space and $A \subseteq X$. A *path* in A from x to y is a continuous function $\gamma: [0, 1] \rightarrow A$ with $\gamma(0) = x$ and $\gamma(1) = y$. The set A is *path-connected* if for every pair $x, y \in A$ there is a path in A from x to y .

Theorem 9.11. *Every path-connected set is connected.*

Proof. Let A be path-connected and suppose, for contradiction, that U, V form a separation of A : open sets with $A \subseteq U \cup V$, $U \cap A \neq \emptyset \neq V \cap A$, $U \cap V \cap A = \emptyset$.

Choose $x \in U \cap A$ and $y \in V \cap A$. By assumption there is a continuous path $\gamma: [0, 1] \rightarrow A$ with $\gamma(0) = x$, $\gamma(1) = y$. Consider the image $\gamma([0, 1]) \subseteq A$. The interval $[0, 1]$ is connected (Theorem 9.3) and γ is continuous, so $\gamma([0, 1])$ is connected (Theorem 9.5).

But U and V separate $\gamma([0, 1])$: they cover it (it sits inside A), $x = \gamma(0) \in U \cap \gamma([0, 1])$, $y = \gamma(1) \in V \cap \gamma([0, 1])$, and $U \cap V \cap \gamma([0, 1]) \subseteq U \cap V \cap A = \emptyset$. This contradicts the connectedness of $\gamma([0, 1])$. So there is no separation of A : it is connected. $\square$

Remark 9.12. The converse is false in general: the classical example is the topologist’s sine curve, $\{(x, \sin \frac{1}{x}) : x \in (0, 1]\} \cup (\{0\} \times [-1, 1])$, which is connected but not path-connected (you cannot “walk” to the vertical segment from the oscillating part). For *open* subsets of $\mathbb{R}^n$ the two notions do coincide. In practice, to prove that a concrete set is connected the easiest route is usually to exhibit paths (often segments) and apply Theorem 9.11.

Proposition 9.13 (Convex sets are path-connected). *Every convex set $A \subseteq \mathbb{R}^n$ (in particular: all of $\mathbb{R}^n$, balls, half spaces, budget sets) is path-connected, and therefore connected.*

Proof. Recall: A is *convex* if for all $x, y \in A$ and $t \in [0, 1]$ the point $(1 - t)x + ty$ belongs to A (the segment between two points of the set stays inside).

Given $x, y \in A$, define the “straight line” path

$$\gamma: [0, 1] \rightarrow \mathbb{R}^n, \quad \gamma(t) = (1 - t)x + ty.$$

First, γ takes values in A : that is exactly the definition of convexity. Second, $\gamma(0) = x$ and $\gamma(1) = y$. Third, γ is continuous: for $t, s \in [0, 1]$,

$$\|\gamma(t) - \gamma(s)\| = \|(s - t)x + (t - s)y\| = |t - s| \|y - x\|,$$

which goes to 0 as $t \rightarrow s$ (in fact γ is Lipschitz with constant $\|y - x\|$). So every pair of points of A is joined by a path in A : A is path-connected and, by Theorem 9.11, connected. $\square$

Definition 9.14 (Connected components). Given a metric space X , the *connected component* of a point $x \in X$ is the union of all connected subsets of X that contain x ; it is the maximal connected set containing x . The connected components form a partition of X : every space decomposes in a unique way into its maximal connected “pieces”. For example, $[0, 1] \cup [2, 3]$ has exactly two components, $[0, 1]$ and $[2, 3]$; and the components of $\mathbb{Q}$ are the single points ($\mathbb{Q}$ is *totally disconnected*).

Exam strategy

Typical questions on this topic and how to attack them:

- “*Prove that A is disconnected*”: exhibit the two open sets and check the four conditions, one by one. Look for the gap where you can cut.
- “*Prove that A is connected*”: if $A \subseteq \mathbb{R}$, check that it is an interval and cite Theorem 9.3. If $A \subseteq \mathbb{R}^n$, try convexity (Proposition 9.13) or build explicit paths (sometimes by joining segments: a path from x to z and one from z to y glue into a path from x to y).
- “*Prove that there is x with $f(x) = c$* ”: this is almost surely the intermediate value theorem. Define the right auxiliary function (usually $g = f - c$, or $g(x) = f(x) - c$ for fixed points), evaluate at two points where it has opposite signs, check continuity and a connected domain, and cite Bolzano.
- *Combinations*: “ f continuous on a connected set and taking values only in $\mathbb{Z}$ (or in $\{0, 1\}$) $\implies f$ is constant” is a classic: the image must be an interval inside $\mathbb{Z}$, and the only intervals contained in $\mathbb{Z}$ are single points.

Careful (common mistake)

Frequent mistakes:

- **Applying the intermediate value theorem without connectedness.** On $[0, 1] \cup [2, 3]$, a continuous function can be -1 on the first block and $+1$ on the second *without ever vanishing*: each block goes its own way. Before invoking Bolzano, check that the domain is an interval (or connected).
- **Believing that the continuous image of an open interval is an interval of the same type.** All you get is that it is an *interval*: $f(x) = x^2$ maps $(-1, 1)$ onto $[0, 1)$ (half open), and $\sin$ maps $(0, 2\pi)$ onto $[-1, 1]$ (closed).

- **Confusing connected with convex.** Convex $\implies$ connected, but an annulus in $\mathbb{R}^2$ is connected without being convex. Convexity is a *linear* property (segments), connectedness is a *topological* one (separations).
- **Forgetting the equality case at the endpoints** when using Bolzano for fixed points: first check $g(0) = 0$ or $g(1) = 0$; only after that apply the intermediate value theorem with strict inequalities.

Chapter summary

- **Separation:** two nonempty open sets, disjoint (over the set), that cover it. **Connected** = no separations = the only subsets that are both open and closed are $\emptyset$ and the whole space.
- **In $\mathbb{R}$: connected $\iff$ interval** (the hard direction uses the supremum argument and the completeness of $\mathbb{R}$).
- The **continuous image of a connected set is connected** (pull the separation back with preimages).
- **Intermediate value theorem:** f continuous on a connected set, with real values, attains every value between two of its values. Bolzano case: a sign change $\implies$ a root. Application: every continuous $f: [0, 1] \rightarrow [0, 1]$ has a fixed point (via $g(x) = f(x) - x$).
- **Path-connected $\implies$ connected;** the converse fails (topologist's sine curve). **Convex $\implies$ path-connected** (segments). Connected components: the maximal pieces.
- In economics: the intermediate value theorem gives existence of equilibrium prices, roots of excess demand, internal rates of return, and it is the one dimensional ancestor of Brouwer and Kakutani.

9.6 Exam-style exercises [P3]

Exercise 9.15. Determine, with justification, whether the following sets are connected: (a) $A = \mathbb{Q} \cap [0, 1]$; (b) $B = \{(x, y) \in \mathbb{R}^2 : xy = 1\}$; (c) $C = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\} \cup \{(x, y) : (x - 1)^2 + y^2 \leq 1\}$.

Solution. (a) **Not connected.** Use an irrational number in $(0, 1)$, for example $\frac{\sqrt{2}}{2}$. The open sets $U = (-\infty, \frac{\sqrt{2}}{2})$ and $V = (\frac{\sqrt{2}}{2}, \infty)$ cover A (no rational equals $\frac{\sqrt{2}}{2}$), they are disjoint, and both meet A (for instance $0 \in U \cap A$, $1 \in V \cap A$). This is a separation. (Alternative: A is not an interval, since $0, 1 \in A$ but $\frac{\sqrt{2}}{2} \notin A$; by Theorem 9.3, it is not connected.)

(b) **Not connected.** B is the hyperbola $y = 1/x$, with one branch in the first quadrant and another in the third. Separation: $U = \{(x, y) : x > 0\}$ and $V = \{(x, y) : x < 0\}$, both open (preimages of open sets under the continuous projection $(x, y) \mapsto x$), disjoint, and they

cover B (there are no points of B with $x = 0$, since $xy = 1$ forbids it), and each one meets B : $(1, 1) \in U \cap B$, $(-1, -1) \in V \cap B$.

(c) **It is connected.** C is the union of two closed disks, with centers $(0, 0)$ and $(1, 0)$ and radius 1, which intersect (for example $(\frac{1}{2}, 0)$ belongs to both: $\frac{1}{4} \leq 1$ and $\frac{1}{4} \leq 1$). Each disk is convex, hence path-connected (Proposition 9.13). Given any two points of C , you travel from the first one to the common point $(\frac{1}{2}, 0)$ along a segment inside its disk, and from there to the second point along a segment inside the other disk; joining the two paths gives a continuous path in C . So C is path-connected and, by Theorem 9.11, connected. (A useful general rule: the union of two connected sets with a point in common is connected.) $\square$

Exercise 9.16. Let X be a connected metric space and $f: X \rightarrow \mathbb{R}$ continuous with $f(x) \neq 0$ for every $x \in X$. Prove that f has constant sign: either $f(x) > 0$ for every x , or $f(x) < 0$ for every x .

Solution. Suppose, for contradiction, that f takes both signs: there are $a, b \in X$ with $f(a) < 0 < f(b)$. Since X is connected and f is continuous, the intermediate value theorem (Theorem 9.6) with the intermediate value $c = 0$ (note that $f(a) < 0 < f(b)$) guarantees that there is $x \in X$ with $f(x) = 0$. This contradicts the assumption $f(x) \neq 0$ for every x . Therefore f cannot take both signs, and since it never vanishes, it is either always positive or always negative.

(An instructive topological restatement: $\{f > 0\}$ and $\{f < 0\}$ are open (preimages of $(0, \infty)$ and $(-\infty, 0)$), disjoint, and they cover X because f is never 0; if both were nonempty, they would separate X . Connectedness forces one of the two to be empty.) $\square$

Exercise 9.17. Prove that there is no continuous and surjective function $f: [0, 1] \rightarrow (0, 1) \cup (2, 3)$. Then either construct a continuous and surjective function $g: [0, 1] \rightarrow [0, 1] \cup [2, 3]$ or explain why it cannot exist either.

Solution. *First part.* $[0, 1]$ is an interval, hence connected. If f were continuous and surjective, its image $f([0, 1]) = (0, 1) \cup (2, 3)$ would be connected (Theorem 9.5). But $(0, 1) \cup (2, 3)$ is disconnected: the open sets $U = (0, 1)$ and $V = (2, 3)$ (the two pieces themselves) separate it: they are open, nonempty, disjoint and they cover it. Contradiction: such an f does not exist.

Second part. No such g can exist either, by the same argument: $[0, 1] \cup [2, 3]$ is disconnected (Example 9.2: separate it with $(-\frac{1}{2}, \frac{3}{2})$ and $(\frac{3}{2}, \frac{7}{2})$), while the continuous image of the connected set $[0, 1]$ must be connected. The moral: connectedness is an *invariant* under continuous surjections; a connected set is never continuously squashed onto a disconnected one. (What does exist is a continuous function from $[0, 1]$ onto $[0, 1]$, or even onto $[0, 1]^2$ (Peano curves), but never onto a space made of two pieces.) $\square$

Exercise 9.18. A hiker climbs a hill along a trail: she leaves the base at 8:00 and reaches the top at 18:00. The next day she walks down along the same trail, leaving the top at 8:00 and reaching the base at 18:00. Prove that there is a time of day at which she was at exactly the same point of the trail on both days. (Formalize it with the intermediate value theorem.)

Solution. Parametrize time by $t \in [8, 18]$ (hours) and the position on the trail by its distance to the base, normalized to $[0, L]$. Let $u(t)$ be the position on the day of the climb and $d(t)$ the

position on the day of the descent; both are continuous functions (physical motion does not jump), with

$$u(8) = 0, \quad u(18) = L, \quad d(8) = L, \quad d(18) = 0.$$

Define the auxiliary function $h(t) = u(t) - d(t)$, continuous on the (connected) interval $[8, 18]$. Evaluate at the endpoints:

$$h(8) = u(8) - d(8) = 0 - L = -L < 0, \quad h(18) = u(18) - d(18) = L - 0 = L > 0.$$

By Bolzano's theorem (Corollary 9.7), there is $t^* \in (8, 18)$ with $h(t^*) = 0$, that is, $u(t^*) = d(t^*)$: at time t^* she was at the same point of the trail on both days. (An equivalent picture: imagine the two hikers walking *on the same day*, one going up and the other going down; since they walk the same trail in opposite directions, they must meet.) This is the exact template of the equilibrium argument: two continuous processes that “cross” must agree at some intermediate point. $\square$

Exercise 9.19. Let $f: [0, 2] \rightarrow \mathbb{R}$ be continuous with $f(0) = f(2)$. Prove that there are $x, y \in [0, 2]$ with $y - x = 1$ and $f(x) = f(y)$ (two points at distance exactly 1 where f takes the same value).

Solution. Define $g: [0, 1] \rightarrow \mathbb{R}$ by $g(x) = f(x+1) - f(x)$; it is well defined (if $x \in [0, 1]$, then $x+1 \in [1, 2] \subseteq [0, 2]$) and continuous (a composition and a difference of continuous functions). You are looking for a root of g : if $g(x^*) = 0$, then $f(x^*+1) = f(x^*)$ and the points $x = x^*$, $y = x^*+1$ solve the problem.

Evaluate g at the endpoints:

$$g(0) = f(1) - f(0), \quad g(1) = f(2) - f(1) = f(0) - f(1),$$

where in the last equality we used the assumption $f(2) = f(0)$. Therefore

$$g(1) = -g(0).$$

Case 1: $g(0) = 0$. Then $x^* = 0$ works. Case 2: $g(0) \neq 0$. Then $g(0)$ and $g(1) = -g(0)$ have strictly opposite signs, and by Bolzano (Corollary 9.7) on the connected interval $[0, 1]$ there is $x^* \in (0, 1)$ with $g(x^*) = 0$. In both cases you are done.

(Comment: this is the “universal chord theorem” in its simplest case; the same trick of the shifted function $g(x) = f(x+h) - f(x)$ shows up in many exam problems. The assumption $f(0) = f(2)$ plays the role of a “closed cycle”: what goes up must come down, and the shifted difference must change sign.) $\square$

Chapter 10

Contraction mappings and the Banach fixed point theorem [P2]

Week 7 of the course. Bibliography: Moroşanu [2], Ch. 1; Ok [4], Chs. C–E.

What is this used for in Microeconomics?

The two most important existence results in microeconomics *are* fixed point theorems. The existence of Nash equilibrium in mixed strategies is proved by applying Brouwer or Kakutani to the best response correspondence (game theory, Mas-Colell, Whinston and Green, ch. 8), and the existence of Walrasian equilibrium is obtained by applying Kakutani to the excess demand correspondence (general equilibrium, ch. 17). Both theorems are stated at the end of this chapter. The Banach theorem that is proved here in detail is the perfect toy model for understanding what a fixed point is and why it exists, and contractions show up again in repeated games and in recursive models, where discounting makes the operator that defines the continuation value a contraction. The priority investment for micro is the idea of a fixed point and the statements of Brouwer and Kakutani; the proof of Banach is a priority of the analysis course.

10.1 Motivation: solving equations of the form $x = f(x)$ [P2]

A huge number of problems in mathematics and in economics reduce to solving an equation of the form

$$x = f(x),$$

that is, to finding a point that the function f leaves *fixed*. Some examples:

- **Nonlinear equations.** Solving $\cos(x) = x$ is exactly finding a fixed point of $f(x) = \cos(x)$. There is no closed formula: you need a method that *guarantees* that the solution exists, that it is unique, and that you can approximate it.
- **Systems of linear equations.** A system $Ax = b$ can be rewritten as $x = (I - A)x + b =: f(x)$, and solved by iteration.
- **Integral and differential equations.** The existence and uniqueness of solutions of

ordinary differential equations (Picard–Lindelöf theorem) is proved by finding a fixed point of an integral operator.

- **Dynamic programming.** The Bellman equation

$$v(x) = \max_a \{u(x, a) + \beta v(g(x, a))\}$$

is a fixed point equation, not in $\mathbb{R}^n$, but in a *space of functions*: the unknown is the whole value function v . This is why this chapter is central in the mathematical training of an economist: the Banach fixed point theorem is *the* theorem that guarantees that the value function exists and is unique.

The strategy will always be the same: iterate. You start from any candidate x_0 , compute $x_1 = f(x_0)$, then $x_2 = f(x_1)$, and so on. The central question is: when does this sequence converge, and does it converge to a solution? The answer is given by the notion of a *contraction*.

Intuition

Imagine that f is a photocopier that shrinks everything to 70%. If you apply the photocopier again and again to a map of Peru, the copies get smaller and smaller and “converge” to a single point. That point is the fixed point: the only place on the map that matches itself in the shrunken copy. A contraction does exactly that to distances: it *pulls* points closer together by a fixed proportion smaller than 1, and that is why the iteration “spirals in” towards a single point.

10.2 Fixed points and contractions [P3]

Definition 10.1 (Fixed point). Let X be a set and $f: X \rightarrow X$ a function. A point $x^* \in X$ is a **fixed point** of f if

$$f(x^*) = x^*.$$

Example 10.2. 1. $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$ has two fixed points: you solve $x^2 = x$, that is $x(x - 1) = 0$, so $x^* = 0$ and $x^* = 1$.

2. $f(x) = x + 1$ has no fixed point in $\mathbb{R}$: the equation $x + 1 = x$ has no solution.

3. $f(x) = x$ (the identity) has infinitely many fixed points: every point of $\mathbb{R}$.

These three examples show that, without extra assumptions, there may be several fixed points, none, or infinitely many. You need conditions on f that guarantee *existence and uniqueness*.

Definition 10.3 (Contraction mapping). Let (X, d) be a metric space. A function $f: X \rightarrow X$ is a **contraction mapping**, or simply a **contraction**, if there is a constant $k \in [0, 1)$ such that

$$d(f(x), f(y)) \leq k d(x, y) \quad \text{for all } x, y \in X.$$

The constant k is called the **contraction constant** (or contraction modulus).

Remark 10.4. It is essential that the *same* constant $k < 1$ works for *all* pairs of points. The weaker condition

$$d(f(x), f(y)) < d(x, y) \quad \text{for all } x \neq y$$

(functions satisfying it are called weakly contractive, or strictly nonexpansive) is **not** enough to guarantee the existence of a fixed point, as you will see in Example 10.7.

Proposition 10.5. *Every contraction is continuous (in fact, uniformly continuous and Lipschitz).*

Proof. Let f be a contraction with constant k . Given $\varepsilon > 0$, take $\delta = \varepsilon$ (if $k = 0$, any δ works; if $k > 0$ you can even take $\delta = \varepsilon/k$). If $d(x, y) < \delta$, then

$$d(f(x), f(y)) \leq k d(x, y) < k \delta \leq \delta = \varepsilon,$$

using $k < 1$. Since δ does not depend on the point x , the continuity is uniform. Recall that a function is *Lipschitz* if $d(f(x), f(y)) \leq L d(x, y)$ for some constant $L \geq 0$; a contraction is exactly a Lipschitz function with constant $L < 1$. $\square$

Example 10.6 (A contraction on $\mathbb{R}$). Let $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{1}{2}x + 3$. Then, with the usual metric $d(x, y) = |x - y|$:

$$|f(x) - f(y)| = \left| \frac{1}{2}x + 3 - \frac{1}{2}y - 3 \right| = \frac{1}{2} |x - y|.$$

It is a contraction with $k = \frac{1}{2}$. Its fixed point is found by solving $x = \frac{1}{2}x + 3$, which gives $x^* = 6$. Look at what happens when you iterate from $x_0 = 0$: you get $x_1 = 3$, $x_2 = 4.5$, $x_3 = 5.25$, $x_4 = 5.625$, $\dots$, approaching 6 and cutting the error in half at each step.

Example 10.7 (Pulling points closer is not enough: $f(x) = x + 1/x$). Let $X = [1, \infty)$ with the usual metric (a complete metric space, since it is a closed subset of $\mathbb{R}$) and let

$$f(x) = x + \frac{1}{x}.$$

Step 1: f maps X into X . If $x \geq 1$ then $f(x) = x + 1/x \geq x \geq 1$, so $f: X \rightarrow X$ is well defined.

Step 2: f pulls points strictly closer. For $1 \leq x < y$:

$$|f(y) - f(x)| = \left| y - x + \frac{1}{y} - \frac{1}{x} \right| = \left| (y - x) - \frac{y - x}{xy} \right| = (y - x) \left| 1 - \frac{1}{xy} \right| < y - x,$$

where we used that $x, y \geq 1$ with $x \neq y$ implies $xy > 1$, so $0 < 1 - \frac{1}{xy} < 1$. That is, f pulls *every* pair of distinct points closer.

Step 3: but f has no fixed point. The equation $x = x + 1/x$ implies $1/x = 0$, which has no solution.

Step 4: why does this contradict nothing? Because f is **not** a contraction: the factor $1 - \frac{1}{xy}$ approaches 1 as $x, y \rightarrow \infty$, so there is no *uniform* $k < 1$ that works for all pairs. This example shows that the uniformity of k in Definition 10.3 is not a technical whim: it is the assumption that makes the whole theorem work.

Careful (common mistake)

In exams it is a classic to ask for an example of a function that satisfies $d(f(x), f(y)) < d(x, y)$ for all $x \neq y$ but has no fixed point. Memorize $f(x) = x + 1/x$ on $[1, \infty)$. The common mistake is to believe that “pulling points closer” already means being a contraction: **it does not**; you need a constant $k < 1$ that works for all pairs at once.

10.3 The Banach fixed point theorem [P3]

This is one of the most important theorems of the whole course and one of the favorites in exams: its proof uses, in a chain, almost everything seen so far (Cauchy sequences, completeness, continuity, uniqueness of the limit). It helps to learn it as a story in four acts.

Theorem 10.8 (Banach fixed point theorem, or contraction mapping principle). *Let (X, d) be a nonempty **complete** metric space, and let $f: X \rightarrow X$ be a **contraction** with constant $k \in [0, 1)$. Then:*

1. *f has exactly one fixed point $x^* \in X$;*
2. *for any starting point $x_0 \in X$, the sequence of iterates defined by $x_{n+1} = f(x_n)$ converges to x^* .*

Proof. Fix an arbitrary starting point $x_0 \in X$ and define the sequence of iterates $x_1 = f(x_0)$, $x_2 = f(x_1)$, and in general $x_{n+1} = f(x_n)$. The proof has four steps.

Step (i): the distances between consecutive terms shrink geometrically. We claim that for every $n \geq 0$:

$$d(x_{n+1}, x_n) \leq k^n d(x_1, x_0).$$

We prove it by induction on n .

Base case ($n = 0$): the inequality says $d(x_1, x_0) \leq k^0 d(x_1, x_0) = d(x_1, x_0)$, which is trivially true.

Inductive step: suppose $d(x_{n+1}, x_n) \leq k^n d(x_1, x_0)$ (the induction hypothesis). Then

$$d(x_{n+2}, x_{n+1}) = d(f(x_{n+1}), f(x_n)) \leq k d(x_{n+1}, x_n) \leq k \cdot k^n d(x_1, x_0) = k^{n+1} d(x_1, x_0),$$

where the first inequality is the contraction property applied to the pair (x_{n+1}, x_n) and the second is the induction hypothesis. This completes the induction.

Step (ii): the sequence (x_n) is Cauchy. Let $m > n \geq 0$. The idea is to go from x_n to x_m by “hopping” through all the intermediate terms and using the triangle inequality repeatedly:

$$d(x_m, x_n) \leq d(x_m, x_{m-1}) + d(x_{m-1}, x_{m-2}) + \cdots + d(x_{n+1}, x_n).$$

Applying Step (i) to each summand:

$$d(x_m, x_n) \leq (k^{m-1} + k^{m-2} + \cdots + k^n) d(x_1, x_0) = k^n (1 + k + k^2 + \cdots + k^{m-1-n}) d(x_1, x_0).$$

The sum in parentheses is a partial geometric sum with ratio k ; since $0 \leq k < 1$, it is bounded by the full geometric series:

$$1 + k + \cdots + k^{m-1-n} \leq \sum_{j=0}^{\infty} k^j = \frac{1}{1-k}.$$

Therefore, for all $m > n$:

$$d(x_m, x_n) \leq \frac{k^n}{1-k} d(x_1, x_0). \quad (*)$$

Now check the definition of a Cauchy sequence. Given $\varepsilon > 0$: since $k^n \rightarrow 0$ (because $0 \leq k < 1$), there is $N \in \mathbb{N}$ such that for every $n \geq N$ you have $\frac{k^n}{1-k} d(x_1, x_0) < \varepsilon$. (If $d(x_1, x_0) = 0$ then x_0 is already a fixed point and there is nothing to prove in this step.) Then, for all $m, n \geq N$ (say $m > n$ without loss of generality), the bound $(*)$ gives $d(x_m, x_n) < \varepsilon$. So (x_n) is Cauchy.

Step (iii): the limit exists and is a fixed point. This is where the **completeness** assumption enters: since (X, d) is complete and (x_n) is Cauchy, there is $x^* \in X$ such that $x_n \rightarrow x^*$. It remains to see that $f(x^*) = x^*$. Since f is continuous (every contraction is) and $x_n \rightarrow x^*$, the sequential characterization of continuity gives

$$f(x_n) \rightarrow f(x^*).$$

But $f(x_n) = x_{n+1}$, and the sequence (x_{n+1}) is the same sequence (x_n) shifted by one place, so it also converges to x^* . By the *uniqueness of the limit* in metric spaces, the two limits must agree:

$$f(x^*) = x^*.$$

Step (iv): uniqueness of the fixed point. Suppose x^* and y^* are both fixed points. Then

$$d(x^*, y^*) = d(f(x^*), f(y^*)) \leq k d(x^*, y^*),$$

so $(1-k)d(x^*, y^*) \leq 0$. Since $1-k > 0$, necessarily $d(x^*, y^*) \leq 0$, and since every distance is nonnegative, $d(x^*, y^*) = 0$, that is $x^* = y^*$.

Finally, note that the starting point x_0 was arbitrary: *every* sequence of iterates converges to the unique fixed point x^* . □

Intuition

The skeleton of the proof, one line per act: (i) each step of the iteration is geometrically shorter than the previous one; (ii) that is why far away terms are arbitrarily close to each other (geometric sum): the sequence is Cauchy; (iii) completeness now does its job: Cauchy $\Rightarrow$ convergent, and the continuity of f turns the limit into a fixed point; (iv) two fixed points would be at distance $d \leq kd$ with $k < 1$, which is impossible unless $d = 0$. **Learn these four steps by heart:** it is the most frequently asked proof of this part of the course.

10.3.1 Error bounds: how far you still are

The Banach theorem does not only say that the iteration converges: it says *how fast*. This turns it into a numerical algorithm with guarantees.

Proposition 10.9 (Error bounds). *With the assumptions and notation of Theorem 10.8, for every $n \geq 1$ the following hold:*

1. **A priori bound:**

$$d(x_n, x^*) \leq \frac{k^n}{1-k} d(x_1, x_0).$$

2. **A posteriori bound:**

$$d(x_n, x^*) \leq \frac{k}{1-k} d(x_n, x_{n-1}).$$

Proof. (1) Start from inequality (*) in Step (ii) of the previous proof: for $m > n$,

$$d(x_m, x_n) \leq \frac{k^n}{1-k} d(x_1, x_0).$$

Fix n and let $m \rightarrow \infty$. Since $x_m \rightarrow x^*$ and the distance function is continuous in each variable (a consequence of the triangle inequality), the left hand side converges to $d(x^*, x_n)$. The right hand side does not depend on m . Taking limits in the inequality (limits preserve non strict inequalities) you get the a priori bound.

(2) Consider the iteration that *starts* at $y_0 := x_{n-1}$. Its iterates are $y_1 = f(x_{n-1}) = x_n$, $y_2 = x_{n+1}$, and so on, and it converges to the same fixed point x^* (the theorem holds for any starting point). Applying the a priori bound to this new iteration with “ n ” equal to 1:

$$d(y_1, x^*) \leq \frac{k^1}{1-k} d(y_1, y_0), \quad \text{that is} \quad d(x_n, x^*) \leq \frac{k}{1-k} d(x_n, x_{n-1}). \quad \square$$

Remark 10.10. What is each bound for? The *a priori* one is used **before** computing: it lets you decide how many iterations n guarantee an error below a given tolerance, knowing only k and the first distance $d(x_1, x_0)$. The *a posteriori* one is used **during** the computation: after computing x_n , it measures the last step $d(x_n, x_{n-1})$ and bounds the true error; it is usually much sharper and it is the typical stopping rule of algorithms (for example, in value function iteration in computational macroeconomics you stop when $\|v_n - v_{n-1}\|_\infty$ is small, and this bound is what justifies it).

10.3.2 The three assumptions are necessary

The theorem has three assumptions: (a) X complete, (b) $f(X) \subseteq X$ (the function maps X into itself), (c) f a contraction with $k < 1$. Each one is indispensable.

Example 10.11 (Without completeness it fails). Let $X = (0, 1)$ with the usual metric, which is **not** complete (the sequence $(1/n)_{n \geq 2}$ is Cauchy in X but its limit 0 does not belong to X). Let $f(x) = x/2$. It is a contraction with $k = 1/2$ and it maps X into X : if $0 < x < 1$ then $0 < x/2 < 1$. However, its only candidate fixed point is $x = 0 \notin X$: the iteration $x_n = x_0/2^n$ “wants” to converge to 0, but 0 was left outside the space. Without completeness, the Cauchy sequence of Step (ii) may have no limit *inside* X .

Example 10.12 (With $k = 1$ it fails). Let $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x + 1$ (a translation). It satisfies $|f(x) - f(y)| = |x - y|$, that is, the contraction condition with $k = 1$ (it is an *isometry*). $\mathbb{R}$ is complete, but f has no fixed point. And Example 10.7 ($f(x) = x + 1/x$ on $[1, \infty)$) shows that even the strict condition $d(f(x), f(y)) < d(x, y)$, pointwise but without a uniform constant, is not enough either.

Example 10.13 (Without invariance $f(X) \subseteq X$ the statement does not even make sense). Let $X = [0, 1]$ and $f(x) = x/2 + 3$. It is a contraction with $k = 1/2$, but $f(0) = 3 \notin X$: you cannot even iterate inside X . In applications (for example, when restricting the Bellman operator to a set of bounded, increasing or concave functions), checking that f leaves invariant the chosen space is a mandatory step that must not be skipped.

10.3.3 Practical test through the derivative

How do you check in practice that a concrete function is a contraction? For differentiable functions on an interval, the tool is the mean value theorem.

Proposition 10.14 (Derivative test). *Let $I \subseteq \mathbb{R}$ be a closed interval and $f: I \rightarrow I$ differentiable, and suppose there is $k < 1$ such that*

$$|f'(x)| \leq k \quad \text{for all } x \in I.$$

Then f is a contraction on I with constant k .

Proof. Let $x, y \in I$ with $x < y$. By the mean value theorem (which you can apply because f is differentiable on the interval $[x, y] \subseteq I$), there is an intermediate point $c \in (x, y)$ such that

$$f(y) - f(x) = f'(c) (y - x).$$

Taking absolute values:

$$|f(y) - f(x)| = |f'(c)| \cdot |y - x| \leq k |y - x|,$$

using the uniform bound on the derivative. This is exactly the contraction condition. $\square$

Careful (common mistake)

The condition $|f'(x)| < 1$ at each point is not enough: you need a **uniform** bound $|f'(x)| \leq k < 1$. For $f(x) = x + 1/x$ on $[1, \infty)$ you have $f'(x) = 1 - 1/x^2 \in [0, 1)$ at each point, but $\sup_x |f'(x)| = 1$: that is why it is not a contraction. On a *compact* interval $[a, b]$, if $|f'| < 1$ pointwise and f' is continuous, then the supremum is attained and is smaller than 1 (Weierstrass theorem), and the problem disappears. Telling these two scenarios apart is a typical exam question.

Example 10.15 (A complete numerical example: $\cos(x) = x$). Let us prove that the equation $\cos(x) = x$ has exactly one solution in $\mathbb{R}$ and estimate how many iterations are needed to approximate it with error below 10^{-3} .

Step 1: reduce to an invariant interval. Any solution satisfies $x = \cos(x) \in [-1, 1]$; and if $x \in [-1, 1]$, to be a solution it needs $x = \cos x \in [\cos 1, 1] \subseteq [0.54, 1]$. Let us work on $I = [0.5, 1]$, which is closed in $\mathbb{R}$ and therefore a complete metric space. Check that $f(x) = \cos(x)$ maps I into I : $\cos$ is decreasing on $[0.5, 1]$, so for $x \in I$,

$$\cos(1) \leq \cos(x) \leq \cos(0.5), \quad \text{with } \cos(1) \approx 0.5403, \cos(0.5) \approx 0.8776,$$

so $f(I) \subseteq [0.54, 0.88] \subseteq I$. ✓

Step 2: contraction through the derivative. $f'(x) = -\sin(x)$, and for $x \in [0.5, 1]$, since $\sin$ is increasing there:

$$|f'(x)| = \sin(x) \leq \sin(1) \approx 0.8415 =: k < 1.$$

By Proposition 10.14, f is a contraction on I with $k \approx 0.8415$.

Step 3: apply Banach. I complete, $f: I \rightarrow I$ a contraction: there is a unique solution $x^* \in I$ of $\cos x = x$ (and it is the only one in all of $\mathbb{R}$ by Step 1). Its numerical value is $x^* \approx 0.7391$ (the so called Dottie number).

Step 4: number of iterations with the a priori bound. Starting from $x_0 = 1$: $x_1 = \cos(1) \approx 0.5403$, so $d(x_1, x_0) \approx 0.4597$. You want

$$\frac{k^n}{1-k} d(x_1, x_0) < 10^{-3} \quad \Longleftrightarrow \quad k^n < \frac{10^{-3}(1-k)}{d(x_1, x_0)} \approx \frac{10^{-3} \cdot 0.1585}{0.4597} \approx 3.45 \times 10^{-4}.$$

Taking logarithms: $n > \frac{\ln(3.45 \times 10^{-4})}{\ln(0.8415)} \approx \frac{-7.97}{-0.1726} \approx 46.2$. That is, $n = 47$ iterations *guarantee* the required error. (In practice convergence is faster; the a posteriori bound would let you stop earlier.)

Example 10.16 (Application to linear systems: simplified Jacobi iteration). You want to solve $x = Ax + b$ where A is an $n \times n$ matrix and $b \in \mathbb{R}^n$. Define $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $f(x) = Ax + b$, on $\mathbb{R}^n$ with the max norm $\|x\|_\infty = \max_i |x_i|$ (recall that $\mathbb{R}^n$ is complete with any norm). Then

$$\|f(x) - f(y)\|_\infty = \|A(x - y)\|_\infty \leq \|A\|_\infty \|x - y\|_\infty,$$

where $\|A\|_\infty = \max_i \sum_j |a_{ij}|$ (the largest absolute row sum; the inequality is checked directly by bounding each component of $A(x - y)$). Conclusion: if

$$\max_i \sum_j |a_{ij}| =: k < 1,$$

then f is a contraction and the system $x = Ax + b$ has a unique solution, which you can approximate by iterating $x_{m+1} = Ax_m + b$ from any x_0 . This is the seed of the iterative methods of numerical linear algebra and of the *Neumann series* $(I - A)^{-1} = \sum_{j \geq 0} A^j$, which you will study when we discuss linear operators.

Example 10.17 (A simple integral equation). Consider the integral equation (with unknown

φ , a continuous function):

$$\varphi(t) = g(t) + \lambda \int_0^1 K(t, s) \varphi(s) ds, \quad t \in [0, 1],$$

with given continuous g and K and $M := \max_{t,s} |K(t, s)|$. Define the operator T on the space $C[0, 1]$ (continuous functions with the sup norm $\|\varphi\|_\infty = \sup_t |\varphi(t)|$, which is complete, as proved in the chapter on Banach spaces) by

$$(T\varphi)(t) = g(t) + \lambda \int_0^1 K(t, s) \varphi(s) ds.$$

For $\varphi, \psi \in C[0, 1]$ and each t :

$$\begin{aligned} |(T\varphi)(t) - (T\psi)(t)| &= \left| \lambda \int_0^1 K(t, s) (\varphi(s) - \psi(s)) ds \right| \\ &\leq |\lambda| M \int_0^1 |\varphi(s) - \psi(s)| ds \leq |\lambda| M \|\varphi - \psi\|_\infty. \end{aligned}$$

Taking the supremum over t : $\|T\varphi - T\psi\|_\infty \leq |\lambda| M \|\varphi - \psi\|_\infty$. If $|\lambda| < 1/M$, T is a contraction and the integral equation has a unique continuous solution. Note the pattern: the fixed point lives in a *space of functions*, exactly like the value function in dynamic programming.

Theorem 10.18 (Variant: some iterate is a contraction). *Let (X, d) be complete and $f: X \rightarrow X$ such that some iterate $f^m = f \circ f \circ \dots \circ f$ (m times) is a contraction. Then f has a unique fixed point x^* , and the iterates of f from any starting point converge to x^* .*

Remark 10.19. We will not prove this variant in detail; the idea is that f^m has a unique fixed point x^* by Banach, and since $f(x^*)$ is also a fixed point of f^m (because $f^m(f(x^*)) = f(f^m(x^*)) = f(x^*)$), uniqueness forces $f(x^*) = x^*$. This version is useful, for example, for the Volterra integral operator, whose iterates gain factors $1/n!$ that eventually become smaller than 1 even if λ is large.

10.4 Central application in economics: the Bellman equation [P3]

This section connects the Banach theorem with the core of macroeconomics and dynamic decision theory, following Ok [4] and the Stokey–Lucas tradition.

10.4.1 The space where the problem lives

Let S be a set of states (for example, capital levels). Write $B(S)$ for the set of bounded functions $v: S \rightarrow \mathbb{R}$, with the **sup norm**

$$\|v\|_\infty = \sup_{x \in S} |v(x)|.$$

It is proved (in the chapter on Banach spaces) that $(B(S), \|\cdot\|_\infty)$ is a **complete** normed space: the perfect setting for applying Banach. The distance between two candidate value functions is “their largest discrepancy across all states”.

10.4.2 Blackwell conditions

Checking by hand that an operator between functions is a contraction can be awkward. Blackwell isolated two conditions that are easy to check and that guarantee it.

Theorem 10.20 (Blackwell sufficient conditions). *Let $T: B(S) \rightarrow B(S)$ be an operator satisfying:*

1. **Monotonicity:** *if $v, w \in B(S)$ satisfy $v(x) \leq w(x)$ for every $x \in S$, then $(Tv)(x) \leq (Tw)(x)$ for every $x \in S$.*
2. **Discounting:** *there is $\beta \in [0, 1)$ such that for every $v \in B(S)$ and every constant $c \geq 0$,*

$$T(v + c)(x) \leq (Tv)(x) + \beta c \quad \text{for all } x \in S,$$

where $v + c$ denotes the function $x \mapsto v(x) + c$.

Then T is a contraction on $(B(S), \|\cdot\|_\infty)$ with constant β .

Proof. Let $v, w \in B(S)$ be arbitrary. The starting point is the obvious pointwise inequality:

$$v(x) = w(x) + (v(x) - w(x)) \leq w(x) + \|v - w\|_\infty \quad \text{for all } x \in S,$$

because $v(x) - w(x) \leq |v(x) - w(x)| \leq \sup_y |v(y) - w(y)|$. That is, as functions, $v \leq w + c$ with the constant $c := \|v - w\|_\infty \geq 0$.

Now apply the two properties in a chain:

$$Tv \leq T(w + c) \leq Tw + \beta c,$$

where the first inequality uses *monotonicity* (applied to $v \leq w + c$; note that $w + c \in B(S)$) and the second uses *discounting*. Rearranging, for every $x \in S$:

$$(Tv)(x) - (Tw)(x) \leq \beta \|v - w\|_\infty.$$

Swapping the roles of v and w (the argument is symmetric) you also get $(Tw)(x) - (Tv)(x) \leq \beta \|v - w\|_\infty$. The two inequalities together say that

$$|(Tv)(x) - (Tw)(x)| \leq \beta \|v - w\|_\infty \quad \text{for all } x \in S,$$

and taking the supremum over x :

$$\|Tv - Tw\|_\infty \leq \beta \|v - w\|_\infty.$$

Since $\beta < 1$, T is a contraction with constant β . □

10.4.3 The Bellman operator is a contraction

Consider the standard dynamic problem: in state x , the agent chooses an action a in a feasible set $\Gamma(x) \neq \emptyset$, receives instantaneous utility $u(x, a)$ (assumed bounded) and the state becomes

$g(x, a)$; the future is discounted with factor $\beta \in [0, 1)$. The **Bellman operator** acts on $v \in B(S)$ as follows:

$$(Tv)(x) = \sup_{a \in \Gamma(x)} \{u(x, a) + \beta v(g(x, a))\}.$$

Proposition 10.21. *The Bellman operator satisfies the Blackwell conditions and therefore is a contraction on $B(S)$ with constant β .*

Proof. **Monotonicity.** If $v \leq w$ pointwise, then for each x and each action $a \in \Gamma(x)$:

$$u(x, a) + \beta v(g(x, a)) \leq u(x, a) + \beta w(g(x, a)) \leq (Tw)(x),$$

where the last inequality holds because $(Tw)(x)$ is the supremum over actions of the expression with w . Taking the supremum over a on the left hand side (the supremum of a function bounded by a constant is bounded by that constant): $(Tv)(x) \leq (Tw)(x)$.

Discounting. For a constant $c \geq 0$:

$$\begin{aligned} T(v + c)(x) &= \sup_{a \in \Gamma(x)} \{u(x, a) + \beta(v(g(x, a)) + c)\} \\ &= \sup_{a \in \Gamma(x)} \{u(x, a) + \beta v(g(x, a))\} + \beta c = (Tv)(x) + \beta c, \end{aligned}$$

where we pulled the constant βc out of the supremum (adding a constant to a family of numbers shifts its supremum by that constant). Discounting holds with equality, so in particular the required inequality holds.

By Theorem 10.20, T is a contraction with constant β . □

Link with economics

Let us put everything together. $B(S)$ with the sup norm is complete; the Bellman operator T is a contraction with modulus $\beta < 1$. The Banach fixed point theorem delivers, in one shot, the three pillars of dynamic programming:

1. **Existence:** there is a value function v^* that solves the Bellman equation $v^* = Tv^*$.
2. **Uniqueness:** it is the only bounded solution; there are no “spurious value functions”.
3. **Algorithm with guarantees:** *value function iteration*, $v_{n+1} = Tv_n$ from any initial guess v_0 (even $v_0 \equiv 0$), converges to v^* in sup norm, with a priori error $\|v_n - v^*\|_\infty \leq \frac{\beta^n}{1-\beta} \|v_1 - v_0\|_\infty$. This is why, with $\beta = 0.95$, convergence is slow (k close to 1), and with $\beta = 0.5$ it is very fast: the contraction modulus *is* the discount factor.

There is also a useful refinement: if you prove that T preserves a *closed* subset of $B(S)$ (for example, the increasing functions, or the concave ones), that subset is complete and Banach applies there, so v^* inherits those properties. This is how you prove that the value function of the neoclassical growth model is increasing and concave.

Finally, not every fixed point in economics comes from contractions: the existence of *general equilibrium* (Arrow–Debreu) and of *Nash equilibrium* use the fixed point theorems

of **Brouwer** (every continuous function from a compact convex set into itself has a fixed point) and of **Kakutani** (the version for correspondences). These guarantee existence but *not* uniqueness and no algorithm; we state them here only as an overview, without proof.

Exam strategy

How this topic is usually examined:

1. “State and prove the Banach fixed point theorem.” Reproduce the 4 steps: geometric bound by induction $\rightarrow$ Cauchy via the geometric sum $\rightarrow$ completeness + continuity $\rightarrow$ uniqueness. Do not forget to say **where** completeness is used (Step iii) and where $k < 1$ is used (Steps ii and iv).
2. “Prove that equation [such] has a unique solution in [such interval].” Recipe: (a) choose a closed interval I (complete); (b) check $f(I) \subseteq I$ by evaluating at the endpoints and using monotonicity; (c) bound $|f'| \leq k < 1$ on I ; (d) cite Banach. The three checks earn points separately: write all of them.
3. “Give a counterexample if [assumption] is dropped.” Have these ready: $(0, 1)$ with $x/2$ (no completeness); $x + 1$ on $\mathbb{R}$ and $x + 1/x$ on $[1, \infty)$ (no $k < 1$).
4. In questions applied to Bellman: check *explicitly* monotonicity and discounting of Blackwell; they are two short checks that give the contraction without computations with double suprema.

Chapter summary

- Fixed point: $f(x^*) = x^*$. Contraction: $d(f(x), f(y)) \leq k d(x, y)$ with **one and the same** $k < 1$ for all pairs.
- **Banach theorem**: X complete + $f: X \rightarrow X$ a contraction $\Rightarrow$ there is a unique fixed point, and the iterates converge to it from any starting point.
- Proof in 4 steps: geometric induction; Cauchy (geometric sum); completeness + continuity; uniqueness through $(1 - k)d \leq 0$.
- Error bounds: a priori $\frac{k^n}{1-k} d(x_1, x_0)$; a posteriori $\frac{k}{1-k} d(x_n, x_{n-1})$.
- Practical test: $|f'| \leq k < 1$ uniformly $\Rightarrow$ contraction (mean value theorem).
- Mandatory counterexamples: $x/2$ on $(0, 1)$; $x + 1$ on $\mathbb{R}$; $x + 1/x$ on $[1, \infty)$.
- Blackwell: monotonicity + discounting $\Rightarrow$ contraction with constant β . The Bellman operator satisfies them: the value function exists, is unique and is computed by iterating.

10.5 Exam-style exercises [P3]

Exercise 10.22. Prove that the function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = \frac{1}{4} \sin(x) + 7$ has exactly one fixed point in $\mathbb{R}$.

Solution. $\mathbb{R}$ with the usual metric is complete, and f is defined from $\mathbb{R}$ into $\mathbb{R}$, so all that is left is to check that it is a contraction. f is differentiable with $f'(x) = \frac{1}{4} \cos(x)$, so

$$|f'(x)| = \frac{1}{4} |\cos(x)| \leq \frac{1}{4} < 1 \quad \text{for all } x \in \mathbb{R}.$$

By the derivative test (mean value theorem: for $x < y$ there is c with $f(y) - f(x) = f'(c)(y - x)$, and taking absolute values $|f(y) - f(x)| \leq \frac{1}{4} |y - x|$), f is a contraction with $k = \frac{1}{4}$. By the Banach fixed point theorem, f has a unique fixed point in $\mathbb{R}$, and moreover $x_{n+1} = f(x_n)$ converges to it from any x_0 . $\square$

Exercise 10.23. Let (X, d) be a complete metric space and $f: X \rightarrow X$ a contraction with constant k . Starting from x_0 , you compute $x_1 = f(x_0)$ with $d(x_1, x_0) = 2$. If $k = \frac{1}{3}$, how many iterations guarantee that $d(x_n, x^*) < 10^{-2}$, where x^* is the fixed point?

Solution. Use the a priori bound:

$$d(x_n, x^*) \leq \frac{k^n}{1 - k} d(x_1, x_0) = \frac{(1/3)^n}{1 - 1/3} \cdot 2 = \frac{(1/3)^n}{2/3} \cdot 2 = 3 \cdot (1/3)^n.$$

You want $3 \cdot (1/3)^n < 10^{-2}$, that is $(1/3)^n < \frac{1}{300}$, or $3^n > 300$. Since $3^5 = 243 < 300$ and $3^6 = 729 > 300$, $n = 6$ is enough. Answer: **6 iterations guarantee the required error.** (Note the general technique: isolate k^n and take logarithms or compare powers.) $\square$

Exercise 10.24. Give an example of a metric space (X, d) and a function $f: X \rightarrow X$ such that $d(f(x), f(y)) < d(x, y)$ for all $x \neq y$, but f has no fixed point. Explain why your example does not contradict the Banach theorem.

Solution. Take $X = [1, \infty)$ with the usual metric (complete, since it is closed in $\mathbb{R}$) and $f(x) = x + \frac{1}{x}$. First, $f(X) \subseteq X$: if $x \geq 1$, then $f(x) \geq x \geq 1$. Second, for $1 \leq x < y$:

$$|f(y) - f(x)| = \left| (y - x) \left(1 - \frac{1}{xy} \right) \right| < y - x,$$

since $xy > 1$ implies $0 < 1 - \frac{1}{xy} < 1$. Third, f has no fixed point: $x = x + \frac{1}{x}$ would imply $\frac{1}{x} = 0$, which is impossible.

There is no contradiction with Banach because **f is not a contraction**: you would need $k < 1$ with $|f(y) - f(x)| \leq k |y - x|$ for *all* pairs, but the factor $1 - \frac{1}{xy}$ tends to 1 as $x, y \rightarrow \infty$; for any proposed $k < 1$ there are pairs with a factor larger than k . The Banach assumption requires uniform contraction, and here exactly that uniformity fails. $\square$

Exercise 10.25. Prove that the system

$$x = \frac{1}{4}x + \frac{1}{5}y + 1, \quad y = \frac{1}{6}x + \frac{1}{3}y - 2$$

has a unique solution $(x^*, y^*) \in \mathbb{R}^2$.

Solution. Write the system as $z = f(z)$ with $z = (x, y) \in \mathbb{R}^2$ and $f(z) = Az + b$, where

$$A = \begin{pmatrix} 1/4 & 1/5 \\ 1/6 & 1/3 \end{pmatrix}, \quad b = (1, -2).$$

Use $\mathbb{R}^2$ with the max norm $\|z\|_\infty = \max(|x|, |y|)$; with any norm, $\mathbb{R}^2$ is complete. For $z, w \in \mathbb{R}^2$, each component of $A(z - w)$ is bounded:

$$|[A(z - w)]_1| \leq \frac{1}{4}|z_1 - w_1| + \frac{1}{5}|z_2 - w_2| \leq \left(\frac{1}{4} + \frac{1}{5}\right) \|z - w\|_\infty = \frac{9}{20} \|z - w\|_\infty,$$

$$|[A(z - w)]_2| \leq \left(\frac{1}{6} + \frac{1}{3}\right) \|z - w\|_\infty = \frac{1}{2} \|z - w\|_\infty.$$

Therefore

$$\|f(z) - f(w)\|_\infty = \|A(z - w)\|_\infty \leq \max\left(\frac{9}{20}, \frac{1}{2}\right) \|z - w\|_\infty = \frac{1}{2} \|z - w\|_\infty.$$

So f is a contraction with $k = \frac{1}{2} < 1$ on the complete space $(\mathbb{R}^2, \|\cdot\|_\infty)$. By Banach, there is a unique fixed point (x^*, y^*) , which is the unique solution of the system. (Note the general test: it is enough that the largest absolute row sum of A be smaller than 1.) $\square$

Exercise 10.26. Let $T: B(S) \rightarrow B(S)$ be a monotone operator (if $v \leq w$ pointwise then $Tv \leq Tw$) satisfying $T(v + c) \leq Tv + \beta c$ for every constant $c \geq 0$, with $\beta \in [0, 1)$. Prove that $\|Tv - Tw\|_\infty \leq \beta \|v - w\|_\infty$ for all $v, w \in B(S)$, and explain what you conclude for the Bellman equation.

Solution. Let $c = \|v - w\|_\infty$. For each $x \in S$:

$$v(x) = w(x) + (v(x) - w(x)) \leq w(x) + |v(x) - w(x)| \leq w(x) + c,$$

that is, $v \leq w + c$ as functions. Applying monotonicity and then discounting:

$$Tv \leq T(w + c) \leq Tw + \beta c \implies (Tv)(x) - (Tw)(x) \leq \beta c \quad \forall x.$$

By symmetry (repeating the argument with $w \leq v + c$): $(Tw)(x) - (Tv)(x) \leq \beta c$ for every x . Combining, $|(Tv)(x) - (Tw)(x)| \leq \beta c$ for every x , and taking the supremum:

$$\|Tv - Tw\|_\infty \leq \beta \|v - w\|_\infty.$$

Conclusion: T is a contraction with constant β on the complete space $(B(S), \|\cdot\|_\infty)$. If T is the Bellman operator (which satisfies monotonicity and discounting, as checked in the text), the Banach theorem guarantees that the Bellman equation $v = Tv$ has a unique bounded solution v^* (the value function) and that value function iteration $v_{n+1} = Tv_n$ converges to v^* in sup norm from any initial guess, at a geometric rate with ratio β . $\square$

Chapter 11

Vector spaces, bases and normed spaces [P2]

Week 8 of the course (the last one before the midterm exam). Bibliography: Alabiso and Weiss [1], Chs. 2 and 5; Moroşanu [2], Ch. 2.

What is this used for in Microeconomics?

The *linear and convex* structure built here runs through the canonical syllabus from start to finish, and that is why the section on vector spaces is [P1]. Convex preferences and convex production sets are basic assumptions of demand theory and production theory (Mas-Colell, Whinston and Green, chs. 3 and 5); mixing lotteries with probabilities is exactly taking convex combinations, and the set of lotteries is convex (expected utility, ch. 6); mixed strategies live in the simplex, which is convex and compact (ch. 8); and the welfare theorems separate convex sets with a price hyperplane (ch. 16). Basis and dimension come back when counting how many assets are needed to complete markets, and function spaces are the formal home of contracts, even if the course handles them informally.

11.1 Motivation [P2]

So far we have worked with metric spaces: sets where all you can do is *measure distances*. But the spaces that show up in analysis and in economics have more structure: you can *add* their elements and *multiply them by numbers*. A consumption plan plus another consumption plan is a consumption plan; twice a utility function is a function. When a set admits these operations with the usual rules of algebra, you have a **vector space**; when in addition every vector has a *length* that is compatible with those operations, you have a **normed space**. This chapter builds both concepts slowly, because they are the language of the whole second half of the course: Banach spaces, linear operators and Hilbert spaces.

The key conceptual jump, and the one to digest well, is that *vectors* will no longer be only arrows in $\mathbb{R}^n$: they will be **functions** and **infinite sequences**. A function $f \in C[a, b]$ is a point of a space; a consumption sequence $(c_0, c_1, c_2, \dots)$ is a point of a space. Everything you know

about convergence and completeness will apply to these objects.

11.2 Vector spaces [P1]

Definition 11.1 (Vector space). A **vector space** over $\mathbb{R}$ is a set V equipped with two operations:

$$+ : V \times V \rightarrow V \quad (\text{vector addition}), \quad \cdot : \mathbb{R} \times V \rightarrow V \quad (\text{scalar multiplication}),$$

which satisfy, for all $u, v, w \in V$ and $\alpha, \beta \in \mathbb{R}$:

1. (Associativity) $(u + v) + w = u + (v + w)$.
2. (Commutativity) $u + v = v + u$.
3. (Zero element) There is $0 \in V$ such that $v + 0 = v$ for every v .
4. (Additive inverse) For each v there is $-v \in V$ with $v + (-v) = 0$.
5. (Distributivity I) $\alpha(u + v) = \alpha u + \alpha v$.
6. (Distributivity II) $(\alpha + \beta)v = \alpha v + \beta v$.
7. (Mixed associativity) $\alpha(\beta v) = (\alpha\beta)v$.
8. (Unit) $1 \cdot v = v$.

The elements of V are called **vectors** and those of $\mathbb{R}$, **scalars**. If the scalars are taken in $\mathbb{C}$ you speak of a complex vector space; in this course we work over $\mathbb{R}$ unless stated otherwise (complex Hilbert spaces appear in physics and in some parts of spectral econometrics, but the real theory is enough for us).

Intuition

The eight axioms say one single thing: in V you can do linear algebra “as usual”. What matters is not memorizing the list, but knowing how to *check it* in concrete examples, where the axioms are almost always inherited from $\mathbb{R}$ by working “coordinate by coordinate” or “pointwise”. The nontrivial question in the examples is never associativity: it is whether the operations are *well defined*, that is, whether adding two elements of the set gives a result that *still belongs* to the set.

Example 11.2 (The fundamental examples). 1. $\mathbb{R}^n$ with componentwise addition and scalar multiplication: the model for all the others.

2. **Function spaces.** Let $[a, b] \subseteq \mathbb{R}$ be an interval. Define

$$C[a, b] := \{f : [a, b] \rightarrow \mathbb{R} : f \text{ is continuous}\},$$

with *pointwise* operations: $(f + g)(t) := f(t) + g(t)$ and $(\alpha f)(t) := \alpha f(t)$. The operations are well defined because *the sum of continuous functions is continuous and a scalar multiple of a continuous function is continuous* (seen in the chapter on continuity). The zero vector

is the identically zero function. Axioms 1 to 8 are checked pointwise, inheriting them from $\mathbb{R}$: for example, $(f+g)(t) = f(t)+g(t) = g(t)+f(t) = (g+f)(t)$ for every t , so $f+g = g+f$.

3. **Polynomials.** $\mathcal{P} := \{p: \mathbb{R} \rightarrow \mathbb{R} : p \text{ is a polynomial}\}$ and, for each n , the set $\mathcal{P}_n$ of polynomials of degree $\leq n$. Both are vector spaces with the pointwise operations (the sum of two polynomials of degree $\leq n$ has degree $\leq n$). Careful: the set of polynomials of degree *exactly* n is NOT a vector space: x^n and $-x^n + x$ have degree n , but their sum has degree 1; it would also be missing the zero vector.
4. **Sequence spaces.** Write $x = (x_1, x_2, x_3, \dots)$ for a real sequence, with termwise operations. Define:

$$\ell^1 := \left\{x : \sum_{i=1}^{\infty} |x_i| < \infty\right\} \quad (\text{absolutely summable sequences}),$$

$$\ell^2 := \left\{x : \sum_{i=1}^{\infty} x_i^2 < \infty\right\} \quad (\text{square summable sequences}),$$

$$\ell^\infty := \left\{x : \sup_i |x_i| < \infty\right\} \quad (\text{bounded sequences}),$$

$$c_0 := \left\{x : \lim_{i \rightarrow \infty} x_i = 0\right\} \quad (\text{sequences tending to } 0),$$

$$c_{00} := \{x : x_i = 0 \text{ except for finitely many } i\} \quad (\text{finitely supported sequences}).$$

Let us check that ℓ^1 is a vector space (the delicate point is that the sum stays inside): if $x, y \in \ell^1$, then for each N ,

$$\sum_{i=1}^N |x_i + y_i| \leq \sum_{i=1}^N |x_i| + \sum_{i=1}^N |y_i| \leq \sum_{i=1}^{\infty} |x_i| + \sum_{i=1}^{\infty} |y_i| < \infty,$$

and since the partial sums of $\sum |x_i + y_i|$ are bounded and increasing, the series converges: $x + y \in \ell^1$. For ℓ^2 , the delicate point is settled with the inequality $(x_i + y_i)^2 \leq 2x_i^2 + 2y_i^2$ (which comes from $(x_i - y_i)^2 \geq 0$): summing, $x + y \in \ell^2$. For ℓ^∞ , $\sup_i |x_i + y_i| \leq \sup_i |x_i| + \sup_i |y_i|$. For c_0 , the limit of the sum is the sum of the limits. In all cases the scalar multiple is immediate. The strict inclusions

$$c_{00} \subsetneq \ell^1 \subsetneq \ell^2 \subsetneq c_0 \subsetneq \ell^\infty$$

hold. (For example, $x_i = 1/i$ is in ℓ^2 but not in ℓ^1 (the harmonic series diverges); $x_i = 1/\ln(i+1)$ tends to 0 but is not square summable; the constant sequence $(1, 1, 1, \dots)$ is bounded but does not tend to 0.)

11.2.1 Subspaces

Definition 11.3 (Vector subspace). Let V be a vector space. A nonempty subset $W \subseteq V$ is a **vector subspace** if, with the operations inherited from V , W is itself a vector space.

Proposition 11.4 (Subspace test). *A nonempty $W \subseteq V$ is a subspace if and only if it is closed under the operations, that is:*

1. $u, v \in W \implies u + v \in W$;
2. $\alpha \in \mathbb{R}, v \in W \implies \alpha v \in W$.

(Equivalently, in a single check: $\alpha u + \beta v \in W$ for all $u, v \in W, \alpha, \beta \in \mathbb{R}$.)

Proof. ($\implies$) If W is a vector space with the inherited operations, then in particular the operations applied to elements of W give results in W : that is exactly (1) and (2).

($\Leftarrow$) Suppose (1) and (2). Axioms 1, 2, 5, 6, 7, 8 are identities between elements: since they hold for all elements of V , they hold in particular for those of W (they are inherited). All you have to produce is the zero element and the inverses *inside* W . Since $W \neq \emptyset$, there is some $w \in W$; by (2) with $\alpha = 0$: $0 = 0 \cdot w \in W$ (that $0 \cdot w$ is the zero vector follows from the axioms: $0w = (0 + 0)w = 0w + 0w$, and adding the inverse of $0w$ to both sides leaves $0 = 0w$). And for each $v \in W$, by (2) with $\alpha = -1$: $-v = (-1)v \in W$ (again, $(-1)v$ is the inverse because $v + (-1)v = (1 + (-1))v = 0v = 0$). So W satisfies all the axioms. $\square$

Example 11.5. 1. $\mathcal{P}_n \subseteq \mathcal{P} \subseteq C[a, b]$: the polynomials (restricted to $[a, b]$) form a subspace of the continuous functions, since sums and multiples of polynomials are polynomials.

2. $W = \{f \in C[a, b] : f(a) = 0\}$ is a subspace of $C[a, b]$: if $f(a) = g(a) = 0$ then $(f+g)(a) = 0$ and $(\alpha f)(a) = 0$. On the other hand, $\{f : f(a) = 1\}$ is **not** one: it does not contain the zero function, and $(f+g)(a) = 2 \neq 1$.
3. In $\mathbb{R}^2$, every line through the origin is a subspace; a line that does not pass through the origin is not (it lacks 0).
4. c_{00} is a subspace of all the ℓ^p and of c_0 .

Careful (common mistake)

An instant check that saves time in an exam: *if $0 \notin W$, then W is not a subspace*. It is the first thing to look at. A second common mistake: checking only the sum and forgetting scalar multiplication (or the other way around). The set $\{(x, y) \in \mathbb{R}^2 : x, y \geq 0\}$ (the positive quadrant, so natural in economics) is closed under sums but not under multiplication by -1 : it is not a subspace, it is a *cone*.

11.3 Linear combinations, independence, bases and dimension [P3]

Definition 11.6 (Linear combination and spanning set). Let V be a vector space and $S = \{v_1, \dots, v_m\} \subseteq V$ finite. A **linear combination** of S is any vector of the form

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_m v_m, \quad \alpha_i \in \mathbb{R}.$$

The set of all linear combinations of S is called the **span** of S , written $\text{span}(S)$; it is always a subspace of V (check it with the subspace test: a sum of linear combinations is a linear combination, and so is a multiple). We say that S **spans** V if $\text{span}(S) = V$. For infinite sets

S , $\text{span}(S)$ is defined as the set of *finite* linear combinations of elements of S : in pure linear algebra there are no infinite sums (for those you will need norms and series).

Definition 11.7 (Linear independence). The vectors $v_1, \dots, v_m$ are **linearly independent** if the only way to write the zero vector as a linear combination of them is with all coefficients equal to zero:

$$\alpha_1 v_1 + \dots + \alpha_m v_m = 0 \implies \alpha_1 = \dots = \alpha_m = 0.$$

An infinite set is linearly independent if every finite subset of it is. If they are not independent, they are called **linearly dependent**: one of them is a linear combination of the others.

Intuition

Independence means “nobody is redundant”: no vector of the set contributes a direction that was already available by combining the others. A spanning set means “together they are enough to build any vector”. A *basis* is the perfect balance: enough to span, none to spare.

Definition 11.8 (Basis and dimension). A **basis** (also called a Hamel basis) of V is a set $B \subseteq V$ that is linearly independent and spans V . If V admits a finite basis with n elements, it can be proved (we will not do it here) that *all* bases have exactly n elements; that common number is the **dimension** of V , written $\dim V = n$, and we say that V is **finite dimensional**. If no finite set spans V , we say that V is **infinite dimensional**.

Example 11.9. 1. The **standard basis** of $\mathbb{R}^n$: $e_1 = (1, 0, \dots, 0)$, $e_2 = (0, 1, 0, \dots, 0)$, $\dots$, $e_n = (0, \dots, 0, 1)$. It spans because $x = (x_1, \dots, x_n) = \sum_i x_i e_i$, and it is independent because $\sum_i \alpha_i e_i = (\alpha_1, \dots, \alpha_n) = 0$ forces all $\alpha_i = 0$. So $\dim \mathbb{R}^n = n$.

2. $\{1, x, x^2, \dots, x^n\}$ is a basis of $\mathcal{P}_n$, so $\dim \mathcal{P}_n = n + 1$ (watch out for the $+1$: it counts the constant term).

3. In $\mathbb{R}^2$, $\{(1, 1), (1, -1)\}$ is another basis: bases are not unique, but their cardinality is.

Proposition 11.10 (The monomials are linearly independent). *For every n , the functions $1, x, x^2, \dots, x^n$ are linearly independent in $C[a, b]$ (with $a < b$).*

Proof. Suppose that $\alpha_0 \cdot 1 + \alpha_1 x + \dots + \alpha_n x^n = 0$ as a function, that is,

$$p(x) := \alpha_0 + \alpha_1 x + \dots + \alpha_n x^n = 0 \quad \text{for every } x \in [a, b].$$

You must conclude that all the α_i are zero. Here is a direct argument: a nonzero polynomial of degree $\leq n$ has at most n real roots (a consequence of the factor theorem: each root r contributes a factor $(x - r)$, and the degree limits the number of factors). But our p vanishes at *every* point of the interval $[a, b]$, and there are infinitely many of them. A polynomial with infinitely many roots cannot be nonzero; so p is the zero polynomial, that is, $\alpha_0 = \alpha_1 = \dots = \alpha_n = 0$.

(An alternative argument, worth knowing: evaluating p and its successive derivatives at an interior point, $p \equiv 0$ implies $p(c) = p'(c) = \dots = p^{(n)}(c) = 0$, and from there you solve for the coefficients one by one.) $\square$

Theorem 11.11 (Infinite dimension of the classical spaces). $C[a, b]$, $\mathcal{P}$ and the sequence spaces $\ell^1, \ell^2, \ell^\infty, c_0, c_{00}$ are infinite dimensional.

Proof. The strategy is the same in all cases: exhibit linearly independent families with as many elements as you want. If V had finite dimension n , every set of $n + 1$ vectors would be linearly dependent (a standard result of linear algebra: a space spanned by n vectors has no room for $n + 1$ independent ones). So it is enough to find, for each $m \in \mathbb{N}$, an independent set of m elements.

In $C[a, b]$ and $\mathcal{P}$: the set $\{1, x, \dots, x^{m-1}\}$ has m elements and is linearly independent by Proposition 11.10. Since m is arbitrary, no finite dimension is possible.

In the sequence spaces: let $e^{(i)}$ be the sequence with a 1 in position i and zeros elsewhere. Each $e^{(i)}$ belongs to c_{00} , and therefore to all the other spaces (which contain it). The family $\{e^{(1)}, \dots, e^{(m)}\}$ is linearly independent: if $\alpha_1 e^{(1)} + \dots + \alpha_m e^{(m)} = 0$, the resulting sequence is $(\alpha_1, \alpha_2, \dots, \alpha_m, 0, 0, \dots)$, and for it to be the zero sequence forces $\alpha_i = 0$ for every i . Again, m is arbitrary. $\square$

Remark 11.12. In ℓ^2 , the infinite family $\{e^{(i)}\}_{i \in \mathbb{N}}$ is independent but is **not** a Hamel basis: the sequence $x = (1, \frac{1}{2}, \frac{1}{4}, \dots) \in \ell^2$ is not a *finite* linear combination of the $e^{(i)}$. It will be a “basis” in a richer sense (a Schauder or Hilbert orthonormal basis) once we have series that converge in norm: that is exactly the gain from adding topology to algebra, and you will see it in the chapter on Hilbert spaces.

11.4 Normed spaces [P3]

Definition 11.13 (Norm and normed space). Let V be a vector space over $\mathbb{R}$. A **norm** on V is a function $\|\cdot\| : V \rightarrow \mathbb{R}$ satisfying, for all $u, v \in V$ and $\alpha \in \mathbb{R}$:

1. (Nonnegativity and separation) $\|v\| \geq 0$, and $\|v\| = 0 \iff v = 0$;
2. (Absolute homogeneity) $\|\alpha v\| = |\alpha| \|v\|$;
3. (Triangle inequality) $\|u + v\| \leq \|u\| + \|v\|$.

The pair $(V, \|\cdot\|)$ is called a **normed vector space**.

Proposition 11.14 (The norm induces a metric). If $(V, \|\cdot\|)$ is a normed space, then

$$d(u, v) := \|u - v\|$$

is a metric on V . Moreover, this metric is:

1. **translation invariant:** $d(u + w, v + w) = d(u, v)$ for every w ;
2. **homogeneous:** $d(\alpha u, \alpha v) = |\alpha| d(u, v)$.

Proof. We check the three metric axioms. (*Nonnegativity and separation*): $d(u, v) = \|u - v\| \geq 0$, and $d(u, v) = 0 \iff \|u - v\| = 0 \iff u - v = 0 \iff u = v$, using the separation property of the norm. (*Symmetry*): $d(v, u) = \|v - u\| = \|(-1)(u - v)\| = |-1| \|u - v\| = d(u, v)$, using

homogeneity with $\alpha = -1$. (*Triangle*): add and subtract the intermediate point:

$$d(u, w) = \|u - w\| = \|(u - v) + (v - w)\| \leq \|u - v\| + \|v - w\| = d(u, v) + d(v, w),$$

by the triangle inequality of the norm.

For the extra properties: $d(u + w, v + w) = \|(u + w) - (v + w)\| = \|u - v\| = d(u, v)$; and $d(\alpha u, \alpha v) = \|\alpha(u - v)\| = |\alpha| \|u - v\| = |\alpha| d(u, v)$. $\square$

Remark 11.15. An important consequence: **every normed space is a metric space**, so everything you have learned (open sets, closed sets, convergence, Cauchy, completeness, compactness, continuity) applies automatically, always taking $d(u, v) = \|u - v\|$. The converse is false: not every metric comes from a norm. For example, the discrete metric on $\mathbb{R}$ is not homogeneous ($d(2u, 0) = 1 \neq 2 = 2d(u, 0)$ for $u \neq 0$), so it cannot come from a norm. The two extra properties in Proposition 11.14 are exactly the fingerprint of the metrics that do come from norms.

11.4.1 Examples of normed spaces, with verification

Example 11.16 (The p norms on $\mathbb{R}^n$ (review)). On $\mathbb{R}^n$:

$$\|x\|_1 = \sum_{i=1}^n |x_i|, \quad \|x\|_2 = \left(\sum_{i=1}^n x_i^2 \right)^{1/2}, \quad \|x\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

All three are norms (checked in the chapter on $\mathbb{R}^n$; the triangle inequality for $\|\cdot\|_2$ is the Minkowski inequality, which in turn follows from Cauchy–Schwarz), and they are equivalent to each other:

$$\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1 \leq n \|x\|_\infty.$$

Example 11.17 (The sup norm on $C[a, b]$). On $C[a, b]$ define the **sup norm** (also called the uniform norm):

$$\|f\|_\infty := \sup_{t \in [a, b]} |f(t)|.$$

Well defined: $|f|$ is continuous on the compact set $[a, b]$, so it attains its maximum (Weierstrass theorem); the supremum is finite and is in fact a maximum. *Checking the axioms:*

1. $\|f\|_\infty \geq 0$ is clear. If $\|f\|_\infty = 0$ then $|f(t)| \leq 0$ for every t , so $f(t) = 0$ for every t : f is the zero function. And conversely.
2. $\|\alpha f\|_\infty = \sup_t |\alpha| |f(t)| = |\alpha| \sup_t |f(t)| = |\alpha| \|f\|_\infty$ (a nonnegative constant comes out of the supremum).
3. For each t : $|f(t) + g(t)| \leq |f(t)| + |g(t)| \leq \|f\|_\infty + \|g\|_\infty$. The right hand side is an upper bound of $|(f + g)(t)|$ that does not depend on t ; taking the supremum over t : $\|f + g\|_\infty \leq \|f\|_\infty + \|g\|_\infty$.

Convergence in this norm is exactly the **uniform convergence** of the chapter on sequences of functions: $\|f_n - f\|_\infty \rightarrow 0$ means that the worst error over the whole interval goes to zero. This is *the* central example of the course.

Example 11.18 (The integral norm on $C[a, b]$). On the same set $C[a, b]$ define another norm:

$$\|f\|_1 := \int_a^b |f(t)| \, dt.$$

Verification: (2) and (3) follow from linearity and monotonicity of the integral ($|f + g| \leq |f| + |g|$ pointwise, and integrating preserves inequalities). The delicate axiom is separation in (1): if $\int_a^b |f| = 0$, is $f \equiv 0$? Yes, *thanks to continuity*: if there were t_0 with $|f(t_0)| = \varepsilon > 0$, by continuity there would be a subinterval $[t_0 - \delta, t_0 + \delta] \cap [a, b]$ of length $\geq \delta > 0$ where $|f| > \varepsilon/2$; the integral would then be $\geq \delta\varepsilon/2 > 0$, a contradiction. (For merely integrable functions this axiom fails (a function that is zero except at one point has zero integral without being the zero function) and that is why the L^p spaces of measure theory work with equivalence classes; in this course it is enough to know this as a remark.)

Example 11.19 (Norms on the sequence spaces).

$$\|x\|_1 := \sum_{i=1}^{\infty} |x_i| \text{ on } \ell^1; \quad \|x\|_2 := \left(\sum_{i=1}^{\infty} x_i^2 \right)^{1/2} \text{ on } \ell^2; \quad \|x\|_{\infty} := \sup_i |x_i| \text{ on } \ell^{\infty}, c_0.$$

Checking the triangle inequality on ℓ^1 : for each N , $\sum_{i \leq N} |x_i + y_i| \leq \sum_{i \leq N} |x_i| + \sum_{i \leq N} |y_i| \leq \|x\|_1 + \|y\|_1$; letting $N \rightarrow \infty$ gives $\|x + y\|_1 \leq \|x\|_1 + \|y\|_1$.

Checking it on ℓ^2 : first, for finite sums the Cauchy–Schwarz inequality $\left| \sum_{i \leq N} x_i y_i \right| \leq (\sum_{i \leq N} x_i^2)^{1/2} (\sum_{i \leq N} y_i^2)^{1/2}$ holds, as proved in the chapter on $\mathbb{R}^n$. Then

$$\sum_{i \leq N} (x_i + y_i)^2 = \sum_{i \leq N} x_i^2 + 2 \sum_{i \leq N} x_i y_i + \sum_{i \leq N} y_i^2 \leq \|x\|_2^2 + 2 \|x\|_2 \|y\|_2 + \|y\|_2^2 = (\|x\|_2 + \|y\|_2)^2,$$

and taking square roots and then $N \rightarrow \infty$: $\|x + y\|_2 \leq \|x\|_2 + \|y\|_2$.

In general, for $1 \leq p < \infty$, $\|x\|_p := (\sum_i |x_i|^p)^{1/p}$ is a norm on $\ell^p := \{x : \sum |x_i|^p < \infty\}$; the triangle inequality is the **Minkowski inequality**, which follows from the **Hölder inequality**:

$$\sum_i |x_i y_i| \leq \|x\|_p \|y\|_q, \quad \frac{1}{p} + \frac{1}{q} = 1.$$

We state both without a general proof (the cases $p = 1, 2$, which are the ones the course uses, were proved above; Hölder with $p = 2$ is Cauchy–Schwarz).

Example 11.20 (Two norms that are NOT equivalent on the same space). In finite dimension all norms are equivalent; in infinite dimension they are not. The canonical example: on $C[0, 1]$ compare $\|\cdot\|_{\infty}$ and $\|\cdot\|_1$.

One inequality always holds:

$$\|f\|_1 = \int_0^1 |f(t)| \, dt \leq \int_0^1 \|f\|_{\infty} \, dt = \|f\|_{\infty}.$$

But there is **no** constant C with $\|f\|_{\infty} \leq C \|f\|_1$ for every f . Build the family of spike functions: for each $n \geq 2$, let $f_n \in C[0, 1]$ be the piecewise linear function that equals 1 at $t = 0$, decreases

linearly to 0 at $t = 1/n$, and equals 0 on $[1/n, 1]$ (a triangle of height 1 and base $1/n$). Then:

$$\|f_n\|_\infty = 1 \quad \text{and} \quad \|f_n\|_1 = \text{area of the triangle} = \frac{1}{2} \cdot \frac{1}{n} \cdot 1 = \frac{1}{2n} \xrightarrow{n \rightarrow \infty} 0.$$

If such a C existed with $\|f\|_\infty \leq C \|f\|_1$, you would have $1 \leq C/(2n)$ for every n , which is absurd. Conclusion: the two norms generate *different* notions of convergence on $C[0, 1]$: $f_n \rightarrow 0$ in the integral norm (the area of the spike vanishes) but $f_n \not\rightarrow 0$ uniformly (the height of the spike stays at 1). You should know how to draw and reproduce this example.

Careful (common mistake)

“Equivalence of norms” means that there are constants $c, C > 0$ with $c\|x\|_a \leq \|x\|_b \leq C\|x\|_a$ for every x . In infinite dimension, having an inequality in one direction only (such as $\|f\|_1 \leq \|f\|_\infty$ above) is frequent and is NOT equivalence. In an exam, to disprove equivalence the standard technique is to exhibit a sequence (f_n) with $\|f_n\|_a$ bounded (or $\rightarrow 0$) and $\|f_n\|_b$ constant or divergent: the triangular spikes are your template.

11.4.2 Basic topological properties of a normed space

Proposition 11.21 (Reverse triangle inequality). *In every normed space, $|\|u\| - \|v\|| \leq \|u - v\|$.*

Proof. By the triangle inequality, $\|u\| = \|(u - v) + v\| \leq \|u - v\| + \|v\|$, so $\|u\| - \|v\| \leq \|u - v\|$. Swapping roles, $\|v\| - \|u\| \leq \|v - u\| = \|u - v\|$. The two inequalities together give the required absolute value. $\square$

Proposition 11.22 (Continuity of the operations). *In a normed space $(V, \|\cdot\|)$, if $u_n \rightarrow u$ and $v_n \rightarrow v$ (convergence in the norm) and $\alpha_n \rightarrow \alpha$ in $\mathbb{R}$, then:*

1. $u_n + v_n \rightarrow u + v$ (addition is continuous);
2. $\alpha_n v_n \rightarrow \alpha v$ (scalar multiplication is continuous);
3. $\|u_n\| \rightarrow \|u\|$ (the norm is continuous).

Proof. (1) Add and subtract, and apply the triangle inequality:

$$\|(u_n + v_n) - (u + v)\| = \|(u_n - u) + (v_n - v)\| \leq \|u_n - u\| + \|v_n - v\| \rightarrow 0 + 0 = 0.$$

(2) The usual trick for products: add and subtract the cross term $\alpha_n v$:

$$\|\alpha_n v_n - \alpha v\| = \|\alpha_n v_n - \alpha_n v + \alpha_n v - \alpha v\| \leq |\alpha_n| \|v_n - v\| + |\alpha_n - \alpha| \|v\|.$$

The second term goes to 0 because $\alpha_n \rightarrow \alpha$ and $\|v\|$ is constant. For the first one: (α_n) converges in $\mathbb{R}$, so it is bounded, say $|\alpha_n| \leq M$; then $|\alpha_n| \|v_n - v\| \leq M \|v_n - v\| \rightarrow 0$.

(3) By the reverse triangle inequality (Proposition 11.21): $|\|u_n\| - \|u\|| \leq \|u_n - u\| \rightarrow 0$. $\square$

Proposition 11.23 (Balls are convex). *In a normed space, every open ball $B(x_0, r) = \{x : \|x - x_0\| < r\}$ is a **convex set**: if $x, y \in B(x_0, r)$ and $t \in [0, 1]$, then $tx + (1-t)y \in B(x_0, r)$. The same holds for closed balls.*

Proof. Write the intermediate point after subtracting the center, using that $x_0 = tx_0 + (1-t)x_0$:

$$\begin{aligned} \|tx + (1-t)y - x_0\| &= \|t(x - x_0) + (1-t)(y - x_0)\| \\ &\leq t\|x - x_0\| + (1-t)\|y - x_0\| < tr + (1-t)r = r, \end{aligned}$$

where we used the triangle inequality, homogeneity (with $t, 1-t \geq 0$) and that both points are at distance less than r from the center. (If $t \in \{0, 1\}$ the claim is trivial and the strict inequality is preserved.) $\square$

Remark 11.24. The convexity of balls is another fingerprint of norms: in a general metric space balls can have arbitrary shapes, but the balls of a norm are always convex and symmetric about the center. Geometrically: the unit ball of $\|\cdot\|_1$ in $\mathbb{R}^2$ is a diamond, the one of $\|\cdot\|_2$ is a disk, and the one of $\|\cdot\|_\infty$ is a square; all three are convex.

Theorem 11.25 (Equivalence of norms in finite dimension). *If V is a finite dimensional vector space, then all norms on V are equivalent: for any norms $\|\cdot\|_a, \|\cdot\|_b$ there are constants $0 < c \leq C$ such that*

$$c\|x\|_a \leq \|x\|_b \leq C\|x\|_a \quad \text{for every } x \in V.$$

As a consequence, in finite dimension the notions of convergence, open and closed set, Cauchy, completeness and compactness do not depend on the chosen norm.

Sketch of the proof. Fixing a basis $\{v_1, \dots, v_n\}$ of V , everything reduces to comparing an arbitrary norm $\|\cdot\|$ with the “Euclidean norm of coordinates” $\|x\|_2 := (\sum \alpha_i^2)^{1/2}$ where $x = \sum \alpha_i v_i$. One side is easy:

$$\|x\| = \left\| \sum_i \alpha_i v_i \right\| \leq \sum_i |\alpha_i| \|v_i\| \leq \left(\max_i \|v_i\| \right) \sum_i |\alpha_i| \leq C \|x\|_2,$$

using the triangle inequality, homogeneity and the comparison between $\|\cdot\|_1$ and $\|\cdot\|_2$ on $\mathbb{R}^n$. This inequality also implies that $\|\cdot\| : (V, \|\cdot\|_2) \rightarrow \mathbb{R}$ is continuous. For the other side you use compactness: the unit sphere $S = \{x : \|x\|_2 = 1\}$ is closed and bounded in coordinates, hence compact (Heine–Borel through the identification with $\mathbb{R}^n$); the continuous function $x \mapsto \|x\|$ attains a minimum c on S , and $c > 0$ because the zero vector is not in S and the norm separates. By homogeneity, $\|x\| \geq c\|x\|_2$ for every x . Note the moral: the proof depends on the compactness of the sphere, which is *exactly* what fails in infinite dimension. $\square$

Link with economics

Where does an economist live inside this chapter?

- **Infinite consumption streams.** In growth models and in intertemporal choice, a consumption plan is a sequence $c = (c_0, c_1, c_2, \dots)$. The natural space is ℓ^∞ (bounded consumption) with the sup norm: two plans are close if they differ little

in every period, including the far future. That ℓ^∞ is infinite dimensional is not a curiosity: it is the reason why separation theorems and price existence results (the infinite dimensional version of the second welfare theorem) require functional analysis and not just linear algebra.

- **Spaces of policies and value functions.** A policy rule $g: S \rightarrow A$ or a value function $v \in B(S)$ are vectors of a function space with the sup norm. The value function iteration of the previous chapter is a sequence *in this normed space*, and its convergence is convergence in $\|\cdot\|_\infty$ (uniform).
- **The positive quadrant is not a subspace.** Consumption sets ($c_t \geq 0$) are *convex cones*, not subspaces. That is why economic theory uses both convexity (Proposition 11.23 is the first step) and the linear structure: separation of convex sets = prices that support allocations.

Exam strategy

Typical questions on this chapter and how to attack them:

1. “Determine whether W is a subspace.” Check in this order: (a) is $0 \in W$? (if not, you are done); (b) is it closed under addition?; (c) is it closed under scalars (including negative ones)? Write the three steps explicitly.
2. “Prove that $\|\cdot\|$ is a norm.” There are three axioms; the one that earns points is almost always **separation** (as in the integral norm: use continuity to go from “zero integral” to “zero function”) or the **triangle inequality** (go through the pointwise or coordinatewise inequality and then take the supremum, the sum or the integral).
3. “Are these two norms equivalent?” On $\mathbb{R}^n$: yes, always (cite Theorem 11.25, and if constants are requested, exhibit them). On $C[0, 1]$ or ℓ^p : suspect that they are not, and build spikes or truncate sequences.
4. “Prove that V is infinite dimensional.” Magic sentence: “for each m I exhibit a linearly independent set of m elements”, and use monomials or the canonical sequences $e^{(i)}$.

Chapter summary

- Vector space: a set with addition and scalar multiplication satisfying 8 axioms; the key examples are $\mathbb{R}^n$, $C[a, b]$, $\mathcal{P}_n$ and the sequence spaces $c_{00} \subsetneq \ell^1 \subsetneq \ell^2 \subsetneq c_0 \subsetneq \ell^\infty$.
- Subspace: nonempty + closed under addition and scalars. Quick test: if $0 \notin W$, it is not a subspace.
- Basis = independent + spanning; dimension = the common cardinality of the bases. $\dim \mathbb{R}^n = n$, $\dim \mathcal{P}_n = n + 1$; $C[a, b]$ and the ℓ^p are infinite dimensional (monomials or canonical sequences).

- Norm: separation, homogeneity, triangle inequality. It induces the metric $d(u, v) = \|u - v\|$, which is translation invariant and homogeneous: every normed space is a metric space.
- Key norms: $\|f\|_\infty = \sup |f|$ (uniform convergence) and $\|f\|_1 = \int |f|$ on $C[a, b]$; $\|\cdot\|_p$ on ℓ^p (Minkowski and Hölder).
- $\|f\|_1 \leq \|f\|_\infty$ on $C[0, 1]$, but they are NOT equivalent: triangular spikes of height 1 and base $1/n$.
- Addition, scalar multiplication and the norm are continuous; balls are convex.
- In finite dimension all norms are equivalent (the unit sphere is compact); in infinite dimension they are not.

11.5 Exam-style exercises [P3]

Exercise 11.26. Determine, with justification, which of the following sets are subspaces of the indicated space:

1. $W_1 = \{f \in C[0, 1] : f(0) = f(1)\}$ in $C[0, 1]$;
2. $W_2 = \{f \in C[0, 1] : f(t) \geq 0 \forall t\}$ in $C[0, 1]$;
3. $W_3 = \{x \in \ell^\infty : \lim_i x_i \text{ exists}\}$ in ℓ^∞ .

Solution. (1) It is a subspace. The zero function satisfies $0 = 0$, so $0 \in W_1$ (nonempty). If $f, g \in W_1$: $(f + g)(0) = f(0) + g(0) = f(1) + g(1) = (f + g)(1)$, so $f + g \in W_1$. If $\alpha \in \mathbb{R}$: $(\alpha f)(0) = \alpha f(0) = \alpha f(1) = (\alpha f)(1)$, so $\alpha f \in W_1$.

(2) It is not a subspace. The constant function $f \equiv 1$ is in W_2 , but $(-1) \cdot f \equiv -1$ is not (it takes negative values): W_2 is not closed under multiplication by negative scalars. (It is a convex cone, not a subspace.)

(3) It is a subspace. W_3 is the space usually called c (convergent sequences). The zero sequence converges (to 0), so $0 \in W_3$. If $x, y \in W_3$ with $x_i \rightarrow L$ and $y_i \rightarrow M$, the algebra of limits gives $x_i + y_i \rightarrow L + M$ (the limit exists), so $x + y \in W_3$; and $\alpha x_i \rightarrow \alpha L$, so $\alpha x \in W_3$. Also $W_3 \subseteq \ell^\infty$ because every convergent sequence is bounded. $\square$

Exercise 11.27. Prove that the functions $f_1(x) = e^x$, $f_2(x) = e^{2x}$, $f_3(x) = e^{3x}$ are linearly independent in $C[0, 1]$.

Solution. Suppose $\alpha_1 e^x + \alpha_2 e^{2x} + \alpha_3 e^{3x} = 0$ for every $x \in [0, 1]$. Strategy: produce three linear equations in the α_i by evaluating the identity and its derivatives at $x = 0$ (the identity between functions can be differentiated because the left hand side is differentiable and the right hand side is the zero function; the identity holds on an interval, so the derivatives also agree in its

interior and, by continuity, at 0).

$$\begin{aligned} x = 0 : \quad & \alpha_1 + \alpha_2 + \alpha_3 = 0, \\ \text{differentiating and } x = 0 : \quad & \alpha_1 + 2\alpha_2 + 3\alpha_3 = 0, \\ \text{differentiating again and } x = 0 : \quad & \alpha_1 + 4\alpha_2 + 9\alpha_3 = 0. \end{aligned}$$

The coefficient matrix is the Vandermonde matrix

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix},$$

whose determinant is $(2-1)(3-1)(3-2) = 2 \neq 0$. The homogeneous system has only the trivial solution: $\alpha_1 = \alpha_2 = \alpha_3 = 0$. Therefore the three functions are linearly independent. (Alternative method: divide the identity by e^{3x} and let $x \rightarrow \infty$ over $\mathbb{R}$; on $[0, 1]$ the derivative method is the clean one.) $\square$

Exercise 11.28. Prove that $\|f\| := |f(0)| + \sup_{t \in [0,1]} |f'(t)|$ defines a norm on the space $C^1[0, 1]$ of functions with continuous derivative on $[0, 1]$.

Solution. We check the three axioms for $f, g \in C^1[0, 1]$, $\alpha \in \mathbb{R}$. Note first that $\sup_t |f'|$ is finite: f' is continuous on the compact set $[0, 1]$, hence bounded (Weierstrass).

Separation. $\|f\| \geq 0$ is clear (a sum of two nonnegative terms). If $\|f\| = 0$, both terms are zero: $f(0) = 0$ and $f'(t) = 0$ for every t . A derivative that is identically zero on an interval implies that f is constant (a consequence of the mean value theorem), and since $f(0) = 0$, the constant is 0: $f \equiv 0$. Conversely, $\|0\| = 0$.

Homogeneity.

$$\|\alpha f\| = |\alpha f(0)| + \sup_t |\alpha f'(t)| = |\alpha| |f(0)| + |\alpha| \sup_t |f'(t)| = |\alpha| \|f\|,$$

using $(\alpha f)' = \alpha f'$ and that the constant $|\alpha| \geq 0$ comes out of the supremum.

Triangle inequality.

$$\|f + g\| = |f(0) + g(0)| + \sup_t |f'(t) + g'(t)| \leq |f(0)| + |g(0)| + \sup_t |f'| + \sup_t |g'| = \|f\| + \|g\|,$$

where we used the triangle inequality in $\mathbb{R}$ pointwise and that the supremum of a sum is $\leq$ the sum of the suprema. The three axioms hold: $\|\cdot\|$ is a norm. (Remark: without the term $|f(0)|$ it would NOT be a norm: nonzero constants would have “norm” zero and separation would fail. Spotting that term is the point of the exercise.) $\square$

Exercise 11.29. Prove that on $C[0, 1]$ the norms $\|f\|_\infty = \sup_t |f(t)|$ and $\|f\|_1 = \int_0^1 |f(t)| dt$ are not equivalent, but that $\|f\|_1 \leq \|f\|_\infty$ does hold for every f .

Solution. The valid inequality. For every $t \in [0, 1]$, $|f(t)| \leq \|f\|_\infty$. Integrating this inequal-

ity between 0 and 1 (the integral preserves inequalities):

$$\|f\|_1 = \int_0^1 |f(t)| dt \leq \int_0^1 \|f\|_\infty dt = \|f\|_\infty.$$

Non equivalence. If they were equivalent there would be $C > 0$ with $\|f\|_\infty \leq C \|f\|_1$ for every $f \in C[0, 1]$. Build spikes: for $n \geq 2$ let

$$f_n(t) = \begin{cases} 1 - nt, & 0 \leq t \leq 1/n, \\ 0, & 1/n \leq t \leq 1. \end{cases}$$

Each f_n is continuous (the two pieces agree at $t = 1/n$, where both equal 0), and:

$$\|f_n\|_\infty = f_n(0) = 1, \quad \|f_n\|_1 = \int_0^{1/n} (1 - nt) dt = \left[t - \frac{nt^2}{2} \right]_0^{1/n} = \frac{1}{n} - \frac{1}{2n} = \frac{1}{2n}.$$

The supposed inequality would give $1 \leq C \cdot \frac{1}{2n}$, that is $n \leq C/2$ for *every* n , which is absurd. So no such C exists and the norms are not equivalent. $\square$

Exercise 11.30. Let $(V, \|\cdot\|)$ be a normed space. Prove that if $x_n \rightarrow x$ in norm, then $\|x_n\| \rightarrow \|x\|$. Is the converse true? Justify with a proof or a counterexample.

Solution. Direct statement. By the reverse triangle inequality: for each n ,

$$| \|x_n\| - \|x\| | \leq \|x_n - x\|.$$

(Proof of the reverse inequality: $\|x_n\| \leq \|x_n - x\| + \|x\|$ gives $\|x_n\| - \|x\| \leq \|x_n - x\|$; symmetrically for the other sign.) Since $\|x_n - x\| \rightarrow 0$ by assumption, the squeeze theorem in $\mathbb{R}$ gives $|\|x_n\| - \|x\|| \rightarrow 0$, that is $\|x_n\| \rightarrow \|x\|$.

The converse is false. Counterexample in $\mathbb{R}$ (which is a normed space with $\|x\| = |x|$): $x_n = (-1)^n$ and $x = 1$. Then $\|x_n\| = 1 \rightarrow 1 = \|x\|$, but $x_n \not\rightarrow x$ (the odd subsequence equals -1). In general, the norm “forgets the direction”: two vectors can have the same length and still be far apart. The only case where the converse does work is $x = 0$: $\|x_n\| \rightarrow 0$ is indeed equivalent to $x_n \rightarrow 0$, because $\|x_n - 0\| = \|x_n\|$. This distinction ($x \neq 0$ versus $x = 0$) is a subtlety that graders look for. $\square$

Chapter 12

Normed subspaces. Sequences and series in normed spaces [P3]

Week 10 of the course (the first one after the midterm exam). Bibliography: Moroşanu [2], Ch. 2; Pons [3], Chs. 3 and 7; Zorich II [6].

What is this used for in Microeconomics?

This chapter gains weight as soon as the syllabus includes repeated games and long term contracts, because both are built on *discounting*. The payoff of a player in a repeated game is the sum $\sum_t \delta^t u_t$, and the arguments for sustaining collusion and for the *folk theorem* consist of comparing that geometric series with the gain from a deviation (game theory and market power, Mas-Colell, Whinston and Green, Chs. 9 and 12). The same holds for the present value of a payment stream in long term agency relations (Ch. 14). All of that is the section on numerical series: convergence with $\delta < 1$ and bounded utilities, and the sum of the geometric series. The rest (dense subspaces, the Banach criterion through absolutely convergent series) is material specific to the analysis course.

12.1 Motivation [P3]

With the normed space structure already in place, this chapter does two things. First, it studies **subspaces** from the topological point of view: in infinite dimension a phenomenon appears that is impossible in $\mathbb{R}^n$, that of proper subspaces which are *dense*, that is, which get arbitrarily close to every point of the space without filling it. Second, it develops the theory of **series**: infinite sums $\sum_{n=1}^{\infty} x_n$ where the summands are vectors (numbers, functions, sequences). Series are the bridge between algebra (finite sums) and analysis (limits), and they lead to a beautiful theorem that *characterizes* the completeness of a normed space in terms of series: a space is Banach exactly when “absolute convergence implies convergence”. For an economist, series are everyday tools: the present value of an infinite payment stream *is* a series.

12.2 Normed subspaces: closed and dense [P5]

If $(V, \|\cdot\|)$ is a normed space and $W \subseteq V$ is a vector subspace, then $(W, \|\cdot\|)$ is itself a normed space (the restricted norm still satisfies the three axioms). But since V is also a metric space, it makes sense to ask whether W is *closed* in V (it contains the limits of its convergent sequences) or *dense* in V ($\text{cl}(W) = V$).

Intuition

In $\mathbb{R}^3$, the proper subspaces are lines and planes through the origin: “thin” objects, closed, which you would never confuse with the whole space. In infinite dimension this intuition breaks down: a proper subspace can be *dense*, like the polynomials inside the continuous functions. Dense means: for any $f \in C[a, b]$ and any tolerance $\varepsilon > 0$, there is a polynomial at distance less than ε from f in the supremum norm. Polynomials are “not everything”, but they are “everywhere”.

Theorem 12.1 (Weierstrass approximation theorem). *The subspace $\mathcal{P}$ of polynomials is dense in $(C[a, b], \|\cdot\|_\infty)$: for every $f \in C[a, b]$ and every $\varepsilon > 0$ there is a polynomial p such that*

$$\|f - p\|_\infty = \sup_{t \in [a, b]} |f(t) - p(t)| < \varepsilon.$$

Remark 12.2. We will not prove this theorem (a constructive proof uses Bernstein polynomials; see Pons [3]); we will use it as a tool. Two important readings: (a) $\mathcal{P}$ is a proper subspace (e^t is not a polynomial) but dense: its closure is all of $C[a, b]$, so $\mathcal{P}$ is **not closed** in $C[a, b]$ (if it were closed, being dense it would be the whole space). (b) Density depends on the norm: this is a statement about $\|\cdot\|_\infty$, so the approximation is *uniform* on the whole interval.

Example 12.3 (c_{00} is dense in c_0 , but not closed; and it is not dense in ℓ^∞). Recall: c_{00} are the sequences with finitely many nonzero terms, c_0 those that tend to 0, both inside ℓ^∞ with $\|x\|_\infty = \sup_i |x_i|$.

(a) c_{00} is dense in c_0 . Let $x = (x_1, x_2, \dots) \in c_0$ and $\varepsilon > 0$. Since $x_i \rightarrow 0$, there is N such that $|x_i| < \varepsilon$ for all $i > N$. Define the truncation $x^{(N)} := (x_1, \dots, x_N, 0, 0, \dots) \in c_{00}$. Then

$$\|x - x^{(N)}\|_\infty = \sup_{i > N} |x_i| \leq \varepsilon.$$

Since ε is arbitrary, every point of c_0 is a limit of truncations: $\text{cl}(c_{00}) \supseteq c_0$. (In fact there is equality, by part (c).)

(b) c_{00} is not closed in c_0 (nor in ℓ^∞): the sequence of truncations of $x = (1, \frac{1}{2}, \frac{1}{3}, \dots) \in c_0$ lies in c_{00} and converges to $x \notin c_{00}$ (it has infinitely many nonzero terms). A proper dense subspace is never closed.

(c) c_{00} is NOT dense in ℓ^∞ . Take the constant sequence $u = (1, 1, 1, \dots) \in \ell^\infty$. For any $y \in c_{00}$, y has only finitely many nonzero terms, so there is i_0 with $y_{i_0} = 0$, and then

$$\|u - y\|_\infty \geq |u_{i_0} - y_{i_0}| = 1.$$

No element of c_{00} gets closer to u than distance 1: u is not in the closure. The lesson: density is relative to the ambient space and to the norm; the closure of c_{00} in $(\ell^\infty, \|\cdot\|_\infty)$ is exactly c_0 .

Proposition 12.4 (Finite dimensional subspaces are closed). *If $(V, \|\cdot\|)$ is a normed space and $W \subseteq V$ is a finite dimensional subspace, then W is closed in V .*

Proof idea. Let $(w_n) \subseteq W$ with $w_n \rightarrow v \in V$; we must show $v \in W$. Fix a basis $\{u_1, \dots, u_k\}$ of W and write $w_n = \sum_j \alpha_j^{(n)} u_j$. By the equivalence of norms in finite dimension, the norm of V restricted to W is equivalent to the maximum norm of the coordinates; since (w_n) is Cauchy in V (because it converges), its coordinate vectors $(\alpha_1^{(n)}, \dots, \alpha_k^{(n)})$ form a Cauchy sequence in $\mathbb{R}^k$, which converges to some $(\alpha_1, \dots, \alpha_k)$. By continuity of the operations, $w_n \rightarrow \sum_j \alpha_j u_j \in W$, and by uniqueness of the limit, $v = \sum_j \alpha_j u_j \in W$. $\square$

Remark 12.5. The contrast you should keep: *finite dimension* $\Rightarrow$ closed, always; *infinite dimension* $\Rightarrow$ it may be closed (for example, c_0 is closed in ℓ^∞) or dense and not closed (polynomials in $C[a, b]$, c_{00} in c_0).

12.3 Review: numerical series [P3]

Before adding vectors, let us recall how numbers are added. Everything in this section happens in $\mathbb{R}$.

Definition 12.6 (Series). Given a sequence $(a_n)_{n \geq 1}$ in $\mathbb{R}$, the **series** $\sum_{n=1}^\infty a_n$ is the sequence of **partial sums**

$$s_N := \sum_{n=1}^N a_n = a_1 + a_2 + \dots + a_N.$$

We say that the series **converges** with sum s if $s_N \rightarrow s$ as $N \rightarrow \infty$; in that case we write $\sum_{n=1}^\infty a_n = s$. If (s_N) does not converge, the series **diverges**.

Intuition

A series is nothing more than a sequence in disguise: the sequence of its partial sums. The whole theory of sequences (limits, Cauchy, monotonicity) is reused. The question “does the series converge?” is the question “does the sequence s_N converge?”.

Example 12.7 (The geometric series). For $r \in \mathbb{R}$, the geometric series $\sum_{n=0}^\infty r^n$ converges if and only if $|r| < 1$, with sum $\frac{1}{1-r}$.

Proof. The key algebraic identity (multiply and telescope): $(1-r)(1+r+\dots+r^N) = 1-r^{N+1}$. If $r \neq 1$,

$$s_N = \sum_{n=0}^N r^n = \frac{1-r^{N+1}}{1-r}.$$

If $|r| < 1$, then $r^{N+1} \rightarrow 0$ (proved in the chapter on sequences), so $s_N \rightarrow \frac{1}{1-r}$. If $|r| \geq 1$, the general term r^n does not tend to 0, and by the necessary condition (Proposition 12.8) the series diverges (for $r = 1$ directly $s_N = N + 1 \rightarrow \infty$).

Proposition 12.8 (Necessary condition for convergence). *If $\sum a_n$ converges, then $a_n \rightarrow 0$.*

Proof. Let s be the sum. Then $a_N = s_N - s_{N-1}$, and since both sequences of partial sums tend to s :

$$\lim_N a_N = \lim_N s_N - \lim_N s_{N-1} = s - s = 0. \quad \square$$

Theorem 12.9 (The converse is false: the harmonic series diverges). *The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ has general term tending to 0, but it diverges.*

Proof. We group the terms into dyadic blocks (Oresme's argument, also called the "Cauchy blocks" argument). Consider the partial sum up to $N = 2^k$ and group:

$$s_{2^k} = 1 + \frac{1}{2} + \underbrace{\left(\frac{1}{3} + \frac{1}{4}\right)}_{2 \text{ terms}} + \underbrace{\left(\frac{1}{5} + \cdots + \frac{1}{8}\right)}_{4 \text{ terms}} + \cdots + \underbrace{\left(\frac{1}{2^{k-1}+1} + \cdots + \frac{1}{2^k}\right)}_{2^{k-1} \text{ terms}}.$$

In each block, all terms are greater than or equal to the last one: the block ending at $\frac{1}{2^j}$ has 2^{j-1} terms, each one $\geq \frac{1}{2^j}$, so the block adds up to at least $2^{j-1} \cdot \frac{1}{2^j} = \frac{1}{2}$. With k blocks (starting from the one ending at $\frac{1}{2}$):

$$s_{2^k} \geq 1 + k \cdot \frac{1}{2}.$$

As $k \rightarrow \infty$, the right hand side tends to ∞ ; the sequence of partial sums is unbounded, so it does not converge: the series diverges. Note that $\frac{1}{n} \rightarrow 0$, so "general term going to zero" does **not** imply convergence. $\square$

Careful (common mistake)

Proposition 12.8 only serves to *prove divergence* (if $a_n \not\rightarrow 0$, the series diverges; this is the "general term test"). Using it backwards (" $a_n \rightarrow 0$, therefore it converges") is perhaps the most heavily penalized mistake in exams on series. The required counterexample is the harmonic series: learn the block proof, which is short and elegant.

12.3.1 Convergence tests

For series with nonnegative terms, the partial sums are increasing, so they *converge if and only if they are bounded* (monotone convergence theorem). The comparison tests follow from this.

Proposition 12.10 (Comparison test). *Let $0 \leq a_n \leq b_n$ for all n (or from some index on).*

1. *If $\sum b_n$ converges, then $\sum a_n$ converges.*
2. *If $\sum a_n$ diverges, then $\sum b_n$ diverges.*

Proof. (1) The partial sums satisfy $\sum_{n \leq N} a_n \leq \sum_{n \leq N} b_n \leq \sum_{n=1}^{\infty} b_n =: M < \infty$. The sequence of partial sums of $\sum a_n$ is increasing (terms ≥ 0) and bounded by M , so it converges. (2) is the contrapositive of (1). $\square$

Theorem 12.11 (Ratio test). *Let (a_n) with $a_n \neq 0$ and suppose that $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ exists.*

1. *If $L < 1$, the series $\sum a_n$ converges absolutely.*
2. *If $L > 1$, the series diverges.*

3. If $L = 1$, the test is inconclusive.

Proof. (1) Choose r with $L < r < 1$ (possible because $L < 1$). By the definition of limit with $\varepsilon = r - L > 0$, there is N such that for all $n \geq N$: $|a_{n+1}/a_n| < r$, that is, $|a_{n+1}| < r |a_n|$. Iterating this inequality from N (immediate induction):

$$|a_{N+k}| < r^k |a_N| \quad (k = 0, 1, 2, \dots).$$

Then the tail of the series is compared with a geometric one: $\sum_{k \geq 0} |a_{N+k}| \leq |a_N| \sum_{k \geq 0} r^k = \frac{|a_N|}{1-r} < \infty$, and by the comparison test (Proposition 12.10) the series $\sum |a_n|$ converges (the first N terms do not affect convergence).

(2) If $L > 1$, there is N such that $|a_{n+1}| > |a_n|$ for $n \geq N$: the sequence $|a_n|$ is increasing from $|a_N| > 0$, so $a_n \not\rightarrow 0$ and the series diverges by the necessary condition.

(3) $\sum \frac{1}{n}$ diverges and $\sum \frac{1}{n^2}$ converges, and in both $L = 1$: the test cannot decide. $\square$

Theorem 12.12 (Root test). *If $L = \lim_n |a_n|^{1/n}$ exists, then: $L < 1$ implies absolute convergence; $L > 1$ implies divergence; $L = 1$ is inconclusive. (The proof is similar to the one for the ratio test, comparing with r^n .)*

Theorem 12.13 (Leibniz test for alternating series). *If (b_n) is decreasing with $b_n \rightarrow 0$, then the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} b_n = b_1 - b_2 + b_3 - \dots$ converges. Moreover, the error when you truncate is bounded by the first omitted term: $|s - s_N| \leq b_{N+1}$.*

Remark 12.14 (Proof idea). The even partial sums s_{2k} increase and the odd ones s_{2k+1} decrease, with $s_{2k} \leq s_{2k+1}$; both subsequences are monotone and bounded, so they converge, and their difference $b_{2k+1} \rightarrow 0$ forces the same limit. Example: the alternating harmonic series $1 - \frac{1}{2} + \frac{1}{3} - \dots$ converges (to $\ln 2$), although the harmonic series diverges.

Definition 12.15 (Absolute and conditional convergence). The series $\sum a_n$ **converges absolutely** if $\sum |a_n|$ converges. If $\sum a_n$ converges but $\sum |a_n|$ diverges, we say that it converges **conditionally**.

Proposition 12.16. *In $\mathbb{R}$, absolute convergence implies convergence.*

Proof. We use the Cauchy criterion for the partial sums ($\mathbb{R}$ is complete, so “being Cauchy” is the same as converging). Given $\varepsilon > 0$, since $\sum |a_n|$ converges, its partial sums are Cauchy: there is N such that for $m > n \geq N$, $\sum_{j=n+1}^m |a_j| < \varepsilon$. Then, by the triangle inequality (extended to finite sums):

$$|s_m - s_n| = \left| \sum_{j=n+1}^m a_j \right| \leq \sum_{j=n+1}^m |a_j| < \varepsilon.$$

So (s_N) is Cauchy in $\mathbb{R}$ and converges. $\square$

Remark 12.17 (Rearrangements: Riemann’s theorem). An *absolutely* convergent series can be rearranged at will without changing its sum. In contrast, the **Riemann rearrangement theorem** states that a *conditionally* convergent series can be rearranged so that it converges to *any* prescribed real number, or so that it diverges. The intuition: in a conditional series, the

positive terms add up to $+\infty$ and the negative ones to $-\infty$; by rearranging you can control how much positive and how much negative you take at each stage, steering the sum wherever you want. Only the statement and the intuition are part of the course; the practical lesson is: *with conditional series, the order of the summands matters.*

12.4 Series in normed spaces [P4]

Now the summands are vectors of a normed space $(X, \|\cdot\|)$: functions, sequences, and so on. The definition is the same, replacing the absolute value by the norm.

Definition 12.18. Let (x_n) be a sequence in a normed space X . The series $\sum_{n=1}^{\infty} x_n$ **converges** to $s \in X$ if the sequence of partial sums $s_N = \sum_{n=1}^N x_n$ converges to s in the norm, that is, $\|s_N - s\| \rightarrow 0$. The series is **absolutely convergent** (also called norm summable) if the numerical series of the norms

$$\sum_{n=1}^{\infty} \|x_n\|$$

converges in $\mathbb{R}$.

Remark 12.19. Note the subtlety: $\sum \|x_n\|$ is a series of nonnegative real numbers, so all the tests of the previous section apply to it. The deep question is: does absolute convergence imply the convergence of the vector series $\sum x_n$? In $\mathbb{R}$ it does (Proposition 12.16), and the proof used the completeness of $\mathbb{R}$. The next theorem shows that this implication is *exactly* completeness.

Theorem 12.20 (Characterization of Banach spaces by series). *Let $(X, \|\cdot\|)$ be a normed space. The following are equivalent:*

1. X is complete (that is, it is a Banach space);
2. every absolutely convergent series in X is convergent in X .

Proof. (1) $\Rightarrow$ (2). Suppose X is complete and let $\sum x_n$ satisfy $\sum \|x_n\| < \infty$. We will show that the partial sums $s_N = \sum_{n \leq N} x_n$ form a Cauchy sequence. Let $\varepsilon > 0$. The partial sums of the numerical series $\sum \|x_n\|$ converge, so they are Cauchy in $\mathbb{R}$: there is N_0 such that for $m > n \geq N_0$,

$$\sum_{j=n+1}^m \|x_j\| < \varepsilon.$$

Then, by the triangle inequality applied to the finite sum $s_m - s_n = x_{n+1} + \cdots + x_m$:

$$\|s_m - s_n\| = \left\| \sum_{j=n+1}^m x_j \right\| \leq \sum_{j=n+1}^m \|x_j\| < \varepsilon.$$

So (s_N) is Cauchy in X ; by completeness, it converges to some $s \in X$: the series converges.

(2) $\Rightarrow$ (1). Suppose that every absolutely convergent series converges, and let (y_n) be an arbitrary Cauchy sequence in X ; we must prove that it converges. The strategy: extract a subsequence that “advances in geometrically small steps”, write it as a telescoping series, and use the hypothesis.

Step 1: extracting the subsequence. We apply the Cauchy definition with $\varepsilon = 2^{-1}$: there is n_1 such that $\|y_m - y_n\| < 2^{-1}$ for all $m, n \geq n_1$. Then with $\varepsilon = 2^{-2}$: there is $n_2 > n_1$ such that $\|y_m - y_n\| < 2^{-2}$ for $m, n \geq n_2$. Continuing (with $\varepsilon = 2^{-k}$ at step k , always choosing $n_{k+1} > n_k$), we obtain indices $n_1 < n_2 < \dots$ such that

$$\|y_{n_{k+1}} - y_{n_k}\| < 2^{-k} \quad (k = 1, 2, \dots),$$

since both indices n_{k+1}, n_k are $\geq n_k$.

Step 2: the telescoping series. Define the “steps” $x_1 := y_{n_1}$ and $x_{k+1} := y_{n_{k+1}} - y_{n_k}$ for $k \geq 1$. The telescoping partial sum collapses:

$$\sum_{j=1}^K x_j = y_{n_1} + (y_{n_2} - y_{n_1}) + \dots + (y_{n_K} - y_{n_{K-1}}) = y_{n_K}.$$

Moreover the series $\sum x_j$ is absolutely convergent:

$$\sum_{j=1}^{\infty} \|x_j\| \leq \|y_{n_1}\| + \sum_{k=1}^{\infty} 2^{-k} = \|y_{n_1}\| + 1 < \infty,$$

comparing with the geometric series. By hypothesis (2), the series converges: that is, there exists $y := \lim_K \sum_{j \leq K} x_j = \lim_K y_{n_K}$. We have proved that the subsequence (y_{n_k}) converges to y .

Step 3: the whole sequence converges. It is a general fact (seen in the chapter on completeness) that if a Cauchy sequence has a subsequence converging to y , then the whole sequence converges to y . Let us recall the argument: given $\varepsilon > 0$, take N (by Cauchy) with $\|y_m - y_n\| < \varepsilon/2$ for $m, n \geq N$, and an index of the subsequence $n_k \geq N$ with $\|y_{n_k} - y\| < \varepsilon/2$. For $n \geq N$:

$$\|y_n - y\| \leq \|y_n - y_{n_k}\| + \|y_{n_k} - y\| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

So $y_n \rightarrow y$. Every Cauchy sequence converges: X is complete. $\square$

Intuition

The direction (2) $\Rightarrow$ (1) hides a very reusable idea: a Cauchy sequence can be “discretized” into a subsequence whose consecutive jumps decrease geometrically ($< 2^{-k}$), and a sequence is the same thing as the series of its jumps (telescoping). So “summing series” and “completing Cauchy sequences” are two sides of the same coin. This 2^{-k} technique appears again in many proofs in functional analysis and measure theory.

Careful (common mistake)

In infinite dimension, if X is **not** complete, a series can be absolutely convergent and not convergent in X (its “sum” escapes from the space). Example: in $(c_{00}, \|\cdot\|_{\infty})$, the series $\sum_k 2^{-k} e^{(k)}$ (where $e^{(k)}$ is the canonical sequence) satisfies $\sum_k \|2^{-k} e^{(k)}\|_{\infty} = \sum_k 2^{-k} = 1 < \infty$, but its partial sums converge to $(2^{-1}, 2^{-2}, 2^{-3}, \dots)$, which has infinitely many nonzero terms: the limit is in c_0 , not in c_{00} . Never write “absolutely convergent $\Rightarrow$

convergent” without having stated that the space is Banach.

12.4.1 Examples and applications

Example 12.21 (The Weierstrass M-test as an absolutely convergent series). Let $\sum_{n=1}^{\infty} f_n$ be a series of functions $f_n \in C[a, b]$, and suppose there are constants $M_n \geq 0$ such that

$$\|f_n\|_{\infty} = \sup_{t \in [a, b]} |f_n(t)| \leq M_n \quad \text{and} \quad \sum_{n=1}^{\infty} M_n < \infty.$$

Then $\sum \|f_n\|_{\infty} \leq \sum M_n < \infty$ by comparison: the series is *absolutely convergent* in the space $(C[a, b], \|\cdot\|_{\infty})$. This space is Banach (the completeness of $C[a, b]$ with the uniform norm is proved in the chapter on Banach spaces; it is the theorem “a uniform limit of continuous functions is continuous” plus the completeness of $\mathbb{R}$). By Theorem 12.20, the series converges in the uniform norm to a function $f \in C[a, b]$: *the series of functions converges uniformly and its sum is continuous*. This is exactly the **Weierstrass M-test** from the chapter on convergence of functions, now seen conceptually: it is nothing more than “absolute convergence $\Rightarrow$ convergence” in the Banach space $C[a, b]$.

Concrete example: $\sum_{n=1}^{\infty} \frac{\cos(nt)}{n^2}$ in $C[0, 2\pi]$: since $\left\| \frac{\cos(n\cdot)}{n^2} \right\|_{\infty} \leq \frac{1}{n^2}$ and $\sum \frac{1}{n^2}$ converges, the series defines a continuous function on $[0, 2\pi]$ (a uniformly convergent Fourier series).

Example 12.22 (A preview: the Neumann series). Let A be an $n \times n$ matrix (or, later, a bounded linear operator) with $\|A\| < 1$ for some submultiplicative matrix norm ($\|A^k\| \leq \|A\|^k$). The series

$$\sum_{k=0}^{\infty} A^k = I + A + A^2 + \cdots$$

is absolutely convergent in the space of matrices (which is finite dimensional, hence Banach): $\sum_k \|A^k\| \leq \sum_k \|A\|^k = \frac{1}{1-\|A\|}$, a geometric series. By Theorem 12.20 it converges, and its sum is exactly $(I - A)^{-1}$ (the **Neumann series**): multiplying the partial sum by $(I - A)$ gives $(I - A)(I + A + \cdots + A^K) = I - A^{K+1} \rightarrow I$. We will return to this with full rigor in the chapter on operators; record the pattern: *the numerical geometric series of Example 12.7 generalizes to matrices and operators thanks to completeness*.

Link with economics

Present value is a series. A stream of payments or utilities $(u_0, u_1, u_2, \dots)$ discounted with factor $\beta \in (0, 1)$ has present value

$$V = \sum_{t=0}^{\infty} \beta^t u_t.$$

If the stream is bounded ($|u_t| \leq M$, that is, $u \in \ell^{\infty}$), the series converges *absolutely* by comparison with the geometric series: $\sum_t |\beta^t u_t| \leq M \sum_t \beta^t = \frac{M}{1-\beta}$. This bound $\frac{M}{1-\beta}$ is exactly the one that guarantees that the value function of a dynamic problem with bounded utility is finite and bounded: the “ $\frac{1}{1-\beta}$ ” that shows up everywhere in dynamic

macroeconomics is the sum of a geometric series.

Multipliers as geometric series. In the elementary Keynesian model, an initial spending of 1 generates successive consumptions $c, c^2, c^3, \dots$ (with marginal propensity to consume $c \in (0, 1)$): the total effect is $\sum_{k \geq 0} c^k = \frac{1}{1-c}$, the multiplier. In input-output analysis (Leontief), the total demand generated by a final demand d is $(I - A)^{-1}d = \sum_k A^k d$: the Neumann series of Example 12.22, where $A^k d$ are the indirect requirements of the k -th round. The condition $\|A\| < 1$ is the mathematical version of “the economy is productive”.

Exam strategy

Battle guide for this chapter:

1. “*Is W closed/dense in X ?*” For **not closed**: exhibit a sequence in W converging (in the norm of X) to a point outside W (truncations, Taylor or Bernstein polynomials). For **dense**: given $x \in X$ and ε , build explicitly $w \in W$ with $\|x - w\| < \varepsilon$ (truncating is almost always the construction). For **not dense**: find a point at distance $\geq \delta > 0$ from all of W (such as $(1, 1, 1, \dots)$ against c_{00}).
2. “*Study the convergence of the series...*” Order of attack: (i) does the general term go to 0? if not, it diverges and you are done; (ii) are the terms positive? try comparison with a geometric series or a p -series, or the ratio or root test if there are factorials or powers; (iii) is it alternating? Leibniz; (iv) ALWAYS distinguish absolute from conditional if you are asked to.
3. “*Prove that if X is Banach, every absolutely convergent series converges*” (or the full equivalence): reproduce the proof of Theorem 12.20; in the converse direction, do not forget the three steps: subsequence with jumps 2^{-k} , telescoping, and “Cauchy + convergent subsequence $\Rightarrow$ convergent”.
4. In any use of “absolute $\Rightarrow$ convergent”, **state the completeness of the space** before concluding: this is the point the grader is looking for.

Chapter summary

- Finite dimensional subspace $\Rightarrow$ closed. In infinite dimension there are *dense* proper subspaces: polynomials in $(C[a, b], \|\cdot\|_\infty)$ (Weierstrass), c_{00} in c_0 . Dense + proper $\Rightarrow$ not closed.
- Series = sequence of partial sums. Geometric: converges iff $|r| < 1$, with sum $\frac{1}{1-r}$. Necessary condition: $a_n \rightarrow 0$ (not sufficient: the harmonic series!).
- Tests: comparison, ratio ($L < 1$ converges, $L > 1$ diverges, $L = 1$ nothing), root, Leibniz for alternating series. Absolute versus conditional; Riemann: conditional series can be rearranged to any sum.
- In a normed space: absolutely convergent means $\sum \|x_n\| < \infty$.
- **Star theorem**: X is Banach $\iff$ every absolutely convergent series converges.

(Forward: triangle inequality + Cauchy. Backward: subsequence with jumps $< 2^{-k}$ + telescoping.)

- Applications: the Weierstrass M-test = the theorem in $C[a, b]$; the Neumann series $(I - A)^{-1} = \sum A^k$ if $\|A\| < 1$; present value $\sum \beta^t u_t$ converges if u is bounded and $\beta \in (0, 1)$, with bound $\frac{M}{1-\beta}$.

12.5 Exam-style exercises [P3]

Exercise 12.23. Study the convergence of the series: (a) $\sum_{n=1}^{\infty} \frac{n!}{n^n}$; (b) $\sum_{n=1}^{\infty} \frac{n}{2n+1}$; (c) $\sum_{n=2}^{\infty} \frac{(-1)^n}{\ln n}$, stating in (c) whether the convergence is absolute or conditional.

Solution. (a) **Converges** (ratio test). With $a_n = n!/n^n > 0$:

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \frac{(n+1)n^n}{(n+1)^{n+1}} = \left(\frac{n}{n+1}\right)^n = \frac{1}{\left(1+\frac{1}{n}\right)^n} \rightarrow \frac{1}{e} < 1,$$

using the fundamental limit $(1 + 1/n)^n \rightarrow e \approx 2.718$. Since $L = 1/e < 1$, the series converges (absolutely).

(b) **Diverges** (general term test): $\frac{n}{2n+1} \rightarrow \frac{1}{2} \neq 0$, so the necessary condition fails and the series diverges. (No other test is needed.)

(c) **Converges conditionally.** *Convergence:* $b_n = \frac{1}{\ln n}$ is positive, decreasing (since $\ln n$ increases) and $b_n \rightarrow 0$; by the Leibniz test, the alternating series converges. *Not absolute:* we compare $\sum \frac{1}{\ln n}$ with the harmonic series: for $n \geq 2$, $\ln n < n$, so $\frac{1}{\ln n} > \frac{1}{n} > 0$; since $\sum \frac{1}{n}$ diverges, by comparison $\sum \frac{1}{\ln n}$ diverges. The convergence is conditional. $\square$

Exercise 12.24. Prove that c_0 is a closed subspace of $(\ell^\infty, \|\cdot\|_\infty)$.

Solution. We already know that c_0 is a vector subspace (algebra of limits). For “closed” we use the sequential characterization: let $(x^{(n)})$ be a sequence of elements of c_0 with $x^{(n)} \rightarrow x$ in the supremum norm, where $x \in \ell^\infty$; we will prove $x \in c_0$, that is, $x_i \rightarrow 0$ as $i \rightarrow \infty$.

Let $\varepsilon > 0$. By convergence in norm, there is n_0 such that

$$\|x^{(n_0)} - x\|_\infty = \sup_i |x_i^{(n_0)} - x_i| < \frac{\varepsilon}{2}.$$

Since $x^{(n_0)} \in c_0$, there is I such that $|x_i^{(n_0)}| < \frac{\varepsilon}{2}$ for all $i > I$. Then, for all $i > I$:

$$|x_i| \leq |x_i - x_i^{(n_0)}| + |x_i^{(n_0)}| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

So $x_i \rightarrow 0$, that is, $x \in c_0$. Every limit of elements of c_0 stays in c_0 : it is closed. (The structure of the argument, “ $\varepsilon/2$ from the uniform approximation + $\varepsilon/2$ from the property of the approximating element”, is identical to the one in “a uniform limit of continuous functions is continuous”: recognizing this pattern is worth gold.) $\square$

Exercise 12.25. Let X be a Banach space and $(x_n) \subseteq X$ with $\|x_n\| \leq \frac{5}{3^n}$ for all n . Prove that $\sum_{n=1}^{\infty} x_n$ converges in X and bound the norm of its sum.

Solution. *Absolute convergence:* by comparison,

$$\sum_{n=1}^{\infty} \|x_n\| \leq \sum_{n=1}^{\infty} \frac{5}{3^n} = 5 \cdot \frac{1/3}{1 - 1/3} = \frac{5}{2} < \infty,$$

using the geometric series with ratio $\frac{1}{3}$ (starting at $n = 1$: $\sum_{n \geq 1} r^n = \frac{r}{1-r}$). The series is absolutely convergent; since X is Banach, the characterization theorem (“absolute $\Rightarrow$ convergent in complete spaces”) guarantees that $\sum x_n$ converges to some $s \in X$.

Bound on the sum: for each N , the triangle inequality gives $\left\| \sum_{n \leq N} x_n \right\| \leq \sum_{n \leq N} \|x_n\| \leq \frac{5}{2}$. Letting $N \rightarrow \infty$ and using the continuity of the norm ($s_N \rightarrow s$ implies $\|s_N\| \rightarrow \|s\|$):

$$\|s\| = \lim_N \|s_N\| \leq \frac{5}{2}. \quad \square$$

Exercise 12.26. Prove that the series of functions $\sum_{n=1}^{\infty} \frac{t^n}{n^2}$ converges uniformly on $[-1, 1]$ and that its sum is a continuous function on $[-1, 1]$.

Solution. We work in the Banach space $(C[-1, 1], \|\cdot\|_{\infty})$. Each $f_n(t) = \frac{t^n}{n^2}$ is continuous, and for $t \in [-1, 1]$:

$$|f_n(t)| = \frac{|t|^n}{n^2} \leq \frac{1}{n^2} =: M_n.$$

The numerical series $\sum M_n = \sum \frac{1}{n^2}$ converges (it is the p -series with $p = 2 > 1$; if you are asked to justify it: $\frac{1}{n^2} \leq \frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}$ for $n \geq 2$ and the telescoping series converges). Therefore

$$\sum_{n=1}^{\infty} \|f_n\|_{\infty} \leq \sum_{n=1}^{\infty} M_n < \infty:$$

the series is absolutely convergent in $C[-1, 1]$. Since $C[-1, 1]$ with the supremum norm is complete, the series converges in that norm, that is, it *converges uniformly* on $[-1, 1]$, and its sum (a uniform limit of the partial sums, which are continuous) is continuous. (This is the Weierstrass M-test with $M_n = 1/n^2$.) $\square$

Exercise 12.27. Let $(X, \|\cdot\|)$ be a normed space in which every absolutely convergent series is convergent. Prove that X is complete.

Solution. Let (y_n) be a Cauchy sequence in X ; we must produce its limit.

Step 1 (subsequence with geometric jumps). For each $k \in \mathbb{N}$, applying the Cauchy definition with $\varepsilon = 2^{-k}$ we get N_k such that $\|y_m - y_n\| < 2^{-k}$ if $m, n \geq N_k$. We choose indices $n_1 < n_2 < \dots$ with $n_k \geq N_k$ (for example, $n_k = \max(N_1, \dots, N_k) + k$ guarantees strict growth). Then, since $n_{k+1} > n_k \geq N_k$:

$$\|y_{n_{k+1}} - y_{n_k}\| < 2^{-k}.$$

Step 2 (absolutely convergent telescoping series). Define $x_1 = y_{n_1}$ and $x_{k+1} =$

$y_{n_{k+1}} - y_{n_k}$. Then

$$\sum_{j=1}^{\infty} \|x_j\| \leq \|y_{n_1}\| + \sum_{k=1}^{\infty} 2^{-k} = \|y_{n_1}\| + 1 < \infty,$$

so $\sum x_j$ is absolutely convergent and, by hypothesis, it converges to some $y \in X$. But its partial sums telescope: $\sum_{j=1}^K x_j = y_{n_K}$. So $y_{n_K} \rightarrow y$: the subsequence converges.

Step 3 (Cauchy + convergent subsequence $\Rightarrow$ the whole sequence converges).

Given $\varepsilon > 0$: by Cauchy there is N with $\|y_m - y_n\| < \varepsilon/2$ for $m, n \geq N$; by the convergence of the subsequence there is $n_k \geq N$ with $\|y_{n_k} - y\| < \varepsilon/2$. For all $n \geq N$:

$$\|y_n - y\| \leq \|y_n - y_{n_k}\| + \|y_{n_k} - y\| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Therefore $y_n \rightarrow y \in X$. Every Cauchy sequence converges: X is complete (Banach). $\square$

Chapter 13

Linear transformations and bounded operators [P3]

Week 11 of the course (“Linear transformations. Bounded linear operators. Continuity”). Bibliography: Alabiso and Weiss [1], Chs. 2 and 5; Moroşanu [2], Ch. 3.

What is this used for in Microeconomics?

Medium priority for Microeconomics, with one important exception: the section on linear functionals and the dual space. A price vector p is exactly a linear functional on the space of goods: the valuation $x \mapsto p \cdot x$ that turns bundles into expenditure. That price–quantity duality (quantities in X , prices in X^*) is the language in which Mas-Colell, Whinston and Green state the welfare theorems (Chs. 15 to 18), and the linearity of the valuation shows up again when you compute expected payments in mechanism design and auctions (Ch. 23). It is a relative of the *convex duality* of demand and production (expenditure, cost and profit functions, Chs. 3 and 5), where an object is also described by the prices that support it. The rest of the chapter (operator norm, bounded = continuous) is the background that supports those results.

13.1 Motivation [P3]

So far we have studied *spaces* (metric, normed) and the *continuous functions* between them. In linear algebra, the natural functions between vector spaces are those that respect the linear structure: the **linear transformations**. The central question of this chapter is:

When is a linear transformation between normed spaces continuous?

In finite dimension the answer is reassuring: *always*. But in infinite dimension (which is where functional analysis lives: spaces of functions, of sequences, of stochastic processes) *discontinuous* linear transformations appear, and telling the continuous ones from the discontinuous ones becomes the basic technical problem. The answer is one of the most important theorems of the course: for a linear transformation,

$$\text{continuous} \iff \text{bounded},$$

and “boundedness” is measured by a number, the **operator norm**, which turns the set of operators into a new normed space. This idea of “spaces whose points are functions or operators” is the essence of functional analysis.

Link with economics

Linear transformations are everywhere in economics:

- The **Leontief input-output model**: if $x \in \mathbb{R}^n$ is the vector of gross outputs and A the matrix of technical coefficients, the intermediate demand is Ax , a linear transformation of output. The question “which output supports a final demand d ?” is to solve $(I - A)x = d$, a problem about linear operators.
- The **conditional expectation** $Y \mapsto E[Y \mid \mathcal{G}]$ is a linear operator on the space of integrable random variables: $E[aY + bZ \mid \mathcal{G}] = aE[Y \mid \mathcal{G}] + bE[Z \mid \mathcal{G}]$. The whole theory of econometric projections (OLS, best predictors) rests on seeing it as a bounded operator.
- **Prices as linear functionals**: if $x \in \mathbb{R}^n$ is a bundle of goods and p the price vector, the expenditure $p \cdot x$ is linear in x . In models with infinitely many goods (goods dated in time or contingent on states), a price system is a *continuous linear functional* on the space of bundles; the quantity–price duality of general equilibrium theory is exactly the duality $X - X^*$ that we will define at the end of the chapter.

13.2 Linear transformations: definition and examples [P3]

Definition 13.1 (Linear transformation). Let X and Y be vector spaces over $\mathbb{R}$. A function $T : X \rightarrow Y$ is a **linear transformation**, also called a **linear operator**, if for all $x, z \in X$ and all $\lambda \in \mathbb{R}$:

1. $T(x + z) = T(x) + T(z)$ (additivity),
2. $T(\lambda x) = \lambda T(x)$ (homogeneity).

When $Y = \mathbb{R}$, we say that T is a **linear functional**. It is customary to write Tx instead of $T(x)$.

Intuition

“Linear” means that T *respects* the two operations of the vector space: adding first and transforming afterwards gives the same result as transforming first and adding afterwards. Geometrically, T sends lines through the origin to lines (or to the origin), and it sends the origin to the origin: taking $\lambda = 0$ in homogeneity, $T(0) = 0$ *always*. This last observation is a very useful quick check: if $T(0) \neq 0$, the function cannot be linear (this is why $f(x) = x + 1$ is *not* linear in the algebraic sense, even though its graph is a straight line: it is *affine*).

Example 13.2 (Matrices: the case $\mathbb{R}^n \rightarrow \mathbb{R}^m$). Every $m \times n$ matrix A defines $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ by $Tx = Ax$ (matrix-vector product). Linearity is the distributivity of the matrix product:

$A(x + z) = Ax + Az$ and $A(\lambda x) = \lambda Ax$. Conversely, every linear transformation $\mathbb{R}^n \rightarrow \mathbb{R}^m$ has this form: if $e_1, \dots, e_n$ is the canonical basis, the matrix whose columns are $Te_1, \dots, Te_n$ represents T , because writing $x = \sum_{j=1}^n x_j e_j$ and using linearity,

$$Tx = T\left(\sum_{j=1}^n x_j e_j\right) = \sum_{j=1}^n x_j Te_j.$$

In finite dimension, “linear operator” and “matrix” are the same thing.

Example 13.3 (Integral operator on $C[a, b]$). Let $X = C[a, b]$ be the space of continuous functions on $[a, b]$ with the supremum norm $\|f\|_\infty = \sup_{t \in [a, b]} |f(t)|$. Define $T : C[a, b] \rightarrow C[a, b]$ by

$$(Tf)(t) = \int_a^t f(s) ds.$$

T is linear because the integral is linear: $\int_a^t (f + g) = \int_a^t f + \int_a^t g$ and $\int_a^t \lambda f = \lambda \int_a^t f$. More generally, if $k : [a, b] \times [a, b] \rightarrow \mathbb{R}$ is continuous (the *kernel*),

$$(Tf)(t) = \int_a^b k(t, s) f(s) ds$$

defines a linear operator, called an **integral operator**. These operators are the continuous model of “multiplying by a matrix”: the kernel $k(t, s)$ plays the role of the entry a_{ij} , and the integral the role of the sum $\sum_j a_{ij} x_j$.

Example 13.4 (Derivative operator). Let $\mathcal{P}[0, 1]$ be the space of polynomials on $[0, 1]$ (a subspace of $C[0, 1]$), and $D : \mathcal{P}[0, 1] \rightarrow \mathcal{P}[0, 1]$ given by $Dp = p'$. It is linear: $(p + q)' = p' + q'$ and $(\lambda p)' = \lambda p'$. We will see that this operator, so natural, is the canonical example of a *discontinuous* linear operator: differentiating is an unstable operation (uniformly small functions can have huge derivatives).

Example 13.5 (Shift operators on ℓ^p). Let ℓ^p ($1 \leq p \leq \infty$) be the space of sequences $x = (x_1, x_2, \dots)$ with $\|x\|_p < \infty$. Define:

- the **right shift** $S_r(x_1, x_2, x_3, \dots) = (0, x_1, x_2, \dots)$,
- the **left shift** $S_\ell(x_1, x_2, x_3, \dots) = (x_2, x_3, x_4, \dots)$.

Both are linear (they act coordinate by coordinate). Note that S_r is injective but not surjective, and S_ℓ is surjective but not injective: in infinite dimension, injectivity and surjectivity of a linear operator are *not* equivalent (unlike square matrices). In economics and time series, S_ℓ is the *lag operator* applied to paths: it shifts time.

Example 13.6 (Evaluation functional). Fix $t_0 \in [a, b]$. The **evaluation functional** $\delta_{t_0} : C[a, b] \rightarrow \mathbb{R}$ given by $\delta_{t_0}(f) = f(t_0)$ is linear: $(f + g)(t_0) = f(t_0) + g(t_0)$. Evaluating at a point is the simplest linear operation on a space of functions.

13.2.1 Kernel and range

Definition 13.7 (Kernel and range). Let $T : X \rightarrow Y$ be linear. The **kernel** (or null space) of T is

$$\operatorname{Ker} T = \{x \in X : Tx = 0\},$$

and the **range** (or image) of T is

$$\operatorname{Im} T = \{Tx : x \in X\} = T(X) \subseteq Y.$$

Proposition 13.8. $\operatorname{Ker} T$ is a vector subspace of X , and $\operatorname{Im} T$ is a vector subspace of Y .

Proof. Recall the subspace criterion: a nonempty subset S of a vector space is a subspace if it is closed under sums and under scalar multiples.

Kernel. First, $\operatorname{Ker} T \neq \emptyset$ because $T(0) = 0$ (seen in the Intuition after Definition 13.1), so $0 \in \operatorname{Ker} T$. Let $x, z \in \operatorname{Ker} T$ and $\lambda \in \mathbb{R}$. Then, using the linearity of T :

$$T(x + z) = Tx + Tz = 0 + 0 = 0, \quad T(\lambda x) = \lambda Tx = \lambda \cdot 0 = 0.$$

Therefore $x + z \in \operatorname{Ker} T$ and $\lambda x \in \operatorname{Ker} T$: the kernel is a subspace.

Range. First, $0 = T(0) \in \operatorname{Im} T$, so it is not empty. Let $y_1, y_2 \in \operatorname{Im} T$: by the definition of range there are $x_1, x_2 \in X$ with $Tx_1 = y_1$ and $Tx_2 = y_2$. Then

$$y_1 + y_2 = Tx_1 + Tx_2 = T(x_1 + x_2) \in \operatorname{Im} T, \quad \lambda y_1 = \lambda Tx_1 = T(\lambda x_1) \in \operatorname{Im} T.$$

In each case we exhibited an element of X (namely $x_1 + x_2$ and λx_1) whose image is the desired vector; that is exactly what belonging to $\operatorname{Im} T$ means. $\square$

Remark 13.9. $\operatorname{Ker} T$ measures injectivity: T is injective if and only if $\operatorname{Ker} T = \{0\}$. Indeed, if T is injective and $Tx = 0 = T0$, then $x = 0$. Conversely, if $\operatorname{Ker} T = \{0\}$ and $Tx = Tz$, linearity gives $T(x - z) = 0$, so $x - z \in \operatorname{Ker} T = \{0\}$, that is, $x = z$. This trick (“move injectivity to the kernel by subtracting”) is used constantly.

13.3 Bounded operators [P4]

From here on, X and Y are *normed* spaces, with norms that we will both denote by $\|\cdot\|$ (the context tells you in which space each one is evaluated).

Definition 13.10 (Bounded operator). A linear transformation $T : X \rightarrow Y$ is **bounded** if there is a constant $M \geq 0$ such that

$$\|Tx\| \leq M \|x\| \quad \text{for all } x \in X.$$

Intuition

A bounded operator *cannot stretch too much*: the length of the output $\|Tx\|$ is controlled by a fixed multiple M of the length of the input $\|x\|$. The maximum stretching (the

“amplification factor” of the operator) is uniformly controlled in all directions. Think of M as a bound on the “leverage” of the operator: small inputs produce proportionally small outputs.

Careful (common mistake)

“Bounded” here does **not** mean that the range of T is a bounded set (that is the notion of bounded function from calculus, and it would be absurd for linear operators). In fact, if $T \neq 0$, its range is *never* a bounded set: if $Tx_0 \neq 0$, then $\|T(\lambda x_0)\| = |\lambda| \|Tx_0\| \rightarrow \infty$ as $\lambda \rightarrow \infty$. What is bounded is the *ratio* $\|Tx\| / \|x\|$, that is, the relative stretching. In an exam, confusing these two notions invalidates the problem: the correct definition is $\|Tx\| \leq M \|x\|$ for all x .

13.3.1 The fundamental theorem: continuous $\iff$ bounded

Theorem 13.11 (Characterization of continuity). *Let $T : X \rightarrow Y$ be a linear transformation between normed spaces. The following are equivalent:*

- (i) T is continuous at some point $x_0 \in X$;
- (ii) T is continuous at every point of X ;
- (iii) T is bounded.

Intuition

Linearity is such a rigid property that continuity at a *single point* spreads to the whole space. The reason: the behavior of T near any point is a *translated copy* of its behavior near 0, because $T(x_0 + h) - T(x_0) = Th$. So it is enough to understand what happens near the origin; and near the origin, “not blowing up” (continuity) is the same as “stretching by at most a fixed factor” (boundedness), because by homogeneity the behavior on a small sphere determines the behavior at all scales.

Proof. We will prove the cycle (ii) $\Rightarrow$ (i) $\Rightarrow$ (iii) $\Rightarrow$ (ii), which establishes the three equivalences.

(ii) $\Rightarrow$ (i). Trivial: if T is continuous at every point, in particular it is continuous at some point.

(i) $\Rightarrow$ (iii). This is the implication with content; let us go slowly. Suppose T is continuous at x_0 . By the ε - δ definition of continuity applied with $\varepsilon = 1$ (any fixed value works; we choose 1 for convenience): there is $\delta > 0$ such that

$$\|z - x_0\| \leq \delta \implies \|Tz - Tx_0\| \leq 1. \quad (*)$$

Step 1: move the information to the origin. Let $h \in X$ be any vector with $\|h\| \leq \delta$. Taking $z = x_0 + h$ we have $\|z - x_0\| = \|h\| \leq \delta$, and by (*) and linearity:

$$\|Th\| = \|T(x_0 + h) - T(x_0)\| = \|Tz - Tx_0\| \leq 1.$$

Intermediate conclusion: *on the ball of radius δ centered at 0, the images have norm at most 1.* Note how linearity erased every reference to x_0 : this is what the Intuition announced.

Step 2: rescale to an arbitrary x . Now let $x \in X$ with $x \neq 0$ (if $x = 0$, the inequality $\|T0\| = 0 \leq M \cdot 0$ is automatic). We want to apply Step 1, but x may be huge; the trick is to *normalize* it so that it fits inside the ball of radius δ . Define

$$h = \frac{\delta}{2\|x\|} x.$$

Let us check that this h qualifies for Step 1: its norm is

$$\|h\| = \frac{\delta}{2\|x\|} \|x\| = \frac{\delta}{2} \leq \delta,$$

(we use $\delta/2$ instead of δ to work with some slack; with δ it would also work with $\leq$). Then Step 1 gives $\|Th\| \leq 1$. But by homogeneity of T and of the norm:

$$\|Th\| = \left\| T\left(\frac{\delta}{2\|x\|}x\right) \right\| = \frac{\delta}{2\|x\|} \|Tx\|.$$

Combining: $\frac{\delta}{2\|x\|} \|Tx\| \leq 1$, and solving for $\|Tx\|$:

$$\|Tx\| \leq \frac{2}{\delta} \|x\|.$$

Since x was arbitrary, T is bounded with constant $M = 2/\delta$. Note the quantitative lesson: the smaller the δ that continuity needs, the larger the boundedness constant: they are two sides of the same number.

(iii) $\Rightarrow$ (ii). Suppose $\|Tx\| \leq M\|x\|$ for all x , with $M > 0$ (if $M = 0$ then $T = 0$, which is obviously continuous). Let $x_0 \in X$ be arbitrary and $\varepsilon > 0$. Choose $\delta = \varepsilon/M$. If $\|x - x_0\| < \delta$, then, using linearity and then boundedness applied to the vector $x - x_0$:

$$\|Tx - Tx_0\| = \|T(x - x_0)\| \leq M\|x - x_0\| < M\delta = M \cdot \frac{\varepsilon}{M} = \varepsilon.$$

This is exactly continuity at x_0 . Since x_0 was arbitrary, T is continuous on all of X . In fact we proved something stronger: the same δ works for every point, that is, T is *uniformly* continuous (even Lipschitz with constant M). $\square$

Exam strategy

This theorem is an absolute exam classic, in two formats:

1. “*Prove that linear + continuous at 0 implies bounded.*” The proof is Step 1 and Step 2 above; practice the normalization trick $h = \frac{\delta}{2\|x\|}x$ until you can reproduce it without looking. Where you lose points: forgetting to treat the case $x = 0$ separately (the normalization divides by $\|x\|$).
2. “*Decide whether the following operator is bounded and compute its norm.*” Two-

move strategy: (a) prove a bound $\|Tx\| \leq M \|x\|$ (this gives $\|T\| \leq M$); (b) exhibit a concrete x (or a sequence x_n) where the bound is attained or approached (this gives $\|T\| \geq M$). Without step (b) you only have one inequality.

13.4 The norm of an operator [P4]

Definition 13.12 (Operator norm). Let $T : X \rightarrow Y$ be linear and bounded. The **operator norm** of T is

$$\|T\| = \inf \{M \geq 0 : \|Tx\| \leq M \|x\| \text{ for all } x \in X\},$$

that is, the *best* (smallest) possible boundedness constant.

Proposition 13.13 (Characterizations of the operator norm). *Let $T : X \rightarrow Y$ be linear and bounded, with $X \neq \{0\}$. Then*

$$\|T\| = \sup_{\|x\| \leq 1} \|Tx\| = \sup_{\|x\|=1} \|Tx\| = \sup_{x \neq 0} \frac{\|Tx\|}{\|x\|},$$

and moreover the fundamental inequality holds:

$$\|Tx\| \leq \|T\| \|x\| \quad \text{for all } x \in X.$$

Proof. For short, write

$$a = \sup_{\|x\| \leq 1} \|Tx\|, \quad b = \sup_{\|x\|=1} \|Tx\|, \quad c = \sup_{x \neq 0} \frac{\|Tx\|}{\|x\|}.$$

We will prove $b \leq a$, then $a \leq c$, then $c \leq b$ (so that $a = b = c$), and finally that this common value is $\|T\|$.

$b \leq a$: the supremum over the sphere $\{\|x\| = 1\}$ is taken over a set contained in the ball $\{\|x\| \leq 1\}$; a supremum over a smaller set cannot exceed the supremum over the larger one.

$a \leq c$: let x with $\|x\| \leq 1$. If $x = 0$, then $\|Tx\| = 0 \leq c$. If $x \neq 0$, then

$$\|Tx\| = \frac{\|Tx\|}{\|x\|} \cdot \|x\| \leq \frac{\|Tx\|}{\|x\|} \cdot 1 \leq c,$$

where we used $\|x\| \leq 1$ in the first inequality and the definition of c as a supremum in the second. Taking the supremum over such x : $a \leq c$.

$c \leq b$: let $x \neq 0$ and set $u = x / \|x\|$, the *normalization* of x . Then $\|u\| = \|x\| / \|x\| = 1$, and by homogeneity of T and of the norm:

$$\frac{\|Tx\|}{\|x\|} = \left\| \frac{1}{\|x\|} Tx \right\| = \left\| T\left(\frac{x}{\|x\|}\right) \right\| = \|Tu\| \leq b.$$

Taking the supremum over $x \neq 0$: $c \leq b$. This proves $a = b = c$.

The common value is $\|T\|$. First: for every $x \neq 0$, the definition of c gives $\|Tx\| \leq c \|x\|$ (and trivially at $x = 0$); that is, c is one of the admissible constants M in Definition 13.12, so

$\|T\| \leq c$ (an infimum does not exceed any element of the set). Second: if M is any admissible constant, then for $x \neq 0$, $\frac{\|Tx\|}{\|x\|} \leq M$; taking the supremum, $c \leq M$. Since this holds for every admissible M , c is a lower bound of the set, so $c \leq \|T\|$ (the infimum is the *greatest* of the lower bounds). We conclude $c = \|T\|$.

The fundamental inequality. We have just seen that $\|T\| = c$ satisfies $\|Tx\| \leq c\|x\| = \|T\| \|x\|$ for all x : the infimum in the definition *is attained* as a valid constant. (This detail deserves attention: in general an infimum need not belong to the set, but here it does, thanks to the previous argument.) $\square$

Intuition

$\|T\|$ is the *maximum stretching factor* of T : look at all unit directions u (the sphere $\|u\| = 1$), apply T , and take the largest resulting size. For a matrix, it is “how much the matrix stretches in its worst direction”. The inequality $\|Tx\| \leq \|T\| \|x\|$ is used dozens of times per page in functional analysis: it is the number one tool for bounding.

Proposition 13.14 ($B(X, Y)$ is a normed space). *Let $B(X, Y)$ be the set of all bounded linear operators from X to Y (alternative notation: $L(X, Y)$; and $B(X) = B(X, X)$). With the pointwise operations $(T + S)x = Tx + Sx$, $(\lambda T)x = \lambda Tx$, the set $B(X, Y)$ is a vector space, and the operator norm is a norm on it. Moreover, the operator norm is **submultiplicative**: if $T \in B(X, Y)$ and $S \in B(Y, Z)$, then $ST \in B(X, Z)$ (where ST denotes the composition $S \circ T$) and*

$$\|ST\| \leq \|S\| \|T\|.$$

Proof. *Vector space:* if T, S are bounded and linear, $T + S$ is linear (direct computation) and bounded, since by the triangle inequality in Y :

$$\|(T + S)x\| = \|Tx + Sx\| \leq \|Tx\| + \|Sx\| \leq (\|T\| + \|S\|) \|x\|. \quad (\dagger)$$

Similarly $\|(\lambda T)x\| = |\lambda| \|Tx\| \leq |\lambda| \|T\| \|x\|$, so λT is bounded. The vector space axioms are inherited pointwise from Y .

Norm: we check the three axioms.

1. *Nonnegativity and separation:* $\|T\| \geq 0$ is clear (a supremum of nonnegative quantities). If $\|T\| = 0$, then $\|Tx\| \leq \|T\| \|x\| = 0$ for all x , so $Tx = 0$ for all x , that is, $T = 0$ (the zero operator). Conversely $\|0\| = 0$ is immediate.
2. *Homogeneity:* using the supremum characterization,

$$\|\lambda T\| = \sup_{\|x\| \leq 1} \|\lambda Tx\| = \sup_{\|x\| \leq 1} |\lambda| \|Tx\| = |\lambda| \sup_{\|x\| \leq 1} \|Tx\| = |\lambda| \|T\|,$$

where we pulled the constant $|\lambda| \geq 0$ out of the supremum (standard property of the supremum: $\sup(\alpha A) = \alpha \sup A$ for $\alpha \geq 0$).

3. *Triangle inequality:* line $(\dagger)$ says that $\|T\| + \|S\|$ is a valid boundedness constant for $T + S$; since $\|T + S\|$ is the *smallest* valid constant, $\|T + S\| \leq \|T\| + \|S\|$.

Submultiplicativity: ST is linear as a composition of linear maps. For any $x \in X$, we apply the fundamental inequality twice, first with S (to the vector $Tx \in Y$) and then with T :

$$\|(ST)x\| = \|S(Tx)\| \leq \|S\| \|Tx\| \leq \|S\| \|T\| \|x\|.$$

So $\|S\| \|T\|$ is a valid boundedness constant for ST , and by minimality $\|ST\| \leq \|S\| \|T\|$. $\square$

Remark 13.15. The inequality $\|ST\| \leq \|S\| \|T\|$ can be strict: for example, in $\mathbb{R}^2$ with S = the projection onto the x axis and T = the projection onto the y axis, we get $ST = 0$ (norm 0) but $\|S\| \|T\| = 1$.

13.5 Computing operator norms: worked examples [P4]

Example 13.16 (A 2×2 matrix with the maximum norm). Consider $\mathbb{R}^2$ with the norm $\|x\|_\infty = \max\{|x_1|, |x_2|\}$, and the operator $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by the matrix

$$A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}, \quad Tx = Ax.$$

We claim that $\|T\| = 4$, the *maximum of the row sums of absolute values*: row 1 gives $|1| + |2| = 3$, row 2 gives $|3| + |-1| = 4$.

Upper bound. For any x with $\|x\|_\infty \leq 1$ (that is, $|x_1| \leq 1$ and $|x_2| \leq 1$):

$$|(Ax)_1| = |x_1 + 2x_2| \leq |x_1| + 2|x_2| \leq 1 + 2 = 3,$$

$$|(Ax)_2| = |3x_1 - x_2| \leq 3|x_1| + |x_2| \leq 3 + 1 = 4.$$

So $\|Ax\|_\infty = \max\{|(Ax)_1|, |(Ax)_2|\} \leq 4$, and by the supremum characterization, $\|T\| \leq 4$.

The bound is attained. We need a unit vector x that activates the worst row (row 2) with all summands aligned in sign. Take $x^* = (1, -1)$: its coordinates have absolute value 1, so $\|x^*\|_\infty = 1$, and

$$Ax^* = \begin{pmatrix} 1 \cdot 1 + 2 \cdot (-1) \\ 3 \cdot 1 - (-1) \end{pmatrix} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}, \quad \|Ax^*\|_\infty = 4.$$

Therefore $\|T\| \geq 4$, and we conclude $\|T\| = 4$. The general pattern: with the ∞ norm on both input and output, the norm of a matrix is $\max_i \sum_j |a_{ij}|$, and the maximizer is built by choosing $x_j = \text{sign}(a_{i^*j})$ in the winning row i^* .

Example 13.17 (Integral operator with a kernel). Let $T : C[0, 1] \rightarrow C[0, 1]$, with the supremum norm on both sides, be given by

$$(Tf)(t) = \int_0^1 k(t, s)f(s) ds,$$

with k continuous. Then T is bounded, with

$$\|T\| \leq \max_{t \in [0, 1]} \int_0^1 |k(t, s)| ds.$$

Indeed, for $\|f\|_\infty \leq 1$ and each fixed t :

$$|(Tf)(t)| = \left| \int_0^1 k(t, s) f(s) ds \right| \leq \int_0^1 |k(t, s)| |f(s)| ds \leq \int_0^1 |k(t, s)| ds \cdot \|f\|_\infty,$$

where we first moved the absolute value inside the integral (triangle inequality for integrals) and then bounded $|f(s)| \leq \|f\|_\infty \leq 1$. Taking the supremum in t and then in f gives the bound. (In fact equality holds, but the inequality is enough for boundedness.)

Concrete case: $(Tf)(t) = \int_0^t f(s) ds$ corresponds to $k(t, s) = 1$ if $s \leq t$, and 0 otherwise; then $\int_0^1 |k(t, s)| ds = t \leq 1$ and $\|T\| \leq 1$. Taking $f \equiv 1$ (which satisfies $\|f\|_\infty = 1$) gives $(Tf)(t) = t$, with $\|Tf\|_\infty = 1$; so $\|T\| = 1$.

Example 13.18 (The evaluation functional). $\delta_{t_0} : (C[a, b], \|\cdot\|_\infty) \rightarrow \mathbb{R}$, $\delta_{t_0}(f) = f(t_0)$. For every f :

$$|\delta_{t_0}(f)| = |f(t_0)| \leq \sup_t |f(t)| = \|f\|_\infty,$$

so $\|\delta_{t_0}\| \leq 1$. With $f \equiv 1$: $|\delta_{t_0}(f)| = 1 = \|f\|_\infty$, so $\|\delta_{t_0}\| = 1$.

Example 13.19 (The shifts have norm 1). In ℓ^p with $1 \leq p < \infty$: for the right shift,

$$\|S_r x\|_p^p = 0^p + |x_1|^p + |x_2|^p + \cdots = \|x\|_p^p,$$

that is, S_r is an *isometry* ($\|S_r x\| = \|x\|$ always), and therefore $\|S_r\| = 1$. For the left shift:

$$\|S_\ell x\|_p^p = |x_2|^p + |x_3|^p + \cdots \leq \|x\|_p^p,$$

so $\|S_\ell\| \leq 1$; and with $x = (0, 1, 0, 0, \dots)$ we get $S_\ell x = (1, 0, 0, \dots)$ and $\|S_\ell x\|_p = 1 = \|x\|_p$, so $\|S_\ell\| = 1$. Note that S_ℓ has norm 1 but it *crushes* the first coordinate: the operator norm is the worst case, not every case.

13.6 Finite dimension: every linear operator is bounded [P3]

Theorem 13.20. *If X is a finite dimensional normed space and Y is any normed space, then every linear transformation $T : X \rightarrow Y$ is bounded (and therefore continuous).*

Proof. Let $\{e_1, \dots, e_n\}$ be a basis of X . Every $x \in X$ is written uniquely as $x = \sum_{j=1}^n x_j e_j$ with coordinates $x_j \in \mathbb{R}$. The strategy: bound $\|Tx\|$ by the coordinates, and the coordinates by $\|x\|$ using the equivalence of norms.

Step 1: bound by the coordinates. By linearity and the triangle inequality:

$$\|Tx\| = \left\| \sum_{j=1}^n x_j T e_j \right\| \leq \sum_{j=1}^n |x_j| \|T e_j\| \leq \left(\max_{1 \leq j \leq n} \|T e_j\| \right) \sum_{j=1}^n |x_j| = C_1 \|x\|_1,$$

where $C_1 = \max_j \|T e_j\|$ (a fixed number: there are only n basis vectors) and $\|x\|_1 = \sum_j |x_j|$ is the “coordinate one norm” with respect to the basis.

Step 2: compare $\|\cdot\|_1$ with the norm of X . Here we use the theorem on the **equivalence of norms in finite dimension** (seen in the chapter on normed spaces): in a finite dimensional

vector space, all norms are equivalent; in particular, there is $C_2 > 0$ such that

$$\|x\|_1 \leq C_2 \|x\| \quad \text{for all } x \in X.$$

(A reminder of why: on the sphere $\{x : \|x\|_1 = 1\}$, which is compact in the coordinate topology by Heine–Borel in $\mathbb{R}^n$, the continuous function $x \mapsto \|x\|$ attains a minimum m , and $m > 0$ because the zero vector is not on the sphere; by homogeneity, $\|x\| \geq m \|x\|_1$ for all $x \in X$, which is the desired inequality with $C_2 = 1/m$.)

Conclusion. Chaining the two steps:

$$\|Tx\| \leq C_1 \|x\|_1 \leq C_1 C_2 \|x\|,$$

so T is bounded with $M = C_1 C_2$. By Theorem 13.11, T is continuous. $\square$

Remark 13.21. The theorem explains why continuity is never discussed in linear algebra courses: in $\mathbb{R}^n$ it is free. All the subtlety of this chapter is genuinely *infinite dimensional*.

13.6.1 The essential counterexample: differentiation is not bounded

Example 13.22 (The derivative operator is not bounded). Let $X = \mathcal{P}[0, 1]$ be the space of polynomials with the supremum norm $\|p\|_\infty = \sup_{t \in [0, 1]} |p(t)|$, and $D : X \rightarrow X$, $Dp = p'$. We claim that D is **not** bounded.

Consider the sequence of monomials $p_n(t) = t^n$, $n \in \mathbb{N}$. Their norm:

$$\|p_n\|_\infty = \sup_{t \in [0, 1]} |t^n| = 1 \quad (\text{the supremum is attained at } t = 1).$$

Their derivative is $Dp_n(t) = nt^{n-1}$, with norm

$$\|Dp_n\|_\infty = \sup_{t \in [0, 1]} |nt^{n-1}| = n \quad (\text{again at } t = 1).$$

If there were M with $\|Dp\| \leq M \|p\|$ for every polynomial p , we would have in particular

$$n = \|Dp_n\|_\infty \leq M \|p_n\|_\infty = M \quad \text{for all } n,$$

which is absurd: no real number M exceeds every natural number. Therefore D is not bounded and, by Theorem 13.11, it is *not continuous* (at any point).

Reading of the phenomenon: the polynomials $q_n = p_n/n$ satisfy $\|q_n\|_\infty = 1/n \rightarrow 0$ (they tend to zero uniformly), but $\|Dq_n\|_\infty = 1$ for all n : inputs that become arbitrarily small whose derivatives do not become small. Differentiation amplifies fast oscillations without limit. This is the deep reason why differential equations are analytically delicate and why in econometrics “differencing the data” amplifies noise.

Careful (common mistake)

Theorem 13.20 and Example 13.22 do not contradict each other: $\mathcal{P}[0, 1]$ has *infinite* dimension (the family $\{1, t, t^2, \dots\}$ is linearly independent). In an exam, if you are asked for an example of a discontinuous linear operator, you *must* go to infinite dimension; looking for one in $\mathbb{R}^n$ is wasted time.

13.7 Linear functionals and the dual space [P2]

Definition 13.23 (Dual space). Let X be a normed space. The **dual space** of X is

$$X^* = B(X, \mathbb{R}) = \{f : X \rightarrow \mathbb{R} \mid f \text{ linear and bounded}\},$$

with the operator norm $\|f\| = \sup_{\|x\| \leq 1} |f(x)|$. Its elements are called **continuous (or bounded) linear functionals**.

Example 13.24 (The integral as a functional on $C[a, b]$). Let $\varphi : C[a, b] \rightarrow \mathbb{R}$ be given by $\varphi(f) = \int_a^b f(t) dt$, with the supremum norm on $C[a, b]$. It is linear (linearity of the integral). Let us compute its exact norm.

Upper bound:

$$|\varphi(f)| = \left| \int_a^b f(t) dt \right| \leq \int_a^b |f(t)| dt \leq \int_a^b \|f\|_\infty dt = (b - a) \|f\|_\infty,$$

so $\|\varphi\| \leq b - a$.

It is attained: with $f \equiv 1$, $\|f\|_\infty = 1$ and $\varphi(f) = b - a$. Therefore $\|\varphi\| = b - a$.

Example 13.25 (Functionals on $\mathbb{R}^n$). Every linear functional on $\mathbb{R}^n$ has the form $f(x) = \sum_{j=1}^n a_j x_j = a \cdot x$ for a unique vector $a \in \mathbb{R}^n$ (take $a_j = f(e_j)$ and use linearity). That is, $(\mathbb{R}^n)^*$ “is” $\mathbb{R}^n$: each functional is identified with a vector of coefficients. In economics: every price system p is the expenditure functional $x \mapsto p \cdot x$. The infinite dimensional version of this identification (for Hilbert spaces) is the Riesz representation theorem, the topic of week 14.

Remark 13.26 (The dual is always complete). A remarkable fact that we will prove in the chapter on Banach spaces: $B(X, Y)$ is complete whenever Y is. Since $\mathbb{R}$ is complete, it follows that *the dual X^* of any normed space is a Banach space*, even if X itself is not complete. This is why economic theorists so often work “in the dual” (with prices instead of quantities): it is analytically firmer ground.

Exam strategy

Typical questions on this chapter and how to attack them:

- “Prove that T is linear.” Check the two properties in full detail; do not forget that they must hold for *all* scalars, including negative ones and zero.
- “Prove that T is bounded and find $\|T\|$.” Two-step ritual: bound + witness (an element or a sequence that attains it). If the supremum is not attained, use a

sequence x_n with $\|x_n\| = 1$ and $\|Tx_n\| \rightarrow M$.

- “Give a discontinuous linear operator.” The derivative operator on polynomials with the sup norm (Example 13.22); memorize the two lines of computation with t^n .
- “Is $\text{Ker } T$ closed?” If T is continuous, yes: $\text{Ker } T = T^{-1}(\{0\})$ is the preimage of a closed set under a continuous function. If T is not continuous, it need not be. This topology–algebra link is a frequent question.

Chapter summary

- T linear: $T(x+z) = Tx + Tz$, $T(\lambda x) = \lambda Tx$; always $T(0) = 0$. $\text{Ker } T$ and $\text{Im } T$ are subspaces; T is injective $\iff \text{Ker } T = \{0\}$.
- T bounded: $\exists M, \|Tx\| \leq M \|x\| \ \forall x$. **Central theorem:** linear $\Rightarrow$ [continuous at one point $\iff$ continuous at every point $\iff$ bounded].
- Operator norm: $\|T\| = \sup_{\|x\| \leq 1} \|Tx\| = \sup_{\|x\|=1} \|Tx\| = \sup_{x \neq 0} \|Tx\| / \|x\|$; key inequalities $\|Tx\| \leq \|T\| \|x\|$ and $\|ST\| \leq \|S\| \|T\|$.
- Finite dimension $\Rightarrow$ every linear operator is bounded. Infinite dimension: the derivative operator (with t^n) is linear and not bounded.
- $B(X, Y)$ is a normed space; $X^* = B(X, \mathbb{R})$ is the dual, and it is always Banach. Prices = continuous linear functionals.

13.8 Exam-style exercises [P3]

Exercise 13.27. Let $T : (C[0, 1], \|\cdot\|_\infty) \rightarrow \mathbb{R}$ be defined by

$$T(f) = \int_0^1 t f(t) dt.$$

Prove that T is linear and bounded, and compute $\|T\|$.

Solution. *Linearity.* For $f, g \in C[0, 1]$ and $\lambda \in \mathbb{R}$:

$$T(f+g) = \int_0^1 t(f(t) + g(t)) dt = \int_0^1 t f(t) dt + \int_0^1 t g(t) dt = T(f) + T(g),$$

$$T(\lambda f) = \int_0^1 t \lambda f(t) dt = \lambda \int_0^1 t f(t) dt = \lambda T(f),$$

using the linearity of the integral.

Boundedness (upper bound). For any f :

$$|T(f)| = \left| \int_0^1 t f(t) dt \right| \leq \int_0^1 |t| |f(t)| dt \leq \|f\|_\infty \int_0^1 t dt = \frac{1}{2} \|f\|_\infty.$$

(First the triangle inequality for integrals; then $|f(t)| \leq \|f\|_\infty$ and $|t| = t$ on $[0, 1]$; finally

$\int_0^1 t \, dt = 1/2$.) Therefore T is bounded and $\|T\| \leq 1/2$.

The bound is attained. We look for a unit f that makes the integrand as large as possible with constant sign: since the weight t is nonnegative on $[0, 1]$, take $f \equiv 1$. Then $\|f\|_\infty = 1$ and

$$T(f) = \int_0^1 t \, dt = \frac{1}{2},$$

so $\|T\| \geq |T(f)| / \|f\|_\infty = 1/2$.

Conclusion: $\|T\| = \frac{1}{2}$. □

Exercise 13.28. Let $T : \ell^\infty \rightarrow \ell^\infty$ be defined by

$$T(x_1, x_2, x_3, \dots) = \left(x_1, \frac{x_2}{2}, \frac{x_3}{3}, \dots\right).$$

Prove that T is linear and bounded, compute $\|T\|$, and decide whether there is $x \neq 0$ such that $\|Tx\|_\infty = \|T\| \|x\|_\infty$.

Solution. *Linearity.* Coordinate by coordinate: the n -th coordinate of $T(x+y)$ is $(x_n+y_n)/n = x_n/n + y_n/n$, which is the sum of the n -th coordinates of Tx and Ty ; similarly with scalars.

Boundedness. For every $x \in \ell^\infty$ and every n :

$$\left|\frac{x_n}{n}\right| = \frac{|x_n|}{n} \leq |x_n| \leq \|x\|_\infty,$$

because $n \geq 1$. Taking the supremum over n : $\|Tx\|_\infty \leq \|x\|_\infty$, so $\|T\| \leq 1$.

Norm. Take $x = e_1 = (1, 0, 0, \dots)$: $\|e_1\|_\infty = 1$ and $Te_1 = (1, 0, 0, \dots)$, with $\|Te_1\|_\infty = 1$. So $\|T\| \geq 1$ and therefore $\|T\| = 1$.

Is it attained? Yes: $x = e_1$ itself satisfies $\|Te_1\|_\infty = 1 = \|T\| \|e_1\|_\infty$. (Any x whose supremum “lives” in the first coordinate works; on the other hand, if you restrict the operator to the coordinates $n \geq 2$, the norm would be $1/2$ and it would also be attained. Compare with the shift: a maximizer does not always exist, here it does.) □

Exercise 13.29. Let X be a normed space and $f : X \rightarrow \mathbb{R}$ a linear functional that is *not identically zero*. Prove:

- (a) If f is continuous, then $\text{Ker } f$ is a proper closed subspace of X .
- (b) If f is not continuous, then for every $x \in X$ and every $\varepsilon > 0$ there is $z \in \text{Ker } f$ with $\|x - z\| < \varepsilon$ (that is, $\text{Ker } f$ is *dense* in X).

Solution. (a) $\text{Ker } f$ is a subspace by Proposition 13.8. It is proper because $f \neq 0$ implies that there is x_1 with $f(x_1) \neq 0$, that is, $x_1 \notin \text{Ker } f$. For closedness, we use the sequential criterion: let $(z_n) \subset \text{Ker } f$ with $z_n \rightarrow z$ in X ; we must show that $z \in \text{Ker } f$. Since f is continuous, $f(z_n) \rightarrow f(z)$. But $f(z_n) = 0$ for all n (they are in the kernel), and the limit of the constant sequence 0 is 0; by uniqueness of the limit in $\mathbb{R}$, $f(z) = 0$, that is, $z \in \text{Ker } f$. (One line alternative: $\text{Ker } f = f^{-1}(\{0\})$ is the preimage of the closed set $\{0\}$ under the continuous function f .)

(b) Since f is not continuous, by Theorem 13.11 it is not bounded: for each $n \in \mathbb{N}$ there is $u_n \in X$ with

$$|f(u_n)| > n \|u_n\|,$$

and necessarily $u_n \neq 0$ (if $u_n = 0$ both sides would be 0). Normalizing $v_n = u_n / \|u_n\|$ we have $\|v_n\| = 1$ and $|f(v_n)| > n \rightarrow \infty$.

Now let $x \in X$ be arbitrary. The idea: correct x by subtracting a suitable multiple of v_n to kill the value of f . Define

$$z_n = x - \frac{f(x)}{f(v_n)} v_n.$$

(The quotient makes sense because $f(v_n) \neq 0$ for large n , since $|f(v_n)| > n \geq 1$.) We check that $z_n \in \text{Ker } f$ using linearity:

$$f(z_n) = f(x) - \frac{f(x)}{f(v_n)} f(v_n) = f(x) - f(x) = 0.$$

And the distance to x :

$$\|x - z_n\| = \left\| \frac{f(x)}{f(v_n)} v_n \right\| = \frac{|f(x)|}{|f(v_n)|} \|v_n\| = \frac{|f(x)|}{|f(v_n)|} < \frac{|f(x)|}{n} \xrightarrow{n \rightarrow \infty} 0.$$

Given $\varepsilon > 0$, it is enough to take $n > |f(x)|/\varepsilon$ and $z = z_n$. *Lesson to remember:* the kernel of a linear functional is either closed (continuous case) or dense (discontinuous case); never anything in between. It is an elegant dichotomy and a frequent exam question. $\square$

Exercise 13.30. On $\mathbb{R}^2$ consider the norms $\|x\|_1 = |x_1| + |x_2|$ and $\|x\|_\infty = \max\{|x_1|, |x_2|\}$. Let $T : (\mathbb{R}^2, \|\cdot\|_1) \rightarrow (\mathbb{R}^2, \|\cdot\|_\infty)$ be the identity $Tx = x$. Compute $\|T\|$ and the norm of the identity in the reverse direction, $S : (\mathbb{R}^2, \|\cdot\|_\infty) \rightarrow (\mathbb{R}^2, \|\cdot\|_1)$, $Sx = x$. How does this relate to the equivalence of norms?

Solution. *Norm of T .* We must find $\sup\{\|x\|_\infty : \|x\|_1 \leq 1\}$. For every x :

$$\|x\|_\infty = \max\{|x_1|, |x_2|\} \leq |x_1| + |x_2| = \|x\|_1,$$

(the maximum of two nonnegative numbers does not exceed their sum), so $\|T\| \leq 1$. With $x = (1, 0)$: $\|x\|_1 = 1$ and $\|x\|_\infty = 1$, so $\|T\| = 1$.

Norm of S . Now we look for $\sup\{\|x\|_1 : \|x\|_\infty \leq 1\}$. For every x :

$$\|x\|_1 = |x_1| + |x_2| \leq \|x\|_\infty + \|x\|_\infty = 2\|x\|_\infty,$$

(each summand is dominated by the maximum), so $\|S\| \leq 2$. With $x = (1, 1)$: $\|x\|_\infty = 1$ and $\|x\|_1 = 2$, so $\|S\| = 2$.

Relation with the equivalence of norms. We have proved exactly the inequalities

$$\|x\|_\infty \leq \|x\|_1 \leq 2\|x\|_\infty \quad (x \in \mathbb{R}^2),$$

which are the *definition* of the two norms being equivalent, with the *best possible constants* (1 and 2, attained by the witnesses we exhibited). In general: the optimal equivalence constants

between two norms are the operator norms of the two crossed identities. Saying “the norms are equivalent” is saying “both identities are bounded operators”. $\square$

Exercise 13.31. Let $T : (C[0, 1], \|\cdot\|_1) \rightarrow \mathbb{R}$ be the evaluation functional $T(f) = f(0)$, where $\|f\|_1 = \int_0^1 |f(t)| dt$. Prove that T is linear but **not** bounded. (Hint: build “spike” functions at $t = 0$ with small integral.)

Solution. *Linearity:* $(f + g)(0) = f(0) + g(0)$ and $(\lambda f)(0) = \lambda f(0)$; immediate.

Not bounded. We must exhibit functions where $|f(0)|$ is huge compared with $\int_0^1 |f|$. We build triangular spikes: for each $n \in \mathbb{N}$, let

$$f_n(t) = \begin{cases} n(1 - nt) & \text{if } 0 \leq t \leq \frac{1}{n}, \\ 0 & \text{if } \frac{1}{n} < t \leq 1. \end{cases}$$

Each f_n is continuous (at $t = 1/n$ both pieces equal 0; direct check), with $f_n(0) = n$.

We compute its $\|\cdot\|_1$ norm. Since $f_n \geq 0$:

$$\|f_n\|_1 = \int_0^{1/n} n(1 - nt) dt = n \left[t - \frac{nt^2}{2} \right]_0^{1/n} = n \left(\frac{1}{n} - \frac{n}{2} \cdot \frac{1}{n^2} \right) = n \cdot \frac{1}{2n} = \frac{1}{2}.$$

(The area of a triangle with base $1/n$ and height n : $\frac{1}{2} \cdot \frac{1}{n} \cdot n = \frac{1}{2}$, consistent.)

If T were bounded with constant M , we would have for every n :

$$n = |T(f_n)| = |f_n(0)| \leq M \|f_n\|_1 = \frac{M}{2},$$

impossible since $n \rightarrow \infty$. Therefore T is not bounded.

Comment: compare with Example 13.18: the *same* formula $f \mapsto f(0)$ is a functional of norm 1 with $\|\cdot\|_\infty$, and unbounded with $\|\cdot\|_1$. The boundedness of an operator depends on the *norms* you choose, not only on the formula. In an exam, before deciding whether an operator is bounded, identify precisely the norms on the domain and on the codomain. $\square$

Chapter 14

Banach spaces [P3]

Week 12 of the course (“Banach spaces”). Bibliography: Moroşanu [2], Chs. 2–3; Alabiso and Weiss [1], Ch. 5.

What is this used for in Microeconomics?

From this chapter, what comes back strongly in Microeconomics is a single thing, but it is a gem: the Hahn–Banach theorem in its geometric version (separating hyperplane for convex sets), presented in the overview section. The Second Welfare Theorem (Mas-Colell, Whinston and Green, Ch. 16) is literally an application of separation: if preferences and technologies are convex, every Pareto optimum can be decentralized with a price system. The same geometry describes production sets through the prices that support them (Ch. 5) and supports the separation arguments that appear in social choice and welfare economics (Chs. 21 and 22). The completeness proofs and the Neumann series (Leontief) are general background: they matter more in macroeconomics and numerical methods than in microeconomics.

14.1 Motivation [P3]

A normed space gives us geometry (distances, sizes); but to do *analysis* (take limits, sum series, solve equations by successive approximations) we need the limits that “should exist” to *really exist*. That is **completeness**: every Cauchy sequence converges *inside* the space. A complete normed space is called a **Banach space**, in honor of Stefan Banach, and it is the natural habitat of functional analysis.

Why do we insist so much on completeness? Because almost all the existence theorems of analysis (Banach fixed point, convergent series, inversion of operators, existence of solutions of differential and integral equations) have the same architecture:

build a sequence of approximations, prove that it is Cauchy, and invoke completeness to get the limit, which is the object you were looking for.

Without completeness, the sequence of approximations can “point at a hole” of the space and the method dies. This chapter catalogues the classical Banach spaces (with full completeness

proofs, which are standard exam exercises), shows important counterexamples, and proves two structural results: $B(X, Y)$ is Banach if Y is, and the Neumann series for inverting operators.

Intuition

Think of $\mathbb{Q}$ versus $\mathbb{R}$: in $\mathbb{Q}$, the sequence 1, 1.4, 1.41, 1.414, ... is Cauchy but its “natural” limit $\sqrt{2}$ is not in the space: $\mathbb{Q}$ has holes. $\mathbb{R}$ is $\mathbb{Q}$ “with the holes filled in”. A Banach space is a normed space without holes. The question “is this space of functions complete?” is always answered in the same way: take an *arbitrary* Cauchy sequence, build a candidate limit (usually pointwise or coordinate by coordinate), and check two things: that the candidate *belongs to the space* and that the convergence is *in the norm of the space*. The three steps (candidate, membership, convergence) are the skeleton of all the proofs in this chapter.

Definition 14.1 (Banach space). A normed space $(X, \|\cdot\|)$ is a **Banach space** if it is complete with respect to the induced metric $d(x, y) = \|x - y\|$; that is, if every Cauchy sequence in X converges to an element of X .

Recall: (x_n) is **Cauchy** if for every $\varepsilon > 0$ there is N such that $\|x_n - x_m\| < \varepsilon$ for all $n, m \geq N$; and every convergent sequence is Cauchy (the converse is precisely completeness).

14.2 The classical catalogue, with proofs [P5]

14.2.1 $\mathbb{R}^n$ is Banach

Theorem 14.2. $\mathbb{R}^n$, with any of its norms (all equivalent), is a Banach space.

Proof. We work with the norm $\|\cdot\|_\infty$ (by equivalence of norms, the Cauchy property and convergence do not depend on the norm chosen in $\mathbb{R}^n$). Let $(x^{(k)})_{k \in \mathbb{N}}$ be Cauchy in $\mathbb{R}^n$; write $x^{(k)} = (x_1^{(k)}, \dots, x_n^{(k)})$.

Step 1 (the coordinates are Cauchy). For each fixed index $j \in \{1, \dots, n\}$:

$$\left| x_j^{(k)} - x_j^{(m)} \right| \leq \max_{1 \leq i \leq n} \left| x_i^{(k)} - x_i^{(m)} \right| = \|x^{(k)} - x^{(m)}\|_\infty,$$

so the numerical sequence $(x_j^{(k)})_k$ inherits the Cauchy property from the vector sequence.

Step 2 (candidate). By the completeness of $\mathbb{R}$ (the fundamental property of the reals, seen in the chapter on completeness), each coordinate converges: there is $x_j = \lim_k x_j^{(k)}$. Candidate: $x = (x_1, \dots, x_n) \in \mathbb{R}^n$; membership is automatic.

Step 3 (convergence in norm). Given $\varepsilon > 0$, for each j there is K_j with $|x_j^{(k)} - x_j| < \varepsilon$ if $k \geq K_j$. Taking $K = \max \{K_1, \dots, K_n\}$ (possible because there are *finitely* many indices), for $k \geq K$:

$$\|x^{(k)} - x\|_\infty = \max_j |x_j^{(k)} - x_j| < \varepsilon.$$

So $x^{(k)} \rightarrow x$ in $\mathbb{R}^n$. □

Remark 14.3. The delicate point of the argument is taking the *maximum of finitely many* K_j . In sequence spaces (infinitely many coordinates) that trick fails and you have to work harder: see the proofs for ℓ^∞ and ℓ^1 .

14.2.2 $C[a, b]$ with the supremum norm is Banach

Theorem 14.4. *The space $C[a, b]$ of continuous functions $f : [a, b] \rightarrow \mathbb{R}$, with the norm $\|f\|_\infty = \sup_{t \in [a, b]} |f(t)|$, is a Banach space.*

Proof. Let (f_n) be Cauchy in $(C[a, b], \|\cdot\|_\infty)$. We follow the skeleton: candidate, membership, convergence.

Step 1 (pointwise candidate). Fix $t \in [a, b]$. Then

$$|f_n(t) - f_m(t)| \leq \sup_{s \in [a, b]} |f_n(s) - f_m(s)| = \|f_n - f_m\|_\infty,$$

so $(f_n(t))_n$ is a Cauchy sequence of real numbers. By the completeness of $\mathbb{R}$, it converges; we define

$$f(t) = \lim_{n \rightarrow \infty} f_n(t) \quad (t \in [a, b]).$$

This defines a function $f : [a, b] \rightarrow \mathbb{R}$, the *pointwise* limit.

Step 2 (the convergence is uniform). This is the heart of the proof: we need $f_n \rightarrow f$ in the supremum norm, not only point by point. Let $\varepsilon > 0$. By Cauchy, there is N such that

$$\|f_n - f_m\|_\infty < \frac{\varepsilon}{2} \quad \text{for all } n, m \geq N.$$

Fix $n \geq N$ and an arbitrary $t \in [a, b]$. For all $m \geq N$:

$$|f_n(t) - f_m(t)| \leq \|f_n - f_m\|_\infty < \frac{\varepsilon}{2}.$$

Now we let $m \rightarrow \infty$ with n and t fixed: since $f_m(t) \rightarrow f(t)$, and the absolute value is continuous, the inequality passes to the limit (with $\leq$):

$$|f_n(t) - f(t)| \leq \frac{\varepsilon}{2} < \varepsilon.$$

Since this holds for *every* $t \in [a, b]$ with the *same* N (independent of t : this is the key to uniformity), taking the supremum in t :

$$\|f_n - f\|_\infty \leq \frac{\varepsilon}{2} < \varepsilon \quad \text{for all } n \geq N.$$

That is, $f_n \rightarrow f$ uniformly.

Step 3 (membership: f is continuous). We invoke the theorem from the chapter on convergence of functions: *the uniform limit of continuous functions is continuous*. (A reminder of the $\varepsilon/3$ argument: for t close to t_0 ,

$$|f(t) - f(t_0)| \leq |f(t) - f_n(t)| + |f_n(t) - f_n(t_0)| + |f_n(t_0) - f(t_0)|,$$

the first and third terms are small by uniform convergence, the middle one by continuity of f_n .) Therefore $f \in C[a, b]$, and Step 2 gives $f_n \rightarrow f$ in the norm of the space. We conclude that $C[a, b]$ is complete. $\square$

Careful (common mistake)

The order of the steps matters: *pointwise* convergence alone would not be enough, because the pointwise limit of continuous functions can be discontinuous (the classical example $f_n(t) = t^n$ on $[0, 1]$). It is *uniformity*, inherited from the fact that the sequence is Cauchy in the supremum norm, that saves the continuity of the limit. In the exam, do not forget the step “let $m \rightarrow \infty$ with n fixed”: it is the technical move that turns “uniformly Cauchy” into “uniformly convergent”.

14.2.3 ℓ^∞ is Banach

Recall: ℓ^∞ is the space of bounded sequences $x = (x_1, x_2, \dots)$ with $\|x\|_\infty = \sup_j |x_j|$.

Theorem 14.5. $(\ell^\infty, \|\cdot\|_\infty)$ is a Banach space.

Proof. Let $(x^{(n)})_n$ be Cauchy in ℓ^∞ , with $x^{(n)} = (x_1^{(n)}, x_2^{(n)}, \dots)$.

Step 1 (coordinate by coordinate candidate). For each fixed j , $|x_j^{(n)} - x_j^{(m)}| \leq \|x^{(n)} - x^{(m)}\|_\infty$, so $(x_j^{(n)})_n$ is Cauchy in $\mathbb{R}$ and converges to a limit x_j . Candidate: $x = (x_1, x_2, \dots)$.

Step 2 (estimate uniform in j). Let $\varepsilon > 0$ and N such that $\|x^{(n)} - x^{(m)}\|_\infty < \varepsilon/2$ for $n, m \geq N$. Fix $n \geq N$ and an arbitrary j . For all $m \geq N$:

$$|x_j^{(n)} - x_j^{(m)}| < \frac{\varepsilon}{2}.$$

Letting $m \rightarrow \infty$ (with n, j fixed) and using $x_j^{(m)} \rightarrow x_j$:

$$|x_j^{(n)} - x_j| \leq \frac{\varepsilon}{2} \quad \text{for every } j \text{ and every } n \geq N. \quad (*)$$

Uniformity in j is crucial and it comes for free from the same argument as in $C[a, b]$: the N does not depend on j .

Step 3 (membership). Is x in ℓ^∞ , that is, is it bounded? Yes: fixing $n = N$ in $(*)$ and using the triangle inequality in $\mathbb{R}$:

$$|x_j| \leq |x_j - x_j^{(N)}| + |x_j^{(N)}| \leq \frac{\varepsilon}{2} + \|x^{(N)}\|_\infty \quad \text{for every } j,$$

and the right hand side is a finite constant (independent of j), because $x^{(N)} \in \ell^\infty$. So x is bounded: $x \in \ell^\infty$.

Step 4 (convergence in norm). Taking the supremum over j in $(*)$:

$$\|x^{(n)} - x\|_\infty = \sup_j |x_j^{(n)} - x_j| \leq \frac{\varepsilon}{2} < \varepsilon \quad (n \geq N),$$

that is, $x^{(n)} \rightarrow x$ in ℓ^∞ . The four steps complete the proof. $\square$

14.2.4 ℓ^1 is Banach

Recall: ℓ^1 is the space of absolutely summable sequences, with $\|x\|_1 = \sum_{j=1}^{\infty} |x_j| < \infty$.

Theorem 14.6. $(\ell^1, \|\cdot\|_1)$ is a Banach space.

Proof. Let $(x^{(n)})$ be Cauchy in ℓ^1 .

Step 1 (candidate). For each j , $|x_j^{(n)} - x_j^{(m)}| \leq \sum_i |x_i^{(n)} - x_i^{(m)}| = \|x^{(n)} - x^{(m)}\|_1$ (one summand does not exceed the sum of nonnegative terms), so each coordinate is Cauchy in $\mathbb{R}$ and converges: $x_j = \lim_n x_j^{(n)}$. Candidate $x = (x_j)_j$.

Step 2 (estimate with partial sums). Let $\varepsilon > 0$ and N with $\|x^{(n)} - x^{(m)}\|_1 < \varepsilon/2$ for $n, m \geq N$. The new difficulty: the norm is a *series*, and we cannot pass to the limit inside an infinite series without justification. The standard trick: work with *finite* partial sums. Fix $n \geq N$ and any integer J . For $m \geq N$:

$$\sum_{j=1}^J |x_j^{(n)} - x_j^{(m)}| \leq \|x^{(n)} - x^{(m)}\|_1 < \frac{\varepsilon}{2}.$$

The left hand side is a *finite sum*, so we can let $m \rightarrow \infty$ term by term (the limit of a finite sum is the sum of the limits):

$$\sum_{j=1}^J |x_j^{(n)} - x_j| \leq \frac{\varepsilon}{2}.$$

Since this holds for *every* J , and the partial sums of the series of nonnegative terms $\sum_j |x_j^{(n)} - x_j|$ are uniformly bounded by $\varepsilon/2$, the series converges and

$$\sum_{j=1}^{\infty} |x_j^{(n)} - x_j| \leq \frac{\varepsilon}{2} \quad \text{for every } n \geq N. \quad (**)$$

Step 3 (membership). With $n = N$ fixed, the triangle inequality for series (coordinate by coordinate and then summing):

$$\sum_{j=1}^{\infty} |x_j| \leq \sum_{j=1}^{\infty} |x_j - x_j^{(N)}| + \sum_{j=1}^{\infty} |x_j^{(N)}| \leq \frac{\varepsilon}{2} + \|x^{(N)}\|_1 < \infty,$$

so $x \in \ell^1$.

Step 4 (convergence). $(**)$ says exactly that $\|x^{(n)} - x\|_1 \leq \varepsilon/2 < \varepsilon$ for $n \geq N$, that is, $x^{(n)} \rightarrow x$ in ℓ^1 . $\square$

Exam strategy

The completeness proofs for $C[a, b]$, ℓ^∞ and ℓ^1 are *the* typical long questions of this part of the course. They all share the skeleton (candidate $\rightarrow$ membership $\rightarrow$ convergence), and each one has *one* technical move of its own that you must master:

- $C[a, b]$: letting $m \rightarrow \infty$ with n fixed, and the $\varepsilon/3$ argument for the continuity of the limit;

- ℓ^∞ : the estimate (*) uniform in j , and bounding the candidate with a fixed term of the sequence;
- ℓ^1 : truncating to finite partial sums *before* passing to the limit in m , and then letting $J \rightarrow \infty$.

A classic penalized mistake: passing to the limit “inside” the supremum or the series without the truncation or uniformity step. Always write in which variable you take the limit and what stays fixed.

14.2.5 Closed subspaces and the example c_0

Theorem 14.7. *Let X be a Banach space and $M \subseteq X$ a vector subspace. Then: M (with the inherited norm) is a Banach space if and only if M is closed in X .*

Proof. ($\Leftarrow$) Suppose M is closed. Let (x_n) be Cauchy in M . In particular it is Cauchy in X (the norm is the same), and since X is complete, $x_n \rightarrow x$ for some $x \in X$. But M is closed and the x_n are in M ; by the sequential characterization of closed sets, the limit x belongs to M . So every Cauchy sequence of M converges in M : M is complete.

($\Rightarrow$) Suppose M is complete. Let $x \in X$ be a point in the closure of M : there is $(x_n) \subset M$ with $x_n \rightarrow x$ in X . Every convergent sequence is Cauchy, so (x_n) is Cauchy in M ; by the completeness of M , it converges to some $y \in M$. But the limit of a sequence in a normed space is unique, and $x_n \rightarrow x$ as well; so $x = y \in M$. This shows that M contains all of its closure points: M is closed. $\square$

Example 14.8 (c_0 is Banach). Let $c_0 = \{x \in \ell^\infty : \lim_{j \rightarrow \infty} x_j = 0\}$, the sequences converging to zero, with the supremum norm. We claim that c_0 is *closed* in ℓ^∞ , and by Theorem 14.7 (with $X = \ell^\infty$, which is Banach by Theorem 14.5), c_0 is Banach.

Proof that it is closed. Let $(x^{(n)}) \subset c_0$ with $x^{(n)} \rightarrow x$ in ℓ^∞ ; we must show that $x_j \rightarrow 0$ as $j \rightarrow \infty$. An $\varepsilon/2$ argument: given $\varepsilon > 0$, choose n with $\|x^{(n)} - x\|_\infty < \varepsilon/2$. Since $x^{(n)} \in c_0$, there is J such that $|x_j^{(n)}| < \varepsilon/2$ for $j \geq J$. Then, for $j \geq J$:

$$|x_j| \leq |x_j - x_j^{(n)}| + |x_j^{(n)}| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

So $x_j \rightarrow 0$, that is, $x \in c_0$. (Note the structure: *uniform* convergence in j lets you transfer the asymptotic property of the approximating element to the limit.)

14.3 Counterexamples: normed spaces that are not complete [P5]

Example 14.9 ($C[0, 1]$ with the integral norm is **not** complete). Consider $X = C[0, 1]$ but now with the norm $\|f\|_1 = \int_0^1 |f(t)| dt$. We build a Cauchy sequence with no continuous limit: *ramps*

approximating a step. For $n \geq 3$, define

$$f_n(t) = \begin{cases} 0 & 0 \leq t \leq \frac{1}{2}, \\ n\left(t - \frac{1}{2}\right) & \frac{1}{2} < t < \frac{1}{2} + \frac{1}{n}, \\ 1 & \frac{1}{2} + \frac{1}{n} \leq t \leq 1. \end{cases}$$

Each f_n is continuous (the pieces match: at $t = 1/2$ it equals 0 on both sides; at $t = 1/2 + 1/n$ it equals 1 on both sides) and it rises linearly from 0 to 1 on the small interval $(\frac{1}{2}, \frac{1}{2} + \frac{1}{n})$.

(f_n) is Cauchy in $\|\cdot\|_1$. For $m > n$, the functions f_n and f_m agree outside $(\frac{1}{2}, \frac{1}{2} + \frac{1}{n})$ (both equal 0 to the left and 1 to the right of their ramps; the ramp of f_m is contained in that of f_n), and on that interval both take values in $[0, 1]$, so $|f_n - f_m| \leq 1$ there. Therefore

$$\|f_n - f_m\|_1 = \int_{1/2}^{1/2+1/n} |f_n - f_m| dt \leq \frac{1}{n} \xrightarrow{n \rightarrow \infty} 0,$$

and given $\varepsilon > 0$ it is enough to take $N > 1/\varepsilon$ so that $\|f_n - f_m\|_1 < \varepsilon$ if $m > n \geq N$. It is Cauchy.

It has no limit in X . Intuitively, f_n converges (in $\|\cdot\|_1$) to the step function $H(t) = 0$ for $t \leq 1/2$, 1 for $t > 1/2$, which is not continuous. Formally, suppose by contradiction that there is $f \in C[0, 1]$ with $\|f_n - f\|_1 \rightarrow 0$. For any $0 < \delta < 1/2$ and $n > 1/\delta$ (so that the ramp stays inside $(\frac{1}{2}, \frac{1}{2} + \delta)$):

$$\int_0^{1/2} |f| dt = \int_0^{1/2} |f - f_n| dt \leq \|f - f_n\|_1 \rightarrow 0,$$

(we use that $f_n \equiv 0$ on $[0, \frac{1}{2}]$), and since the left hand side does not depend on n , $\int_0^{1/2} |f| = 0$; as f is continuous, this forces $f \equiv 0$ on $[0, \frac{1}{2}]$. Similarly, on $[\frac{1}{2} + \delta, 1]$ we have $f_n \equiv 1$ for large n , and

$$\int_{1/2+\delta}^1 |f - 1| dt \leq \|f - f_n\|_1 \rightarrow 0 \implies f \equiv 1 \text{ on } [\frac{1}{2} + \delta, 1],$$

for every $\delta > 0$, so $f \equiv 1$ on $(\frac{1}{2}, 1]$. But then f equals 0 at $\frac{1}{2}$ and 1 at points arbitrarily close to the right of $\frac{1}{2}$: f is discontinuous at $\frac{1}{2}$, a contradiction. Therefore $(C[0, 1], \|\cdot\|_1)$ is **not** complete.

Remark 14.10. The same vector space $C[0, 1]$ is complete with $\|\cdot\|_\infty$ (Theorem 14.4) and incomplete with $\|\cdot\|_1$: completeness is a property of the pair (space, norm), not of the set. The “filling” of the holes of $(C[0, 1], \|\cdot\|_1)$ is the Lebesgue space $L^1[0, 1]$, studied in measure theory; for this course it is enough to know that it exists and that this is its role.

Example 14.11 (c_{00} is not complete). Let c_{00} be the space of sequences with finitely many nonzero terms (eventually zero), with the supremum norm. The sequence

$$x^{(n)} = \left(1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}, 0, 0, \dots\right)$$

is Cauchy: for $m > n$, $\|x^{(n)} - x^{(m)}\|_\infty = \frac{1}{n+1} \rightarrow 0$. Its only possible limit in ℓ^∞ is $x = (1, \frac{1}{2}, \frac{1}{3}, \dots)$ (coordinate by coordinate convergence plus uniform convergence: $\|x^{(n)} - x\|_\infty = \frac{1}{n+1} \rightarrow 0$), but $x \notin c_{00}$ because it has infinitely many nonzero terms. So the Cauchy sequence

does not converge *inside* c_{00} : it is not complete. Equivalently, c_{00} is not closed in ℓ^∞ (in fact its closure is c_0).

14.4 $B(X, Y)$ is Banach when Y is Banach [P5]

Theorem 14.12. *Let X be a normed space and Y a **Banach** space. Then $B(X, Y)$, the space of bounded linear operators from X to Y with the operator norm, is a Banach space.*

Proof. Let (T_n) be a Cauchy sequence in $B(X, Y)$: for every $\varepsilon > 0$ there is N with $\|T_n - T_m\| < \varepsilon$ if $n, m \geq N$. We follow the usual skeleton.

Step 1 (pointwise candidate). Fix $x \in X$. By the fundamental inequality of the operator norm:

$$\|T_n x - T_m x\| = \|(T_n - T_m)x\| \leq \|T_n - T_m\| \|x\|,$$

so $(T_n x)_n$ is Cauchy in Y . This is where we use the hypothesis: since Y is Banach, this limit exists. We define

$$Tx = \lim_{n \rightarrow \infty} T_n x \quad (x \in X).$$

Step 2 (T is linear). Using the continuity of addition and scalar multiplication in Y (algebra of limits in normed spaces):

$$T(x + z) = \lim_n T_n(x + z) = \lim_n (T_n x + T_n z) = \lim_n T_n x + \lim_n T_n z = Tx + Tz,$$

and similarly $T(\lambda x) = \lim_n \lambda T_n x = \lambda Tx$.

Step 3 (T is bounded). Every Cauchy sequence is bounded: there is C with $\|T_n\| \leq C$ for all n . Then, for each x :

$$\|T_n x\| \leq \|T_n\| \|x\| \leq C \|x\|,$$

and letting $n \rightarrow \infty$ (the norm is continuous: if $y_n \rightarrow y$ then $\|y_n\| \rightarrow \|y\|$):

$$\|Tx\| \leq C \|x\|.$$

So $T \in B(X, Y)$: the candidate belongs to the space.

Step 4 (convergence in the operator norm). Let $\varepsilon > 0$ and N with $\|T_n - T_m\| < \varepsilon/2$ for $n, m \geq N$. Fix $n \geq N$ and x with $\|x\| \leq 1$. For all $m \geq N$:

$$\|T_n x - T_m x\| \leq \|T_n - T_m\| \|x\| < \frac{\varepsilon}{2}.$$

Letting $m \rightarrow \infty$ with n, x fixed (and using $T_m x \rightarrow Tx$ together with the continuity of the norm):

$$\|T_n x - Tx\| \leq \frac{\varepsilon}{2}.$$

Taking the supremum over $\|x\| \leq 1$:

$$\|T_n - T\| = \sup_{\|x\| \leq 1} \|(T_n - T)x\| \leq \frac{\varepsilon}{2} < \varepsilon \quad (n \geq N).$$

Therefore $T_n \rightarrow T$ in $B(X, Y)$, and the space is complete. Note that the completeness of X was *never* used: only that of Y , where the limits live. $\square$

Corollary 14.13. *For any normed space X (complete or not), its dual $X^* = B(X, \mathbb{R})$ is a Banach space.*

Proof. Apply Theorem 14.12 with $Y = \mathbb{R}$, which is complete. $\square$

14.5 The Neumann series: inverting operators [P4]

In a Banach space, *series* of vectors (or operators) behave well; recall the criterion from the chapter on series in normed spaces: *in a Banach space, every absolutely convergent series* ($\sum \|u_k\| < \infty$) *is convergent*. This is the tool that lets us copy the numerical geometric series $\frac{1}{1-r} = \sum_{k \geq 0} r^k$ (valid for $|r| < 1$) into the world of operators.

Theorem 14.14 (Neumann series). *Let X be a Banach space and $T \in B(X)$ with $\|T\| < 1$. Then the operator $I - T$ (where I is the identity) is invertible, its inverse is a bounded operator given by the series*

$$(I - T)^{-1} = \sum_{k=0}^{\infty} T^k = I + T + T^2 + T^3 + \cdots,$$

(convergent in the operator norm), and the estimate

$$\|(I - T)^{-1}\| \leq \frac{1}{1 - \|T\|}$$

holds.

Proof. Step 1 (the series converges). By submultiplicativity of the operator norm, $\|T^k\| \leq \|T\|^k$ (by induction: $\|T^k\| = \|T \cdot T^{k-1}\| \leq \|T\| \|T^{k-1}\|$). Since $r = \|T\| < 1$:

$$\sum_{k=0}^{\infty} \|T^k\| \leq \sum_{k=0}^{\infty} r^k = \frac{1}{1-r} < \infty,$$

the series $\sum T^k$ is *absolutely convergent* in $B(X)$. Now, X is Banach, so $B(X) = B(X, X)$ is Banach (Theorem 14.12); by the absolute convergence criterion, the series converges in the operator norm to an operator $S \in B(X)$:

$$S = \sum_{k=0}^{\infty} T^k = \lim_{n \rightarrow \infty} S_n, \quad S_n = \sum_{k=0}^n T^k.$$

Step 2 (S is the inverse). We compute with the partial sums (telescoping sum):

$$(I - T)S_n = (I - T)(I + T + \cdots + T^n) = I + T + \cdots + T^n - (T + T^2 + \cdots + T^{n+1}) = I - T^{n+1},$$

and similarly $S_n(I - T) = I - T^{n+1}$. Now we let $n \rightarrow \infty$: since $\|T^{n+1}\| \leq r^{n+1} \rightarrow 0$, we have $T^{n+1} \rightarrow 0$ in $B(X)$; and since the composition of operators is continuous with respect to the

operator norm (a consequence of submultiplicativity: $\|AB_n - AB\| \leq \|A\| \|B_n - B\|$), we can pass to the limit in both identities:

$$(I - T)S = I, \quad S(I - T) = I.$$

This says exactly that S is the inverse (on both sides) of $I - T$: $(I - T)^{-1} = S \in B(X)$.

Step 3 (the estimate). By the triangle inequality for series (the norm of the limit is the limit of the norms of the partial sums, and these are bounded by the series of norms):

$$\|(I - T)^{-1}\| = \left\| \sum_{k=0}^{\infty} T^k \right\| \leq \sum_{k=0}^{\infty} \|T^k\| \leq \sum_{k=0}^{\infty} \|T\|^k = \frac{1}{1 - \|T\|}. \quad \square$$

Remark 14.15 (Perturbation and stability). A practical reading of the theorem: *invertible operators are stable under small perturbations*. If A is invertible and E is a “perturbation” with $\|A^{-1}E\| < 1$, then $A + E = A(I + A^{-1}E)$ is invertible (a product of invertible operators, using Neumann with $T = -A^{-1}E$). In applied terms: if a linear model $Ax = b$ has a unique solution and the coefficients are measured with a small enough error, the perturbed model still has a unique solution, and the solution depends continuously on the data. Without this robustness, no model calibrated with data would make sense.

Link with economics

Leontief and the rounds of production. In the input-output model, $A \geq 0$ is the matrix of technical coefficients (a_{ij} = units of good i needed to produce one unit of good j) and d is final demand. Gross output x must cover intermediate plus final demand: $x = Ax + d$, that is, $(I - A)x = d$. If $\|A\| < 1$ in a suitable operator norm (for example, if all the columns sum to less than 1: “productive” industries that spend less than one unit of inputs per unit of output), the Neumann series gives

$$x = (I - A)^{-1}d = d + Ad + A^2d + A^3d + \dots$$

with an exact economic interpretation: to satisfy d you must produce d (round 0), plus the inputs Ad needed to produce d (round 1), plus the inputs A^2d needed to produce Ad (round 2), and so on. The convergence of the series says that the rounds of indirect requirements *die out* geometrically: that is a viable economy. The same scheme $x = (I - \beta A)^{-1}d$ appears in discounted Markov chains, in linear Bellman equations and in production network models (the “multiplier” of a network is a Neumann series).

14.6 Overview: the four big theorems [P1]

The following structural results about Banach spaces are stated here without proof (they go beyond the course), but it is worth knowing them: they are cited throughout the mathematical economics literature and sometimes they are asked as “state and interpret”.

Theorem 14.16 (Uniform boundedness, Banach–Steinhaus). *Let X be Banach, Y normed, and*

$\mathcal{F} \subseteq B(X, Y)$ a family of operators that is pointwise bounded: for each $x \in X$, $\sup_{T \in \mathcal{F}} \|Tx\| < \infty$. Then the family is uniformly bounded: $\sup_{T \in \mathcal{F}} \|T\| < \infty$.

Intuition

“If a family of operators does not blow up at any individual point, then it does not blow up at all.” It turns pointwise information (cheap) into uniform information (valuable). It is used, for example, to prove that pointwise limits of bounded operators are bounded.

Theorem 14.17 (Open mapping). *If X, Y are Banach and $T \in B(X, Y)$ is surjective, then T is an open map: it sends open sets to open sets. In particular, if T is bijective, its inverse T^{-1} is automatically continuous.*

Intuition

In Banach spaces, “solving the equation $Tx = y$ uniquely” implies for free that the solution x depends *continuously* on the datum y : well-posed linear problems are automatically stable.

Theorem 14.18 (Closed graph). *Let X, Y be Banach and $T : X \rightarrow Y$ linear. If the graph $\text{graph } T = \{(x, Tx) : x \in X\}$ is closed in $X \times Y$, then T is bounded.*

Intuition

To prove continuity of a linear T between Banach spaces it is enough to check something apparently weaker: if $x_n \rightarrow x$ and moreover $Tx_n \rightarrow y$, then $y = Tx$. (You do not have to prove that Tx_n converges: that is given to you.) It is a very powerful logical shortcut.

Theorem 14.19 (Hahn–Banach, extension form). *Let X be normed, $M \subseteq X$ a subspace, and $f : M \rightarrow \mathbb{R}$ a bounded linear functional on M . Then there is an extension $F \in X^*$ with $F|_M = f$ and $\|F\| = \|f\|$.*

Intuition

Every functional defined on a part of the space extends to the whole space *without increasing its norm*. Consequences: the dual X^* is “rich” (it separates points: if $x \neq y$, there is $F \in X^*$ with $F(x) \neq F(y)$), and the *geometric forms* of Hahn–Banach exist: two disjoint convex sets (one of them open) can be separated by a hyperplane $\{F = \alpha\}$. A separating hyperplane is, in economic terms, a price system.

Link with economics

Hahn–Banach and the second welfare theorem. The second fundamental theorem of welfare economics says that every Pareto optimal allocation (under convexity and other conditions) can be supported as an equilibrium with transfers through *some* price system. The modern proof is Hahn–Banach geometry: at the optimum, the set of allocations “better for everyone” and the feasible set are disjoint convex sets; the separation theorem delivers a hyperplane $\{x : F(x) = \alpha\}$ between them, and the continuous linear functional F is the price vector that supports the optimum (consumers minimize expenditure and

firms maximize profit at those prices). In economies with infinitely many goods (continuous time, uncertainty), the existence of supporting prices is exactly the applicability of Hahn–Banach in the corresponding space of goods; this is why general equilibrium theorists study functional analysis.

Careful (common mistake)

Frequent mistakes in this part:

1. Saying “ X is complete” without specifying the norm. Completeness depends on the norm: $C[0, 1]$ is Banach with $\|\cdot\|_\infty$ and it is not with $\|\cdot\|_1$.
2. In the subspace criterion: a subspace of a Banach space is Banach *if and only if it is closed*; “being a vector subspace” is not enough (c_{00} is a subspace of ℓ^∞ and it is not complete).
3. In the Neumann series, the hypothesis is $\|T\| < 1$ *and* X Banach; without completeness the series can be Cauchy and fail to converge.
4. Confusing convergence of $\sum T^k$ *in the operator norm* with pointwise convergence $\sum T^k x$; the first one is stronger.

Chapter summary

- **Banach** = normed + complete. Universal skeleton of the completeness proofs: candidate (pointwise or by coordinates) $\rightarrow$ membership in the space $\rightarrow$ convergence in the norm of the space.
- Banach spaces: $\mathbb{R}^n$; $(C[a, b], \|\cdot\|_\infty)$; ℓ^∞ ; ℓ^1 (and all the ℓ^p); c_0 (closed in ℓ^∞). Not complete: $(C[0, 1], \|\cdot\|_1)$ (ramps $\rightarrow$ step), c_{00} .
- A subspace of a Banach space is Banach $\iff$ it is closed.
- $B(X, Y)$ is Banach if Y is Banach; in particular X^* is *always* Banach.
- Neumann series: $\|T\| < 1$ in X Banach $\Rightarrow (I - T)^{-1} = \sum_{k \geq 0} T^k$, with $\|(I - T)^{-1}\| \leq (1 - \|T\|)^{-1}$. Application: Leontief $x = (I - A)^{-1}d =$ sum of the rounds of production.
- Overview: Banach–Steinhaus (pointwise $\Rightarrow$ uniform), open mapping (continuous inverse for free), closed graph (continuity shortcut), Hahn–Banach (extension and separation; supporting prices, second welfare theorem).

14.7 Exam-style exercises [P3]

Exercise 14.20. Let c be the space of *convergent* sequences of real numbers (not necessarily to zero), with the supremum norm. Prove that c is a Banach space. (Hint: prove that c is closed in ℓ^∞ by showing that a uniform limit of convergent sequences is Cauchy as a sequence in j .)

Solution. Every convergent sequence is bounded, so $c \subseteq \ell^\infty$, and c is a vector subspace (the sum and the multiples of convergent sequences are convergent, by the algebra of limits). By Theorem 14.7 it is enough to prove that c is *closed* in ℓ^∞ .

Let $(x^{(n)}) \subset c$ with $x^{(n)} \rightarrow x$ in ℓ^∞ ; we must prove that $x = (x_j)_j$ converges. Since $\mathbb{R}$ is complete, it is enough to show that $(x_j)_j$ is Cauchy as a sequence in the index j . An $\varepsilon/3$ argument: given $\varepsilon > 0$, choose a fixed n with $\|x^{(n)} - x\|_\infty < \varepsilon/3$. The sequence $x^{(n)}$ is convergent, hence Cauchy in j : there is J such that

$$|x_j^{(n)} - x_k^{(n)}| < \frac{\varepsilon}{3} \quad \text{for } j, k \geq J.$$

Then, for $j, k \geq J$, inserting $x_j^{(n)}$ and $x_k^{(n)}$:

$$|x_j - x_k| \leq |x_j - x_j^{(n)}| + |x_j^{(n)} - x_k^{(n)}| + |x_k^{(n)} - x_k| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon,$$

where the first and third terms are bounded by $\|x^{(n)} - x\|_\infty$. So (x_j) is Cauchy in $\mathbb{R}$, hence it converges, hence $x \in c$. Therefore c is closed in the Banach space ℓ^∞ , and it is Banach. $\square$

Exercise 14.21. Let $X = C[0, 1]$ with $\|\cdot\|_\infty$ and $M = \{f \in C[0, 1] : f(0) = 0\}$. Prove that M is a Banach space. Is $M' = \{f \in C[0, 1] : f(0) > 0\}$ one? Justify your answer.

Solution. M is Banach. M is a vector subspace: if $f(0) = g(0) = 0$, then $(f + g)(0) = 0$ and $(\lambda f)(0) = 0$. It is closed: $M = \delta_0^{-1}(\{0\})$, where $\delta_0(f) = f(0)$ is the evaluation functional, which is continuous with respect to $\|\cdot\|_\infty$ (it has norm 1: $|f(0)| \leq \|f\|_\infty$). The preimage of the closed set $\{0\}$ under a continuous function is closed. (Alternative sequential check: if $f_n \rightarrow f$ uniformly and $f_n(0) = 0$, then $|f(0)| = |f(0) - f_n(0)| \leq \|f - f_n\|_\infty \rightarrow 0$, so $f(0) = 0$.) Since $(C[0, 1], \|\cdot\|_\infty)$ is Banach (Theorem 14.4) and M is a closed subspace, Theorem 14.7 gives that M is Banach.

M' is not. M' is not even a vector subspace: it does not contain the zero function ($0(0) = 0 \not> 0$), and it is not closed under negative multiples (if $f(0) > 0$ then $(-f)(0) < 0$). The question “is it Banach?” presupposes “is it a normed space?”, and M' is not. Moreover, as a subset of $C[0, 1]$ it is not closed either: $f_n \equiv 1/n$ is in M' and converges uniformly to the zero function, which is not in M' . *Exam lesson:* before discussing completeness, check the vector space structure; it is an easy point that many students give away. $\square$

Exercise 14.22. Let X be a Banach space and $T \in B(X)$ with $\|T^{k_0}\| < 1$ for some $k_0 \in \mathbb{N}$ (not necessarily $k_0 = 1$). Prove that $I - T$ is invertible. (Hint: factor $I - T^{k_0}$.)

Solution. *Step 1.* Since $\|T^{k_0}\| < 1$ and X is Banach, the Neumann series (Theorem 14.14) applied to the operator T^{k_0} says that $I - T^{k_0}$ is invertible, with inverse $S = \sum_{m=0}^{\infty} (T^{k_0})^m \in B(X)$.

Step 2 (factorization). Polynomial algebra applied to operators (valid because the powers of T commute with each other):

$$I - T^{k_0} = (I - T)Q = Q(I - T), \quad \text{where } Q = I + T + T^2 + \cdots + T^{k_0-1},$$

which is checked by expanding the product (a telescoping sum, as in the Neumann proof but with a finite sum). Note that $Q \in B(X)$ because it is a finite sum of bounded operators.

Step 3 (building the inverse). Multiplying $I - T^{k_0} = (I - T)Q$ by S on the right:

$$I = (I - T^{k_0})S = (I - T)(QS),$$

and from $I - T^{k_0} = Q(I - T)$, multiplying by S on the left:

$$I = S(I - T^{k_0}) = (SQ)(I - T).$$

So QS is a right inverse and SQ is a left inverse of $I - T$. When an element has an inverse on both sides, they coincide: $SQ = SQ(I - T)QS = (SQ(I - T))QS = I \cdot QS = QS$. Therefore $I - T$ is invertible with $(I - T)^{-1} = QS = SQ \in B(X)$.

Comment: this generalization is useful when $\|T\| \geq 1$ but the powers decay (“eventually contractive” operators); in dynamic programming it corresponds to effective discounting that only bites after several periods (k_0 stages), and in Markov chains to Doeblin type conditions. $\square$

Exercise 14.23. Consider the Leontief model with two sectors and coefficient matrix

$$A = \begin{pmatrix} 0.2 & 0.3 \\ 0.4 & 0.1 \end{pmatrix},$$

and work in $(\mathbb{R}^2, \|\cdot\|_\infty)$, where the operator norm of a matrix is the maximum of the row sums of absolute values.

- (a) Check that $\|A\| < 1$ and deduce that $(I - A)$ is invertible.
- (b) Bound $\|(I - A)^{-1}\|$ using the Neumann series.
- (c) For the final demand $d = (10, 20)$, compute the second round approximation $x^{(2)} = d + Ad + A^2d$ and bound the error $\|x - x^{(2)}\|_\infty$, where $x = (I - A)^{-1}d$ is the exact output.

Solution. (a) Row sums of $|a_{ij}|$: row 1: $0.2 + 0.3 = 0.5$; row 2: $0.4 + 0.1 = 0.5$. So $\|A\| = \max\{0.5, 0.5\} = 0.5 < 1$. Since $\mathbb{R}^2$ is Banach, Theorem 14.14 gives that $I - A$ is invertible with $(I - A)^{-1} = \sum_{k \geq 0} A^k$.

$$(b) \|(I - A)^{-1}\| \leq \frac{1}{1 - \|A\|} = \frac{1}{1 - 0.5} = 2.$$

(c) We compute the rounds. First:

$$Ad = \begin{pmatrix} 0.2 \cdot 10 + 0.3 \cdot 20 \\ 0.4 \cdot 10 + 0.1 \cdot 20 \end{pmatrix} = \begin{pmatrix} 8 \\ 6 \end{pmatrix}.$$

Second:

$$A^2d = A(Ad) = \begin{pmatrix} 0.2 \cdot 8 + 0.3 \cdot 6 \\ 0.4 \cdot 8 + 0.1 \cdot 6 \end{pmatrix} = \begin{pmatrix} 3.4 \\ 3.8 \end{pmatrix}.$$

Then

$$x^{(2)} = d + Ad + A^2d = \begin{pmatrix} 10 + 8 + 3.4 \\ 20 + 6 + 3.8 \end{pmatrix} = \begin{pmatrix} 21.4 \\ 29.8 \end{pmatrix}.$$

Error. The tail of the series:

$$x - x^{(2)} = \sum_{k=3}^{\infty} A^k d, \quad \|x - x^{(2)}\|_{\infty} \leq \sum_{k=3}^{\infty} \|A\|^k \|d\|_{\infty} = \|d\|_{\infty} \cdot \frac{\|A\|^3}{1 - \|A\|}.$$

With $\|d\|_{\infty} = 20$, $\|A\| = 0.5$:

$$\|x - x^{(2)}\|_{\infty} \leq 20 \cdot \frac{0.125}{0.5} = 5.$$

Interpretation: after only two rounds of indirect requirements, the necessary gross output is determined with an error of at most 5 units per sector, and each extra round halves the bound (factor $\|A\| = 0.5$). This is the quantitative content of the geometric convergence of the Neumann series. $\square$

Exercise 14.24. Let X be a normed space. Prove that if every absolutely convergent series in X is convergent, then X is complete. (This is the converse of the criterion used in the Neumann series; together they give: X is Banach $\iff$ [absolutely convergent $\Rightarrow$ convergent].) (Hint: given a Cauchy sequence, extract a subsequence with $\|x_{n_{k+1}} - x_{n_k}\| < 2^{-k}$ and form a telescoping series.)

Solution. Let (x_n) be Cauchy in X ; we must produce a limit.

Step 1 (fast subsequence). We apply the Cauchy definition with $\varepsilon = 2^{-k}$ successively: there is n_1 such that $\|x_n - x_m\| < 2^{-1}$ for $n, m \geq n_1$; there is $n_2 > n_1$ such that $\|x_n - x_m\| < 2^{-2}$ for $n, m \geq n_2$; and in general, having chosen $n_1 < \dots < n_k$, there is $n_{k+1} > n_k$ with $\|x_n - x_m\| < 2^{-(k+1)}$ for $n, m \geq n_{k+1}$. In particular, the consecutive terms of the subsequence satisfy

$$\|x_{n_{k+1}} - x_{n_k}\| < 2^{-k}.$$

Step 2 (telescoping series). Define $u_1 = x_{n_1}$ and $u_{k+1} = x_{n_{k+1}} - x_{n_k}$ for $k \geq 1$. Then the telescoping partial sum is

$$\sum_{i=1}^K u_i = x_{n_1} + \sum_{k=1}^{K-1} (x_{n_{k+1}} - x_{n_k}) = x_{n_K},$$

(all the intermediate terms cancel). Moreover the series is absolutely convergent:

$$\sum_{i=1}^{\infty} \|u_i\| \leq \|x_{n_1}\| + \sum_{k=1}^{\infty} 2^{-k} = \|x_{n_1}\| + 1 < \infty.$$

By hypothesis, the series $\sum u_i$ converges in X ; that is, its partial sums x_{n_K} converge: there is $x \in X$ with $x_{n_K} \rightarrow x$.

Step 3 (from the subsequence to the whole sequence). General fact: a Cauchy sequence with a convergent subsequence converges (to the same limit). Proof: given $\varepsilon > 0$, take N with $\|x_n - x_m\| < \varepsilon/2$ for $n, m \geq N$, and K with $n_K \geq N$ and $\|x_{n_K} - x\| < \varepsilon/2$. For $n \geq N$:

$$\|x_n - x\| \leq \|x_n - x_{n_K}\| + \|x_{n_K} - x\| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

So $x_n \rightarrow x$, and X is complete.

Comment: this equivalence (completeness $\iff$ every absolutely convergent series converges) is the bridge between the theory of sequences and the theory of series, and it is the technical reason why the Neumann series requires a Banach space. It is a very popular theory question because it combines three techniques: fast subsequences, telescoping and the lemma “Cauchy + convergent subsequence $\Rightarrow$ convergent”. $\square$

Chapter 15

Inner product spaces and Hilbert spaces [P4]

Week 13 of the course (“Hilbert spaces”). Bibliography: Alabiso and Weiss [1], Ch. 5; Moroşanu [2], Ch. 4.

What is this used for in Microeconomics?

Honesty first: for Microeconomics this chapter is peripheral. Where it really pays off is in *econometrics* (the orthogonal projection is the OLS regression and the conditional expectation; the residual orthogonal to the regressors is the moment condition) and in *finance* (pricing in L^2). For micro, keep two things: the Cauchy–Schwarz inequality, which shows up in bounds and in correlation coefficients whenever you work with uncertainty and expected utility (Mas-Colell, Whinston and Green, ch. 6), and the geometric vocabulary (orthogonality, projection) that comes back in the Riesz theorem and in the state prices of the next chapter.

15.1 Motivation [P4]

A norm measures *sizes*; but the Euclidean geometry you learned in school has something more: *angles*, and in particular the notion of *perpendicularity*. The algebraic object that encodes angles is the **inner product**, a generalization of the dot product $x \cdot y = \sum x_j y_j$ of $\mathbb{R}^n$. An inner product space that is also complete is called a **Hilbert space**, and it is the most perfect of the spaces of analysis: in it you have the Pythagorean theorem, the parallelogram law, and above all the **projection theorem**: every point has a unique “shadow” on any closed convex set, characterized by an orthogonality condition.

Why should an economist care about this? Because *projecting is estimating*: ordinary least squares (OLS) regression is literally the orthogonal projection of the data vector onto the subspace spanned by the regressors, and the conditional expectation is the projection in the space L^2 of random variables. The projection theorem is the existence and uniqueness theorem for the “best predictor”. Week 14 (Riesz theorem) will use this geometry to identify prices with vectors.

15.2 Inner product: definition and examples [P3]

Definition 15.1 (Inner product). Let H be a vector space over $\mathbb{R}$. An **inner product** on H is a function $\langle \cdot, \cdot \rangle : H \times H \rightarrow \mathbb{R}$ such that, for all $x, y, z \in H$ and $\lambda \in \mathbb{R}$:

(P1) *Bilinearity* (linearity in each argument): $\langle x + z, y \rangle = \langle x, y \rangle + \langle z, y \rangle$ and $\langle \lambda x, y \rangle = \lambda \langle x, y \rangle$ (and the same in the second argument, which here follows from (P2));

(P2) *Symmetry*: $\langle x, y \rangle = \langle y, x \rangle$;

(P3) *Positive definiteness*: $\langle x, x \rangle \geq 0$, with $\langle x, x \rangle = 0$ if and only if $x = 0$.

The pair $(H, \langle \cdot, \cdot \rangle)$ is called an **inner product space** or pre-Hilbert space.

Remark 15.2. Over $\mathbb{C}$ the definition changes: symmetry is replaced by $\langle x, y \rangle = \overline{\langle y, x \rangle}$ (Hermitian symmetry) and linearity is required in one argument only. In this course we work over $\mathbb{R}$, which is the relevant case for economics; the complex formulas differ only in conjugations.

Example 15.3 (The three canonical examples). 1. $\mathbb{R}^n$ with the *dot product*: $\langle x, y \rangle = \sum_{j=1}^n x_j y_j$. (P1) to (P3) are direct computations; for (P3), $\langle x, x \rangle = \sum x_j^2 = 0$ forces each $x_j = 0$.

2. ℓ^2 , the square summable sequences $(\sum_j x_j^2 < \infty)$, with

$$\langle x, y \rangle = \sum_{j=1}^{\infty} x_j y_j.$$

The series converges absolutely: by the elementary inequality $|x_j y_j| \leq \frac{1}{2}(x_j^2 + y_j^2)$ (which comes from $(|x_j| - |y_j|)^2 \geq 0$), the series of absolute values is dominated by $\frac{1}{2}(\|x\|_2^2 + \|y\|_2^2) < \infty$. This is *the* infinite dimensional Hilbert space.

3. $C[a, b]$ with the integral inner product:

$$\langle f, g \rangle = \int_a^b f(t) g(t) dt.$$

(P3): if f is continuous and $\int_a^b f^2 = 0$, then $f \equiv 0$ (if $f(t_0) \neq 0$, by continuity $f^2 > f(t_0)^2/2 > 0$ on a small interval around t_0 , and the integral would be positive). We will see that this space, with this structure, is *not* complete.

Definition 15.4 (Induced norm). In an inner product space we define the **induced norm**

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

In $\mathbb{R}^n$ it is the Euclidean norm $\|x\|_2$; in ℓ^2 , the norm $\|x\|_2 = (\sum x_j^2)^{1/2}$; in $C[a, b]$, the norm $\|f\|_2 = (\int_a^b f^2)^{1/2}$.

For this to be a norm (the triangle inequality is not obvious) you need the most famous inequality in analysis:

15.3 Cauchy–Schwarz and its consequences [P3]

Theorem 15.5 (Cauchy–Schwarz inequality). *In every inner product space,*

$$|\langle x, y \rangle| \leq \|x\| \|y\| \quad \text{for all } x, y \in H,$$

with equality if and only if x and y are linearly dependent (one is a multiple of the other).

Proof. If $y = 0$, both sides are 0 (note that $\langle x, 0 \rangle = \langle x, 0 \cdot 0 \rangle = 0 \langle x, 0 \rangle = 0$) and the inequality holds with equality, since $\{x, 0\}$ is dependent. Suppose $y \neq 0$.

The quadratic polynomial trick. For every $t \in \mathbb{R}$, axiom (P3) gives

$$0 \leq \langle x + ty, x + ty \rangle.$$

We expand the right side using bilinearity (P1) and symmetry (P2), exactly as you expand a binomial:

$$\langle x + ty, x + ty \rangle = \langle x, x \rangle + t \langle x, y \rangle + t \langle y, x \rangle + t^2 \langle y, y \rangle = \|x\|^2 + 2t \langle x, y \rangle + t^2 \|y\|^2.$$

Write $p(t) = \|y\|^2 t^2 + 2 \langle x, y \rangle t + \|x\|^2$. We have shown that $p(t) \geq 0$ for every t : the parabola p (it opens upward, since $\|y\|^2 > 0$) never crosses the axis. A quadratic polynomial that is nonnegative on the whole line has discriminant ≤ 0 (if the discriminant were positive there would be two real roots and the polynomial would be negative between them). Therefore:

$$\Delta = (2 \langle x, y \rangle)^2 - 4 \|y\|^2 \|x\|^2 \leq 0 \iff \langle x, y \rangle^2 \leq \|x\|^2 \|y\|^2,$$

and taking square roots (both sides are nonnegative): $|\langle x, y \rangle| \leq \|x\| \|y\|$.

Equality case. There is equality $\iff \Delta = 0 \iff$ the parabola has a real root t_0 , that is $p(t_0) = \langle x + t_0 y, x + t_0 y \rangle = 0$. By (P3) this is equivalent to $x + t_0 y = 0$, that is $x = -t_0 y$: linear dependence. Conversely, if $x = \lambda y$, both sides equal $|\lambda| \|y\|^2$. $\square$

Intuition

In $\mathbb{R}^2$, $\langle x, y \rangle = \|x\| \|y\| \cos \theta$ where θ is the angle between the vectors; Cauchy–Schwarz simply says $|\cos \theta| \leq 1$. The polynomial proof is a trick worth memorizing: *insert a parameter t , use positivity, look at the discriminant.* In probability, applied to centered random variables with $\langle X, Y \rangle = E[XY]$, it gives $|\text{Cov}(X, Y)| \leq \sigma_X \sigma_Y$: the correlation lies between -1 and 1 *because of* Cauchy–Schwarz.

Proposition 15.6 (The induced norm is a norm). $\|x\| = \sqrt{\langle x, x \rangle}$ satisfies the norm axioms.

Proof. *Nonnegativity and separation:* direct from (P3). *Homogeneity:* $\|\lambda x\| = \sqrt{\langle \lambda x, \lambda x \rangle} = \sqrt{\lambda^2 \langle x, x \rangle} = |\lambda| \|x\|$. *Triangle inequality:* expand and apply Cauchy–Schwarz to the cross term:

$$\|x + y\|^2 = \|x\|^2 + 2 \langle x, y \rangle + \|y\|^2 \leq \|x\|^2 + 2 \|x\| \|y\| + \|y\|^2 = (\|x\| + \|y\|)^2,$$

and taking roots (both sides are nonnegative) $\|x + y\| \leq \|x\| + \|y\|$. $\square$

Proposition 15.7 (Continuity of the inner product). *If $x_n \rightarrow x$ and $y_n \rightarrow y$ (in the induced norm), then $\langle x_n, y_n \rangle \rightarrow \langle x, y \rangle$.*

Proof. We add and subtract the mixed term $\langle x, y_n \rangle$ to separate the two limits, and apply Cauchy–Schwarz to each piece:

$$\begin{aligned} |\langle x_n, y_n \rangle - \langle x, y \rangle| &= |\langle x_n, y_n \rangle - \langle x, y_n \rangle + \langle x, y_n \rangle - \langle x, y \rangle| \\ &\leq |\langle x_n - x, y_n \rangle| + |\langle x, y_n - y \rangle| \\ &\leq \|x_n - x\| \|y_n\| + \|x\| \|y_n - y\|. \end{aligned}$$

Since (y_n) converges, it is bounded ($\|y_n\| \leq C$); the first term goes to $0 \cdot C = 0$ and the second to $\|x\| \cdot 0 = 0$. So the total goes to 0. $\square$

15.4 The parallelogram law: which norms come from an inner product? [P5]

Theorem 15.8 (Parallelogram law). *In every inner product space, the induced norm satisfies*

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2 \quad \text{for all } x, y.$$

Proof. We expand both squares with bilinearity and symmetry:

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y \rangle = \|x\|^2 + 2\langle x, y \rangle + \|y\|^2, \\ \|x - y\|^2 &= \langle x - y, x - y \rangle = \|x\|^2 - 2\langle x, y \rangle + \|y\|^2. \end{aligned}$$

Adding them, the cross terms $\pm 2\langle x, y \rangle$ cancel and what remains is $2\|x\|^2 + 2\|y\|^2$. $\square$

Intuition

Classical geometry: in a parallelogram with sides x and y , the diagonals are $x + y$ and $x - y$; the law says that *the sum of the squares of the diagonals equals the sum of the squares of the four sides*. It is a property that belongs only to the “round” geometry of the inner product: norms whose balls have corners (like $\|\cdot\|_1$ or $\|\cdot\|_\infty$) violate it. That is why it works as a *test*: if a norm fails the parallelogram law for *some* pair x, y , it does not come from any inner product.

Remark 15.9 (Polarization identity). The converse is also true (the Jordan–von Neumann theorem, which we state without proof): if a norm satisfies the parallelogram law for all pairs, then it comes from an inner product, which is recovered through the **polarization identity**:

$$\langle x, y \rangle = \frac{1}{4} \left(\|x + y\|^2 - \|x - y\|^2 \right).$$

(Direct verification by expanding, in the case where the norm does come from an inner product.) The lesson: *the norm knows the inner product*; there is no extra hidden information.

Example 15.10 ($\|\cdot\|_\infty$ on $C[0, 1]$ does not come from an inner product). One pair that violates the parallelogram law is enough. Take $f(t) = 1$ (constant) and $g(t) = t$. We compute the four supremum norms on $[0, 1]$:

$$\begin{aligned}\|f\|_\infty &= 1, & \|g\|_\infty &= 1, \\ (f+g)(t) &= 1+t \Rightarrow \|f+g\|_\infty = 2 & (\text{maximum at } t=1), \\ (f-g)(t) &= 1-t \Rightarrow \|f-g\|_\infty = 1 & (\text{maximum at } t=0).\end{aligned}$$

Then:

$$\|f+g\|_\infty^2 + \|f-g\|_\infty^2 = 4 + 1 = 5, \quad 2\|f\|_\infty^2 + 2\|g\|_\infty^2 = 2 + 2 = 4.$$

Since $5 \neq 4$, the parallelogram law fails: there is no inner product on $C[0, 1]$ whose induced norm is $\|\cdot\|_\infty$.

Example 15.11 ($\|\cdot\|_1$ on $\mathbb{R}^2$ does not come from an inner product). Take $x = (1, 0)$ and $y = (0, 1)$. Then

$$\|x\|_1 = \|y\|_1 = 1, \quad x+y = (1, 1), \quad \|x+y\|_1 = 2, \quad x-y = (1, -1), \quad \|x-y\|_1 = 2.$$

Parallelogram: $4 + 4 = 8$ against $2 + 2 = 4$. It fails ($8 \neq 4$), so $\|\cdot\|_1$ is not induced by any inner product. (The same pair works to rule out $\|\cdot\|_\infty$ on $\mathbb{R}^2$: $1 + 1 = 2 \neq 4$.)

Exam strategy

“Show that this norm does not come from an inner product” is a classic short question. A sure method: (1) state the parallelogram law; (2) choose two *simple* vectors and compute the four norms; (3) check the numerical inequality. Choose vectors with separate supports (like e_1, e_2) or simple functions (1 and t): the computation should take four lines. Do not try to prove “directly” that no inner product exists: the parallelogram test is there for that.

15.5 Orthogonality [P5]

Definition 15.12 (Orthogonality). Two vectors $x, y \in H$ are **orthogonal**, written $x \perp y$, if $\langle x, y \rangle = 0$. A set $\{u_1, u_2, \dots\} \subseteq H$ is **orthonormal** if $\langle u_i, u_j \rangle = 0$ for $i \neq j$ and $\|u_i\| = 1$ for every i .

Theorem 15.13 (Pythagoras). If $x \perp y$, then $\|x+y\|^2 = \|x\|^2 + \|y\|^2$. More generally, if $u_1, \dots, u_n$ are pairwise orthogonal,

$$\|u_1 + \dots + u_n\|^2 = \|u_1\|^2 + \dots + \|u_n\|^2.$$

Proof. Case $n = 2$: $\|x+y\|^2 = \|x\|^2 + 2\langle x, y \rangle + \|y\|^2$, and the cross term is 0 by orthogonality. General case by induction: assuming it holds for $n-1$, let $s = u_1 + \dots + u_{n-1}$. Then $\langle s, u_n \rangle = \sum_{i=1}^{n-1} \langle u_i, u_n \rangle = 0$ (bilinearity and pairwise orthogonality), so $s \perp u_n$ and the case $n = 2$ gives $\|s+u_n\|^2 = \|s\|^2 + \|u_n\|^2$; the induction hypothesis splits $\|s\|^2$. $\square$

Remark 15.14. In probability: if $X \perp Y$ in the sense $E[XY] = 0$ (for example, centered uncorrelated variables), Pythagoras reads $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$: *the variance of the sum of uncorrelated variables is the sum of the variances*. The variance decomposition of regression analysis ($\text{TSS} = \text{ESS} + \text{RSS}$) is Pythagoras in the data space.

15.5.1 Gram–Schmidt: building orthonormal bases

Theorem 15.15 (Gram–Schmidt process). *Let $\{v_1, v_2, \dots\}$ be a linearly independent family (finite or countable) in an inner product space. Defining recursively*

$$w_1 = v_1, \quad w_k = v_k - \sum_{i=1}^{k-1} \frac{\langle v_k, w_i \rangle}{\|w_i\|^2} w_i, \quad u_k = \frac{w_k}{\|w_k\|},$$

you obtain an orthonormal family $\{u_1, u_2, \dots\}$ with $\text{span}\{u_1, \dots, u_k\} = \text{span}\{v_1, \dots, v_k\}$ for every k .

Intuition

At each step, you *subtract from v_k its shadow* on what has already been built (the terms $\frac{\langle v_k, w_i \rangle}{\|w_i\|^2} w_i$ are the projections onto each previous direction); what remains, w_k , is the “new part” of v_k , perpendicular to everything before it. Then you normalize. Check of orthogonality in one step: for $j < k$,

$$\langle w_k, w_j \rangle = \langle v_k, w_j \rangle - \sum_{i=1}^{k-1} \frac{\langle v_k, w_i \rangle}{\|w_i\|^2} \langle w_i, w_j \rangle = \langle v_k, w_j \rangle - \frac{\langle v_k, w_j \rangle}{\|w_j\|^2} \|w_j\|^2 = 0,$$

because by induction $\langle w_i, w_j \rangle = 0$ except when $i = j$. In econometrics, Gram–Schmidt is the *orthogonalization of regressors* (the Frisch–Waugh–Lovell theorem works with residualized regressors, which are exactly the w_k).

Example 15.16 (Numerical Gram–Schmidt in $\mathbb{R}^3$). Let us orthonormalize $v_1 = (1, 1, 0)$, $v_2 = (1, 0, 1)$, $v_3 = (0, 1, 1)$ with the dot product.

Step 1. $w_1 = v_1 = (1, 1, 0)$, $\|w_1\|^2 = 1 + 1 + 0 = 2$; $u_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$.

Step 2. $\langle v_2, w_1 \rangle = 1 \cdot 1 + 0 \cdot 1 + 1 \cdot 0 = 1$.

$$w_2 = v_2 - \frac{1}{2} w_1 = (1, 0, 1) - \left(\frac{1}{2}, \frac{1}{2}, 0\right) = \left(\frac{1}{2}, -\frac{1}{2}, 1\right).$$

Check: $\langle w_2, w_1 \rangle = \frac{1}{2} - \frac{1}{2} + 0 = 0$. $\checkmark$ $\|w_2\|^2 = \frac{1}{4} + \frac{1}{4} + 1 = \frac{3}{2}$; $u_2 = \sqrt{\frac{2}{3}}\left(\frac{1}{2}, -\frac{1}{2}, 1\right) = \frac{1}{\sqrt{6}}(1, -1, 2)$.

Step 3. $\langle v_3, w_1 \rangle = 0 + 1 + 0 = 1$; $\langle v_3, w_2 \rangle = 0 - \frac{1}{2} + 1 = \frac{1}{2}$.

$$w_3 = v_3 - \frac{1}{2} w_1 - \frac{1/2}{3/2} w_2 = (0, 1, 1) - \left(\frac{1}{2}, \frac{1}{2}, 0\right) - \frac{1}{3}\left(\frac{1}{2}, -\frac{1}{2}, 1\right) = \left(-\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right).$$

Checks: $\langle w_3, w_1 \rangle = -\frac{2}{3} + \frac{2}{3} = 0$; $\langle w_3, w_2 \rangle = -\frac{1}{3} - \frac{1}{3} + \frac{2}{3} = 0$. $\checkmark$ $\|w_3\|^2 = 3 \cdot \frac{4}{9} = \frac{4}{3}$; $u_3 = \frac{\sqrt{3}}{2}\left(-\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right) = \frac{1}{\sqrt{3}}(-1, 1, 1)$.

Result: $\{u_1, u_2, u_3\}$ is orthonormal and spans the same space. The same algorithm applied to $\{1, t, t^2, \dots\}$ in $C[-1, 1]$ with $\langle f, g \rangle = \int_{-1}^1 fg$ produces (up to constants) the **Legendre**

polynomials, a classical basis for approximation.

15.6 Hilbert spaces [P4]

Definition 15.17 (Hilbert space). A **Hilbert space** is an inner product space that is *complete* with respect to the induced norm. (That is: Hilbert = inner product + Banach.)

Example 15.18 ($\mathbb{R}^n$ and ℓ^2 are Hilbert; $C[a, b]$ is not). 1. $\mathbb{R}^n$ with the dot product: the induced norm is the Euclidean one, and $\mathbb{R}^n$ is complete with any norm. Hilbert. ✓

2. ℓ^2 is complete; we sketch the proof, which follows the standard skeleton (it is identical to the one for ℓ^1 , replacing sums of absolute values by sums of squares). Let $(x^{(n)})$ be Cauchy in ℓ^2 . (i) *Candidate*: each coordinate satisfies $|x_j^{(n)} - x_j^{(m)}| \leq \|x^{(n)} - x^{(m)}\|_2$, so it is Cauchy in $\mathbb{R}$ and converges to some x_j . (ii) *Truncated estimate*: given $\varepsilon > 0$, let N be such that $\|x^{(n)} - x^{(m)}\|_2^2 < \varepsilon^2$ for $n, m \geq N$; for every finite J , $\sum_{j=1}^J (x_j^{(n)} - x_j^{(m)})^2 < \varepsilon^2$, and letting $m \rightarrow \infty$ (finite sum): $\sum_{j=1}^J (x_j^{(n)} - x_j)^2 \leq \varepsilon^2$; since this holds for every J , the full series satisfies $\sum_{j=1}^\infty (x_j^{(n)} - x_j)^2 \leq \varepsilon^2$. (iii) *Membership and convergence*: from (ii), $x^{(n)} - x \in \ell^2$, so $x = x^{(n)} - (x^{(n)} - x) \in \ell^2$ (sum of elements of ℓ^2 , using the triangle inequality of Minkowski), and $\|x^{(n)} - x\|_2 \leq \varepsilon$ for $n \geq N$. Hilbert. ✓

3. $C[a, b]$ with $\langle f, g \rangle = \int_a^b fg$ is **not** complete: the induced norm $\|f\|_2 = (\int f^2)^{1/2}$ has the same problem as the norm $\|\cdot\|_1$: the sequence of *ramps* approximating the step function (built in the chapter on Banach spaces) is Cauchy in $\|\cdot\|_2$ (the computations are analogous: $\|f_n - f_m\|_2^2 \leq \frac{1}{n} \rightarrow 0$, since the integrand is bounded by 1 and the support has measure $\leq 1/n$) and has no continuous limit. It is an incomplete pre-Hilbert space; its completion is the Lebesgue space $L^2[a, b]$, a central object in econometrics and finance.

15.7 The projection theorem [P4]

This is the main theorem of the chapter. Recall that $C \subseteq H$ is **convex** if for all $u, v \in C$ and $t \in [0, 1]$ you have $tu + (1 - t)v \in C$ (the segment between two points of the set stays inside).

Theorem 15.19 (Projection onto a closed convex set). *Let H be a Hilbert space and $C \subseteq H$ a nonempty, convex and closed set. Then for each $x \in H$ there exists a **unique** point $Px = P_C x \in C$ such that*

$$\|x - Px\| = \text{dist}(x, C) := \inf_{v \in C} \|x - v\|.$$

Px is called the **projection** of x onto C . Moreover, Px is characterized by the **variational inequality**:

$$z = Px \iff z \in C \text{ and } \langle x - z, v - z \rangle \leq 0 \quad \forall v \in C.$$

Intuition

Px is the point of C closest to x : the shadow of x on C . The variational inequality says that the “residual” $x - Px$ makes an obtuse (or right) angle with any direction $v - Px$ pointing into C : from Px , moving inside C can only take you *farther* from x . The three

ingredients are all needed: without closedness the infimum may not be attained; without convexity there may be several closest points (think of the projection onto a circle from its center: every point ties); without completeness the minimizing sequence may have no limit.

Proof. Let $d = \text{dist}(x, C) = \inf_{v \in C} \|x - v\|$, which is finite because $C \neq \emptyset$ (any $v_0 \in C$ gives the bound $d \leq \|x - v_0\|$).

Step 1 (minimizing sequence). By definition of infimum, for each $n \in \mathbb{N}$ there is $v_n \in C$ with

$$d^2 \leq \|x - v_n\|^2 < d^2 + \frac{1}{n}.$$

Step 2 (the parallelogram trick). We claim that (v_n) is Cauchy. We apply the parallelogram law (Theorem 15.8) to the vectors $a = x - v_n$ and $b = x - v_m$:

$$\|a + b\|^2 + \|a - b\|^2 = 2\|a\|^2 + 2\|b\|^2.$$

Here $a + b = 2x - v_n - v_m = 2(x - \frac{v_n + v_m}{2})$ and $a - b = v_m - v_n$. Substituting and solving:

$$\|v_m - v_n\|^2 = 2\|x - v_n\|^2 + 2\|x - v_m\|^2 - 4\left\|x - \frac{v_n + v_m}{2}\right\|^2.$$

This is where *convexity* enters: the midpoint $\frac{v_n + v_m}{2}$ belongs to C (take $t = \frac{1}{2}$), so by definition of d :

$$\left\|x - \frac{v_n + v_m}{2}\right\| \geq d \implies -4\left\|x - \frac{v_n + v_m}{2}\right\|^2 \leq -4d^2.$$

Combining with the bounds of Step 1:

$$\|v_m - v_n\|^2 \leq 2\left(d^2 + \frac{1}{n}\right) + 2\left(d^2 + \frac{1}{m}\right) - 4d^2 = \frac{2}{n} + \frac{2}{m} \xrightarrow{n, m \rightarrow \infty} 0.$$

So (v_n) is Cauchy.

Step 3 (existence). H is complete, so $v_n \rightarrow z$ for some $z \in H$; and C is closed, so $z \in C$. By continuity of the norm, $\|x - z\| = \lim_n \|x - v_n\| = d$: the infimum is attained at z .

Step 4 (uniqueness). If $z, z' \in C$ both satisfy $\|x - z\| = \|x - z'\| = d$, we apply the same parallelogram identity to $a = x - z$, $b = x - z'$:

$$\|z - z'\|^2 = 2d^2 + 2d^2 - 4\left\|x - \frac{z + z'}{2}\right\|^2 \leq 4d^2 - 4d^2 = 0,$$

using again that $\frac{z + z'}{2} \in C$ and therefore its distance to x is $\geq d$. So $\|z - z'\| = 0$, that is $z = z'$.

Step 5 (variational characterization). ($\Rightarrow$) Let $z = Px$ and $v \in C$ be arbitrary. By convexity, for $t \in (0, 1]$ the point $v_t = z + t(v - z) = (1 - t)z + tv$ is in C , so $\|x - v_t\|^2 \geq \|x - z\|^2 = d^2$. We expand the left side:

$$\|x - z - t(v - z)\|^2 = \|x - z\|^2 - 2t\langle x - z, v - z \rangle + t^2\|v - z\|^2.$$

The inequality $\|x - v_t\|^2 \geq \|x - z\|^2$ becomes

$$-2t \langle x - z, v - z \rangle + t^2 \|v - z\|^2 \geq 0 \implies \langle x - z, v - z \rangle \leq \frac{t}{2} \|v - z\|^2,$$

(we divide by $2t > 0$). Letting $t \downarrow 0$, the right side goes to 0 and we are left with $\langle x - z, v - z \rangle \leq 0$. (This limit in the parameter of the convex combination is *the* basic variational argument of optimization theory: perturb toward a feasible point and look at the first derivative.)

($\Leftarrow$) Suppose $z \in C$ with $\langle x - z, v - z \rangle \leq 0$ for every $v \in C$. Then, for any $v \in C$:

$$\begin{aligned} \|x - v\|^2 &= \|(x - z) - (v - z)\|^2 = \|x - z\|^2 - 2 \langle x - z, v - z \rangle + \|v - z\|^2 \\ &\geq \|x - z\|^2 + \|v - z\|^2 \geq \|x - z\|^2, \end{aligned}$$

since $-2 \langle x - z, v - z \rangle \geq 0$. So z minimizes the distance, and by uniqueness $z = Px$. $\square$

15.7.1 Projection onto closed subspaces and orthogonal decomposition

Definition 15.20 (Orthogonal complement). For $M \subseteq H$, the **orthogonal complement** is

$$M^\perp = \{y \in H : \langle y, m \rangle = 0 \text{ for every } m \in M\}.$$

Proposition 15.21. $M^\perp$ is always a closed vector subspace of H , and $M \cap M^\perp \subseteq \{0\}$.

Proof. Subspace: if $y_1, y_2 \in M^\perp$ and $\lambda \in \mathbb{R}$, then for every $m \in M$: $\langle y_1 + y_2, m \rangle = \langle y_1, m \rangle + \langle y_2, m \rangle = 0$ and $\langle \lambda y_1, m \rangle = \lambda \langle y_1, m \rangle = 0$. *Closed:* if $y_n \in M^\perp$ and $y_n \rightarrow y$, continuity of the inner product (Proposition 15.7) gives $\langle y, m \rangle = \lim_n \langle y_n, m \rangle = 0$ for each $m \in M$; so $y \in M^\perp$. *Intersection:* if $x \in M \cap M^\perp$, then x is orthogonal to itself: $\langle x, x \rangle = 0$, so $x = 0$ by (P3). $\square$

Theorem 15.22 (Orthogonal decomposition). Let M be a closed subspace of a Hilbert space H . Then every $x \in H$ is written in a **unique** way as

$$x = m + y, \quad m \in M, \quad y \in M^\perp,$$

with $m = P_M x$ (the projection onto M). We write $H = M \oplus M^\perp$. Moreover, for subspaces the variational inequality becomes an orthogonality equality:

$$z = P_M x \iff z \in M \text{ and } x - z \perp M \text{ (that is } \langle x - z, m \rangle = 0 \forall m \in M).$$

Proof. A subspace is convex (convex combinations are linear combinations) and M is closed and nonempty, so Theorem 15.19 applies: there is a unique $z = P_M x \in M$ with $\langle x - z, v - z \rangle \leq 0$ for every $v \in M$.

From the inequality to the equality. Fix $m \in M$ and use the freedom that M is a subspace: for any $t \in \mathbb{R}$, the vector $v = z + tm$ is in M (sum of elements of M). Substituting in the variational inequality:

$$\langle x - z, (z + tm) - z \rangle = t \langle x - z, m \rangle \leq 0 \quad \text{for every } t \in \mathbb{R}.$$

A real number $\alpha = \langle x - z, m \rangle$ with $t\alpha \leq 0$ for every t (positive and negative) must be 0. So $\langle x - z, m \rangle = 0$ for every $m \in M$: that is, $y := x - z \in M^\perp$. This gives the decomposition $x = z + y$ with $z \in M$, $y \in M^\perp$, and also the stated characterization (the converse is the same computation as in Theorem 15.19, where now the cross term is zero: for $v \in M$, $\|x - v\|^2 = \|x - z\|^2 + \|z - v\|^2 \geq \|x - z\|^2$, pure Pythagoras).

Uniqueness of the decomposition. If $x = m + y = m' + y'$ with $m, m' \in M$, $y, y' \in M^\perp$, then $m - m' = y' - y$ belongs to $M \cap M^\perp = \{0\}$ (Proposition 15.21), so $m = m'$ and $y = y'$. $\square$

Link with economics

Projection = regression. Let $H = L^2$ (random variables with finite second moment, $\langle X, Y \rangle = E[XY]$) and $M = \text{span}\{X_1, \dots, X_k\}$ the subspace spanned by the regressors. The projection $\hat{Y} = P_M Y$ is the *best linear predictor* of Y : it minimizes the mean squared error $E[(Y - \hat{Y})^2]$ over M . The characterization $Y - \hat{Y} \perp M$ is exactly the set of *orthogonality conditions* (normal equations) of OLS: $E[(Y - \hat{Y})X_j] = 0$ for each regressor: “the residual is not correlated with the regressors”. Pythagoras is the variance decomposition. If instead of a linear subspace you project onto the (convex, closed) set of all measurable functions of the regressors, you get the *conditional expectation* $E[Y | X]$: the best predictor without qualification. And in finance, the Hansen–Richard theorem represents asset prices by projecting the stochastic discount factor onto the payoff space: *pricing by projection*. In every case, existence and uniqueness of the estimator is Theorem 15.19, and the first order conditions are the variational inequality. For stationary processes, ℓ^2 and its projections onto the past are the framework of Wold prediction and of ARMA models.

15.7.2 Bessel inequality and abstract Fourier series

Theorem 15.23 (Bessel inequality). *Let $\{u_1, u_2, \dots\}$ be an orthonormal set (finite or countable) in an inner product space H . Then, for every $x \in H$:*

$$\sum_k \langle x, u_k \rangle^2 \leq \|x\|^2.$$

The numbers $\langle x, u_k \rangle$ are called the **Fourier coefficients** of x .

Proof. It is enough to prove it for finite sums (a series of nonnegative terms converges if its partial sums are bounded). Let $s_n = \sum_{k=1}^n \langle x, u_k \rangle u_k$ (the “Fourier approximation” of order n). We compute $\|x - s_n\|^2 \geq 0$ by expanding:

$$\|x - s_n\|^2 = \|x\|^2 - 2\langle x, s_n \rangle + \|s_n\|^2.$$

By bilinearity, $\langle x, s_n \rangle = \sum_{k=1}^n \langle x, u_k \rangle \langle x, u_k \rangle = \sum_{k=1}^n \langle x, u_k \rangle^2$; and by Pythagoras (Theorem 15.13, the vectors $\langle x, u_k \rangle u_k$ are pairwise orthogonal), $\|s_n\|^2 = \sum_{k=1}^n \langle x, u_k \rangle^2 \|u_k\|^2 = \sum_{k=1}^n \langle x, u_k \rangle^2$. Then

$$0 \leq \|x - s_n\|^2 = \|x\|^2 - \sum_{k=1}^n \langle x, u_k \rangle^2 \implies \sum_{k=1}^n \langle x, u_k \rangle^2 \leq \|x\|^2. \quad \square$$

Remark 15.24 (Orthonormal bases and Parseval). An orthonormal set $\{u_k\}$ is an **orthonormal basis** of the Hilbert space H if every $x \in H$ is reconstructed from its coefficients: $x = \sum_k \langle x, u_k \rangle u_k$ (series convergent in the norm of H). In that case Bessel becomes an equality, the **Parseval identity**: $\|x\|^2 = \sum_k \langle x, u_k \rangle^2$. Intuition: an orthonormal basis is an infinite “system of perpendicular coordinates”, and Parseval is infinite dimensional Pythagoras; ℓ^2 is the universal model (every separable infinite dimensional Hilbert space is isometric to ℓ^2 through its Fourier coefficients). Classical example: sines and cosines in $L^2[-\pi, \pi]$, the Fourier series of harmonic analysis. Convergence of $\sum_k \langle x, u_k \rangle u_k$ in a Hilbert space is guaranteed because its partial sums are Cauchy (Pythagoras plus Bessel) and the space is complete.

Careful (common mistake)

Three frequent mistakes:

1. Applying the projection theorem without checking *closed* and *convex*. For finite dimensional subspaces closedness is automatic, but for general subspaces it is not: there are proper dense subspaces (like c_{00} in ℓ^2) where the projection *does not exist* for some points.
2. Writing $H = M \oplus M^\perp$ for M *not* closed. False in general; the correct statement for any subspace is $(M^\perp)^\perp = \text{cl}(M)$ (the closure).
3. Confusing the variational inequality (convex sets: ≤ 0) with the orthogonality equality (subspaces: $= 0$). The equality uses that M is closed under *negative* multiples, something a general convex set does not allow.

Exam strategy

Typical question formats in this part of the course:

1. *Prove Cauchy–Schwarz or the parallelogram law*: expansion computations; memorize the discriminant trick.
2. *“This norm does not come from an inner product”*: parallelogram test with simple vectors.
3. *Prove the projection theorem*: the full proof is long; what is asked most often is Step 2 (parallelogram on the minimizing sequence) and uniqueness. Practice writing the identity $\|v_m - v_n\|^2 = 2\|x - v_n\|^2 + 2\|x - v_m\|^2 - 4\|x - \frac{v_n + v_m}{2}\|^2$ without help.
4. *Compute a concrete projection*: set $x - z \perp$ (generators of M) and solve the linear system; check at the end.

Chapter summary

- Inner product: bilinear, symmetric, positive definite; induced norm $\|x\| = \sqrt{\langle x, x \rangle}$. Examples: $\mathbb{R}^n$, ℓ^2 , $C[a, b]$ with $\int fg$.
- **Cauchy–Schwarz**: $|\langle x, y \rangle| \leq \|x\| \|y\|$ (equality $\iff$ linear dependence). From it:

the triangle inequality and continuity of the inner product.

- **Parallelogram:** $\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$; a test to decide whether a norm comes from an inner product ($\|\cdot\|_\infty, \|\cdot\|_1$: no).
- Orthogonality, Pythagoras, Gram–Schmidt (subtract shadows and normalize).
- **Hilbert** = inner product + complete: $\mathbb{R}^n, \ell^2$ yes; $(C[a, b], \|\cdot\|_2)$ no (ramps; its completion is L^2).
- **Projection theorem:** C nonempty closed convex in a Hilbert space $\Rightarrow$ there is a unique closest point Px , characterized by $\langle x - Px, v - Px \rangle \leq 0 \ \forall v \in C$. For M a closed subspace: $x - P_M x \perp M$ and $H = M \oplus M^\perp$.
- Bessel: $\sum_k \langle x, u_k \rangle^2 \leq \|x\|^2$; with an orthonormal basis, Parseval (equality).
- Economics: projection = OLS and conditional expectation; orthogonality of the residual = normal equations; Pythagoras = variance decomposition.

15.8 Exam style exercises [P4]

Exercise 15.25. In $\mathbb{R}^2$ with the dot product, let $M = \{(t, 2t) : t \in \mathbb{R}\}$ (the line $y = 2x$). Find the projection $P_M x$ of the point $x = (3, 1)$, check the orthogonality of the residual, and verify Pythagoras: $\|x\|^2 = \|P_M x\|^2 + \|x - P_M x\|^2$.

Solution. M is a closed subspace (finite dimension) spanned by $u = (1, 2)$. By Theorem 15.22, $z = P_M x$ is the unique point $z = tu \in M$ with $x - z \perp u$:

$$0 = \langle x - tu, u \rangle = \langle x, u \rangle - t \|u\|^2 \implies t = \frac{\langle x, u \rangle}{\|u\|^2}.$$

We compute: $\langle x, u \rangle = 3 \cdot 1 + 1 \cdot 2 = 5$; $\|u\|^2 = 1 + 4 = 5$; so $t = 1$ and

$$P_M x = 1 \cdot (1, 2) = (1, 2).$$

Orthogonality: $x - P_M x = (3, 1) - (1, 2) = (2, -1)$; $\langle (2, -1), (1, 2) \rangle = 2 - 2 = 0$. ✓

Pythagoras: $\|x\|^2 = 9 + 1 = 10$; $\|P_M x\|^2 = 1 + 4 = 5$; $\|x - P_M x\|^2 = 4 + 1 = 5$; indeed $10 = 5 + 5$. ✓

(Econometric reading: if u is the “regressor” and x the “data”, $t = 1$ is the OLS coefficient $\hat{\beta} = \langle x, u \rangle / \|u\|^2$, the residual $(2, -1)$ is orthogonal to the regressor, and $10 = 5 + 5$ is TSS = ESS + RSS.) □

Exercise 15.26. Prove that in every inner product space the polarization identity holds:

$$\langle x, y \rangle = \frac{1}{4} (\|x + y\|^2 - \|x - y\|^2),$$

and use it to prove: if $T : H \rightarrow H$ is linear and preserves norms ($\|Tx\| = \|x\|$ for every x), then it preserves inner products ($\langle Tx, Ty \rangle = \langle x, y \rangle$ for all x, y).

Solution. *Polarization.* We expand both squares:

$$\|x + y\|^2 = \|x\|^2 + 2\langle x, y \rangle + \|y\|^2, \quad \|x - y\|^2 = \|x\|^2 - 2\langle x, y \rangle + \|y\|^2.$$

Subtracting: $\|x + y\|^2 - \|x - y\|^2 = 4\langle x, y \rangle$, and dividing by 4 you get the identity.

Application. Let T be linear with $\|Tz\| = \|z\|$ for every z . Using polarization twice and the linearity of T (which gives $Tx + Ty = T(x + y)$ and $Tx - Ty = T(x - y)$):

$$\begin{aligned} \langle Tx, Ty \rangle &= \frac{1}{4}(\|Tx + Ty\|^2 - \|Tx - Ty\|^2) = \frac{1}{4}(\|T(x + y)\|^2 - \|T(x - y)\|^2) \\ &= \frac{1}{4}(\|x + y\|^2 - \|x - y\|^2) = \langle x, y \rangle, \end{aligned}$$

where in the second to last step we used norm preservation applied to the vectors $x + y$ and $x - y$. *Lesson:* for linear maps, preserving lengths forces preserving angles; the linear isometries of a Hilbert space are automatically “orthogonal”. $\square$

Exercise 15.27. In $C[0, 1]$ with $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$, let $M = \text{span}\{1, t\}$ (polynomials of degree ≤ 1). Find the projection of $f(t) = t^2$ onto M , that is, the best approximation of t^2 by a line $a + bt$ in the least squares sense. Compute the error $\|f - P_M f\|$.

Solution. M has finite dimension, so it is closed, and we look for $P_M f = a + bt$ with residual orthogonal to the generators:

$$\langle f - (a + bt), 1 \rangle = 0, \quad \langle f - (a + bt), t \rangle = 0.$$

We compute the moments we need: $\int_0^1 1 dt = 1$, $\int_0^1 t dt = \frac{1}{2}$, $\int_0^1 t^2 dt = \frac{1}{3}$, $\int_0^1 t^3 dt = \frac{1}{4}$.

First equation ($\perp 1$):

$$\int_0^1 (t^2 - a - bt) dt = \frac{1}{3} - a - \frac{b}{2} = 0.$$

Second equation ($\perp t$):

$$\int_0^1 (t^3 - at - bt^2) dt = \frac{1}{4} - \frac{a}{2} - \frac{b}{3} = 0.$$

Linear system: from the first one, $a = \frac{1}{3} - \frac{b}{2}$. Substituting in the second:

$$\frac{1}{4} - \frac{1}{2}\left(\frac{1}{3} - \frac{b}{2}\right) - \frac{b}{3} = \frac{1}{4} - \frac{1}{6} + \frac{b}{4} - \frac{b}{3} = \frac{1}{12} - \frac{b}{12} = 0 \implies b = 1,$$

and then $a = \frac{1}{3} - \frac{1}{2} = -\frac{1}{6}$. Therefore

$$P_M f = t - \frac{1}{6}.$$

Error. The residual is $r(t) = t^2 - t + \frac{1}{6}$. Its squared norm:

$$\|r\|^2 = \int_0^1 \left(t^2 - t + \frac{1}{6}\right)^2 dt.$$

We expand the square: $(t^2 - t + \frac{1}{6})^2 = t^4 - 2t^3 + \frac{4}{3}t^2 - \frac{1}{3}t + \frac{1}{36}$. (Detail: $t^4 - 2t^3 + t^2 + \frac{1}{3}t^2 - \frac{1}{3}t + \frac{1}{36}$,

grouping $t^2 + \frac{1}{3}t^2 = \frac{4}{3}t^2$.) Integrating term by term:

$$\|r\|^2 = \frac{1}{5} - \frac{2}{4} + \frac{4}{9} - \frac{1}{6} + \frac{1}{36} = \frac{1}{5} - \frac{1}{2} + \frac{4}{9} - \frac{1}{6} + \frac{1}{36}.$$

Common denominator 180: $\frac{36}{180} - \frac{90}{180} + \frac{80}{180} - \frac{30}{180} + \frac{5}{180} = \frac{1}{180}$. So

$$\|f - P_M f\| = \frac{1}{\sqrt{180}} = \frac{1}{6\sqrt{5}}.$$

(Sanity check: the error is small but positive: t^2 is not a line. The line $t - \frac{1}{6}$ is the “regression” of the parabola.) $\square$

Exercise 15.28. Let H be a Hilbert space and $M \subseteq H$ a subspace (not necessarily closed). Prove that $(M^\perp)^\perp = \text{cl}(M)$, the closure of M . Conclude that M is *dense* in H if and only if $M^\perp = \{0\}$.

Solution. *Inclusion* $\text{cl}(M) \subseteq (M^\perp)^\perp$. First $M \subseteq (M^\perp)^\perp$: if $m \in M$ and $y \in M^\perp$, then $\langle m, y \rangle = 0$, which is what $m \perp M^\perp$ means. Moreover $(M^\perp)^\perp$ is closed (Proposition 15.21, applied to the set $M^\perp$); a closed set containing M contains its closure. So $\text{cl}(M) \subseteq (M^\perp)^\perp$.

Inclusion $(M^\perp)^\perp \subseteq \text{cl}(M)$. Let $x \in (M^\perp)^\perp$. The set $N = \text{cl}(M)$ is a closed subspace (the closure of a subspace is a subspace: limits of sums are sums of limits), so the orthogonal decomposition (Theorem 15.22) writes

$$x = m + y, \quad m \in N, \quad y \in N^\perp.$$

Note that $N^\perp = M^\perp$: certainly $N^\perp \subseteq M^\perp$ (being orthogonal to a larger set is more demanding), and if $y \perp M$ then $y \perp \text{cl}(M)$ by continuity of the inner product (if $m_n \rightarrow m$ with $m_n \in M$, then $\langle y, m \rangle = \lim \langle y, m_n \rangle = 0$). So $y \in M^\perp$. Now we compute $\langle x, y \rangle$ in two ways: on one hand, $x \in (M^\perp)^\perp$ and $y \in M^\perp$ give $\langle x, y \rangle = 0$; on the other,

$$0 = \langle x, y \rangle = \langle m + y, y \rangle = \langle m, y \rangle + \|y\|^2 = 0 + \|y\|^2,$$

(the term $\langle m, y \rangle$ is zero because $m \in N$ and $y \in N^\perp$). So $\|y\| = 0$, that is $x = m \in \text{cl}(M)$.

Conclusion about density. M dense means $\text{cl}(M) = H$. By what we proved, $\text{cl}(M) = (M^\perp)^\perp$. If $M^\perp = \{0\}$, then $(M^\perp)^\perp = \{0\}^\perp = H$ (every vector is orthogonal to the zero vector), so M is dense. Conversely, if M is dense and $y \in M^\perp$, then $y \in \text{cl}(M)^\perp = H^\perp = \{0\}$ (only 0 is orthogonal to the whole space, in particular to itself). So density $\iff M^\perp = \{0\}$. *Typical use:* to prove that a family of functions “generates” a Hilbert space, it is enough to see that nothing nonzero is orthogonal to the whole family (this is how completeness of trigonometric and polynomial systems is proved). $\square$

Exercise 15.29. Let H be a Hilbert space, $\{u_1, \dots, u_n\}$ a finite orthonormal set and $M = \text{span}\{u_1, \dots, u_n\}$. Prove that the projection onto M has the explicit formula

$$P_M x = \sum_{k=1}^n \langle x, u_k \rangle u_k,$$

and that $\text{dist}(x, M)^2 = \|x\|^2 - \sum_{k=1}^n \langle x, u_k \rangle^2$. Interpret the Bessel inequality in the light of this result.

Solution. M is a finite dimensional subspace, therefore closed, and the projection exists and is unique. By the characterization of Theorem 15.22, it is enough to check that the candidate $z = \sum_{k=1}^n \langle x, u_k \rangle u_k$ satisfies: (i) $z \in M$, (ii) $x - z \perp M$.

(i) is clear: z is a linear combination of the u_k .

(ii) Since M is spanned by $u_1, \dots, u_n$, it is enough that $\langle x - z, u_j \rangle = 0$ for each j (orthogonality extends to linear combinations by bilinearity). We compute:

$$\langle z, u_j \rangle = \sum_{k=1}^n \langle x, u_k \rangle \langle u_k, u_j \rangle = \langle x, u_j \rangle,$$

because $\langle u_k, u_j \rangle = 0$ if $k \neq j$ and $= 1$ if $k = j$ (orthonormality): in the sum only the term $k = j$ survives. Then

$$\langle x - z, u_j \rangle = \langle x, u_j \rangle - \langle z, u_j \rangle = 0. \quad \checkmark$$

By uniqueness, $P_M x = z$.

Distance. By the orthogonal decomposition, $x - z \perp z$, and Pythagoras gives $\|x\|^2 = \|z\|^2 + \|x - z\|^2$. Moreover, by Pythagoras applied to the orthogonal summands of z :

$$\|z\|^2 = \sum_{k=1}^n \langle x, u_k \rangle^2 \|u_k\|^2 = \sum_{k=1}^n \langle x, u_k \rangle^2.$$

Since $\text{dist}(x, M) = \|x - z\|$:

$$\text{dist}(x, M)^2 = \|x\|^2 - \sum_{k=1}^n \langle x, u_k \rangle^2.$$

Interpretation of Bessel. The Bessel inequality $\sum_k \langle x, u_k \rangle^2 \leq \|x\|^2$ is exactly the statement $\text{dist}(x, M)^2 \geq 0$: *the energy captured by the Fourier coordinates never exceeds the total energy*, and the gap is precisely the squared error of the best approximation in M . In regression terms: the explained sum does not exceed the total sum, and the difference is the residual sum: Bessel is “ $R^2 \leq 1$ ” in abstract form. $\square$

Chapter 16

The Riesz representation theorem

[P4]

Week 14 of the course. Bibliography: Moroşanu [2], Ch. 4; Alabiso and Weiss [1], Ch. 5. This chapter uses freely the material on Hilbert spaces (inner product, orthogonality, orthogonal projection and the decomposition $H = M \oplus M^\perp$).

What is this used for in Microeconomics?

Low to medium priority for Microeconomics, but with one connection worth knowing: general equilibrium under uncertainty and Arrow–Debreu markets (Mas-Colell, Whinston and Green, ch. 19). There, the finite dimensional version of Riesz is exactly the logic of *state prices*: every linear price functional on contingent payoffs is represented as an inner product against a vector of state prices (or, after normalizing, against a stochastic discount factor), as developed in the economics box of this chapter. The theorem in general Hilbert spaces and Lax–Milgram are mathematical background: useful in macro, finance and partial differential equations, not in day to day micro.

16.1 Motivation [P4]

Throughout this chapter, H denotes a real Hilbert space, that is, a vector space with an inner product $\langle \cdot, \cdot \rangle$ that is complete with respect to the induced norm $\|x\| = \sqrt{\langle x, x \rangle}$.

The central question is: *what does a “linear and continuous measurement” on H look like?* A linear measurement is a continuous linear functional, that is, a function $f: H \rightarrow \mathbb{R}$ that is linear and continuous. Examples of linear measurements in mathematics and in economics:

- In $\mathbb{R}^n$: “the cost of the bundle x at prices p ”, that is, $x \mapsto p_1x_1 + \cdots + p_nx_n$.
- In a function space: “the average value of u ”, that is, $u \mapsto \int_0^1 u(t) dt$.
- In finance: “today’s price of the random payoff x ”, that is, a price functional $\pi(x)$.

The Riesz representation theorem gives a strikingly simple answer: in a Hilbert space, *every* continuous linear measurement is obtained by “taking the inner product against a fixed vector”. There are no other continuous linear functionals. That is, the space of all continuous linear

measurements on H (the dual of H) is essentially a copy of H itself.

Intuition

Think of $\mathbb{R}^2$ with the dot product. A linear functional f assigns a number to each vector, and its level curves $\{x : f(x) = c\}$ are parallel lines. The most natural way to produce such a pattern of parallel lines is to fix a direction y and measure “how far x advances in the direction y ”: that is exactly $x \mapsto \langle x, y \rangle$. Riesz says that in a Hilbert space *there is no other way*: every continuous linear measurement is a “weighted projection” against a fixed vector y , which is also unique and perpendicular to the level curves.

16.2 Review: continuous linear functionals and the dual space [P4]

Definition 16.1 (Continuous linear functional and dual space). Let $(H, \langle \cdot, \cdot \rangle)$ be a real Hilbert space.

1. A **linear functional** is a function $f : H \rightarrow \mathbb{R}$ such that $f(\alpha x + \beta z) = \alpha f(x) + \beta f(z)$ for all $x, z \in H$ and $\alpha, \beta \in \mathbb{R}$.
2. f is **bounded** if there is $M \geq 0$ such that $|f(x)| \leq M \|x\|$ for every $x \in H$. In that case its **norm** is

$$\|f\| = \sup_{x \neq 0} \frac{|f(x)|}{\|x\|} = \sup_{\|x\| \leq 1} |f(x)|.$$

3. The **dual space** H^* is the set of all *continuous* linear functionals on H , with the norm above.

Remark 16.2. Recall a fact proved in the chapter on linear operators: for a linear map between normed spaces, *continuous* $\iff$ *continuous at 0* $\iff$ *bounded*. That is why we use “continuous” and “bounded” as synonyms for linear functionals. The norm $\|f\|$ measures “the largest amplification factor” of the functional: how large $|f(x)|$ can be per unit of length of x .

Example 16.3 (Every fixed vector generates a functional). Fix $y \in H$ and define $f_y : H \rightarrow \mathbb{R}$ by

$$f_y(x) = \langle x, y \rangle.$$

We claim that $f_y \in H^*$ and that $\|f_y\| = \|y\|$.

Linearity: immediate from linearity of the inner product in its first entry: $f_y(\alpha x + \beta z) = \langle \alpha x + \beta z, y \rangle = \alpha \langle x, y \rangle + \beta \langle z, y \rangle = \alpha f_y(x) + \beta f_y(z)$.

Boundedness: by the Cauchy–Schwarz inequality,

$$|f_y(x)| = |\langle x, y \rangle| \leq \|x\| \|y\| \quad \text{for every } x \in H,$$

so f_y is bounded with $\|f_y\| \leq \|y\|$.

The norm is exactly $\|y\|$: if $y = 0$ this is obvious. If $y \neq 0$, evaluate at y itself (the “best possible case” for the Cauchy–Schwarz inequality, which is an equality when the vectors are

parallel):

$$|f_y(y)| = \langle y, y \rangle = \|y\|^2 = \|y\| \cdot \|y\|,$$

so the ratio $|f_y(y)| / \|y\| = \|y\|$ is attained, and therefore $\|f_y\| \geq \|y\|$. Combining: $\|f_y\| = \|y\|$.

Example 16.3 says that the map $y \mapsto f_y$ puts a copy of H inside H^* without distorting norms. The deep question is the converse: are *all* elements of H^* of this form? Riesz answers yes.

16.2.1 The finite dimensional case as a warm up

Proposition 16.4 (In $\mathbb{R}^n$ every linear functional is an inner product). *Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be linear (in finite dimension every linear functional is automatically continuous). Then there is a unique $a \in \mathbb{R}^n$ such that $f(x) = \langle x, a \rangle$ for every $x \in \mathbb{R}^n$, where $\langle x, a \rangle = \sum_{i=1}^n x_i a_i$ is the usual dot product.*

Proof. Step 1 (building the candidate). Let $\{e_1, \dots, e_n\}$ be the canonical basis of $\mathbb{R}^n$ (e_i has a 1 in position i and zeros elsewhere). We define the vector of “coefficients” of the functional:

$$a = (f(e_1), f(e_2), \dots, f(e_n)) \in \mathbb{R}^n.$$

The idea: a linear functional is completely determined by what it does on a basis, and those n numbers are precisely the coordinates of a .

Step 2 (verification). Every $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ is written as $x = \sum_{i=1}^n x_i e_i$. Applying f and using linearity (which lets you pull sums and scalars out):

$$f(x) = f\left(\sum_{i=1}^n x_i e_i\right) = \sum_{i=1}^n x_i f(e_i) = \sum_{i=1}^n x_i a_i = \langle x, a \rangle.$$

Step 3 (uniqueness). If also $f(x) = \langle x, b \rangle$ for every x , then $\langle x, a - b \rangle = 0$ for every $x \in \mathbb{R}^n$. Taking in particular $x = a - b$ we get $\|a - b\|^2 = 0$, and since the norm vanishes only at the zero vector, $a = b$. $\square$

Remark 16.5. In finite dimension the proof was “mechanical” thanks to the canonical basis. In infinite dimension we cannot list a basis so easily, and there are also *non continuous* linear functionals (pathological ones). The magic of the Riesz theorem is that *completeness* of H and *continuity* of f are enough to reproduce the result, using geometry (orthogonality) instead of coordinates.

16.3 Geometric preparation [P4]

We need two results from the chapter on Hilbert spaces. We restate them precisely because they are the exact pieces of the proof.

Theorem 16.6 (Orthogonal decomposition, reminder). *Let H be a Hilbert space and $M \subseteq H$ a closed vector subspace. Then every $x \in H$ is written in a unique way as*

$$x = m + p, \quad m \in M, \quad p \in M^\perp,$$

where $M^\perp = \{p \in H : \langle p, m \rangle = 0 \ \forall m \in M\}$ is the orthogonal complement of M . In symbols: $H = M \oplus M^\perp$. In particular, if $M \neq H$, then $M^\perp \neq \{0\}$.

The last sentence is the one we will use: *if a closed subspace is not the whole space, there is some nonzero vector perpendicular to it*. Why? If $M \neq H$, take $x \in H \setminus M$; its component $p \in M^\perp$ in the decomposition $x = m + p$ cannot be 0 (if it were, $x = m \in M$, a contradiction).

Lemma 16.7 (The kernel of a continuous functional is closed). *Let $f \in H^*$. Then its kernel*

$$N = \text{Ker } f = \{x \in H : f(x) = 0\}$$

is a closed vector subspace of H .

Proof. It is a subspace: if $x, z \in N$ and $\alpha, \beta \in \mathbb{R}$, then by linearity $f(\alpha x + \beta z) = \alpha f(x) + \beta f(z) = 0$, so $\alpha x + \beta z \in N$. Also $f(0) = 0$, so $0 \in N$ and $N \neq \emptyset$.

It is closed: we use the sequential criterion for closedness. Let (x_k) be a sequence in N with $x_k \rightarrow x$ in H . We must show that $x \in N$. Since f is continuous, the convergence $x_k \rightarrow x$ implies $f(x_k) \rightarrow f(x)$. But $f(x_k) = 0$ for every k , so the sequence of images is the constant sequence 0, whose limit is 0. By uniqueness of limits in $\mathbb{R}$, $f(x) = 0$, that is, $x \in N$.

Another way to say it: $N = f^{-1}(\{0\})$ is the preimage of the closed set $\{0\}$ under the continuous function f , and continuous preimages of closed sets are closed. $\square$

Intuition

Geometrically, $\text{Ker } f$ is the “hyperplane through the origin” where the functional vanishes: in $\mathbb{R}^3$ with $f(x) = \langle x, a \rangle$, the kernel is the plane perpendicular to a through 0. The proof of Riesz exploits this in reverse: first it finds the hyperplane N , and then it recovers the perpendicular direction a as the “normal vector” to N . Completeness of H guarantees that this normal direction really exists inside the space.

16.4 The Riesz representation theorem [P4]

Theorem 16.8 (Riesz representation). *Let H be a real Hilbert space and $f \in H^*$ a continuous linear functional. Then there is a unique $y \in H$ such that*

$$f(x) = \langle x, y \rangle \quad \text{for every } x \in H.$$

Moreover, $\|f\| = \|y\|$.

Proof. We organize the proof in five steps: existence (steps 1 to 3), uniqueness (step 4) and equality of norms (step 5).

Step 0 (trivial case). If $f = 0$ (the identically zero functional), take $y = 0$: indeed, $\langle x, 0 \rangle = 0 = f(x)$ for every x . From now on we assume $f \neq 0$, that is, there is some x_0 with $f(x_0) \neq 0$.

Step 1 (the kernel and its perpendicular). Let $N = \text{Ker } f$. By Lemma 16.7, N is a closed subspace of H . Since $f \neq 0$, there is $x_0 \notin N$, so $N \neq H$. By Theorem 16.6 (orthogonal decomposition, which requires N to be *closed* and H to be *complete*: this is where the Hilbert

hypothesis enters), the orthogonal complement is not trivial: $N^\perp \neq \{0\}$. So we choose $z \in N^\perp$ with $\|z\| = 1$ (take any $w \in N^\perp$, $w \neq 0$, and normalize: $z = w / \|w\|$).

Note that $f(z) \neq 0$: if $f(z) = 0$, then z would be in N and also in $N^\perp$, so $\langle z, z \rangle = 0$ (being perpendicular to itself), that is $z = 0$, contradicting $\|z\| = 1$.

Step 2 (the auxiliary vector trick). Here is the heart of the proof. For an arbitrary $x \in H$, form the vector

$$u = f(x)z - f(z)x.$$

Why this vector? Because it is *designed* to fall in the kernel: applying f and using linearity (note that $f(x)$ and $f(z)$ are *numbers*, so they come out as scalars),

$$f(u) = f(f(x)z - f(z)x) = f(x)f(z) - f(z)f(x) = 0.$$

Therefore $u \in N$. Since $z \in N^\perp$, z is perpendicular to u :

$$0 = \langle u, z \rangle = \langle f(x)z - f(z)x, z \rangle = f(x)\langle z, z \rangle - f(z)\langle x, z \rangle.$$

(We used linearity of the inner product in the first entry.) Since $\langle z, z \rangle = \|z\|^2 = 1$, we solve:

$$f(x) = f(z)\langle x, z \rangle = \langle x, f(z)z \rangle.$$

(In the last step we moved the scalar $f(z)$ into the second entry of the inner product, which is allowed in the real case by bilinearity.)

Step 3 (the representative). We define

$$\boxed{y = f(z)z}.$$

The computation of Step 2 shows exactly that $f(x) = \langle x, y \rangle$ for *every* $x \in H$, which is the existence we wanted. Note how concrete the result is: y points in the direction normal to the kernel (z), scaled by the value $f(z)$.

Step 4 (uniqueness). Suppose $y_1, y_2 \in H$ both satisfy

$$f(x) = \langle x, y_1 \rangle = \langle x, y_2 \rangle \quad \text{for every } x \in H.$$

Subtracting, $\langle x, y_1 - y_2 \rangle = 0$ for every $x \in H$. This equality holds for *any* x ; we choose the most informative x possible: $x = y_1 - y_2$. Then

$$\|y_1 - y_2\|^2 = \langle y_1 - y_2, y_1 - y_2 \rangle = 0,$$

and since the norm is positive definite (only the zero vector has zero norm), we conclude $y_1 = y_2$.

Step 5 (equality of norms). We prove the two inequalities.

($\leq$) For every x with $\|x\| \leq 1$, Cauchy-Schwarz gives

$$|f(x)| = |\langle x, y \rangle| \leq \|x\| \|y\| \leq \|y\|.$$

Taking the supremum over such x : $\|f\| \leq \|y\|$.

($\geq$) If $y = 0$ then $f = 0$ and there is nothing to prove. If $y \neq 0$, we evaluate f at the unit vector $x = y/\|y\|$:

$$f\left(\frac{y}{\|y\|}\right) = \left\langle \frac{y}{\|y\|}, y \right\rangle = \frac{\|y\|^2}{\|y\|} = \|y\|.$$

Since $\|f\|$ is the supremum of $|f(x)|$ over unit vectors and we have just exhibited a unit vector where f equals $\|y\|$, it follows that $\|f\| \geq \|y\|$.

Combining both: $\|f\| = \|y\|$. □

Remark 16.9 (Where each hypothesis was used). It is worth auditing the proof:

- *Continuity of f* : so that $N = \text{Ker } f$ is closed (Lemma 16.7).
- *Completeness of H* : for the orthogonal decomposition $H = N \oplus N^\perp$ (the projection theorem requires completeness; without it, the “normal direction” could be “missing” from the space).
- *Inner product*: to speak of orthogonality and to use Cauchy–Schwarz.

If H were only an inner product space that is *not complete*, the theorem can fail: there are continuous functionals with no representative inside the space (the “natural” representative lives in the completion).

Intuition

[Geometric reading of the theorem] The level sets $\{x : f(x) = c\}$ of a nonzero continuous linear functional are hyperplanes parallel to each other, all parallel to the kernel N . The theorem says that there is a vector y , perpendicular to all those hyperplanes, such that f measures “the advance in the direction of y ”, weighted by $\|y\|$. The larger $\|y\|$, the more “sensitive” the measurement: the level hyperplanes are packed closer together. This explains the equality $\|f\| = \|y\|$.

Careful (common mistake)

Three frequent mistakes in exams about this theorem:

1. **Forgetting continuity.** The statement is false for non continuous linear functionals (they exist in infinite dimension, via Hamel bases). If the exam asks for the statement, write “*continuous* (or bounded) linear functional”.
2. **Forgetting completeness.** “Inner product space” is not enough; you need *Hilbert*. In the proof, completeness enters through the projection theorem and the orthogonal decomposition.
3. **Not justifying that $\text{Ker } f$ is closed.** It is not a free fact: it is exactly the continuity of f that guarantees it. In a complete proof, this lemma must be stated and proved (two lines with the sequential criterion).

Exam strategy

Theorem 16.8 is an absolute classic of the final exam, in two formats:

1. “State and prove the Riesz representation theorem.” Learn the proof as a story in 5 steps: (0) case $f = 0$; (1) $N = \text{Ker } f$ closed and $N^\perp \neq \{0\}$, take z of norm one; (2) the auxiliary vector $u = f(x)z - f(z)x \in N$ and the computation with $\langle u, z \rangle = 0$; (3) $y = f(z)z$; (4) uniqueness by evaluating at $y_1 - y_2$; (5) norms with Cauchy–Schwarz and with $x = y/\|y\|$. If you remember the auxiliary vector of step 2, the rest reconstructs itself.
2. “Find the Riesz representative of the concrete functional...” (see Examples 16.10 and 16.11). Method: propose y such that $\langle x, y \rangle$ reproduces $f(x)$, and verify.

16.5 Examples of computing representatives [P4]

Example 16.10 (Representative in ℓ^2). Let $H = \ell^2$ be the space of real sequences $x = (x_1, x_2, \dots)$ with $\sum_k x_k^2 < \infty$, with inner product $\langle x, y \rangle = \sum_k x_k y_k$. Consider the functional

$$f(x) = x_1 + \frac{1}{2}x_2.$$

It is linear: obvious, it is a combination of coordinates. *It is continuous:* by Cauchy–Schwarz in $\mathbb{R}^2$ (or directly), $|f(x)| \leq \sqrt{1 + \frac{1}{4}} \sqrt{x_1^2 + x_2^2} \leq \frac{\sqrt{5}}{2} \|x\|_{\ell^2}$. *Representative:* we look for $y \in \ell^2$ with $\sum_k x_k y_k = x_1 + \frac{1}{2}x_2$ for every x . Matching coefficients,

$$y = (1, \frac{1}{2}, 0, 0, \dots).$$

Verification: $\langle x, y \rangle = x_1 \cdot 1 + x_2 \cdot \frac{1}{2} = f(x)$. Also $\|f\| = \|y\| = \sqrt{1 + \frac{1}{4}} = \frac{\sqrt{5}}{2}$, which confirms the continuity bound we had found (and shows that it was the best possible one).

Example 16.11 (Representative of an integral functional). Let $H = L^2[0, 1]$ (square integrable functions on $[0, 1]$, with $\langle u, v \rangle = \int_0^1 u(t)v(t) dt$; it is enough to think of continuous functions and accept that the complete space is L^2). Consider

$$f(u) = \int_0^1 t u(t) dt.$$

It is linear: the integral is linear. *It is continuous:* by Cauchy–Schwarz for integrals,

$$|f(u)| \leq \left(\int_0^1 t^2 dt \right)^{1/2} \left(\int_0^1 u(t)^2 dt \right)^{1/2} = \frac{1}{\sqrt{3}} \|u\|.$$

Representative: the way f is written already gives it away: $f(u) = \langle u, g \rangle$ with $g(t) = t$. By uniqueness, the Riesz representative is $g(t) = t$, and $\|f\| = \|g\| = \left(\int_0^1 t^2 dt \right)^{1/2} = 1/\sqrt{3}$.

Example 16.12 (A functional where Riesz fails without completeness). Let $V \subset \ell^2$ be the subspace of sequences with *finitely* many nonzero terms (an inner product space, but *not* complete),

and

$$f(x) = \sum_{k=1}^{\infty} \frac{x_k}{k}$$

(the sum is finite for each $x \in V$, so f is well defined; it is linear and continuous because $|f(x)| \leq \|(1/k)\|_{\ell^2} \|x\|$ by Cauchy–Schwarz). The only candidate representative would be $y = (1, \frac{1}{2}, \frac{1}{3}, \dots)$, which is in ℓ^2 but *not* in V (it has infinitely many nonzero terms). Inside V there is no representative: the conclusion of Riesz fails because V is not complete. The representative “lives” in the completion ℓ^2 of V .

16.6 The isometry between H and its dual [P5]

Theorem 16.13 (Self duality of Hilbert spaces). *The map $\Phi: H \rightarrow H^*$ given by $\Phi(y) = f_y$, where $f_y(x) = \langle x, y \rangle$, is a linear bijection that preserves norms (an isometry): $\|\Phi(y)\| = \|y\|$ for every y . In this sense, H^* “is” H : Hilbert spaces are self dual.*

Proof. Linearity of Φ : we must check that $\Phi(\alpha y_1 + \beta y_2) = \alpha \Phi(y_1) + \beta \Phi(y_2)$ as functionals, that is, evaluated at any x :

$$f_{\alpha y_1 + \beta y_2}(x) = \langle x, \alpha y_1 + \beta y_2 \rangle = \alpha \langle x, y_1 \rangle + \beta \langle x, y_2 \rangle = \alpha f_{y_1}(x) + \beta f_{y_2}(x),$$

using bilinearity of the real inner product.

It preserves norms: this is Example 16.3: $\|f_y\| = \|y\|$.

Injectivity: if $\Phi(y_1) = \Phi(y_2)$, then $f_{y_1 - y_2}$ is the zero functional, and $0 = \|f_{y_1 - y_2}\| = \|y_1 - y_2\|$ forces $y_1 = y_2$. (Every linear isometry is injective by this argument.)

Surjectivity: this is exactly the existence part of Theorem 16.8: every $f \in H^*$ is f_y for some y . □

Remark 16.14 (Complex case and reflexivity). 1. If H is a *complex* Hilbert space, everything works the same except for one detail: Φ turns out to be *conjugate linear*: $\Phi(\alpha y) = \bar{\alpha} \Phi(y)$. We only mention this; in the course we work with real spaces.

2. An important consequence (as background only): applying Riesz twice you see that H is *reflexive*: the double dual H^{**} is canonically identified with H . Reflexive spaces have very good properties for optimization (existence of minima of coercive convex functionals).

Remark 16.15 (In general Banach spaces, Riesz fails). In a Banach space without an inner product, the dual is usually a *different* space, larger or simply different. A background example: the dual of $C[a, b]$ (with the supremum norm) is not made of functions but of *signed measures* (the Riesz–Markov theorem): every continuous functional is $u \mapsto \int u d\mu$. And the dual of ℓ^1 is ℓ^∞ , which is not ℓ^1 . Self duality is a privilege of Hilbert spaces only, and it is the technical reason why “everything is easier” in them.

16.7 Two star applications [P5]

16.7.1 The Lax–Milgram theorem

Theorem 16.16 (Lax–Milgram, statement). *Let H be a Hilbert space and $B: H \times H \rightarrow \mathbb{R}$ a bilinear form that is:*

1. continuous (bounded): *there is $C > 0$ with $|B(u, v)| \leq C \|u\| \|v\|$ for all u, v ;*
2. coercive: *there is $\alpha > 0$ with $B(u, u) \geq \alpha \|u\|^2$ for every u .*

Then, for each $f \in H^$ there is a unique $u^* \in H$ such that*

$$B(u^*, v) = f(v) \quad \text{for every } v \in H,$$

and moreover $\|u^\| \leq \|f\|/\alpha$.*

Intuition

Lax–Milgram is “Riesz for skewed inner products”. If B were symmetric, B would be an inner product equivalent to the original one and the theorem would be exactly Riesz with respect to that new product. The merit of Lax–Milgram is that it does *not* require symmetry: it is enough that B is continuous (it does not amplify too much) and coercive (it does not degenerate: it “sees” each vector with strength at least α). Sketch of the proof: (i) for each fixed u , $v \mapsto B(u, v)$ is a continuous functional, so by Riesz there is Au with $B(u, v) = \langle Au, v \rangle$; (ii) A turns out to be a continuous linear operator, and coercivity makes it injective, with closed range, and surjective; (iii) the equation $B(u, v) = f(v) \forall v$ is equivalent to $Au = y_f$ where y_f is the Riesz representative of f , which has the unique solution $u^* = A^{-1}y_f$. This theorem is the basis of the variational formulation of differential equations (finite element methods).

16.7.2 Existence of the adjoint operator

Theorem 16.17 (Existence of the adjoint operator). *Let $T: H \rightarrow H$ be a continuous linear operator on a Hilbert space H . There is a unique continuous linear operator $T^*: H \rightarrow H$, called the adjoint of T , such that*

$$\langle Tx, z \rangle = \langle x, T^*z \rangle \quad \text{for all } x, z \in H.$$

Construction (detailed sketch). Fix $z \in H$ and consider the map

$$g_z: H \rightarrow \mathbb{R}, \quad g_z(x) = \langle Tx, z \rangle.$$

g_z is linear: composition of the linear T with the inner product, which is linear in its first entry.

g_z is continuous: by Cauchy–Schwarz and boundedness of T ,

$$|g_z(x)| = |\langle Tx, z \rangle| \leq \|Tx\| \|z\| \leq \|T\| \|x\| \cdot \|z\|.$$

So $g_z \in H^*$, and by Theorem 16.8 there is a unique vector, which we name T^*z , such that $g_z(x) = \langle x, T^*z \rangle$ for every x . This defines a function $z \mapsto T^*z$ that satisfies the desired identity by construction. Linearity of T^* is checked using uniqueness of the Riesz representative (the representative of $g_{\alpha z_1 + \beta z_2}$ must be $\alpha T^*z_1 + \beta T^*z_2$ because both represent the same functional), and boundedness follows from $\|T^*z\| = \|g_z\| \leq \|T\| \|z\|$. $\square$

Remark 16.18. In $\mathbb{R}^n$ with the dot product, the adjoint of the matrix A is the transpose A^T : $\langle Ax, z \rangle = (Ax)^T z = x^T A^T z = \langle x, A^T z \rangle$. Theorem 16.17 generalizes transposition to infinite dimension, and it *would not exist* without Riesz.

Link with economics

[State prices and the stochastic discount factor] This is perhaps the most famous appearance of Riesz in financial economics (asset pricing, cf. Hansen–Richard type models).

The setting. There is uncertainty: tomorrow a state of nature s occurs. An *asset* is its random payoff x , a random variable. We model the payoff space as $H = L^2(\Omega, \mathbb{P})$: payoffs with finite variance, with inner product $\langle x, z \rangle = E[xz]$ (the expectation of the product). It is a Hilbert space.

The price functional. The market assigns to each payoff x a price today $\pi(x)$. Two economic assumptions: (i) *no arbitrage and the law of one price* $\Rightarrow \pi$ is *linear* (the price of a portfolio is the combination of prices: $\pi(\alpha x + \beta z) = \alpha\pi(x) + \beta\pi(z)$); (ii) a condition ruling out price explosion for small payoffs $\Rightarrow \pi$ is *continuous* in L^2 .

Riesz enters. Since $\pi \in H^*$, Theorem 16.8 guarantees that there is a unique random variable $m \in L^2$ such that

$$\pi(x) = \langle x, m \rangle = E[mx] \quad \text{for every payoff } x.$$

That m is the famous *stochastic discount factor* (SDF) or *pricing kernel*. The formula $p = E[mx]$ is the “fundamental theorem of asset pricing”: *every* linear and continuous price system consists of weighting payoffs state by state ($m(s)$ = price per unit of probability of state s , the “state prices”) and averaging. In consumption models, $m = \beta u'(c_{t+1})/u'(c_t)$: states where consumption is scarce (high marginal utility) have high state prices. A fine point: Riesz gives *existence and uniqueness* of m in the payoff space, but it does *not* guarantee $m > 0$; positivity requires the extra assumption of strict no arbitrage.

Second appearance: regression and conditional expectation. The orthogonal projection in L^2 onto the subspace of functions of the available information is the *conditional expectation* $E[\cdot | \mathcal{G}]$, and the projection onto the span of the regressors is *linear regression*: the OLS coefficients are the solution of the best approximation problem in a Hilbert space. Riesz duality is the reason why “covariance against a factor” ($E[mx]$, an inner product) and “price functional” (a linear measurement) are equivalent languages: it is the same object seen from H or from H^* .

Chapter summary

- H^* = continuous linear functionals on H with $\|f\| = \sup_{\|x\| \leq 1} |f(x)|$.
- **Riesz:** for every $f \in H^*$ there is a unique $y \in H$ with $f(x) = \langle x, y \rangle \forall x$, and $\|f\| = \|y\|$.
- Proof in 5 steps: $f = 0$ trivial; $N = \text{Ker } f$ closed (continuity) and $N^\perp \neq \{0\}$ (completeness plus orthogonal decomposition); $z \in N^\perp$ of norm one; the auxiliary vector $u = f(x)z - f(z)x \in N$ gives $y = f(z)z$; uniqueness with $x = y_1 - y_2$; norms with Cauchy–Schwarz and $x = y/\|y\|$.
- $y \mapsto \langle \cdot, y \rangle$ is a bijective linear isometry $H \cong H^*$: Hilbert spaces are self dual; in general Banach spaces this fails.
- Applications: Lax–Milgram (coercive bilinear forms), existence of the adjoint T^* , and in economics: $p = E[mx]$ (the SDF is the Riesz representative of the price functional).

16.8 Exam style exercises [P4]

Exercise 16.19. State precisely the Riesz representation theorem and prove the *uniqueness* part and the equality $\|f\| = \|y\|$.

Solution. *Statement.* Let H be a real Hilbert space and $f: H \rightarrow \mathbb{R}$ a continuous linear functional. Then there is a unique $y \in H$ such that $f(x) = \langle x, y \rangle$ for every $x \in H$; moreover $\|f\| = \|y\|$.

Uniqueness. If $\langle x, y_1 \rangle = \langle x, y_2 \rangle$ for every x , then $\langle x, y_1 - y_2 \rangle = 0$ for every x . In particular, with $x = y_1 - y_2$: $\|y_1 - y_2\|^2 = 0$, so $y_1 = y_2$ by positive definiteness of the norm.

Norms. ($\leq$): for $\|x\| \leq 1$, Cauchy–Schwarz gives $|f(x)| = |\langle x, y \rangle| \leq \|x\| \|y\| \leq \|y\|$; taking the supremum, $\|f\| \leq \|y\|$. ($\geq$): if $y \neq 0$, the unit vector $x_0 = y/\|y\|$ satisfies $f(x_0) = \langle y, y \rangle / \|y\| = \|y\|$, so $\|f\| \geq |f(x_0)| = \|y\|$. If $y = 0$, then $f = 0$ and both norms are 0. Therefore $\|f\| = \|y\|$. $\square$

Exercise 16.20. Prove that if $f: H \rightarrow \mathbb{R}$ is a continuous linear functional, then $\text{Ker } f$ is a closed subspace, and that if $f \neq 0$ then $(\text{Ker } f)^\perp \neq \{0\}$. (You may use the orthogonal decomposition theorem.)

Solution. *Subspace:* if $x, z \in \text{Ker } f$ and $\alpha, \beta \in \mathbb{R}$, then $f(\alpha x + \beta z) = \alpha f(x) + \beta f(z) = 0$, so $\alpha x + \beta z \in \text{Ker } f$.

Closed: let $x_k \in \text{Ker } f$ with $x_k \rightarrow x$. By continuity of f , $f(x) = \lim_k f(x_k) = \lim_k 0 = 0$, so $x \in \text{Ker } f$. (Equivalently: $\text{Ker } f = f^{-1}(\{0\})$ is the continuous preimage of a closed set.)

Nontrivial complement: since $f \neq 0$, there is x_0 with $f(x_0) \neq 0$, in particular $x_0 \notin \text{Ker } f$, so $N := \text{Ker } f \subsetneq H$. By the orthogonal decomposition theorem (applicable because N is closed and H is complete), $x_0 = m + p$ with $m \in N$, $p \in N^\perp$. If $p = 0$ we would have $x_0 = m \in N$, a contradiction. So $p \neq 0$ and $N^\perp \neq \{0\}$. $\square$

Exercise 16.21. In $H = \ell^2$, consider the functional

$$f(x) = \sum_{k=1}^{\infty} \frac{x_k}{2^k}.$$

(a) Prove that f is well defined, linear and continuous. (b) Find its Riesz representative y and compute $\|f\|$.

Solution. (a) *Well defined and bounded:* let $y = (1/2, 1/4, 1/8, \dots)$, that is $y_k = 2^{-k}$. Since $\sum_k y_k^2 = \sum_k 4^{-k} = \frac{1/4}{1-1/4} = \frac{1}{3} < \infty$, we have $y \in \ell^2$ with $\|y\| = 1/\sqrt{3}$. By the Cauchy–Schwarz inequality in ℓ^2 , the series $\sum_k x_k y_k$ converges absolutely for each $x \in \ell^2$ and

$$|f(x)| = \left| \sum_k x_k y_k \right| \leq \|x\| \|y\| = \frac{1}{\sqrt{3}} \|x\|,$$

which gives well definedness and boundedness (hence continuity). Linearity is the linearity of convergent series.

(b) By construction $f(x) = \langle x, y \rangle$ with $y = (2^{-k})_{k \geq 1}$, and by uniqueness in the Riesz theorem this is *the* representative. Finally

$$\|f\| = \|y\| = \left(\sum_{k=1}^{\infty} 4^{-k} \right)^{1/2} = \sqrt{\frac{1}{3}} = \frac{1}{\sqrt{3}}.$$

□

Exercise 16.22. Let H be a Hilbert space and $T: H \rightarrow H$ linear and continuous. Using the Riesz theorem, prove that there is a unique operator $T^*: H \rightarrow H$ with $\langle Tx, z \rangle = \langle x, T^*z \rangle$ for all x, z , and prove that $\|T^*z\| \leq \|T\| \|z\|$ for every z .

Solution. Fix $z \in H$ and define $g_z(x) = \langle Tx, z \rangle$. It is linear in x (composition of the linear T with the first entry of the inner product) and bounded:

$$|g_z(x)| \leq \|Tx\| \|z\| \leq \|T\| \|x\| \|z\|,$$

using Cauchy–Schwarz and the definition of the operator norm. So $g_z \in H^*$ with $\|g_z\| \leq \|T\| \|z\|$. By Riesz there is a unique vector, denoted T^*z , with $g_z(x) = \langle x, T^*z \rangle$ for every x ; this is exactly $\langle Tx, z \rangle = \langle x, T^*z \rangle$. Uniqueness of the representative gives uniqueness of T^*z for each z (and therefore of the operator T^*). Finally, by the equality of norms in Riesz,

$$\|T^*z\| = \|g_z\| \leq \|T\| \|z\|.$$

(Comment: from here you get $\|T^*\| \leq \|T\|$, and by symmetry of the argument, applying it to T^* , you obtain the equality $\|T^*\| = \|T\|$.) □

Exercise 16.23. (Economic application.) In a market without arbitrage, the payoff space is $H = L^2(\Omega, \mathbb{P})$ with $\langle x, z \rangle = E[xz]$, and the price functional $\pi: H \rightarrow \mathbb{R}$ is linear and continuous. It is known that the riskless asset (constant payoff $x \equiv 1$) has price $\pi(1) = 0.95$. (a) Justify the existence of a unique variable $m \in L^2$ with $\pi(x) = E[mx]$ for every x . (b) Prove that

$E[m] = 0.95$. (c) Show that for every payoff x : $\pi(x) = E[m] E[x] + \text{Cov}(m, x)$, and interpret it economically.

Solution. (a) $(H, \langle \cdot, \cdot \rangle)$ with $\langle x, z \rangle = E[xz]$ is a Hilbert space (it is L^2 of a probability space). Since π is a continuous linear functional on H , the Riesz representation theorem guarantees that there is a unique $m \in L^2$ such that $\pi(x) = \langle x, m \rangle = E[mx]$ for every $x \in H$.

(b) Applying the formula to the constant payoff $x \equiv 1$:

$$0.95 = \pi(1) = E[m \cdot 1] = E[m].$$

(c) By the definition of covariance, $\text{Cov}(m, x) = E[mx] - E[m]E[x]$, so

$$\pi(x) = E[mx] = E[m] E[x] + \text{Cov}(m, x) = 0.95 E[x] + \text{Cov}(m, x).$$

Interpretation: the price of an asset is its expected payoff discounted at the bond price ($0.95 E[x]$) plus a risk adjustment given by the covariance with the factor m . Assets that pay a lot when m is high (“bad” states, with high marginal utility of consumption) have positive covariance with m and are *more expensive*: they work as insurance. Assets that covary negatively with m (they pay in “good” states) are worth *less* than their discounted expected payoff: they require a risk premium. All of this is, mathematically, a reading of the Riesz representative of the price functional. $\square$

Chapter 17

Differentiation in normed spaces and necessary conditions for optimality

[P1]

Week 15 of the course. Bibliography: Zorich II [6], Ch. 10; Moroşanu [2]; Ok [4]. This chapter uses normed spaces, bounded linear operators and their norm.

What is this used for in Microeconomics?

Top priority: almost a whole Microeconomics course is solved with first order conditions (FOC), and this chapter is the rigorous justification of that recipe. The consumer and the firm are “differentiate and set equal to zero” with constraints, and from there come the envelope theorem, Shephard’s lemma and the Slutsky equation (Mas-Colell, Whinston and Green, chs. 2 to 5). Optimal contracts with asymmetric information are characterized with the FOC of the principal subject to participation and incentive constraints (ch. 14); the “first order approach” in moral hazard is exactly about when the FOC of the agent describes its behavior well. And in mechanism design and auctions the Myerson condition is obtained by the same envelope argument (ch. 23). The sections on optimization over convex sets explain corner solutions and active constraints; the one on convexity explains when the FOC is also sufficient.

17.1 Motivation [P1]

In calculus you learned to optimize functions of one or several variables: set the derivative (or the gradient) equal to zero. But many problems in economics ask you to optimize over *function spaces*:

- A planner chooses a whole *consumption path* $c: [0, T] \rightarrow \mathbb{R}$ to maximize $\int_0^T e^{-\rho t} u(c(t)) dt$.
- In optimal growth you choose the capital path $k(\cdot)$.
- In contracts, you choose a whole *scheme* $w(\cdot)$ that assigns a payment to each observed outcome.

The “choice variable” is a point of an infinite dimensional normed space (for example $C[0, T]$). To apply there the recipe “differentiate and set equal to zero” we need: (i) a notion of derivative for functions between normed spaces, and (ii) a theorem saying that at an optimum the derivative vanishes. This chapter builds both. The correct notion of derivative is the Fréchet one, and the conceptual key is to stop thinking “derivative = number or matrix” and start thinking “**derivative = best linear approximation**”.

17.2 Review: the derivative as a linear approximation [P1]

In one variable, $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$. Let us rewrite this definition in a form that does generalize. Saying that the quotient tends to $f'(x)$ is exactly saying that the “error”

$$r(h) = f(x+h) - f(x) - f'(x)h$$

satisfies $\frac{|r(h)|}{|h|} \rightarrow 0$ as $h \rightarrow 0$. That is:

$$f(x+h) = \underbrace{f(x)}_{\text{current value}} + \underbrace{f'(x)h}_{\text{linear part in } h} + \underbrace{r(h)}_{\text{error } o(h)}, \quad \frac{|r(h)|}{|h|} \xrightarrow{h \rightarrow 0} 0.$$

Remark 17.1 (The little-o notation). We write $r(h) = o(h)$ (read “ r is little-o of h ”) to say that $r(h)/|h| \rightarrow 0$ as $h \rightarrow 0$: the error not only goes to zero, it goes to zero *faster than* h , so compared with the linear term it is negligible near $h = 0$. Examples: $h^2 = o(h)$ (since $h^2/|h| = |h| \rightarrow 0$) and $|h|^{3/2} = o(h)$; but $3h \neq o(h)$ (the quotient is the constant 3) and $|h| \neq o(h)$. Working rules: sums of $o(h)$ are $o(h)$; a constant times $o(h)$ is $o(h)$; and if L is bounded linear, $L(o(h))$ is $o(h)$.

Intuition

“ f is differentiable at x ” means: if you zoom in enough near x , the graph of f is indistinguishable from a line (in general, from a linear map). The derivative *is* that linear map, not the number: in one variable the number $f'(x)$ is just a convenient way to describe the linear map $h \mapsto f'(x)h$. This rereading is what travels to any normed space: you only have to require that the error is small *in norm* compared with the norm of the increment.

17.3 The Fréchet derivative [P2]

Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed spaces, $U \subseteq X$ open, and $f: U \rightarrow Y$.

Definition 17.2 (Fréchet derivative). We say that f is **Fréchet differentiable** at $x \in U$ if there is a *bounded* linear operator $L: X \rightarrow Y$ such that

$$\lim_{h \rightarrow 0} \frac{\|f(x+h) - f(x) - L(h)\|_Y}{\|h\|_X} = 0.$$

In that case we write $Df(x) = L$ (the notation $f'(x)$ is also used) and we say that $Df(x)$ is *the*

Fréchet derivative of f at x . Equivalently, with little-o notation:

$$f(x+h) = f(x) + Df(x)h + r(h), \quad \|r(h)\|_Y = o(\|h\|_X).$$

Let us read the definition symbol by symbol:

- h is an *increment*: a vector of X , small in norm, that can point in *any* direction. Since U is open, for $\|h\|$ small the point $x+h$ is still in U and the expression makes sense.
- $L(h)$ is the *linear prediction* of the change in f . We ask L to be linear *and bounded* (= continuous): in infinite dimension there are non continuous linear maps and we do not want them as derivatives (an approximation that amplifies without control does not approximate anything useful).
- The quotient compares the error of the prediction, $\|f(x+h) - f(x) - L(h)\|_Y$, with the size of the increment, $\|h\|_X$. The limit being 0 says: the error is negligible *compared with h itself*, uniformly in the direction of h (the limit is as $\|h\| \rightarrow 0$ with no restriction on direction; this will be the crucial difference with Gâteaux).

Proposition 17.3 (Uniqueness of the derivative). *If L_1 and L_2 are bounded linear operators both satisfying Definition 17.2 at the point x , then $L_1 = L_2$.*

Proof. Let $M = L_1 - L_2$, linear and bounded. Subtracting the two decompositions

$$f(x+h) - f(x) = L_1h + r_1(h) = L_2h + r_2(h), \quad r_i(h) = o(\|h\|),$$

we get $Mh = r_2(h) - r_1(h)$, and by the triangle inequality $\|Mh\| \leq \|r_1(h)\| + \|r_2(h)\| = o(\|h\|)$.

We want to conclude $M = 0$, that is $Mv = 0$ for each fixed $v \in X$, $v \neq 0$. The trick: approach 0 *along the direction v* . Take $h = tv$ with $t > 0$, $t \rightarrow 0$. By linearity $M(tv) = tMv$, so

$$\frac{\|M(tv)\|}{\|tv\|} = \frac{t\|Mv\|}{t\|v\|} = \frac{\|Mv\|}{\|v\|},$$

which is *constant* in t . But by the above that quotient must tend to 0 as $t \rightarrow 0$. A constant that tends to 0 is 0: $\|Mv\| = 0$, that is $Mv = 0$. Since v was arbitrary, $M = 0$ and $L_1 = L_2$. $\square$

Proposition 17.4 (Differentiable $\Rightarrow$ continuous). *If f is Fréchet differentiable at x , then f is continuous at x .*

Proof. We write $f(x+h) - f(x) = Df(x)h + r(h)$ and estimate in norm:

$$\|f(x+h) - f(x)\| \leq \|Df(x)h\| + \|r(h)\| \leq \|Df(x)\| \|h\| + \|r(h)\|,$$

where we used the definition of the operator norm ($\|Lh\| \leq \|L\| \|h\|$; here it is essential that the derivative is a *bounded* operator). As $h \rightarrow 0$: the first term goes to 0 because $\|Df(x)\|$ is a finite constant, and the second goes to 0 because $r(h) = o(\|h\|)$ implies in particular $r(h) \rightarrow 0$. Therefore $f(x+h) \rightarrow f(x)$, which is continuity at x . $\square$

17.3.1 Fundamental examples

Example 17.5 (The derivative of a linear map is the map itself). Let $T: X \rightarrow Y$ be linear and bounded, and $f(x) = Tx$. Then f is differentiable at every x with $Df(x) = T$ (the same T at all points). Indeed, the natural candidate is $L = T$, and the error is

$$r(h) = f(x+h) - f(x) - Th = T(x+h) - Tx - Th = Tx + Th - Tx - Th = 0,$$

using linearity of T . The error is identically zero, and 0 is trivially $o(\|h\|)$. Lesson: a linear map is its own linear approximation, just as in calculus the derivative of ax is a .

Example 17.6 (Quadratic form in a Hilbert space). Let H be a Hilbert space, $A: H \rightarrow H$ bounded linear, and $f(x) = \langle Ax, x \rangle$. Let us compute $Df(x)$ from the definition, which is the method you should master: expand $f(x+h)$, separate what is constant, what is linear in h , and what remains must be $o(\|h\|)$.

$$\begin{aligned} f(x+h) &= \langle A(x+h), x+h \rangle = \langle Ax + Ah, x+h \rangle \\ &= \langle Ax, x \rangle + \langle Ax, h \rangle + \langle Ah, x \rangle + \langle Ah, h \rangle. \end{aligned}$$

(We used bilinearity: the inner product “distributes”.) The first term is $f(x)$. The two middle ones are *linear in h* ; to write them as a single functional of h we use the adjoint ($\langle Ah, x \rangle = \langle h, A^*x \rangle = \langle A^*x, h \rangle$ in the real case):

$$\langle Ax, h \rangle + \langle Ah, x \rangle = \langle Ax + A^*x, h \rangle.$$

The last term is the candidate error: by Cauchy-Schwarz and boundedness of A ,

$$|\langle Ah, h \rangle| \leq \|Ah\| \|h\| \leq \|A\| \|h\|^2,$$

and $\|A\| \|h\|^2 / \|h\| = \|A\| \|h\| \rightarrow 0$: it is $o(\|h\|)$. Conclusion:

$$Df(x)h = \langle (A + A^*)x, h \rangle, \quad \text{that is, the “gradient” is } \nabla f(x) = (A + A^*)x.$$

In $\mathbb{R}^n$ with A a matrix: $\nabla f(x) = (A + A^T)x$, and if A is symmetric, $\nabla f(x) = 2Ax$, the familiar formula from econometrics (the derivative of $x^T Ax$).

Example 17.7 (An integral functional). Let $X = C[a, b]$ with the supremum norm $\|u\|_\infty = \max_{t \in [a, b]} |u(t)|$, and

$$J(u) = \int_a^b u(t)^2 dt.$$

We compute with the definition. For $u, h \in C[a, b]$:

$$J(u+h) = \int_a^b (u(t) + h(t))^2 dt = \int_a^b u^2 dt + 2 \int_a^b u(t)h(t) dt + \int_a^b h(t)^2 dt.$$

The first term is $J(u)$; the second is linear in h ; the third is the error. Candidate:

$$DJ(u)h = 2 \int_a^b u(t) h(t) dt.$$

It is linear in h : linearity of the integral. *It is bounded:* $|DJ(u)h| \leq 2 \int_a^b |u| |h| dt \leq 2(b-a) \|u\|_\infty \|h\|_\infty$. *The error is $o(\|h\|_\infty)$:*

$$\left| \int_a^b h(t)^2 dt \right| \leq (b-a) \|h\|_\infty^2, \quad \frac{(b-a) \|h\|_\infty^2}{\|h\|_\infty} = (b-a) \|h\|_\infty \rightarrow 0.$$

Conclusion: J is Fréchet differentiable at every u , with $DJ(u)h = 2 \int_a^b u h dt$. Note how the derivative at the point u is a *linear functional* on the space of functions; in $L^2[a, b]$, by Riesz, its representative is the function $2u$: the “gradient” of J at u is $\nabla J(u) = 2u$, a perfect analogue of $\frac{d}{dx} x^2 = 2x$.

Example 17.8 (The case $\mathbb{R}^n \rightarrow \mathbb{R}^m$: Jacobian and gradient). If $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is Fréchet differentiable at x , the derivative $Df(x): \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear map between finite dimensional spaces, so it is given by an $m \times n$ matrix: it is exactly the *Jacobian matrix* $[\partial f_i / \partial x_j(x)]_{ij}$. When $m = 1$, $Df(x)$ is a linear functional on $\mathbb{R}^n$ and (as we saw in the chapter on the Riesz representation theorem, or simply by linear algebra) it is represented as a dot product against a vector: that vector is the *gradient* $\nabla f(x)$, and $Df(x)h = \langle \nabla f(x), h \rangle$. A classic warning from multivariable calculus that still applies: the existence of the partial derivatives is *not* enough for Fréchet differentiability; *continuous* partials do guarantee it.

17.4 The Gâteaux derivative and its relation with Fréchet [P3]

Definition 17.9 (Gâteaux derivative). $f: U \rightarrow Y$ is **Gâteaux differentiable** at $x \in U$ if there is a bounded linear operator $L: X \rightarrow Y$ such that for each fixed direction $v \in X$:

$$\lim_{t \rightarrow 0} \frac{f(x + tv) - f(x)}{t} = Lv \quad (t \in \mathbb{R}, t \neq 0).$$

We write $L = D_G f(x)$. The limit on the left side (when it exists for a given v) is called the *directional derivative* in the direction v .

Intuition

Gâteaux looks at the function *ray by ray*: fix a direction v and differentiate the one variable function $t \mapsto f(x + tv)$ at $t = 0$. Fréchet asks for much more: that *one and the same* linear approximation works with uniformly small error *in all directions at once*, even along curves that change direction as they approach x . That is why Fréchet $\Rightarrow$ Gâteaux but not the other way around: a function can behave well on every straight line and still behave badly on a parabola approaching x .

Proposition 17.10 (Fréchet $\Rightarrow$ Gâteaux, with the same derivative). *If f is Fréchet differentiable at x , then it is Gâteaux differentiable at x and $D_G f(x) = Df(x)$.*

Proof. Let $L = Df(x)$ and fix $v \in X$, $v \neq 0$ (the case $v = 0$ is trivial: both sides are 0). We take the particular increment $h = tv$ with $t \rightarrow 0$. The Fréchet definition applied to this h gives

$$f(x + tv) - f(x) = L(tv) + r(tv) = tLv + r(tv), \quad \|r(tv)\| = o(\|tv\|).$$

Dividing by t :

$$\frac{f(x + tv) - f(x)}{t} - Lv = \frac{r(tv)}{t}.$$

We estimate the norm of the remainder:

$$\left\| \frac{r(tv)}{t} \right\| = \frac{\|r(tv)\|}{|t|} = \frac{\|r(tv)\|}{\|tv\|} \cdot \|v\| \xrightarrow{t \rightarrow 0} 0 \cdot \|v\| = 0,$$

because $\|tv\| = |t| \|v\| \rightarrow 0$ and the quotient $\|r(h)\| / \|h\| \rightarrow 0$ by Fréchet. So the directional limit exists and equals Lv for every direction v . $\square$

Example 17.11 (Gâteaux without Fréchet: the classic counterexample). Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be given by

$$f(x_1, x_2) = \begin{cases} \frac{x_1^3}{x_1^2 + x_2^2} & (x_1, x_2) \neq (0, 0), \\ 0 & (x_1, x_2) = (0, 0). \end{cases}$$

At the origin, all directional derivatives exist: for $v = (v_1, v_2) \neq 0$,

$$\frac{f(tv) - f(0)}{t} = \frac{1}{t} \cdot \frac{t^3 v_1^3}{t^2(v_1^2 + v_2^2)} = \frac{v_1^3}{v_1^2 + v_2^2},$$

which does not even depend on t : the limit exists. However, the map $v \mapsto v_1^3/(v_1^2 + v_2^2)$ is *not linear* in v (for example, at $v = (1, 0)$ it equals 1 and at $v = (1, 1)$ it equals 1/2, which is not what linearity would give by combining $(1, 0)$ and $(0, 1)$, where it equals 0). Since every Fréchet derivative produces directional derivatives that are *linear* in the direction (Proposition 17.10), f cannot be Fréchet differentiable at 0; in fact here even Gâteaux differentiability fails in the strict sense of Definition 17.9 (which requires the limit to be *linear* in v). This example shows that “all directional derivatives exist” is strictly weaker than being differentiable.

Careful (common mistake)

In exams: “ f has directional derivatives in all directions” $\nRightarrow$ “ f is differentiable”, and not even $\Rightarrow$ “ f is continuous”. The only valid implication chain is: Fréchet $\Rightarrow$ Gâteaux $\Rightarrow$ directional derivatives. To *refute* Fréchet differentiability, the short path is almost always: compute directional derivatives and see that they do not depend linearly on the direction, or exhibit discontinuity.

17.5 Calculus rules [P2]

Proposition 17.12 (Linearity of the derivative). *If $f, g: U \rightarrow Y$ are differentiable at x and $\alpha, \beta \in \mathbb{R}$, then $\alpha f + \beta g$ is differentiable at x and*

$$D(\alpha f + \beta g)(x) = \alpha Df(x) + \beta Dg(x).$$

Proof. Adding the expansions $f(x+h) = f(x) + Df(x)h + r_f(h)$ and $g(x+h) = g(x) + Dg(x)h + r_g(h)$ weighted by α, β :

$$(\alpha f + \beta g)(x+h) = (\alpha f + \beta g)(x) + (\alpha Df(x) + \beta Dg(x))h + \underbrace{\alpha r_f(h) + \beta r_g(h)}_{=: r(h)}.$$

The operator $\alpha Df(x) + \beta Dg(x)$ is linear and bounded (a combination of bounded ones), and $\|r(h)\| \leq |\alpha| \|r_f(h)\| + |\beta| \|r_g(h)\| = o(\|h\|)$ (a finite combination of $o(\|h\|)$ terms). By uniqueness, that is the derivative. $\square$

Theorem 17.13 (Chain rule). *Let X, Y, Z be normed spaces, $f: U \subseteq X \rightarrow Y$ differentiable at x , and $g: V \subseteq Y \rightarrow Z$ differentiable at $y = f(x)$ (with $f(U) \subseteq V$). Then $g \circ f$ is differentiable at x and*

$$D(g \circ f)(x) = Dg(f(x)) \circ Df(x),$$

the composition of the two linear operators.

Proof. We write the two available expansions. For f at x :

$$f(x+h) = f(x) + Df(x)h + r_1(h), \quad \|r_1(h)\| = o(\|h\|).$$

For g at $y = f(x)$, with increment $k \in Y$:

$$g(y+k) = g(y) + Dg(y)k + r_2(k), \quad \|r_2(k)\| = o(\|k\|).$$

We apply g at the point $f(x+h)$, taking as increment $k = k(h) := Df(x)h + r_1(h)$ (so that $f(x+h) = y + k$):

$$\begin{aligned} (g \circ f)(x+h) &= g(y+k) = g(y) + Dg(y)[Df(x)h + r_1(h)] + r_2(k) \\ &= (g \circ f)(x) + \underbrace{Dg(y) Df(x) h}_{\text{linear part}} + \underbrace{Dg(y) r_1(h) + r_2(k)}_{=: R(h)}. \end{aligned}$$

The candidate $Dg(y) \circ Df(x)$ is linear and bounded (composition of bounded ones, with $\|Dg(y)Df(x)\| \leq \|Dg(y)\| \|Df(x)\|$). What remains is the delicate part: showing that $R(h) = o(\|h\|)$. We do it in pieces.

First term: $\|Dg(y) r_1(h)\| \leq \|Dg(y)\| \|r_1(h)\| = o(\|h\|)$, because a constant times $o(\|h\|)$ is $o(\|h\|)$.

Second term: first we bound the size of k :

$$\|k\| \leq \|Df(x)\| \|h\| + \|r_1(h)\|.$$

Since $\|r_1(h)\| = o(\|h\|)$, for $\|h\|$ small we have $\|r_1(h)\| \leq \|h\|$, and then $\|k\| \leq (\|Df(x)\| + 1)\|h\| =: C\|h\|$. In particular $h \rightarrow 0 \Rightarrow k \rightarrow 0$. Now, given $\varepsilon > 0$, since $r_2(k) = o(\|k\|)$ there is $\delta > 0$ such that $\|k\| < \delta \Rightarrow \|r_2(k)\| \leq \varepsilon\|k\|$. For $\|h\|$ small enough we have $\|k\| \leq C\|h\| < \delta$, and then

$$\|r_2(k)\| \leq \varepsilon\|k\| \leq \varepsilon C\|h\|.$$

That is, $\|r_2(k)\|/\|h\| \leq C\varepsilon$ for small h ; since ε is arbitrary, $\|r_2(k(h))\| = o(\|h\|)$.

Adding up, $R(h) = o(\|h\|)$, and by uniqueness of the derivative (Proposition 17.3), $D(g \circ f)(x) = Dg(f(x)) \circ Df(x)$. $\square$

Remark 17.14. In $\mathbb{R}^n$ the chain rule is “the product of Jacobians”: composition of linear maps corresponds to multiplying matrices. The proof above shows why the abstract statement is the natural one: there is not a single coordinate in the whole argument.

17.6 Necessary conditions for optimality [P1]

Now we harvest. Let $(X, \|\cdot\|)$ be normed, $U \subseteq X$ open, $f: U \rightarrow \mathbb{R}$ (a real valued functional: we optimize numbers).

Definition 17.15 (Local extrema). $x^* \in U$ is a **local minimum** of f if there is $\delta > 0$ such that

$$f(x^*) \leq f(x) \quad \text{for every } x \in U \text{ with } \|x - x^*\| < \delta.$$

It is a **local maximum** if the inequality is $f(x^*) \geq f(x)$. In both cases we say that x^* is a *local extremum*. If the inequality holds for every $x \in U$, the extremum is *global*.

Theorem 17.16 (Generalized Fermat: first order necessary condition). *Let $f: U \rightarrow \mathbb{R}$ be Fréchet differentiable at $x^* \in U$, with U open. If x^* is a local extremum of f , then*

$$Df(x^*) = 0$$

(the zero linear functional: $Df(x^*)v = 0$ for every direction $v \in X$).

Proof. Suppose x^* is a local minimum (the maximum case is identical applying the result to $-f$, whose derivative is $-Df$).

Step 1: reduce to one variable. Fix an arbitrary direction $v \in X$, $v \neq 0$, and consider the real function of a real variable

$$\varphi(t) = f(x^* + tv),$$

defined for small $|t|$ (say $|t| < \delta/\|v\|$, so that $x^* + tv$ stays inside the ball where x^* is a minimum and inside the open set U ; here we use that U is open). Since x^* is a local minimum,

$$\varphi(t) = f(x^* + tv) \geq f(x^*) = \varphi(0) \quad \text{for small } |t|:$$

$t = 0$ is a local minimum of φ on an open interval around 0.

Step 2: φ is differentiable at 0. By Proposition 17.10 (Fréchet $\Rightarrow$ directional),

$$\varphi'(0) = \lim_{t \rightarrow 0} \frac{f(x^* + tv) - f(x^*)}{t} = Df(x^*)v.$$

Step 3: classical Fermat in $\mathbb{R}$. For a real function with an interior local minimum at 0 and differentiable at 0, $\varphi'(0) = 0$. Recall the argument (it is often asked): for small $t > 0$, $\frac{\varphi(t) - \varphi(0)}{t} \geq 0$ (numerator ≥ 0 , denominator > 0), and letting $t \downarrow 0$ the limit gives $\varphi'(0) \geq 0$. For $t < 0$, the quotient is ≤ 0 (numerator ≥ 0 , denominator < 0), and letting $t \uparrow 0$, $\varphi'(0) \leq 0$. So $\varphi'(0) = 0$.

Conclusion. $Df(x^*)v = \varphi'(0) = 0$ for every direction v , that is, $Df(x^*)$ is the zero functional. $\square$

Careful (common mistake)

Two hypotheses that students forget and that break the theorem:

1. **Interiority** (U open, x^* interior). At a boundary point the argument of Step 3 fails because we cannot move in both directions ($t > 0$ and $t < 0$). Example: $f(x) = x$ on $[0, 1]$ has a minimum at $x^* = 0$ with $f'(0) = 1 \neq 0$.
2. **It is a necessary condition, not a sufficient one.** $Df(x^*) = 0$ does not imply an extremum: $f(x) = x^3$ on $\mathbb{R}$ has $f'(0) = 0$ with no extremum at 0 (a saddle or inflection point). Sufficiency requires second order conditions or convexity.

17.6.1 Convex constraints: the variational inequality

In constrained optimization the optimal point is usually on the boundary of the feasible set, and “derivative equal to zero” is replaced by an *inequality*.

Definition 17.17 (Convex set). $K \subseteq X$ is **convex** if for all $x, z \in K$ and $\lambda \in [0, 1]$ you have $\lambda x + (1 - \lambda)z \in K$: the segment between two points of K stays inside K .

Theorem 17.18 (Variational inequality). *Let $K \subseteq X$ be convex and f Fréchet differentiable at $x^* \in K$. If x^* minimizes f over K (locally), then*

$$Df(x^*)(z - x^*) \geq 0 \quad \text{for every } z \in K.$$

Proof. Fix $z \in K$. By convexity, for $t \in [0, 1]$ the point

$$x_t = x^* + t(z - x^*) = (1 - t)x^* + tz$$

belongs to K (it is a point of the segment from x^* to z). Since x^* is a (local) minimum over K and $x_t \rightarrow x^*$ as $t \rightarrow 0$, for small $t > 0$ we have $f(x_t) \geq f(x^*)$, that is

$$\frac{f(x^* + t(z - x^*)) - f(x^*)}{t} \geq 0.$$

(Numerator ≥ 0 and $t > 0$.) Letting $t \downarrow 0$, the left side tends to the directional derivative in the direction $v = z - x^*$, which by Fréchet differentiability is $Df(x^*)(z - x^*)$. Limits preserve non strict inequalities, so $Df(x^*)(z - x^*) \geq 0$. $\square$

Intuition

$Df(x^*)(z - x^*)$ is the rate of change of f when you start walking from x^* toward z . The inequality says: *from the optimum, moving toward any feasible point cannot decrease f to first order*. If x^* is interior to K , for each direction v both v and $-v$ are “directions toward feasible points”, and the two inequalities $Df(x^*)v \geq 0$, $-Df(x^*)v \geq 0$ force $Df(x^*)v = 0$: we recover Fermat. The variational inequality is the parent form of the Kuhn–Tucker conditions.

17.6.2 Convexity: when the necessary becomes sufficient

Definition 17.19 (Convex function). Let $K \subseteq X$ be convex. $f: K \rightarrow \mathbb{R}$ is **convex** if for all $x, z \in K$ and $\lambda \in [0, 1]$:

$$f(\lambda x + (1 - \lambda)z) \leq \lambda f(x) + (1 - \lambda)f(z).$$

(f is *concave* if $-f$ is convex; to maximize concave functions, everything that follows applies with reversed inequalities.)

Lemma 17.20 (The graph lies above the tangent). *If f is convex on the open convex set U and Fréchet differentiable at $x \in U$, then*

$$f(z) \geq f(x) + Df(x)(z - x) \quad \text{for every } z \in U.$$

Proof. By convexity, for $t \in (0, 1]$:

$$f(x + t(z - x)) = f((1 - t)x + tz) \leq (1 - t)f(x) + tf(z).$$

We subtract $f(x)$ from both sides and divide by $t > 0$:

$$\frac{f(x + t(z - x)) - f(x)}{t} \leq f(z) - f(x).$$

The left side tends, as $t \downarrow 0$, to the directional derivative $Df(x)(z - x)$ (by differentiability). Taking the limit, $Df(x)(z - x) \leq f(z) - f(x)$, which is the desired inequality rearranged. $\square$

Theorem 17.21 (Sufficiency of the FOC under convexity). *Let f be convex and Fréchet differentiable on the open convex set U .*

1. *If $Df(x^*) = 0$, then x^* is a global minimum of f on U .*
2. *More generally, if $K \subseteq U$ is convex and $Df(x^*)(z - x^*) \geq 0$ for every $z \in K$, then x^* minimizes f over K .*

Proof. (1) By Lemma 17.20, for any $z \in U$:

$$f(z) \geq f(x^*) + Df(x^*)(z - x^*) = f(x^*) + 0 = f(x^*),$$

which is exactly global minimality.

(2) The same: $f(z) \geq f(x^*) + Df(x^*)(z - x^*) \geq f(x^*)$ for every $z \in K$, using the variational hypothesis in the second inequality. $\square$

Remark 17.22 (Second order conditions in $\mathbb{R}^n$). In $\mathbb{R}^n$, if f is twice continuously differentiable and $\nabla f(x^*) = 0$: if the Hessian matrix $Hf(x^*) = [\partial^2 f / \partial x_i \partial x_j]$ is positive definite, x^* is a strict local minimum; if x^* is a local minimum, then $Hf(x^*)$ is positive semidefinite (second order necessary condition). We only state this: the machinery of second Fréchet derivatives exists (they are bilinear maps) but goes beyond this course.

Exam strategy

Typical questions in this part of the course:

1. *Compute Df from the definition* for a concrete functional (quadratic and integral ones like Examples 17.6 and 17.7). Method in 3 steps: expand $f(x + h)$; identify the term that is linear in h (check that it is linear and bounded); bound the rest and show that it is $o(\|h\|)$ by dividing by $\|h\|$.
2. *Prove generalized Fermat* (Theorem 17.16): the reduction $\varphi(t) = f(x^* + tv)$ is the idea to remember; the rest is one variable calculus.
3. *State precisely* uniqueness of the derivative, differentiable $\Rightarrow$ continuous, or Fréchet $\Rightarrow$ Gâteaux, and prove one of them.
4. Distinguish with counterexamples: directional derivatives without differentiability (Example 17.11), FOC without an extremum (x^3), boundary extremum without FOC ($f(x) = x$ on $[0, 1]$).

17.7 Application: calculus of variations and Euler–Lagrange [P3]

We close with the historical example that motivated the whole theory: optimizing an integral functional over a space of functions. Let

$$J(u) = \int_a^b F(t, u(t), u'(t)) dt,$$

where $F(t, u, p)$ is a smooth function of three variables, and we minimize over the functions $u \in C^1[a, b]$ with fixed endpoints $u(a) = A$, $u(b) = B$.

Computation (heuristic but careful) of the directional derivative. The admissible directions are functions $h \in C^1[a, b]$ with $h(a) = h(b) = 0$ (so as not to move the fixed endpoints: $u + th$ must still satisfy the boundary conditions). We differentiate under the integral sign the function $\tau \mapsto J(u + \tau h)$ at $\tau = 0$, using the chain rule in the integrand:

$$DJ(u)h = \left. \frac{d}{d\tau} \right|_{\tau=0} J(u + \tau h) = \int_a^b \left[F_u(t, u, u') h(t) + F_p(t, u, u') h'(t) \right] dt,$$

where F_u, F_p are the partial derivatives of F with respect to its second and third arguments.

The term with h' is in the way; we turn it into a term with h by *integrating by parts* (here the boundary conditions $h(a) = h(b) = 0$ kill the boundary term):

$$\int_a^b F_p h' dt = [F_p h]_a^b - \int_a^b \frac{d}{dt}(F_p) h dt = - \int_a^b \frac{d}{dt}(F_p) h dt.$$

Therefore

$$DJ(u)h = \int_a^b \left[F_u - \frac{d}{dt} F_p \right] h(t) dt.$$

If u^* is a minimum (with fixed endpoints), Theorem 17.16 (in its directional version on the subspace of admissible h) requires $DJ(u^*)h = 0$ for *every* admissible h . A classical lemma (the fundamental lemma of the calculus of variations: if $\int_a^b g h = 0$ for every continuous h vanishing at the endpoints, then $g \equiv 0$) lets you conclude the **Euler–Lagrange equation**:

$$F_u(t, u^*(t), (u^*)'(t)) = \frac{d}{dt} F_p(t, u^*(t), (u^*)'(t)) \quad t \in [a, b].$$

The “first order condition” in a function space is a *differential equation*: this is the conceptual jump of the chapter.

Link with economics

[FOC in function spaces: the Euler equations]

- *Intertemporal consumption.* The consumer chooses the path $c(\cdot)$ that maximizes $\int_0^T e^{-\rho t} u(c(t)) dt$ subject to the wealth dynamics $a'(t) = ra(t) + y(t) - c(t)$. Substituting $c = ra + y - a'$, the problem is to minimize (the negative of) a functional $J(a) = \int F(t, a, a') dt$: the Euler–Lagrange equation produces the *consumption Euler equation* $\frac{u''(c)c'}{u'(c)} = \rho - r$, that is, the Ramsey–Keynes rule: consumption grows when $r > \rho$. The “FOC” of dynamic macro *is* a Fermat theorem in a normed space.
- *Discrete and infinite.* In sequence problems $\{c_t\}_{t=0}^\infty \in \ell^\infty$, the Fréchet derivative of the objective $\sum \beta^t u(c_t)$ coordinate by coordinate gives the Euler equations $u'(c_t) = \beta(1+r)u'(c_{t+1})$, plus a transversality condition (the analogue of the boundary term from integration by parts, which in infinite horizon does not vanish for free).
- *Convexity = decreasing returns.* If u is concave (decreasing marginal utility) and the constraints are convex, the objective is concave, and Theorem 17.21 guarantees that the FOC is not only necessary but *sufficient* and the optimum is *global*. That is why in economics, with concavity, “solving the FOC” closes the problem; without concavity the FOC only gives candidates.
- *Variational inequality.* With nonnegativity constraints ($c_t \geq 0$, a convex set), the condition $Df(x^*)(z - x^*) \geq 0$ of Theorem 17.18 is exactly the form of the Kuhn–Tucker conditions with corner solutions: if $c_t^* = 0$, marginal utility can be lower than the shadow price (the inequality is not tight).

Chapter summary

- **Fréchet:** $Df(x)$ is the bounded linear operator with $f(x+h) = f(x) + Df(x)h + o(\|h\|)$. It is unique; its existence implies continuity.
- **Gâteaux:** derivatives along rays, $\lim_t [f(x+tv) - f(x)]/t = Lv$. Fréchet $\Rightarrow$ Gâteaux; the converse is false (counterexample $x_1^3/(x_1^2 + x_2^2)$).
- **Calculus:** linear $\Rightarrow Df = T$ constant; quadratic $\langle Ax, x \rangle \Rightarrow \nabla f = (A + A^*)x$; in $\mathbb{R}^n$, $Df(x)$ = Jacobian or gradient; chain rule $D(g \circ f)(x) = Dg(f(x)) \circ Df(x)$.
- **Generalized Fermat:** interior local extremum plus differentiable $\Rightarrow Df(x^*) = 0$. Proof: reduce to $\varphi(t) = f(x^* + tv)$.
- Over a convex set K : minimum $\Rightarrow Df(x^*)(z - x^*) \geq 0 \forall z \in K$ (variational inequality).
- f convex: the graph lies above the tangent, and the FOC becomes *sufficient* for a *global* minimum.
- Euler–Lagrange: the FOC of $J(u) = \int F(t, u, u')$ is $F_u = \frac{d}{dt} F_p$; in economics, the consumption Euler equations.

17.8 Exam style exercises [P1]

Exercise 17.23. Let H be a real Hilbert space, $b \in H$ fixed and $f(x) = \|x\|^2 - 2\langle b, x \rangle$. (a) Compute $Df(x)$ from the definition. (b) Find all critical points and classify them using convexity.

Solution. (a) We expand using $\|x+h\|^2 = \langle x+h, x+h \rangle = \|x\|^2 + 2\langle x, h \rangle + \|h\|^2$ (bilinearity and symmetry of the inner product):

$$f(x+h) = \|x\|^2 + 2\langle x, h \rangle + \|h\|^2 - 2\langle b, x \rangle - 2\langle b, h \rangle = f(x) + 2\langle x-b, h \rangle + \|h\|^2.$$

The term $h \mapsto 2\langle x-b, h \rangle$ is linear and bounded (Cauchy–Schwarz: $|2\langle x-b, h \rangle| \leq 2\|x-b\|\|h\|$). The rest is $\|h\|^2$, and $\|h\|^2/\|h\| = \|h\| \rightarrow 0$, so it is $o(\|h\|)$. By uniqueness of the derivative,

$$Df(x)h = 2\langle x-b, h \rangle, \quad \text{that is } \nabla f(x) = 2(x-b).$$

(b) $Df(x) = 0$ as a functional $\iff \langle x-b, h \rangle = 0$ for every $h \iff$ (taking $h = x-b$) $x = b$. Unique critical point: $x^* = b$. *Classification:* f is convex, since $f(x) = \|x-b\|^2 - \|b\|^2$ (completing the square: $\|x-b\|^2 = \|x\|^2 - 2\langle b, x \rangle + \|b\|^2$) and $x \mapsto \|x-b\|^2$ is convex (composition of the norm, which is convex, with the square, which is convex and increasing on $[0, \infty)$; or directly by the parallelogram identity). By sufficiency under convexity (Theorem 17.21), $x^* = b$ is a *global* minimum, with value $f(b) = -\|b\|^2$. $\square$

Exercise 17.24. Prove that if $f: U \rightarrow Y$ is Fréchet differentiable at x , then (a) the derivative is unique, and (b) f is continuous at x .

Solution. (a) Let L_1, L_2 be two derivatives. Subtracting the expansions, $\|(L_1 - L_2)h\| \leq \|r_1(h)\| + \|r_2(h)\| = o(\|h\|)$. Fix $v \neq 0$ and put $h = tv$, $t \rightarrow 0^+$. By linearity,

$$\frac{\|(L_1 - L_2)(tv)\|}{\|tv\|} = \frac{\|(L_1 - L_2)v\|}{\|v\|},$$

constant in t ; but it must tend to 0, so it equals 0: $(L_1 - L_2)v = 0$ for every v , that is, $L_1 = L_2$.

(b) From $f(x+h) - f(x) = Df(x)h + r(h)$:

$$\|f(x+h) - f(x)\| \leq \|Df(x)\| \|h\| + \|r(h)\| \rightarrow 0 \quad (h \rightarrow 0),$$

since $\|Df(x)\| < \infty$ (the derivative is bounded by definition) and $r(h) = o(\|h\|) \Rightarrow r(h) \rightarrow 0$. This is continuity at x . $\square$

Exercise 17.25. Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = \frac{x_1^2 x_2}{x_1^4 + x_2^2}$ if $(x_1, x_2) \neq (0, 0)$ and $f(0, 0) = 0$. (a) Compute the directional derivatives of f at $(0, 0)$ in every direction $v = (v_1, v_2)$. (b) Prove that f is not continuous at $(0, 0)$ (consider the curve $x_2 = x_1^2$) and conclude that f is not Fréchet differentiable at $(0, 0)$.

Solution. (a) For $v = (v_1, v_2)$ with $v_2 \neq 0$:

$$\frac{f(tv_1, tv_2) - 0}{t} = \frac{1}{t} \cdot \frac{t^3 v_1^2 v_2}{t^4 v_1^4 + t^2 v_2^2} = \frac{v_1^2 v_2}{t^2 v_1^4 + v_2^2} \xrightarrow{t \rightarrow 0} \frac{v_1^2 v_2}{v_2^2} = \frac{v_1^2}{v_2}.$$

If $v_2 = 0$ (and $v_1 \neq 0$): $f(tv_1, 0) = 0$, so the directional derivative is 0. Conclusion: all directional derivatives exist at the origin (and equal v_1^2/v_2 or 0).

(b) On the curve $x_2 = x_1^2$, with $x_1 = s \rightarrow 0$:

$$f(s, s^2) = \frac{s^2 \cdot s^2}{s^4 + s^4} = \frac{s^4}{2s^4} = \frac{1}{2} \quad \text{for every } s \neq 0,$$

which does not tend to $f(0, 0) = 0$: f is discontinuous at the origin. Since Fréchet differentiable $\Rightarrow$ continuous (Proposition 17.4), f cannot be Fréchet differentiable at $(0, 0)$, *despite* having all directional derivatives. (Note also that $v \mapsto v_1^2/v_2$ is not even linear.) $\square$

Exercise 17.26. Let U be open in a normed space X and $f: U \rightarrow \mathbb{R}$ Fréchet differentiable. Prove the generalized Fermat theorem: if $x^* \in U$ is a local maximum of f , then $Df(x^*) = 0$.

Solution. Fix a direction $v \in X$, $v \neq 0$. Since U is open and x^* is a local maximum, there is $\varepsilon > 0$ such that for $|t| < \varepsilon$: $x^* + tv \in U$ and $f(x^* + tv) \leq f(x^*)$. Define $\varphi(t) = f(x^* + tv)$ on $(-\varepsilon, \varepsilon)$; then φ has a local maximum at $t = 0$ and is differentiable at 0 with $\varphi'(0) = Df(x^*)v$ (Fréchet implies the directional derivative, with the same linear map: divide the Fréchet expansion with $h = tv$ by t and pass to the limit). For $t > 0$: $\frac{\varphi(t) - \varphi(0)}{t} \leq 0$, so $\varphi'(0) \leq 0$ letting $t \downarrow 0$. For $t < 0$: $\frac{\varphi(t) - \varphi(0)}{t} \geq 0$ (numerator ≤ 0 , denominator < 0), so $\varphi'(0) \geq 0$. Therefore $Df(x^*)v = \varphi'(0) = 0$. Since v is arbitrary, $Df(x^*)$ is the zero functional. $\square$

Exercise 17.27. (Economic application.) A two product monopolist has profit

$$\pi(q_1, q_2) = p_1 q_1 + p_2 q_2 - C(q_1, q_2), \quad p_1 = 10 - q_1, \quad p_2 = 8 - q_2, \quad C(q) = q_1^2 + q_1 q_2 + q_2^2.$$

(a) Write the first order condition using the gradient and solve it. (b) Check by concavity that the point you found is a global maximum on $\mathbb{R}^2$ (ignoring for simplicity the nonnegativity constraints, and checking at the end that they hold).

Solution. (a) The profit is

$$\pi(q_1, q_2) = (10 - q_1)q_1 + (8 - q_2)q_2 - q_1^2 - q_1 q_2 - q_2^2 = 10q_1 + 8q_2 - 2q_1^2 - 2q_2^2 - q_1 q_2.$$

$\mathbb{R}^2$ is open and π is differentiable (a polynomial), so at an interior maximum the Fermat theorem requires $\nabla\pi(q^*) = 0$:

$$\frac{\partial\pi}{\partial q_1} = 10 - 4q_1 - q_2 = 0, \quad \frac{\partial\pi}{\partial q_2} = 8 - 4q_2 - q_1 = 0.$$

From the first one, $q_2 = 10 - 4q_1$; substituting in the second: $8 - 4(10 - 4q_1) - q_1 = 0 \Rightarrow 8 - 40 + 16q_1 - q_1 = 0 \Rightarrow 15q_1 = 32 \Rightarrow q_1^* = \frac{32}{15}$, and $q_2^* = 10 - \frac{128}{15} = \frac{22}{15}$. Both are positive, so the interior solution is admissible.

(b) The Hessian of π is constant:

$$H\pi = \begin{pmatrix} -4 & -1 \\ -1 & -4 \end{pmatrix}.$$

Its leading principal minors: $-4 < 0$ and $\det = 16 - 1 = 15 > 0$, so $H\pi$ is negative definite at every point: π is strictly concave on $\mathbb{R}^2$. For a differentiable concave function, the condition $\nabla\pi(q^*) = 0$ is *sufficient* for a *global* maximum (Theorem 17.21 applied to $-\pi$, which is convex): the graph of a concave function lies below its tangent, so $\pi(q) \leq \pi(q^*) + \langle \nabla\pi(q^*), q - q^* \rangle = \pi(q^*)$ for every q . Therefore $(q_1^*, q_2^*) = (\frac{32}{15}, \frac{22}{15})$ is the global maximum of profit. $\square$

Chapter 18

Semicontinuity, correspondences and the maximum theorem [P1]

Weeks 15–16 of the course (semicontinuity and applications). Bibliography: Ok [4], chapters on semicontinuity and correspondences; Moroşanu [2], Ch. 6. This is the most “economic” chapter of the course: it ends with the Berge maximum theorem, the tool that makes comparative statics rigorous in consumer theory and in dynamic programming.

What is this used for in Microeconomics?

This is the most important chapter of the whole notebook for Microeconomics: its theorems are literally the ones that Mas-Colell, Whinston and Green use in demand theory (ch. 3) and in chapters 15 to 18 on general equilibrium. The budget correspondence, demand and the best response of a game *are* correspondences; the Berge maximum theorem is what guarantees that indirect utility is continuous and that demand behaves well (upper hemicontinuous) when prices and wealth change. The Kakutani fixed point theorem, applied to the best response correspondence, is the standard proof of existence of Nash equilibrium and of its Bayesian version (ch. 8), and the same argument gives the existence of Walrasian equilibrium (ch. 17). Semicontinuity also shows up when you maximize value functions with jumps, as in principal-agent problems where a constraint becomes binding (ch. 14).

18.1 Motivation [P1]

Two uncomfortable facts of economic practice:

1. *Important functions are not always continuous.* Value functions, profit functions with fixed costs, participation indicator functions: they have jumps. But they often keep “the good half” of continuity: the half that guarantees that the maximum (or the minimum) exists. That half is *semicontinuity*.
2. *Feasible sets move with the parameters.* The budget set $B(p, w)$ depends on prices and income; the set of feasible actions of a rival depends on my action. A rule that assigns a *set* to each parameter is a *correspondence*, and you need a notion of “continuity” for sets

that move.

The final prize of the chapter, the **Berge maximum theorem**, combines both things: if the objective is continuous and the feasible set moves continuously, then the optimal value moves continuously and the optimal solutions do not “jump outside”. It is the theorem that answers: does demand depend “well” on prices? is the Bellman value function continuous?

18.2 Limit superior and limit inferior of a sequence [P2]

Before semicontinuity you need to master $\limsup$ and $\liminf$, because the definitions are written with them. We work with *bounded* real sequences (the general case allows the values $\pm\infty$ with the same ideas).

Definition 18.1 (Limit superior and limit inferior). Let (a_n) be a bounded sequence in $\mathbb{R}$. For each n define

$$s_n = \sup_{k \geq n} a_k, \quad i_n = \inf_{k \geq n} a_k,$$

the supremum and the infimum of the “tail” of the sequence from index n on. The sequence (s_n) is decreasing (when you cut the tail, the supremum can only go down or stay) and bounded, so it converges; in the same way (i_n) is increasing and bounded. We define

$$\limsup a_n = \lim_{n \rightarrow \infty} s_n = \inf_n \sup_{k \geq n} a_k, \quad \liminf a_n = \lim_{n \rightarrow \infty} i_n = \sup_n \inf_{k \geq n} a_k.$$

Intuition

$\limsup a_n$ is “the asymptotic ceiling”: the largest value that the sequence keeps approaching infinitely often; $\liminf$ is the asymptotic floor. Unlike the limit, they *always exist* (for bounded sequences). An oscillating sequence has no limit, but it does have a ceiling and a floor. A useful description: $\limsup a_n$ is the *largest* limit of convergent subsequences of (a_n) , and $\liminf$ is the smallest one.

Example 18.2. 1. $a_n = (-1)^n$: the tails always contain 1 and -1 , so $s_n = 1$ and $i_n = -1$ for every n . Then $\limsup = 1$, $\liminf = -1$. There is no limit (the two differ).

2. $a_n = (-1)^n(1 + \frac{1}{n})$: the even subsequence goes to 1 and the odd one to -1 ; $\limsup = 1$, $\liminf = -1$.

3. $a_n = \frac{1}{n}$: $\limsup = \liminf = 0 = \lim a_n$.

Proposition 18.3 (Basic properties). For (a_n) bounded:

1. $\liminf a_n \leq \limsup a_n$.
2. (a_n) converges to $L \iff \liminf a_n = \limsup a_n = L$.
3. If $a_n \leq b_n$ for every n , then $\limsup a_n \leq \limsup b_n$ and $\liminf a_n \leq \liminf b_n$.
4. There is a subsequence $a_{n_k} \rightarrow \limsup a_n$ (and another one going to the $\liminf$).

Proof (sketch of parts 1, 2 and 4). (1) $i_n \leq s_n$ for each n (the infimum of a set does not exceed its supremum); taking limits preserves $\leq$.

(2) ($\Rightarrow$) If $a_n \rightarrow L$, given $\varepsilon > 0$, the whole tail from some N on lives in $(L - \varepsilon, L + \varepsilon)$; then $L - \varepsilon \leq i_n \leq s_n \leq L + \varepsilon$ for $n \geq N$, and both limits stay at distance $\leq \varepsilon$ from L for every ε : they are equal to L . ($\Leftarrow$) If $i_n \rightarrow L$ and $s_n \rightarrow L$, the sandwich $i_n \leq a_n \leq s_n$ forces $a_n \rightarrow L$.

(4) By definition of supremum, for each n there is $k_n \geq n$ with $a_{k_n} > s_n - \frac{1}{n}$; also $a_{k_n} \leq s_{k_n} \leq s_n$. Since $s_n \rightarrow \limsup a_n$, the sandwich gives $a_{k_n} \rightarrow \limsup a_n$, and the indices k_n can be chosen strictly increasing. $\square$

18.3 Semicontinuous functions [P1]

Let (X, d) be a metric space and $f: X \rightarrow \mathbb{R}$.

Definition 18.4 (Semicontinuity, sequential version). 1. f is **upper semicontinuous** (*usc*) at x_0 if for every sequence $x_n \rightarrow x_0$:

$$\limsup_{n \rightarrow \infty} f(x_n) \leq f(x_0).$$

2. f is **lower semicontinuous** (*lsc*) at x_0 if for every sequence $x_n \rightarrow x_0$:

$$\liminf_{n \rightarrow \infty} f(x_n) \geq f(x_0).$$

f is usc (resp. lsc) if it is so at every point of X .

Intuition

Continuity requires $f(x_n) \rightarrow f(x_0)$: the limit value is *exactly* the value at the limit point. Upper semicontinuity relaxes this to an inequality: as you approach x_0 , the values of f may *drop* suddenly, but they cannot *jump above* $f(x_0)$; the value at the point sits “at the same height or above” what the neighbors suggest. Mental picture with a step function: if at the jump point the function takes the *high* value, it is usc there; if it takes the *low* value, it is lsc. “The solid dot is on top” = usc; “the solid dot is at the bottom” = lsc.

Example 18.5 (Steps and indicator functions). 1. $f(x) = 1$ if $x \leq 0$, $f(x) = 0$ if $x > 0$. At $x_0 = 0$: take any $x_n \rightarrow 0$. The values $f(x_n)$ are 0 or 1, so $\limsup f(x_n) \leq 1 = f(0)$: *usc at 0* (and at the other points it is locally constant, hence continuous). It is not lsc at 0: with $x_n = 1/n \rightarrow 0$, $\liminf f(x_n) = 0 < 1 = f(0)$.

2. $g(x) = 0$ if $x \leq 0$, $g(x) = 1$ if $x > 0$ (the jump with the “solid dot” at the bottom). It is lsc at 0 ($\liminf g(x_n) \geq 0 = g(0)$ trivially) but not usc ($x_n = 1/n$ gives $\limsup = 1 > 0$).

3. The indicator function of the closed set $\{0\}$: $f(0) = 1$, $f(x) = 0$ if $x \neq 0$. It is usc: at 0, $\limsup f(x_n) \leq 1$; away from 0 it is locally 0. In general: *the indicator function of a closed set is usc; the one of an open set is lsc.*

Theorem 18.6 (Description by level sets). *Let $f: X \rightarrow \mathbb{R}$. The following are equivalent:*

1. f is usc on X .
2. For every $c \in \mathbb{R}$, the upper level set $\{x \in X : f(x) \geq c\}$ is closed.
3. For every $c \in \mathbb{R}$, $\{x \in X : f(x) < c\}$ is open.

In the same way, f is lsc $\iff \{x : f(x) \leq c\}$ is closed for every $c \iff \{x : f(x) > c\}$ is open for every c .

Proof. (2) $\iff$ (3) is immediate: $\{f < c\}$ is the complement of $\{f \geq c\}$, and a set is open if and only if its complement is closed.

(1) $\Rightarrow$ (2). Let $F_c = \{x : f(x) \geq c\}$ and show that it is closed with the sequential criterion: take $x_n \in F_c$ with $x_n \rightarrow x_0$; you have to prove $x_0 \in F_c$. Since $f(x_n) \geq c$ for every n , Proposition 18.3(3) gives $\limsup f(x_n) \geq c$ (the lim sup of a sequence that is $\geq c$ is $\geq c$). By usc at x_0 :

$$c \leq \limsup_n f(x_n) \leq f(x_0),$$

so $x_0 \in F_c$.

(2) $\Rightarrow$ (1). By contradiction: suppose f is not usc at some x_0 , that is, there is $x_n \rightarrow x_0$ with $\limsup f(x_n) > f(x_0)$. Choose a number c strictly between the two: $f(x_0) < c < \limsup f(x_n)$. By Proposition 18.3(4) there is a subsequence with $f(x_{n_k}) \rightarrow \limsup f(x_n) > c$, so $f(x_{n_k}) \geq c$ for large k : those x_{n_k} belong to F_c and converge to x_0 . Since F_c is closed, $x_0 \in F_c$, that is $f(x_0) \geq c$, which contradicts $f(x_0) < c$.

The lsc case is symmetric (apply the above to $-f$: note that f is lsc $\iff -f$ is usc, and $\{-f \geq -c\} = \{f \leq c\}$). $\square$

Careful (common mistake)

Memorize the correct pairing, which is very often confused:

$$\text{usc} \leftrightarrow \{f \geq c\} \text{ closed}, \quad \text{lsc} \leftrightarrow \{f \leq c\} \text{ closed}.$$

Memory rule: *usc keeps the high part*: the sets where f stays high ($\geq c$) are closed (they contain their limits). Always check it with the step function of Example 18.5(1): there $\{f \geq 1\} = (-\infty, 0]$, closed, as it must be for a usc function.

Proposition 18.7 (Continuous = usc + lsc). f is continuous at $x_0 \iff f$ is usc and lsc at x_0 .

Proof. ($\Rightarrow$) If f is continuous and $x_n \rightarrow x_0$, then $f(x_n) \rightarrow f(x_0)$, and by Proposition 18.3(2) $\limsup f(x_n) = \liminf f(x_n) = f(x_0)$; in particular the two inequalities of Definition 18.4 hold.

($\Leftarrow$) If both hold, then for every $x_n \rightarrow x_0$:

$$f(x_0) \leq \liminf f(x_n) \leq \limsup f(x_n) \leq f(x_0),$$

so all the inequalities are equalities: $\liminf f(x_n) = \limsup f(x_n) = f(x_0)$, which by Proposition 18.3(2) means $f(x_n) \rightarrow f(x_0)$: sequential continuity, which in metric spaces is the same as continuity. $\square$

18.3.1 The generalized Weierstrass theorem

The reason semicontinuity exists: it is *exactly* what you need for maxima to exist.

Theorem 18.8 (Generalized Weierstrass theorem). *Let K be a nonempty compact metric space (for example, a closed and bounded subset of $\mathbb{R}^n$) and $f: K \rightarrow \mathbb{R}$ usc. Then f is bounded above and attains its maximum: there is $x^* \in K$ with $f(x^*) = \sup_{x \in K} f(x)$. Symmetrically, a lsc function on a compact set attains its minimum.*

Proof. Let $M = \sup_{x \in K} f(x) \in \mathbb{R} \cup \{+\infty\}$ (we do not know yet that it is finite).

Step 1: a maximizing sequence. By definition of supremum, there is a sequence (x_n) in K with $f(x_n) \rightarrow M$ (if $M < \infty$, take x_n with $f(x_n) > M - 1/n$; if $M = +\infty$, take $f(x_n) > n$). This is called a *maximizing sequence*.

Step 2: compactness. Since K is compact (sequentially compact in metric spaces), (x_n) has a convergent subsequence $x_{n_k} \rightarrow x^* \in K$.

Step 3: usc finishes the job. By upper semicontinuity at x^* :

$$f(x^*) \geq \limsup_k f(x_{n_k}) = \lim_k f(x_{n_k}) = M,$$

where the sequence $f(x_{n_k})$ has limit M because it is a subsequence of one that goes to M . But also $f(x^*) \leq M$ (definition of supremum). Then $M = f(x^*) \in \mathbb{R}$ (finite, because f takes real values) and the supremum is attained at x^* .

For the minimum with f lsc, apply the above to $-f$, which is usc. □

Remark 18.9. The proof shows the division of labor: *compactness* produces the candidate point x^* ; *upper semicontinuity* guarantees that the value at the candidate does not “collapse” with respect to the supremum. Full continuity is a luxury you do not need in order to *maximize*: usc is enough. In the same way, to *minimize* lsc is enough. In optimization and in economics this refinement is not pedantry: value functions and many objective functions with natural discontinuities are only semicontinuous.

Link with economics

Existence of optima with fixed costs. The profit of a firm that pays a fixed cost F only if it produces: $\pi(q) = pq - c(q) - F \cdot \mathbf{1}\{q > 0\}$. At $q = 0$ there is an *upward* jump (if you do not produce, you save the fixed cost): π is usc but not continuous. Theorem 18.8 still guarantees that an optimal output exists in $[0, \bar{q}]$. Without the concept of usc, you could not even claim that the firm problem has a solution.

18.4 Correspondences [P1]

Definition 18.10 (Correspondence). Let X, Y be metric spaces. A **correspondence** (also called a set-valued map) $\Gamma: X \rightrightarrows Y$ is a rule that assigns to each $x \in X$ a subset $\Gamma(x) \subseteq Y$. Its **graph** is

$$\text{graph}(\Gamma) = \{(x, y) \in X \times Y : y \in \Gamma(x)\}.$$

We say that Γ has *nonempty / compact / convex / closed values* if each set $\Gamma(x)$ has that property.

Example 18.11 (Correspondences in economics). 1. *Budget correspondence*: to each vector of prices and income $(p, w) \in \mathbb{R}_{++}^n \times \mathbb{R}_+$ it assigns

$$B(p, w) = \{x \in \mathbb{R}_+^n : \langle p, x \rangle \leq w\},$$

the set of affordable bundles. It has nonempty values ($0 \in B$), compact values (closed and bounded in $\mathbb{R}^n$ if $p \gg 0$) and convex values.

2. *Best response* in a game: to each profile of rival strategies s_{-i} it assigns the set $BR_i(s_{-i})$ of optimal strategies of player i (there may be several: that is why it is a correspondence and not a function).
3. *Marshallian demand*: $x(p, w) = \arg \max \{u(x) : x \in B(p, w)\}$. If there are utility ties, it is multivalued.

18.4.1 Hemicontinuity

What does it mean for $\Gamma(x)$ to “move continuously” with x ? There are two possible pathologies, and each one has its own antidote:

- The set *explodes* suddenly: at the limit, new points appear far from the neighboring sets. This is ruled out by *upper* hemicontinuity.
- The set *collapses* suddenly: at the limit, points disappear that the neighboring sets should approximate. This is ruled out by *lower* hemicontinuity.

Definition 18.12 (Hemicontinuity, sequential versions). Let $\Gamma: X \rightrightarrows Y$ have nonempty and compact values.

1. Γ is **upper hemicontinuous** (*uhc*) at x_0 if for every sequence $x_n \rightarrow x_0$ and every choice $y_n \in \Gamma(x_n)$, the sequence (y_n) has a convergent subsequence whose limit belongs to $\Gamma(x_0)$.
2. Γ is **lower hemicontinuous** (*lhc*) at x_0 if for every sequence $x_n \rightarrow x_0$ and every target point $y \in \Gamma(x_0)$, there is a sequence $y_n \in \Gamma(x_n)$ with $y_n \rightarrow y$.
3. Γ is **continuous** at x_0 if it is *uhc* and *lhc* at x_0 .

Intuition

uhc looks “from outside in”: anything you pick in the sets that approach must accumulate *inside* the limit set. The neighbors cannot announce points that accumulate *outside* $\Gamma(x_0)$: when you move the parameter away from the point, the set may shrink suddenly, but it cannot grow suddenly. *lhc* looks “from inside out”: every point of the limit set must be *reachable* from the neighboring sets. The set may grow suddenly, but it cannot shrink suddenly. Short summary: **uhc = does not explode; lhc = does not collapse**.

Example 18.13 (The two standard counterexamples). Define correspondences from $\mathbb{R}$ to $[0, 1]$:

1. *Collapse when you move (uhc holds, lhc fails):*

$$\Gamma_1(x) = \begin{cases} \{0\} & x \neq 0, \\ [0, 1] & x = 0. \end{cases}$$

The set *grows* exactly at $x = 0$, and it is worth checking with the definitions which of the two properties fails. *uhc at 0*: take $x_n = 1/n \rightarrow 0$ and $y_n = 0 \in \Gamma_1(x_n)$; every subsequence goes to $0 \in [0, 1] = \Gamma_1(0)$. The condition holds with these choices and with any other one, because for $x_n \neq 0$ the only possible choice is $y_n = 0$. So Γ_1 is *uhc*. What about *lhc*? Take the target $y = 1 \in \Gamma_1(0)$ and $x_n = 1/n$: the only available sequences are $y_n = 0 \not\rightarrow 1$. *lhc fails*: the set collapses as soon as you leave 0 (seen from 0, its points cannot be approximated). This example shows that “growing suddenly at a point” = “collapsing when you move” violates *lhc*, not *uhc*.

2. *Explosion (uhc fails, lhc holds):*

$$\Gamma_2(x) = \begin{cases} [0, 1] & x \neq 0, \\ \{0\} & x = 0. \end{cases}$$

uhc at 0: with $x_n = 1/n$ choose $y_n = 1 \in \Gamma_2(x_n)$; every subsequence goes to $1 \notin \{0\} = \Gamma_2(0)$. *uhc fails*: the neighbors announce points (all of $[0, 1]$) that the limit set does not contain: seen from outside, the set at 0 is smaller than what the limits suggest, so the points “explode outside” $\Gamma_2(0)$. *lhc at 0*: the only target is $y = 0 \in \Gamma_2(0)$, and $y_n = 0 \in \Gamma_2(x_n)$ works. So *lhc* holds.

Practical rule for exams: to *refute uhc*, exhibit $x_n \rightarrow x_0$, $y_n \in \Gamma(x_n)$ with $y_n \rightarrow y \notin \Gamma(x_0)$. To *refute lhc*, exhibit a target $y \in \Gamma(x_0)$ that no sequence $y_n \in \Gamma(x_n)$ can reach.

Remark 18.14 (Closed graph and *uhc*). We say that Γ has a *closed graph* if $\text{graph}(\Gamma)$ is closed in $X \times Y$; in sequential terms: $x_n \rightarrow x$, $y_n \in \Gamma(x_n)$, $y_n \rightarrow y$ imply $y \in \Gamma(x)$. The link with *uhc*:

1. If Γ is *uhc* with closed values, then it has a closed graph. *Proof*: let $x_n \rightarrow x$, $y_n \in \Gamma(x_n)$, $y_n \rightarrow y$. By *uhc*, some subsequence y_{n_k} converges to a point of $\Gamma(x)$; but every subsequence of y_n converges to y (a convergent sequence drags its subsequences along), so that point is y : $y \in \Gamma(x)$.
2. The converse is false in general, but it holds with a boundedness condition: if Γ has a closed graph and its range is contained in a compact set (for example Y compact), then Γ is *uhc*. *Proof*: let $x_n \rightarrow x$ and $y_n \in \Gamma(x_n)$. Since the y_n live in a compact set, there is a subsequence $y_{n_k} \rightarrow y$; the closed graph gives $y \in \Gamma(x)$, which is exactly the *uhc* condition. (Counterexample without boundedness: $\Gamma(x) = \{1/x\}$ if $x \neq 0$, $\Gamma(0) = \{0\}$: the graph is closed in $\mathbb{R} \times \mathbb{R}$, but with $x_n = 1/n$, $y_n = n$ there is no convergent subsequence and Γ is not *uhc* at 0.)

In economic practice the relevant spaces are usually compact, and “closed graph” is the most convenient test for *uhc*.

18.5 The Berge maximum theorem [P1]

The setting: an agent chooses y in the feasible set $\Gamma(x)$, which depends on the parameter x (prices, income, state), in order to maximize an objective $f(x, y)$. Define:

$$V(x) = \max_{y \in \Gamma(x)} f(x, y) \quad (\text{value function}),$$

$$\sigma(x) = \arg \max_{y \in \Gamma(x)} f(x, y) = \{y \in \Gamma(x) : f(x, y) = V(x)\} \quad (\text{solution correspondence}).$$

Theorem 18.15 (Berge maximum theorem). *Let X, Y be metric spaces, $f: X \times Y \rightarrow \mathbb{R}$ continuous, and $\Gamma: X \rightrightarrows Y$ a continuous correspondence (uhc and lhc) with nonempty and compact values. Then:*

1. V is well defined (the maximum is attained) and is continuous;
2. σ has nonempty and compact values, and is uhc.

Proof. We organize the argument in four steps.

Step 1: V and σ are well defined. For each x , the function $y \mapsto f(x, y)$ is continuous on the nonempty compact set $\Gamma(x)$; by the Weierstrass theorem (a particular case of Theorem 18.8) it attains its maximum. So $V(x) \in \mathbb{R}$ and $\sigma(x) \neq \emptyset$. Also $\sigma(x) = \{y \in \Gamma(x) : f(x, y) \geq V(x)\}$ is the intersection of the compact set $\Gamma(x)$ with a closed set (an upper level set of the continuous function $f(x, \cdot)$, Theorem 18.6), hence it is compact (closed inside a compact set).

Step 2: V is usc. Let $x_n \rightarrow x$; we want $\limsup V(x_n) \leq V(x)$. By Proposition 18.3(4), take a subsequence that realizes the lim sup: $V(x_{n_k}) \rightarrow \limsup_n V(x_n)$. For each k choose a maximizer $y_{n_k} \in \sigma(x_{n_k}) \subseteq \Gamma(x_{n_k})$, so that $V(x_{n_k}) = f(x_{n_k}, y_{n_k})$. By **uhc** of Γ applied to $x_{n_k} \rightarrow x$, there is a further subsequence (which we rename in the same way, to avoid drowning in indices) with $y_{n_k} \rightarrow y^0 \in \Gamma(x)$. By continuity of f in the pair:

$$V(x_{n_k}) = f(x_{n_k}, y_{n_k}) \longrightarrow f(x, y^0).$$

But $V(x_{n_k})$ was already converging to $\limsup V(x_n)$, so $\limsup_n V(x_n) = f(x, y^0)$. And since $y^0 \in \Gamma(x)$ is a feasible point at x (not necessarily optimal), $f(x, y^0) \leq V(x)$. Conclusion: $\limsup V(x_n) \leq V(x)$.

Step 3: V is lsc. Let $x_n \rightarrow x$; we want $\liminf V(x_n) \geq V(x)$. Take a maximizer at the limit point: $y^* \in \sigma(x)$, that is $f(x, y^*) = V(x)$. By **lhc** of Γ applied to the target $y^* \in \Gamma(x)$, there is a sequence $y_n \in \Gamma(x_n)$ with $y_n \rightarrow y^*$. Since y_n is *feasible* at x_n (although perhaps not optimal), the definition of maximum gives

$$V(x_n) \geq f(x_n, y_n) \quad \text{for every } n.$$

Taking $\liminf$ on both sides and using that $f(x_n, y_n) \rightarrow f(x, y^*)$ by continuity of f (a convergent sequence has $\liminf = \lim$):

$$\liminf_n V(x_n) \geq \lim_n f(x_n, y_n) = f(x, y^*) = V(x).$$

Step 4: V continuous and σ uhc. Steps 2 and 3 together with Proposition 18.7 give the

continuity of V . For the uhc of σ : let $x_n \rightarrow x$ and $y_n \in \sigma(x_n)$ be arbitrary. Since $\sigma(x_n) \subseteq \Gamma(x_n)$ and Γ is uhc, there is a subsequence $y_{n_k} \rightarrow y^0 \in \Gamma(x)$. It remains to see that y^0 is *optimal* at x . By optimality of y_{n_k} : $f(x_{n_k}, y_{n_k}) = V(x_{n_k})$. Take limits on both sides: the left one goes to $f(x, y^0)$ (continuity of f), the right one to $V(x)$ (continuity of V , which we have just proved!). Then $f(x, y^0) = V(x)$ with $y^0 \in \Gamma(x)$: exactly $y^0 \in \sigma(x)$. This is the uhc condition for σ . $\square$

Remark 18.16 (Which hypothesis gives which conclusion). The architecture of the proof is a study map in itself:

uhc of $\Gamma \Rightarrow V$ usc (the values do not jump up at the limit);
 lhc of $\Gamma \Rightarrow V$ lsc (the values do not collapse at the limit);
 both + continuity of $f \Rightarrow V$ continuous and σ uhc.

Note what Berge does *not* conclude: σ need **not** be lhc or continuous; solutions may “merge” or disappear because of a tie when you move the parameter (think of a consumer who is indifferent between two bundles: a price perturbation breaks the tie and demand “jumps”, but always *inside* the set of limit optima, never outside: that is uhc).

Corollary 18.17 (When the solution is unique). *Under the hypotheses of Theorem 18.15, if in addition $\sigma(x)$ is a singleton for each x (for example, because $f(x, \cdot)$ is strictly quasiconcave and $\Gamma(x)$ is convex), then the function $x \mapsto$ (the unique element of $\sigma(x)$) is continuous.*

Proof. Call $g(x)$ the unique maximizer. Let $x_n \rightarrow x$; we prove $g(x_n) \rightarrow g(x)$ with the subsequence trick: it is enough to see that *every* subsequence of $(g(x_n))$ has in turn a further subsequence that converges to $g(x)$ (this forces the convergence of the whole sequence: if it did not converge to $g(x)$, some subsequence would stay at distance $\geq \varepsilon$, and its convergent further subsequence going to $g(x)$ would give a contradiction). Given any subsequence, by uhc of σ (Berge) there is a further subsequence $g(x_{n_k}) \rightarrow y^0 \in \sigma(x) = \{g(x)\}$, that is $y^0 = g(x)$. Done. $\square$

Exam strategy

This chapter pays off in the final exam in three ways:

1. *Exact definitions*: usc/lsc (sequential and by level sets, with the correct pairing), sequential uhc/lhc. Practice writing them from memory with all the quantifiers.
2. *Generalized Weierstrass* (Theorem 18.8): a short proof that is asked very often: maximizing sequence + compactness + usc. Learn it as a three line script.
3. *Berge*: either state it with surgical precision (do not forget “nonempty and compact values”!), or prove *one* of the two semicontinuity properties of V . Remember the split: usc of V uses **uhc** + choosing maximizers y_n ; lsc of V uses **lhc** + chasing a limit maximizer y^* with feasible points $y_n \rightarrow y^*$ and the inequality $V(x_n) \geq f(x_n, y_n)$.
4. *Counterexamples*: the two of Definition 18.12/Example 18.13 (explosion and collapse) and the usc/lsc steps. Having counterexamples ready is worth easy points.

18.6 Economic applications [P1]

Link with economics

[Marshallian demand and indirect utility] Let $u: \mathbb{R}_+^n \rightarrow \mathbb{R}$ be continuous (utility) and let the budget correspondence be $B(p, w) = \{x \in \mathbb{R}_+^n : \langle p, x \rangle \leq w\}$ on the domain $p \gg 0$, $w \geq 0$. One proves (a classical exercise) that B is *continuous* with nonempty, compact and convex values: *uhc* follows from the closed graph ($\langle p_n, x_n \rangle \leq w_n$ passes to the limit) plus local boundedness; *lhc* needs the “shrink the bundle” argument: given $x \in B(p, w)$, approximate it with $x_n = \lambda_n x$ for factors $\lambda_n \uparrow 1$ adjusted to (p_n, w_n) ; here you use $w > 0$ or $p \gg 0$ to have some slack. Applying Berge (Theorem 18.15) with $f(p, w, x) = u(x)$:

- The *indirect utility* $v(p, w) = \max \{u(x) : x \in B(p, w)\}$ is **continuous** in (p, w) .
- *Demand* $x(p, w) = \arg \max$ is **uhc** with compact values: small price changes do not make the optimal bundles jump to something far from the original optimal ones.
- If u is *strictly quasiconcave* (strictly convex indifference curves), the maximizer on the convex set $B(p, w)$ is unique (two different optima would give, by strict quasiconcavity, a feasible midpoint that is strictly better: a contradiction), and by Corollary 18.17 demand is a *continuous function*. This is the rigorous foundation of the “well behaved” demand curves of the textbooks.

Link with economics

[Dynamic programming and best response]

- *Bellman value function*. In the problem $V(k) = \max_{k' \in \Gamma(k)} \{F(k, k') + \beta V(k')\}$ (optimal growth: $\Gamma(k) = [0, f(k)]$ is continuous with compact values, F continuous), Berge is the gear that makes the fixed point argument work: the Bellman operator $(Tv)(k) = \max_{k' \in \Gamma(k)} \{F(k, k') + \beta v(k')\}$ maps continuous bounded functions into continuous bounded functions: the continuity of Tv is exactly conclusion (1) of Berge. With that, T is a contraction on $(C_b, \|\cdot\|_\infty)$ (Blackwell conditions) and the Banach fixed point theorem delivers a unique continuous V , and Berge again delivers the optimal policy $g(k)$ *uhc* (continuous if there is strict concavity).
- *Nash equilibrium*. In a game with compact and convex strategy spaces S_i and continuous payoffs u_i (quasiconcave in the own strategy), Berge implies that each best response correspondence $BR_i(s_{-i})$ is *uhc* with nonempty, compact values (and convex by quasiconcavity). The product $BR(s) = BR_1 \times \cdots \times BR_N$ inherits all of that, and the **Kakutani fixed point theorem**, if $K \subset \mathbb{R}^n$ is compact, convex and nonempty and $\Phi: K \rightrightarrows K$ has a closed graph (*uhc*) with nonempty convex values, then there is $s^* \in \Phi(s^*)$, produces a fixed point $s^* \in BR(s^*)$: a *Nash equilibrium*. The full chain Weierstrass $\rightarrow$ Berge $\rightarrow$ Kakutani is the skeleton of the proof of Nash (1950), and it is the flagship example of why this chapter is part of the mathematical training of an economist.

Chapter summary

- $\limsup a_n = \lim_n \sup_{k \geq n} a_k$ (asymptotic ceiling), $\liminf$ (floor); convergence $\iff$ they coincide.
- **usc**: $\limsup f(x_n) \leq f(x_0) \iff \{f \geq c\}$ closed. **lsc**: $\liminf f(x_n) \geq f(x_0) \iff \{f \leq c\}$ closed. Continuous = usc + lsc.
- **Generalized Weierstrass**: usc on a compact set attains a maximum (maximizing sequence + subsequence + usc); lsc attains a minimum.
- Correspondences: **uhc** = every $y_n \in \Gamma(x_n)$ has a subsequence with limit in $\Gamma(x_0)$ (“does not explode”); **lhc** = every $y \in \Gamma(x_0)$ is reached by feasible points $y_n \in \Gamma(x_n)$ (“does not collapse”). uhc (closed values) $\Rightarrow$ closed graph; the converse holds with range in a compact set.
- **Berge**: f continuous + Γ continuous, nonempty, compact $\Rightarrow V$ continuous and σ uhc compact (with uniqueness: continuous solution). Split: uhc gives V usc; lhc gives V lsc.
- Economics: existence of demand and its continuity, continuous indirect utility, well defined Bellman operator, best response uhc + Kakutani = existence of Nash equilibrium.

18.7 Exam style exercises [P1]

Exercise 18.18. Compute $\limsup$ and $\liminf$ of: (a) $a_n = (-1)^n + \frac{1}{n}$; (b) $b_n = \sin(\frac{n\pi}{2})$; (c) $c_n = \frac{n}{n+1} \cos(n\pi)$. In each case say whether the sequence converges.

Solution. (a) The even subsequence is $1 + \frac{1}{2m} \rightarrow 1$; the odd one is $-1 + \frac{1}{2m+1} \rightarrow -1$. Every term belongs to one of the two, so the subsequential limits are $\{1, -1\}$: $\limsup = 1$, $\liminf = -1$. It does not converge.

(b) b_n cycles through $1, 0, -1, 0$. Subsequential limits: $\{1, 0, -1\}$. $\limsup = 1$, $\liminf = -1$. It does not converge.

(c) $\cos(n\pi) = (-1)^n$, so $c_n = (-1)^n \frac{n}{n+1}$: the even terms go to 1, the odd ones to -1 . $\limsup = 1$, $\liminf = -1$; it does not converge. $\square$

Exercise 18.19. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 1 - x^2$ if $x \neq 0$ and $f(0) = 2$. (a) Prove that f is usc at 0 but not lsc at 0. (b) Check the level set description by computing $\{f \geq c\}$ for $c = \frac{3}{2}$ and for $c = \frac{1}{2}$, and verifying that they are closed. (c) Does f attain its maximum on $[-1, 1]$? And its infimum on $(0, 1)$?

Solution. (a) usc at 0: if $x_n \rightarrow 0$, the values $f(x_n)$ are $1 - x_n^2 \rightarrow 1$ (if $x_n \neq 0$) or 2 (if $x_n = 0$). In the worst case, $\limsup f(x_n) \leq 2 = f(0)$: the condition holds (in fact the $\limsup$ is 1 or 2 depending on the sequence, always ≤ 2). Not lsc at 0: with $x_n = \frac{1}{n}$, $\liminf f(x_n) = \lim(1 - \frac{1}{n^2}) = 1 < 2 = f(0)$.

(b) For $c = \frac{3}{2}$: $f(x) \geq \frac{3}{2}$ requires $x = 0$ (since $1 - x^2 \leq 1 < \frac{3}{2}$ for $x \neq 0$) and $f(0) = 2 \geq \frac{3}{2}$: the set is $\{0\}$, closed. For $c = \frac{1}{2}$: $x = 0$ works, and for $x \neq 0$: $1 - x^2 \geq \frac{1}{2} \iff x^2 \leq \frac{1}{2} \iff |x| \leq \frac{1}{\sqrt{2}}$. The set is $[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}]$, closed. Consistent with usc.

(c) *Maximum on* $[-1, 1]$: yes; $[-1, 1]$ is compact and f is usc, so by the generalized Weierstrass theorem it attains the maximum; indeed $\sup f = 2 = f(0)$, attained. *Infimum on* $(0, 1)$: on that interval $f(x) = 1 - x^2$, with infimum 0 (as $x \rightarrow 1$), which is *not* attained: $(0, 1)$ is not compact (it is not closed), and the theorem does not apply. A good reminder that compactness is essential. $\square$

Exercise 18.20. Prove the generalized Weierstrass theorem: if K is a nonempty compact metric space and $f: K \rightarrow \mathbb{R}$ is usc, then f attains its maximum on K .

Solution. Let $M = \sup_K f \in \mathbb{R} \cup \{+\infty\}$. Choose a maximizing sequence $(x_n) \subseteq K$: if $M < \infty$, with $f(x_n) > M - 1/n$; if $M = +\infty$, with $f(x_n) > n$. In both cases $f(x_n) \rightarrow M$. By (sequential) compactness of K , there is a subsequence $x_{n_k} \rightarrow x^* \in K$. The sequence $f(x_{n_k})$, a subsequence of one that goes to M , also goes to M ; in particular $\limsup_k f(x_{n_k}) = M$. By upper semicontinuity of f at x^* :

$$M = \limsup_k f(x_{n_k}) \leq f(x^*).$$

Since also $f(x^*) \leq M$ by definition of supremum, $f(x^*) = M$; in particular $M < \infty$ (because $f(x^*) \in \mathbb{R}$) and the maximum is attained at x^* . $\square$

Exercise 18.21. Consider the correspondence $\Gamma: [0, 2] \rightarrow \mathbb{R}$ given by

$$\Gamma(x) = \begin{cases} [0, 1] & \text{if } x \in [0, 1), \\ [0, 2] & \text{if } x \in [1, 2]. \end{cases}$$

Study the upper and lower hemicontinuity of Γ at the point $x_0 = 1$.

Solution. *uhc at 1*: let $x_n \rightarrow 1$ and $y_n \in \Gamma(x_n)$ be arbitrary. All possible values satisfy $y_n \in [0, 2]$, which is compact, so there is a subsequence $y_{n_k} \rightarrow y \in [0, 2]$. Since $\Gamma(1) = [0, 2]$, we always have $y \in \Gamma(1)$: Γ is **uhc at 1**. (The set “grows” at 1; growing at the point does not hurt uhc.)

lhc at 1: take the target $y = 2 \in \Gamma(1)$ and the sequence $x_n = 1 - \frac{1}{n} \rightarrow 1$, which approaches *from the left*. Any feasible choice satisfies $y_n \in \Gamma(x_n) = [0, 1]$, so that $y_n \leq 1$ for every n and $y_n \rightarrow 2$ is impossible. Γ is **not lhc at 1**: the point 2 of the limit set cannot be approximated from the left: seen from the left, the set “collapses” from $[0, 2]$ to $[0, 1]$.

Conclusion: Γ is uhc but not lhc at $x_0 = 1$ (and therefore it is not continuous there). $\square$

Exercise 18.22. (Berge in action.) Consider the consumer problem with two goods:

$$V(w) = \max_{(x_1, x_2)} x_1 x_2 \quad \text{subject to} \quad x_1 + x_2 \leq w, \quad x_1 \geq 0, \quad x_2 \geq 0,$$

with parameter $w \in [1, 10]$ (fixed prices $p_1 = p_2 = 1$). (a) Check that the hypotheses of the Berge maximum theorem hold. (b) Solve explicitly for $V(w)$ and the solution correspondence $\sigma(w)$, and confirm the conclusions of the theorem.

Solution. (a) The objective $f(w, x) = x_1 x_2$ is a polynomial: continuous in all variables. The feasible correspondence $\Gamma(w) = \{x \in \mathbb{R}_+^2 : x_1 + x_2 \leq w\}$ has *nonempty* values ($0 \in \Gamma(w)$) and *compact* values (closed because it is defined with $\leq$ inequalities of continuous functions, and bounded: $0 \leq x_i \leq w \leq 10$). *uhc*: if $w_n \rightarrow w$, $x^n \in \Gamma(w_n)$, then x^n lives in the compact set $[0, 10]^2$; a subsequence converges to x , and passing to the limit in $x_1^n + x_2^n \leq w_n$ and $x_i^n \geq 0$ you get $x \in \Gamma(w)$ (closed graph + compact range $\Rightarrow$ uhc, Remark 18.14). *lhc*: given $x \in \Gamma(w)$ and $w_n \rightarrow w$ with $w \geq 1 > 0$, define $x^n = \frac{w_n}{w} x$ (scale the bundle to the available income): then $x_1^n + x_2^n = \frac{w_n}{w} (x_1 + x_2) \leq \frac{w_n}{w} w = w_n$, that is $x^n \in \Gamma(w_n)$, and $x^n \rightarrow x$ because $\frac{w_n}{w} \rightarrow 1$. The hypotheses hold.

(b) For fixed w , the maximum of $x_1 x_2$ with $x_1 + x_2 \leq w$: since the objective is increasing in each variable (when the other one is positive), the constraint binds, $x_2 = w - x_1$, and you maximize $h(x_1) = x_1(w - x_1)$ on $[0, w]$: $h'(x_1) = w - 2x_1 = 0$ gives $x_1 = w/2$, a maximum by concavity of h ($h'' = -2 < 0$). Then

$$\sigma(w) = \left\{ \left(\frac{w}{2}, \frac{w}{2} \right) \right\}, \quad V(w) = \frac{w^2}{4}.$$

Confirmation of Berge: $V(w) = w^2/4$ is continuous on $[1, 10]$; σ is a singleton (the Cobb–Douglas utility is strictly quasiconcave in the interior) and $w \mapsto (w/2, w/2)$ is a continuous function, as Corollary 18.17 predicts. Everything fits. $\square$

Appendix A

Midterm review (weeks 1–8) [P2]

Midterm review. It gathers the exact definitions, the theorems with their hypotheses, the recurring proof techniques and a solved mock exam for weeks 1 to 8.

This appendix condenses **everything that can be examined up to week 8**: the space $\mathbb{R}^n$ and its norms, sequences and continuity, metric spaces, basic topology, completeness, convergence of sequences of functions, compactness, connectedness, the Banach fixed point theorem and the first elements of normed vector spaces. Use it in the week before the midterm: first the “cheat sheet” of definitions, then the theorems with their exact hypotheses, after that the proof techniques and, at the end, the full mock exam.

What is this used for in Microeconomics?

This appendix condenses the compactness + continuity + fixed point kit that supports the existence of optima and equilibria in all of microeconomic theory: Weierstrass is “demand exists”, the fixed point is “equilibrium exists” and the topology of open and closed sets is the language in which Mas-Colell, Whinston and Green define continuous preferences and budget sets (ch. 1 to 3). Reviewing it before you start Microeconomics works as a crash course in the mathematics that choice and demand theory, and later general equilibrium (ch. 15 to 18), take for granted.

Exam strategy

The midterm of a course like this one almost always combines: (i) a question asking you to *state with precision* definitions or theorems; (ii) a short theoretical proof seen in class or a direct variant of it; (iii) two or three “concrete” problems (check that something is a metric or a norm, study uniform convergence, apply the Banach fixed point theorem, decide compactness). The last section of this appendix has a mock exam with that format.

A.1 Cheat sheet of exact definitions [P2]

Memorize these definitions *word by word, with their quantifiers*. Stating a definition badly usually invalidates the whole problem.

Norm. Let V be a real vector space. A function $\|\cdot\| : V \rightarrow \mathbb{R}$ is a norm if: (N1) $\|x\| \geq 0$ and

$\|x\| = 0 \iff x = 0$; (N2) $\|\lambda x\| = |\lambda| \|x\|$ for every $\lambda \in \mathbb{R}$, $x \in V$; (N3) $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality).

Usual norms on $\mathbb{R}^n$. $\|x\|_1 = \sum_{i=1}^n |x_i|$, $\|x\|_2 = (\sum_{i=1}^n x_i^2)^{1/2}$, $\|x\|_\infty = \max_{1 \leq i \leq n} |x_i|$.

Equivalent norms. $\|\cdot\|_a$ and $\|\cdot\|_b$ on V are equivalent if there are constants $c, C > 0$ such that $c\|x\|_a \leq \|x\|_b \leq C\|x\|_a$ for every $x \in V$.

Metric. $d: X \times X \rightarrow \mathbb{R}$ is a metric if: (M1) $d(x, y) \geq 0$ and $d(x, y) = 0 \iff x = y$; (M2) $d(x, y) = d(y, x)$; (M3) $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in X$.

Equivalent metrics. d_1 and d_2 on X are *topologically equivalent* if they generate the same open sets (equivalently: the same convergent sequences, with the same limits). They are *strongly equivalent* if there are $c, C > 0$ with $c d_1(x, y) \leq d_2(x, y) \leq C d_1(x, y)$ for all x, y . Strong $\Rightarrow$ topological; the converse is false.

Open ball. $B(x_0, r) = \{x \in X : d(x, x_0) < r\}$, with $r > 0$.

Open set. $A \subseteq X$ is open if for every $x \in A$ there is $r > 0$ such that $B(x, r) \subseteq A$.

Closed set. $F \subseteq X$ is closed if its complement $X \setminus F$ is open. *Sequential description:* F is closed $\iff$ for every sequence $(x_n) \subseteq F$ with $x_n \rightarrow x$, you have $x \in F$.

Interior, closure, boundary. $\text{int}(A)$ is the largest open set contained in A (the points x with some ball $B(x, r) \subseteq A$); the closure $\text{cl}(A)$ is the smallest closed set that contains A (the points x such that every ball $B(x, r)$ meets A); the boundary is $\partial A = \text{cl}(A) \setminus \text{int}(A)$.

Accumulation point. x is an accumulation point of A if every ball $B(x, r)$ contains some point of A different from x .

Convergence of a sequence. $x_n \rightarrow x$ in (X, d) if for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that $n \geq N \Rightarrow d(x_n, x) < \varepsilon$.

Cauchy sequence. (x_n) is Cauchy if for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that $m, n \geq N \Rightarrow d(x_m, x_n) < \varepsilon$.

Complete metric space. (X, d) is complete if every Cauchy sequence in X converges to a point of X .

Dense set. $D \subseteq X$ is dense if $\text{cl}(D) = X$; equivalently, for every $x \in X$ and every $\varepsilon > 0$ there is $y \in D$ with $d(x, y) < \varepsilon$.

Continuity at a point (ε - δ). $f: (X, d_X) \rightarrow (Y, d_Y)$ is continuous at x_0 if for every $\varepsilon > 0$ there is $\delta > 0$ such that $d_X(x, x_0) < \delta \Rightarrow d_Y(f(x), f(x_0)) < \varepsilon$.

Sequential criterion for continuity. f is continuous at $x_0 \iff$ for every sequence $x_n \rightarrow x_0$ you have $f(x_n) \rightarrow f(x_0)$.

Global description. $f: X \rightarrow Y$ is continuous $\iff$ the preimage $f^{-1}(V)$ of every open set $V \subseteq Y$ is open in X ($\iff$ the preimage of every closed set is closed).

Lipschitz function. f is Lipschitz with constant $L \geq 0$ if $d_Y(f(x), f(y)) \leq L d_X(x, y)$ for all x, y . Lipschitz $\Rightarrow$ uniformly continuous $\Rightarrow$ continuous.

Contraction. $f: X \rightarrow X$ is a contraction if there is $k \in [0, 1)$ such that $d(f(x), f(y)) \leq k d(x, y)$ for all $x, y \in X$. The constant must be strictly smaller than 1 and uniform in x, y .

Fixed point. $x^* \in X$ is a fixed point of $f: X \rightarrow X$ if $f(x^*) = x^*$.

Pointwise convergence. $f_n \rightarrow f$ pointwise on A if for each $x \in A$ and every $\varepsilon > 0$ there is $N = N(\varepsilon, x)$ such that $n \geq N \Rightarrow |f_n(x) - f(x)| < \varepsilon$.

Uniform convergence. $f_n \rightarrow f$ uniformly on A if for every $\varepsilon > 0$ there is $N = N(\varepsilon)$ (the same one for all x) such that $n \geq N \Rightarrow \sup_{x \in A} |f_n(x) - f(x)| < \varepsilon$. Practical criterion: $f_n \rightarrow f$ uniformly $\iff M_n := \sup_{x \in A} |f_n(x) - f(x)| \rightarrow 0$.

Bounded and totally bounded set. A is bounded if $\text{diam}(A) = \sup_{x, y \in A} d(x, y) < \infty$. A is totally bounded if for every $\varepsilon > 0$ there is a finite number of balls of radius ε that cover A .

Sequential compactness. $K \subseteq X$ is sequentially compact if every sequence in K has a subsequence that converges to a point of K . In metric spaces, this is equivalent to compactness by coverings (every open cover has a finite subcover) and also to: complete + totally bounded.

Connected space. X is connected if it cannot be written as a union $X = A \cup B$ with A, B open, nonempty and disjoint. The connected subsets of $\mathbb{R}$ are exactly the intervals.

Vector space, basis, dimension. A basis of V is a linearly independent set that spans V ; the dimension is the cardinality of any basis. A normed space is a pair $(V, \|\cdot\|)$; every normed space is a metric space with $d(x, y) = \|x - y\|$.

Careful (common mistake)

Three confusions that cost a lot of points: (1) In the Cauchy condition you compare *two terms of the sequence with each other* ($d(x_m, x_n)$), not a term with the limit: the definition of Cauchy does not mention any limit. (2) In uniform convergence, N depends *only* on ε ; if your estimate gives an N that depends on x , you only proved pointwise convergence. (3) A contraction requires *one and the same* constant $k < 1$ for all pairs of points: having $d(f(x), f(y)) < d(x, y)$ for every $x \neq y$ is *not* enough.

A.2 Key theorems: exact hypotheses and counterexamples [P2]

For each theorem, learn: hypotheses $\rightarrow$ conclusion $\rightarrow$ what fails if you remove one hypothesis. In the exam, *before you use a theorem, state it and check its hypotheses explicitly*.

Equivalence of norms in finite dimension. *Hypotheses:* V a finite dimensional vector space (for example $\mathbb{R}^n$). *Conclusion:* all norms on V are equivalent; in particular they generate the same open sets, the same convergent sequences and the same compact sets. *Counterexample without the hypothesis:* in $C[0, 1]$ (infinite dimension), the norm $\|f\|_\infty = \sup |f|$ and the norm $\|f\|_1 = \int_0^1 |f|$ are not equivalent: $f_n(t) = t^n$ satisfies $\|f_n\|_1 = \frac{1}{n+1} \rightarrow 0$ but $\|f_n\|_\infty = 1$ for every n .

Bolzano–Weierstrass. *Hypotheses:* (x_n) a bounded sequence in $\mathbb{R}^n$. *Conclusion:* there is a convergent subsequence. *Counterexamples:* without boundedness, $x_n = n$ in $\mathbb{R}$ has no convergent subsequence. In infinite dimension the theorem fails even with boundedness: in ℓ^2 , the canonical sequence $e_n = (0, \dots, 0, 1, 0, \dots)$ is bounded ($\|e_n\| = 1$) but $\|e_n - e_m\| = \sqrt{2}$ for $n \neq m$, so no subsequence is Cauchy.

Heine–Borel. *Hypotheses:* $K \subseteq \mathbb{R}^n$ (with any norm, but in $\mathbb{R}^n$!). *Conclusion:* K is compact $\iff K$ is closed and bounded. *Counterexample outside $\mathbb{R}^n$:* the closed unit ball of ℓ^2 is closed and bounded but not compact (the sequence e_n above has no convergent subsequence). In general metric spaces the correct statement is: compact $\iff$ complete and totally bounded.

Cauchy and convergence. Every convergent sequence is Cauchy (always, in any metric space). The converse needs *completeness*. *Counterexample:* in $X = (0, 1)$ with the usual metric, $x_n = 1/n$ is Cauchy but does not converge in X (its “limit” 0 does not belong to the space). In $\mathbb{Q}$, a sequence of rationals that approximates $\sqrt{2}$ is Cauchy with no rational limit.

Complete subspaces. A closed subset of a complete metric space is complete (with the restricted metric). Conversely, a complete subspace of any metric space is closed. *Example phrase:* “closed inside complete is complete; complete inside anything is closed”.

Weierstrass theorem (extreme values). *Hypotheses:* $f: K \rightarrow \mathbb{R}$ continuous and K nonempty compact. *Conclusion:* f attains a maximum and a minimum on K . *Counterexamples:* $f(x) = x$ on $(0, 1)$ (domain not compact because it is not closed: it attains neither sup nor inf); $f(x) = 1/x$ on $(0, 1]$ (not bounded); $f(x) = x$ on $\mathbb{R}$ (domain not bounded).

Continuous image of a compact or connected set. If f is continuous, $f(\text{compact})$ is compact and $f(\text{connected})$ is connected. From this you get the *intermediate value theorem*: if $f: [a, b] \rightarrow \mathbb{R}$ is continuous and $f(a) < c < f(b)$, there is $x \in (a, b)$ with $f(x) = c$ (because $f([a, b])$ is an interval).

Uniform limit of continuous functions. *Hypotheses:* $f_n \rightarrow f$ uniformly and each f_n continuous. *Conclusion:* f is continuous. Moreover, on $[a, b]$: $\int_a^b f_n \rightarrow \int_a^b f$. *Counterexample with pointwise convergence only:* $f_n(x) = x^n$ on $[0, 1]$ converges pointwise to $f(x) = 0$ if $x < 1$, $f(1) = 1$, which is discontinuous; therefore the convergence *cannot* be uniform. For the exchange with the integral: $f_n(x) = 2n^2 x e^{-n^2 x^2}$ on $[0, 1]$ converges pointwise to 0 but $\int_0^1 f_n \rightarrow 1 \neq 0$.

Banach fixed point theorem (contraction mapping principle). *Hypotheses:* (X, d) a complete metric space, nonempty, and $f: X \rightarrow X$ a contraction with constant $k < 1$. *Conclusion:* there is a unique fixed point x^* ; moreover, for any starting point x_0 , the iteration $x_{n+1} = f(x_n)$ converges to x^* with the error bounds

$$d(x_n, x^*) \leq \frac{k^n}{1-k} d(x_1, x_0) \quad (\text{a priori}),$$

$$d(x_n, x^*) \leq \frac{k}{1-k} d(x_n, x_{n-1}) \quad (\text{a posteriori}).$$

Counterexamples: without completeness, $f(x) = x/2$ on $X = (0, 1]$ is a contraction with no fixed point (the candidate 0 is not in X). Without a uniform constant $k < 1$, $f(x) = x + \frac{1}{x}$ on $[1, \infty)$ (complete) satisfies $|f(x) - f(y)| < |x - y|$ for $x \neq y$ but has no fixed point ($f(x) = x$ would give $1/x = 0$). Without $f(X) \subseteq X$, the theorem cannot even be stated: always check that f maps X into X .

Cauchy with a convergent subsequence. If (x_n) is Cauchy and some subsequence $x_{n_j} \rightarrow x$, then $x_n \rightarrow x$. (Model proof in the mock exam, Problem 5.) It is the key piece to prove “compact $\Rightarrow$ complete”.

A.3 The ten arguments that appear most often in exams [P2]

1. **The $\varepsilon/3$ (or $\varepsilon/2$) argument.** To bound $d(a, c)$ insert intermediate terms: $d(a, c) \leq d(a, b) + d(b, c)$. Standard example (the uniform limit of continuous functions is continuous):

$$|f(x) - f(x_0)| \leq \underbrace{|f(x) - f_n(x)|}_{< \varepsilon/3 \text{ (unif.)}} + \underbrace{|f_n(x) - f_n(x_0)|}_{< \varepsilon/3 \text{ (cont. of } f_n)} + \underbrace{|f_n(x_0) - f(x_0)|}_{< \varepsilon/3 \text{ (unif.)}}.$$

First fix a large n using uniformity; then use the continuity of that f_n to choose δ .

2. **Negating a definition with quantifiers.** The negation of “ $x_n \rightarrow x$ ” is: *there is* $\varepsilon_0 > 0$ such that *for every* N *there is* $n \geq N$ with $d(x_n, x) \geq \varepsilon_0$. Mechanical rule: $\forall \leftrightarrow \exists$ and the final inequality is reversed. To prove that a convergence is *not* uniform, exhibit ε_0 and a sequence of points x_n with $|f_n(x_n) - f(x_n)| \geq \varepsilon_0$.
3. **Sequential description of closed sets.** To prove that F is closed: take $(x_n) \subseteq F$ with $x_n \rightarrow x$ and prove $x \in F$ (usually by passing to the limit in the inequalities that define F , thanks to continuity). Example: $F = \{x : g(x) \leq 0\}$ with g continuous is closed because $g(x) = \lim g(x_n) \leq 0$.
4. **Proving that something is a metric or a norm.** Check the axioms one by one; the only nontrivial one is usually the triangle inequality. Tricks: for $d' = \min\{1, d\}$ or $d' = \frac{d}{1+d}$, use that $t \mapsto \frac{t}{1+t}$ is increasing and subadditive. For norms of type $\|\cdot\|_2$, the triangle inequality *is* the Minkowski inequality, which rests on Cauchy–Schwarz: cite them by name.
5. **Extracting subsequences in compact sets.** If the statement says “ K compact” and there is a sequence, the key step is almost surely: “by sequential compactness, there is $x_{n_j} \rightarrow x \in K$ ”. Then you combine it with continuity or with the Cauchy property (previous argument).
6. **Non compactness: exhibit the bad sequence.** To prove that A is not compact, give explicitly a sequence in A with no subsequence that converges *in* A : either it escapes to infinity ($x_n = n$), or it converges to a point outside A ($1/n$ in $(0, 1]$: every subsequence converges to $0 \notin A$), or it is uniformly separated ($\|e_n - e_m\| = \sqrt{2}$).

7. **The geometric series in the Banach proof.** To see that the iterates are Cauchy: $d(x_{n+1}, x_n) \leq k^n d(x_1, x_0)$ by induction, and for $m > n$,

$$d(x_m, x_n) \leq \sum_{j=n}^{m-1} d(x_{j+1}, x_j) \leq d(x_1, x_0) \sum_{j=n}^{\infty} k^j = \frac{k^n}{1-k} d(x_1, x_0) \xrightarrow{n \rightarrow \infty} 0.$$

This computation *is* the a priori error bound: learn it once, use it twice.

8. **Contraction through the derivative (mean value theorem).** If $f: I \rightarrow I$ with $I \subseteq \mathbb{R}$ a closed interval and $\sup_{x \in I} |f'(x)| = k < 1$, then by the mean value theorem $|f(x) - f(y)| = |f'(\xi)| |x - y| \leq k |x - y|$: f is a contraction. *Do not forget to check $f(I) \subseteq I$ and that I is closed (complete).*
9. **Proving density.** To see that D is dense in X : given $x \in X$ and $\varepsilon > 0$, *construct* $y \in D$ with $d(x, y) < \varepsilon$ (truncate decimals for $\mathbb{Q}$ in $\mathbb{R}$; truncate tails for finitely supported sequences in ℓ^2 ; polynomials by Weierstrass in $C[a, b]$).
10. **Uniform convergence by the supremum criterion.** Compute $M_n = \sup_{x \in A} |f_n(x) - f(x)|$ by optimizing in x (differentiate with respect to x , find the maximum, evaluate). If $M_n \rightarrow 0$: uniform; if $M_n \not\rightarrow 0$: not uniform, and the maximizers x_n give you the bad sequence of argument 2.

A.4 Mock midterm exam (with model solutions) [P2]

Try each problem in **25–30 minutes without looking at the solution**. The solutions are written as model solutions: the phrases in italics point at what the grader expects to see written.

Exercise A.1 (Open and closed sets in $\mathbb{R}^2$). Let $A = \{(x, y) \in \mathbb{R}^2 : y > x^2\}$ and $F = \{(x, y) \in \mathbb{R}^2 : y \geq x^2\}$, with the Euclidean norm.

- Prove that A is open using the definition (with balls).
- Prove that F is closed using the sequential description.

Solution. (a) Define $g: \mathbb{R}^2 \rightarrow \mathbb{R}$, $g(x, y) = y - x^2$. *We claim that g is continuous:* it is a polynomial in the coordinates, and sums and products of continuous functions are continuous. Then

$$A = \{(x, y) : g(x, y) > 0\} = g^{-1}((0, \infty)),$$

which is the preimage of an open subset of $\mathbb{R}$ under a continuous function; *by the global description of continuity, A is open.*

Direct version with balls (in case the exam requires the definition): let $p_0 = (x_0, y_0) \in A$, that is $\varepsilon_0 := y_0 - x_0^2 > 0$. We look for $r > 0$ with $B(p_0, r) \subseteq A$. If $\|(x, y) - (x_0, y_0)\| < r \leq 1$, then $|x - x_0| < r$ and $|y - y_0| < r$, so that

$$\begin{aligned} y - x^2 &> (y_0 - r) - (|x_0| + |x - x_0|)^2 \geq y_0 - r - (|x_0| + r)^2 \\ &= \underbrace{y_0 - x_0^2}_{=\varepsilon_0} - r(1 + 2|x_0| + r) \geq \varepsilon_0 - r(2 + 2|x_0|). \end{aligned}$$

Choosing $r = \min \left\{ 1, \frac{\varepsilon_0}{2(2+2|x_0|)} \right\}$ we get $y - x^2 \geq \varepsilon_0/2 > 0$, so $(x, y) \in A$. Since p_0 was arbitrary, A is open. $\square$

(b) Let $(p_n) = ((x_n, y_n)) \subseteq F$ with $p_n \rightarrow p = (x, y)$ in $\mathbb{R}^2$. Convergence in $\mathbb{R}^2$ implies convergence coordinate by coordinate: $x_n \rightarrow x$ and $y_n \rightarrow y$. Since each $p_n \in F$, you have $y_n \geq x_n^2$ for every n . Passing to the limit and using that non strict inequalities are preserved in the limit (and that $t \mapsto t^2$ is continuous):

$$y = \lim_n y_n \geq \lim_n x_n^2 = \left(\lim_n x_n \right)^2 = x^2.$$

So $p \in F$. By the sequential description of closed sets, F is closed. $\square$

Phrases that must appear: “preimage of an open set under a continuous function”, “convergence coordinate by coordinate”, “the inequalities $\geq$ are preserved in the limit”. $\square$

Exercise A.2 (Completeness). On $X = \mathbb{N} = \{1, 2, 3, \dots\}$ define

$$d(m, n) = \left| \frac{1}{m} - \frac{1}{n} \right|.$$

(a) Check that d is a metric on X .

(b) Prove that (X, d) is not complete.

Solution. (a) Let $\varphi(n) = 1/n$, which is injective on $\mathbb{N}$. Then $d(m, n) = |\varphi(m) - \varphi(n)|$ inherits the axioms of the absolute value on $\mathbb{R}$: (M1) $d \geq 0$; and $d(m, n) = 0 \iff \varphi(m) = \varphi(n) \iff m = n$ by injectivity (this is the point you have to highlight). (M2) symmetry: obvious from the absolute value. (M3) triangle inequality: $|\varphi(m) - \varphi(p)| \leq |\varphi(m) - \varphi(n)| + |\varphi(n) - \varphi(p)|$. So d is a metric. $\square$

(b) Candidate: $x_n = n$. We claim that (x_n) is Cauchy. Given $\varepsilon > 0$, take $N > 2/\varepsilon$. If $m, n \geq N$:

$$d(m, n) = \left| \frac{1}{m} - \frac{1}{n} \right| \leq \frac{1}{m} + \frac{1}{n} \leq \frac{2}{N} < \varepsilon.$$

So (x_n) is Cauchy. Suppose, for contradiction, that $x_n \rightarrow p$ for some $p \in \mathbb{N}$. Then

$$d(x_n, p) = \left| \frac{1}{n} - \frac{1}{p} \right| \xrightarrow{n \rightarrow \infty} \frac{1}{p}.$$

Since $p \in \mathbb{N}$ is fixed, $\frac{1}{p} > 0$; taking $\varepsilon_0 = \frac{1}{2p}$, for large n you get $d(x_n, p) \geq \frac{1}{2p} > 0$, which contradicts $d(x_n, p) \rightarrow 0$. Therefore the Cauchy sequence (x_n) does not converge in X and (X, d) is not complete. $\square$

Lesson for the exam: completeness depends on the metric and on the set; $\mathbb{N}$ with the usual metric is complete (every Cauchy sequence is eventually constant), but with this metric it “behaves like” $\left\{ 1, \frac{1}{2}, \frac{1}{3}, \dots \right\} \subseteq \mathbb{R}$, whose limit point 0 is “missing”. $\square$

Exercise A.3 (Pointwise and uniform convergence). For $n \in \mathbb{N}$ define $f_n: [0, 1] \rightarrow \mathbb{R}$ by $f_n(x) = x^n$ and $g_n: [0, \infty) \rightarrow \mathbb{R}$ by $g_n(x) = \frac{x}{1 + nx}$.

(a) Find the pointwise limit of (f_n) on $[0, 1]$ and prove that the convergence is not uniform on $[0, 1]$, but it is uniform on $[0, a]$ for each $0 < a < 1$.

(b) Prove that $g_n \rightarrow 0$ uniformly on $[0, \infty)$.

Solution. (a) *Pointwise limit:* if $0 \leq x < 1$, $x^n \rightarrow 0$; if $x = 1$, $x^n = 1 \rightarrow 1$. So $f_n \rightarrow f$ pointwise with $f(x) = 0$ on $[0, 1)$ and $f(1) = 1$.

Non uniformity, argument 1 (the fastest one): each f_n is continuous on $[0, 1]$ but f is discontinuous at $x = 1$. If the convergence were uniform, the limit would be continuous (uniform limit theorem); since it is not, the convergence is not uniform. $\square$

Non uniformity, argument 2 (from the definition, in case they ask for it): with $\varepsilon_0 = \frac{1}{2}$ and the points $x_n = (1/2)^{1/n} \in [0, 1)$: $|f_n(x_n) - f(x_n)| = x_n^n = \frac{1}{2} \geq \varepsilon_0$ for every n , so $\sup_{x \in [0, 1]} |f_n - f| \geq \frac{1}{2} \not\rightarrow 0$.

Uniformity on $[0, a]$: there $f \equiv 0$ and

$$M_n = \sup_{x \in [0, a]} |x^n - 0| = a^n \xrightarrow{n \rightarrow \infty} 0 \quad (\text{because } 0 < a < 1),$$

and $M_n \rightarrow 0$ is exactly the supremum criterion for uniform convergence. $\square$

(b) *Pointwise limit:* $g_n(0) = 0$; if $x > 0$, $g_n(x) = \frac{x}{1+nx} \leq \frac{x}{nx} = \frac{1}{n} \rightarrow 0$. In fact this bound is independent of x : for every $x \geq 0$,

$$0 \leq g_n(x) = \frac{x}{1+nx} \leq \frac{x}{nx} = \frac{1}{n} \quad (x > 0), \quad g_n(0) = 0 \leq \frac{1}{n},$$

so $M_n = \sup_{x \geq 0} |g_n(x) - 0| \leq \frac{1}{n} \rightarrow 0$. By the supremum criterion, $g_n \rightarrow 0$ uniformly on $[0, \infty)$. $\square$

Phrase that must appear: “the bound does not depend on x ; taking the supremum, $M_n \rightarrow 0$ ”. $\square$

Exercise A.4 (Banach fixed point with error estimate). Consider the equation $x = \frac{1}{3} \cos x$ in $\mathbb{R}$.

(a) Prove that the equation has a unique solution $x^* \in \mathbb{R}$.

(b) Starting from $x_0 = 0$, how many iterations $x_{n+1} = \frac{1}{3} \cos x_n$ guarantee an error smaller than 10^{-4} ? Use the a priori bound.

Solution. (a) Let $F(x) = \frac{1}{3} \cos x$, defined on $X = \mathbb{R}$, which with the usual metric is complete. Clearly $F(\mathbb{R}) \subseteq [-\frac{1}{3}, \frac{1}{3}] \subseteq \mathbb{R}$, so $F: \mathbb{R} \rightarrow \mathbb{R}$ is well defined as a self map of the space. For the contraction property we use the *mean value theorem*: for $x \neq y$ there is ξ between them with

$$|F(x) - F(y)| = |F'(\xi)| |x - y| = \frac{1}{3} |\sin \xi| |x - y| \leq \frac{1}{3} |x - y|,$$

so F is a contraction with constant $k = \frac{1}{3} < 1$. With the three hypotheses checked (nonempty complete space, F a self map, F a contraction), the Banach fixed point theorem guarantees that there is a unique $x^* \in \mathbb{R}$ with $x^* = \frac{1}{3} \cos x^*$, and that the iterates converge to it from any starting point. $\square$

(b) With $x_0 = 0$: $x_1 = F(0) = \frac{1}{3}$, so $d(x_1, x_0) = \frac{1}{3}$. The a priori bound says

$$|x_n - x^*| \leq \frac{k^n}{1-k} |x_1 - x_0| = \frac{(1/3)^n}{1-1/3} \cdot \frac{1}{3} = \frac{(1/3)^n}{2}.$$

We want $\frac{(1/3)^n}{2} < 10^{-4}$, that is $3^n > \frac{10^4}{2} = 5000$. Since $3^7 = 2187 < 5000$ and $3^8 = 6561 > 5000$, $n = 8$ iterations are enough. $\square$

Typical mistakes: forgetting the factor $\frac{1}{1-k}$; using $d(x_0, x_1)$ without computing it; not checking that F maps the space into itself. $\square$

Exercise A.5 (Short theoretical proof). Let (X, d) be a metric space and (x_n) a Cauchy sequence that has a subsequence $x_{n_j} \rightarrow x \in X$. Prove that $x_n \rightarrow x$. Conclude that every sequentially compact metric space is complete.

Solution. Let $\varepsilon > 0$. Since (x_n) is Cauchy, there is N_1 such that

$$m, n \geq N_1 \implies d(x_m, x_n) < \frac{\varepsilon}{2}.$$

By the convergence of the subsequence, there is J such that $j \geq J \implies d(x_{n_j}, x) < \frac{\varepsilon}{2}$. Since $n_j \rightarrow \infty$, you can choose an index $j^* \geq J$ with $n_{j^*} \geq N_1$ (this small step, *choosing an element of the subsequence with a large index*, is the heart of the proof and must be written down).

Then, for every $n \geq N_1$, using the triangle inequality with the intermediate term $x_{n_{j^*}}$:

$$d(x_n, x) \leq d(x_n, x_{n_{j^*}}) + d(x_{n_{j^*}}, x) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $x_n \rightarrow x$. $\square$

Consequence: let (X, d) be sequentially compact and (x_n) Cauchy in X . By sequential compactness there is a subsequence $x_{n_j} \rightarrow x \in X$; by what we proved, the whole sequence converges to $x \in X$. So every Cauchy sequence converges and X is complete. $\square$

Exercise A.6 (Compactness). Consider in $\mathbb{R}$ (usual metric) the sets

$$K = \{0\} \cup \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}, \quad A = (0, 1].$$

- (a) Prove that K is compact.
- (b) Prove that A is not compact by exhibiting a sequence with no subsequence that converges in A .

Solution. (a) Through Heine–Borel (the shortest way, valid because we are in $\mathbb{R}$): $K \subseteq [0, 1]$ is bounded. Let us see that it is closed: its complement is

$$\mathbb{R} \setminus K = (-\infty, 0) \cup (1, \infty) \cup \bigcup_{n \in \mathbb{N}} \left(\frac{1}{n+1}, \frac{1}{n} \right),$$

a union of open intervals, hence open. (Sequential alternative: if $y_k \in K$ and $y_k \rightarrow y$, then either y_k takes some value $\frac{1}{n}$ infinitely often and $y = \frac{1}{n} \in K$, or y_k runs through values $\frac{1}{n_k}$ with $n_k \rightarrow \infty$ and then $y = 0 \in K$.) Closed and bounded in $\mathbb{R}$: by Heine–Borel, K is compact. $\square$

Through coverings (in case the exam requires it): let $\{U_i\}_{i \in I}$ be an open cover of K . Some U_{i_0} contains 0; since it is open, it contains an interval $(-r, r)$, and since $\frac{1}{n} \rightarrow 0$, all but finitely many points $\frac{1}{n}$ fall inside U_{i_0} . The finitely many remaining ones $\frac{1}{1}, \dots, \frac{1}{m}$ are covered by finitely many $U_{i_1}, \dots, U_{i_m}$. Then $\{U_{i_0}, U_{i_1}, \dots, U_{i_m}\}$ is a finite subcover. $\square$

(b) Consider $x_n = \frac{1}{n} \in A$. In $\mathbb{R}$, $x_n \rightarrow 0$, and every subsequence of a convergent sequence converges to the same limit; so any subsequence $x_{n_j} \rightarrow 0$. But $0 \notin A$. If some subsequence converged to a point $a \in A$, by uniqueness of the limit we would have $a = 0$, and then $0 \in A$, a contradiction. Therefore no subsequence of (x_n) converges in A , and A is not sequentially compact. $\square$

Exam note: it is not enough to say “ A is not closed, hence it is not compact” unless you state Heine–Borel first; if the problem asks for the explicit sequence, give it. $\square$

A.5 Overall strategy for the exam day [P2]

Exam strategy

- **Manage your time:** read the whole exam first (5 min); start with the questions on definitions and statements (safe points) and with the problem you recognize from the exercise lists.
- **State before you use:** every time you apply a theorem, write its name, its hypotheses and check each one in your problem (“ $\mathbb{R}$ is complete”, “ F maps X into X ”, “ $k = 1/3 < 1$ ”). A large part of the score is in that verification, not in the conclusion.
- **Quantifiers in order:** in every ε – N or ε – δ proof, the first line must be “Let $\varepsilon > 0$ ” and at some point the explicit choice of N or δ must appear. If your N depends on x in a proof of uniformity, stop: something is wrong.
- **Minimal counterexamples from memory:** x^n on $[0, 1]$ (uniform vs. pointwise), $(0, 1)$ with $1/n$ (Cauchy with no limit), $f(x) = x/2$ on $(0, 1]$ (contraction with no fixed point), e_n in ℓ^2 (bounded and closed but not compact), $f(x) = x + 1/x$ (shrinks distances without being a contraction).
- **If you get stuck,** write the relevant definition and the outline of the argument ($\varepsilon/3$, subsequence in a compact set, and so on): correct outlines with incomplete details still earn points; a blank page does not.

Appendix B

Final review (weeks 10–16) [P2]

Final review. It gathers the exact definitions, the theorems with their hypotheses, the recurring proof techniques and a solved mock exam for weeks 10 to 16.

This appendix condenses the material of the second half of the course: normed subspaces, series in normed spaces, bounded linear operators, Banach spaces, Hilbert spaces, the Riesz representation theorem, differentiation in normed spaces with optimality conditions, and semi-continuity with correspondences (Berge maximum theorem). The format is the same as in the midterm review: exact definitions $\rightarrow$ theorems with hypotheses $\rightarrow$ techniques $\rightarrow$ solved mock exam.

What is this used for in Microeconomics?

For Microeconomics, the valuable half of this appendix is the optimization part (Fréchet, first order conditions, convexity) and the semicontinuity part with the Berge theorem: together they are the executive summary of the mathematics that Mas-Colell, Whinston and Green use without stopping to prove it, from demand theory (ch. 2 to 5) to general equilibrium (ch. 15 to 18) and mechanism design (ch. 23). The Banach, Hilbert and Riesz part is training with a different destination: econometrics (projection = least squares), macro and finance, with the exception of Hahn–Banach, which is the separation argument behind the second welfare theorem. When you come back to this appendix before Microeconomics, it is better to prioritize what is marked [P1] in chapters 17 and 18.

Exam strategy

In the final exam, the material of the first half *does not disappear*: completeness, compactness and fixed points come back as hypotheses of the new theorems (Banach = *complete* normed space; projection = existence of a *minimum*; Berge = *compactness* of the constraint). Review Appendix A at least once before the final.

B.1 Cheat sheet of exact definitions [P2]

Vector subspace and normed subspace. $M \subseteq V$ is a subspace if $0 \in M$ and $\alpha x + \beta y \in M$ for all $x, y \in M$, $\alpha, \beta \in \mathbb{R}$. Every subspace of a normed space is normed with the restricted

norm. A subspace of a Banach space is Banach $\iff$ it is *closed*.

Convergent series in a normed space. $\sum_{n=1}^{\infty} x_n$ converges if the sequence of partial sums $s_N = \sum_{n=1}^N x_n$ converges in the norm of the space.

Absolute convergence. $\sum x_n$ converges absolutely if the numerical series $\sum_{n=1}^{\infty} \|x_n\|$ converges in $\mathbb{R}$.

Linear operator. $T: V \rightarrow W$ is linear if $T(\alpha x + \beta y) = \alpha Tx + \beta Ty$ for all $x, y \in V$, $\alpha, \beta \in \mathbb{R}$.

Bounded operator. A linear T is bounded if there is $M \geq 0$ such that $\|Tx\|_W \leq M \|x\|_V$ for every $x \in V$. *Careful:* “bounded” here does not mean that the image is bounded (the only linear operator with bounded image is $T = 0$: if $Tx \neq 0$, then $\|T(\lambda x)\| = |\lambda| \|Tx\| \rightarrow \infty$ as $\lambda \rightarrow \infty$); it means *bounded on the unit ball*.

Operator norm.

$$\|T\| = \sup_{x \neq 0} \frac{\|Tx\|}{\|x\|} = \sup_{\|x\| \leq 1} \|Tx\| = \sup_{\|x\|=1} \|Tx\| = \inf \{M \geq 0 : \|Tx\| \leq M \|x\| \ \forall x\}.$$

The fundamental inequality $\|Tx\| \leq \|T\| \|x\|$ and the submultiplicativity $\|ST\| \leq \|S\| \|T\|$ always hold.

Space of operators. $B(X, Y)$ (also written $\mathcal{L}(X, Y)$) is the set of bounded linear operators from X to Y , with the operator norm. It is a normed space; if Y is Banach, $B(X, Y)$ is Banach.

Linear functional and dual space. A linear functional is a linear operator $f: X \rightarrow \mathbb{R}$. The dual space is $X^* = B(X, \mathbb{R})$, with $\|f\| = \sup_{\|x\| \leq 1} |f(x)|$; X^* is *always* Banach (because $\mathbb{R}$ is).

Banach space. A normed space that is *complete* with respect to the metric $d(x, y) = \|x - y\|$.

Neumann series. If X is Banach and $T \in B(X, X)$ with $\|T\| < 1$, then $I - T$ is invertible and $(I - T)^{-1} = \sum_{n=0}^{\infty} T^n$, with $\|(I - T)^{-1}\| \leq \frac{1}{1 - \|T\|}$.

Inner product. $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R}$ such that: (P1) $\langle x, x \rangle \geq 0$ and $\langle x, x \rangle = 0 \iff x = 0$; (P2) $\langle x, y \rangle = \langle y, x \rangle$; (P3) linear in the first variable: $\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$. It induces the norm $\|x\| = \sqrt{\langle x, x \rangle}$.

Cauchy–Schwarz inequality. $|\langle x, y \rangle| \leq \|x\| \|y\|$, with equality $\iff x, y$ are linearly dependent.

Parallelogram law. $\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$. A norm comes from an inner product $\iff$ it satisfies this identity; in that case the product is recovered by *polarization*: $\langle x, y \rangle = \frac{1}{4}(\|x + y\|^2 - \|x - y\|^2)$.

Orthogonality. $x \perp y$ if $\langle x, y \rangle = 0$. Pythagoras: $x \perp y \Rightarrow \|x + y\|^2 = \|x\|^2 + \|y\|^2$. A system $\{e_1, \dots, e_k\}$ is orthonormal if $\langle e_i, e_j \rangle = 0$ for $i \neq j$ and $\|e_i\| = 1$.

Hilbert space. An inner product space that is *complete* with respect to the induced norm. Examples: $\mathbb{R}^n$, ℓ^2 , $L^2[a, b]$. Counterexample: $C[0, 1]$ with $\langle f, g \rangle = \int_0^1 fg$ is a pre-Hilbert space but is not complete.

Convex set. C is convex if $x, y \in C$ and $t \in [0, 1]$ imply $tx + (1 - t)y \in C$.

Orthogonal projection. If H is Hilbert and $C \subseteq H$ is convex, closed and nonempty, then for each $x \in H$ there is a unique $P_C(x) \in C$ with $\|x - P_C(x)\| = \text{dist}(x, C) = \inf_{y \in C} \|x - y\|$. When $C = M$ is a closed subspace, $P_M(x)$ is characterized by $x - P_M(x) \perp M$.

Orthogonal complement. $M^\perp = \{y \in H : \langle y, m \rangle = 0 \ \forall m \in M\}$; it is always a *closed* subspace. If M is a closed subspace of a Hilbert space: $H = M \oplus M^\perp$ (every x is written in a unique way as $x = P_M x + (x - P_M x)$).

Gâteaux derivative. $f: U \subseteq X \rightarrow \mathbb{R}$ is Gâteaux differentiable at x in the direction h if $\lim_{t \rightarrow 0} \frac{f(x+th) - f(x)}{t} =: \delta f(x; h)$ exists; if in addition $h \mapsto \delta f(x; h)$ is linear and continuous, it is called the Gâteaux derivative.

Fréchet derivative. f is Fréchet differentiable at x if there is $A \in B(X, Y)$ such that

$$\lim_{h \rightarrow 0} \frac{\|f(x+h) - f(x) - Ah\|_Y}{\|h\|_X} = 0;$$

you write $f'(x) = A$ (or $Df(x)$). Fréchet $\Rightarrow$ Gâteaux and $\Rightarrow$ continuity at x ; the converses are false.

Gradient in a Hilbert space. If $f: H \rightarrow \mathbb{R}$ is Fréchet differentiable, then $f'(x) \in H^*$ and by Riesz there is a unique $\nabla f(x) \in H$ with $f'(x)h = \langle \nabla f(x), h \rangle$ for every h .

Local minimum and necessary condition. x^* is a local minimum of f if there is $r > 0$ with $f(x^*) \leq f(x)$ for every $x \in B(x^*, r)$. If f is Fréchet differentiable at an *interior* local minimum, then $f'(x^*) = 0$ (the zero functional). On a convex set C : $f'(x^*)(y - x^*) \geq 0$ for every $y \in C$ (variational inequality).

Semicontinuity of functions. $f: X \rightarrow \mathbb{R}$ is upper semicontinuous (usc) at x_0 if $\limsup_{x \rightarrow x_0} f(x) \leq f(x_0)$; lower semicontinuous (lsc) if $\liminf_{x \rightarrow x_0} f(x) \geq f(x_0)$. Continuous $\iff$ usc and lsc. In sequential terms: f is lsc if $x_n \rightarrow x_0$ implies $f(x_0) \leq \liminf f(x_n)$.

Generalized Weierstrass. A lsc function on a nonempty compact set attains its *minimum*; a usc function on a compact set attains its *maximum*.

Correspondence. $\Gamma: X \rightrightarrows Y$ assigns to each $x \in X$ a set $\Gamma(x) \subseteq Y$. It is compact valued if each $\Gamma(x)$ is compact.

Hemicontinuity. (With Γ compact valued, sequential version.) Γ is upper hemicontinuous (uhc) at x if for every $x_n \rightarrow x$ and every $y_n \in \Gamma(x_n)$ there is a subsequence $y_{n_j} \rightarrow y \in \Gamma(x)$. It is lower hemicontinuous (lhc) at x if for every $x_n \rightarrow x$ and every $y \in \Gamma(x)$ there are $y_n \in \Gamma(x_n)$ with $y_n \rightarrow y$. Continuous = uhc + lhc.

Berge maximum theorem. Let $f: X \times \Theta \rightarrow \mathbb{R}$ be continuous and $\Gamma: \Theta \rightrightarrows X$ continuous, compact valued and nonempty valued. Then the value function $v(\theta) = \max_{x \in \Gamma(\theta)} f(x, \theta)$ is continuous, and the correspondence of maximizers $x^*(\theta) = \arg \max_{x \in \Gamma(\theta)} f(x, \theta)$ is uhc, compact valued and nonempty valued.

Blackwell conditions. Let $B(S)$ be the space of bounded functions with the supremum norm and $T: B(S) \rightarrow B(S)$. If T is (i) *monotone* ($f \leq g \Rightarrow Tf \leq Tg$) and (ii) it *discounts* (there is $\beta \in (0, 1)$ with $T(f + c) \leq Tf + \beta c$ for every constant $c \geq 0$), then T is a contraction with modulus β in the supremum norm.

Careful (common mistake)

(1) The operator norm is a *supremum*: proving $\|T\| \leq M$ is one inequality for every x ; proving $\|T\| = M$ also needs an *extremal function or sequence* that attains (or approaches) the value M . (2) $H = M \oplus M^\perp$ needs M *closed* and H *complete*: without that, the projection may not exist. (3) usc/lsc of *functions* and uhc/lhc of *correspondences* are different concepts; in Berge, the value function is continuous but the arg max is only *uhc* (demand may “jump”).

B.2 Key theorems: exact hypotheses and counterexamples [P2]

Bounded $\iff$ continuous (linear operators). For a linear T between normed spaces the following are equivalent: (i) T continuous at some point; (ii) T continuous at 0; (iii) T continuous everywhere (in fact Lipschitz); (iv) T bounded. *Linearity is essential*: for nonlinear functions, boundedness and continuity do not imply each other. *Counterexample of an unbounded linear map*: the derivative $D: (C^1[0, 1], \|\cdot\|_\infty) \rightarrow (C[0, 1], \|\cdot\|_\infty)$, $Df = f'$: with $f_n(t) = \sin(nt)$, $\|f_n\|_\infty \leq 1$ but $\|Df_n\|_\infty = n \rightarrow \infty$.

Finite dimension. If $\dim X < \infty$, *every* linear operator from X is bounded. This is why unbounded counterexamples always live in infinite dimension.

Series criterion for Banach spaces. A normed space X is Banach $\iff$ every absolutely convergent series converges in X . It is the standard tool to prove completeness “through series” and the basis of the Neumann series.

$B(X, Y)$ Banach. If Y is Banach, $B(X, Y)$ is Banach (X only needs to be normed). In particular $X^* = B(X, \mathbb{R})$ is always Banach.

Neumann series. *Hypotheses*: X Banach, $\|T\| < 1$. *Conclusion*: $I - T$ is bijective with bounded inverse $\sum_{n \geq 0} T^n$. *If X is not complete*, the series may fail to converge. Typical application: solving $x = Tx + b$ (integral equations, linear dynamic systems).

Completeness of concrete spaces. $(C[a, b], \|\cdot\|_\infty)$ is Banach (the proof is Problem 2 of the mock exam). ℓ^p and L^p are Banach. *Key counterexample*: $C[0, 1]$ with $\|f\|_1 = \int_0^1 |f|$ is *not* complete: the functions f_n that are 0 on $[0, \frac{1}{2}]$, 1 on $[\frac{1}{2} + \frac{1}{n}, 1]$ and interpolate linearly in between are Cauchy in $\|\cdot\|_1$, but their “limit” is a discontinuous step function, outside $C[0, 1]$.

Parallelogram. The norm $\|\cdot\|_\infty$ on $C[0, 1]$ (or on $\mathbb{R}^2$) does *not* come from an inner product: with $f(t) = 1$ and $g(t) = t$, $\|f + g\|_\infty^2 + \|f - g\|_\infty^2 = 4 + 1 = 5 \neq 4 = 2\|f\|_\infty^2 + 2\|g\|_\infty^2$. This two line computation is a frequent exam question.

Projection theorem. *Hypotheses:* H Hilbert, C convex, closed, nonempty. *Conclusion:* there is a unique minimizer of the distance; for closed subspaces, the description is the orthogonality of the residual. *Counterexamples:* $C = \mathbb{Q} \cap [0, 1]$ in $\mathbb{R}$ (not closed: the infimum may not be attained); in an incomplete pre-Hilbert space the projection onto closed subspaces may fail.

Riesz representation. *Hypotheses:* H Hilbert, $\varphi \in H^*$ (linear and continuous). *Conclusion:* there is a unique $y_\varphi \in H$ with $\varphi(x) = \langle x, y_\varphi \rangle$ for every x , and moreover $\|\varphi\|_{H^*} = \|y_\varphi\|_H$. *Without completeness it fails:* in the pre-Hilbert space c_{00} (sequences with finitely many nonzero terms, with the product of ℓ^2), the functional $\varphi(x) = \sum_n x_n/n$ is continuous but its “representative” $(1/n)_n \notin c_{00}$.

Fréchet vs. Gâteaux. Fréchet $\Rightarrow$ Gâteaux with $\delta f(x; h) = f'(x)h$, and Fréchet $\Rightarrow$ continuous. *Converse false:* $f(x, y) = \frac{x^3 y}{x^6 + y^2}$ (with $f(0, 0) = 0$) has all directional derivatives equal to zero at the origin but it is not even continuous there (look at the curve $y = x^3$).

First order necessary condition. *Hypotheses:* x^* a local minimum, x^* interior to the domain, f Fréchet differentiable at x^* . *Conclusion:* $f'(x^*) = 0$. On a convex set C (without interiority): $f'(x^*)(y - x^*) \geq 0 \ \forall y \in C$. If in addition f is convex, these necessary conditions are also *sufficient* for a global minimum.

Generalized Weierstrass and Berge. lsc + compact $\Rightarrow$ the minimum is attained (lower semicontinuity is enough: continuity is not needed). Berge with its four hypotheses (f continuous; Γ continuous, that is uhc and lhc; Γ compact valued; Γ nonempty valued): if Γ is not lhc, the value function may stop being continuous (only usc remains); if the values are not compact, the maximum may fail to exist.

B.3 The ten arguments that appear most often in the final [P2]

1. **Operator norm in two steps.** Step 1 (upper bound): estimate $\|Tx\| \leq M\|x\|$ for every x ; conclude $\|T\| \leq M$. Step 2 (extremal): find x^* with $\|x^*\| = 1$ and $\|Tx^*\| = M$ (or a sequence x_n with $\|Tx_n\| \rightarrow M$); conclude $\|T\| \geq M$. *Never forget step 2:* without an extremal you only have one inequality.
2. **The continuous = bounded cycle.** To prove that a functional or an operator is continuous, the shortest way is almost always to prove that it is bounded (one inequality with Cauchy–Schwarz or with a sup). To prove that it is *not* continuous, exhibit x_n with $\|x_n\| \leq 1$ and $\|Tx_n\| \rightarrow \infty$.
3. **“Inherited” completeness.** To prove that a space is Banach: either it is closed inside a known Banach space (phrase: “a closed subspace of a complete space is complete”), or you run the standard argument Cauchy $\rightarrow$ pointwise candidate limit $\rightarrow$ convergence in the norm $\rightarrow$ the limit belongs to the space. The three steps must appear.
4. **Series criterion.** If the problem says “prove that every absolutely convergent series in X converges”, it is asking for the nontrivial half of the Banach criterion: partial sums are Cauchy because of the tail of $\sum \|x_n\|$.

5. **Riesz as a representation machine.** Faced with “prove that there is a unique y such that $\varphi(x) = \langle x, y \rangle$ ”: (i) check that φ is linear; (ii) that it is bounded (Cauchy–Schwarz or an explicit bound); (iii) cite Riesz; (iv) identify y by “guessing” and verify; (v) $\|\varphi\| = \|y\|$.
6. **Parallelogram as a test.** For “does this norm come from an inner product?”: test the parallelogram identity with two concrete and simple vectors. If it fails for one pair, the answer is no and the two line computation is the whole solution.
7. **Projection = orthogonality of the residual.** To compute $P_M x$ with $M = \text{span}\{e_1, \dots, e_k\}$: orthonormalize (Gram–Schmidt) and use $P_M x = \sum_i \langle x, e_i \rangle e_i$; or impose that the residual $x - P_M x$ be orthogonal to each generator and solve the system (normal equations). Check at the end: $\langle x - P_M x, e_i \rangle = 0$.
8. **Blackwell in two lines.** To prove that a Bellman operator T is a contraction: monotonicity (the max preserves pointwise inequalities) and discounting ($T(v + c) = Tv + \beta c$ by the linearity of the expectation or by the structure of the problem). Cite Blackwell and then Banach on $(B(S), \|\cdot\|_\infty)$, which is complete.
9. **Differentiating a functional from the definition.** Expand $J(u + h) - J(u)$, separate the part that is *linear in h* (the candidate for $J'(u)h$) and prove that the rest is $o(\|h\|)$ (usually the rest is quadratic: $\|h\|^2 / \|h\| = \|h\| \rightarrow 0$). Write the quotient of the Fréchet definition explicitly.
10. **Checking usc/lsc and applying Berge.** For usc/lsc at a jump point: compute the one sided lim sup and lim inf and compare them with the value of the function. For Berge: list the four hypotheses, check them one by one (compactness of $\Gamma(\theta)$ is usually Heine–Borel), and conclude by citing the theorem; if they ask “demand is continuous”, add the argument that a uhc and *single valued* arg max is a continuous function.

B.4 Mock final exam (with model solutions) [P2]

Exercise B.1 (Norm of an integral operator). Let $T: (C[0, 1], \|\cdot\|_\infty) \rightarrow (C[0, 1], \|\cdot\|_\infty)$ be defined by

$$(Tf)(x) = \int_0^x f(t) dt.$$

Prove that T is linear and bounded, and compute $\|T\|$ exactly.

Solution. *Linearity:* by the linearity of the integral, $T(\alpha f + \beta g) = \alpha Tf + \beta Tg$. Also Tf is continuous (the indefinite integral of a continuous function), so T is well defined.

Upper bound: for every $x \in [0, 1]$,

$$|(Tf)(x)| = \left| \int_0^x f(t) dt \right| \leq \int_0^x |f(t)| dt \leq \int_0^x \|f\|_\infty dt = x \|f\|_\infty \leq \|f\|_\infty.$$

Taking the supremum over x : $\|Tf\|_\infty \leq \|f\|_\infty$ for every f , so T is bounded and $\|T\| \leq 1$.

Extremal: let $f^* \equiv 1$, which satisfies $\|f^*\|_\infty = 1$. Then $(Tf^*)(x) = x$ and $\|Tf^*\|_\infty = \sup_{x \in [0, 1]} x = 1$. By definition of supremum, $\|T\| \geq \|Tf^*\|_\infty = 1$.

Conclusion: $\|T\| = 1$. $\square$

Structure that the grader expects: a bound for every $f \Rightarrow \|T\| \leq 1$; an explicit extremal function $\Rightarrow \|T\| \geq 1$. $\square$

Exercise B.2 ($(C[0, 1], \|\cdot\|_\infty)$ is Banach). Prove that $C[0, 1]$ with the supremum norm is a Banach space.

Solution. Let (f_n) be a Cauchy sequence in $(C[0, 1], \|\cdot\|_\infty)$: given $\varepsilon > 0$ there is N such that

$$m, n \geq N \implies \|f_n - f_m\|_\infty = \sup_{x \in [0, 1]} |f_n(x) - f_m(x)| < \varepsilon. \quad (*)$$

Step 1: candidate limit. For each fixed x , $(*)$ implies $|f_n(x) - f_m(x)| < \varepsilon$, so that $(f_n(x))_n$ is a Cauchy sequence in $\mathbb{R}$, which is complete; therefore it converges. Define $f(x) := \lim_n f_n(x)$.

Step 2: the convergence is uniform. Fix $n \geq N$ and $x \in [0, 1]$. Letting $m \rightarrow \infty$ in $|f_n(x) - f_m(x)| < \varepsilon$ (non strict inequalities are preserved in the limit): $|f_n(x) - f(x)| \leq \varepsilon$. Since this bound holds for every x with the same N , $\sup_x |f_n(x) - f(x)| \leq \varepsilon$ for every $n \geq N$, that is $f_n \rightarrow f$ uniformly, so $\|f_n - f\|_\infty \rightarrow 0$.

Step 3: the limit belongs to the space. f is a uniform limit of continuous functions, and the uniform limit of continuous functions is continuous ($\varepsilon/3$ argument, proved in the chapter on pointwise and uniform convergence); so $f \in C[0, 1]$.

Every Cauchy sequence converges in the space: $(C[0, 1], \|\cdot\|_\infty)$ is Banach. $\square$

The three steps (pointwise candidate, upgrade to uniform, membership of the limit) must appear separated and named: that is the scheme that gets the points. $\square$

Exercise B.3 (Orthogonal projection in L^2). In $H = L^2[0, 1]$ with $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$, let $M = \text{span}\{1, t\}$ (polynomials of degree ≤ 1). Compute the orthogonal projection of $u(t) = t^2$ onto M and the distance $\text{dist}(u, M)$.

Solution. M is a finite dimensional subspace, hence *closed*, and H is Hilbert: the projection theorem guarantees that $P_M u$ exists and is unique.

Step 1: orthogonal basis of M (Gram–Schmidt). Start from $\{1, t\}$. Take $v_1 = 1$. Then

$$v_2 = t - \frac{\langle t, 1 \rangle}{\langle 1, 1 \rangle} \cdot 1 = t - \frac{\int_0^1 t dt}{\int_0^1 dt} = t - \frac{1}{2},$$

and we check: $\langle v_2, v_1 \rangle = \int_0^1 (t - \frac{1}{2}) dt = 0$. Also $\|v_1\|^2 = 1$ and $\|v_2\|^2 = \int_0^1 (t - \frac{1}{2})^2 dt = \frac{1}{12}$.

Step 2: coefficients of the projection.

$$P_M u = \frac{\langle u, v_1 \rangle}{\|v_1\|^2} v_1 + \frac{\langle u, v_2 \rangle}{\|v_2\|^2} v_2,$$

with

$$\langle t^2, 1 \rangle = \int_0^1 t^2 dt = \frac{1}{3}, \quad \langle t^2, t - \frac{1}{2} \rangle = \int_0^1 \left(t^3 - \frac{t^2}{2}\right) dt = \frac{1}{4} - \frac{1}{6} = \frac{1}{12}.$$

Therefore

$$P_M u = \frac{1}{3} \cdot 1 + \frac{1/12}{1/12} \left(t - \frac{1}{2}\right) = t - \frac{1}{2} + \frac{1}{3} = t - \frac{1}{6}.$$

Step 3: verification (orthogonality of the residual). With $r(t) = t^2 - t + \frac{1}{6}$: $\int_0^1 r = \frac{1}{3} - \frac{1}{2} + \frac{1}{6} = 0$ and $\int_0^1 t r(t) dt = \frac{1}{4} - \frac{1}{3} + \frac{1}{12} = 0$: the residual is orthogonal to M , as the description of the projection requires.

Step 4: distance.

$$\text{dist}(u, M)^2 = \|r\|^2 = \int_0^1 \left(t^2 - t + \frac{1}{6}\right)^2 dt = \int_0^1 \left(t^4 - 2t^3 + \frac{4}{3}t^2 - \frac{1}{3}t + \frac{1}{36}\right) dt = \frac{1}{5} - \frac{1}{2} + \frac{4}{9} - \frac{1}{6} + \frac{1}{36} = \frac{1}{180}.$$

$$\text{So } \text{dist}(u, M) = \frac{1}{6\sqrt{5}}. \quad \square$$

Points that are graded: saying why the projection exists (M closed, H Hilbert), a correct Gram–Schmidt, and the *verification* of the orthogonality of the residual. $\square$

Exercise B.4 (Riesz representation). In ℓ^2 define $\varphi(x) = \sum_{n=1}^{\infty} \frac{x_n}{2^n}$ for $x = (x_n)$.

(a) Prove that φ is well defined, linear and continuous.

(b) Find the Riesz representative $y \in \ell^2$ and compute $\|\varphi\|$.

Solution. (a) Let $y = (2^{-n})_{n \geq 1}$. Since $\sum_n (2^{-n})^2 = \sum_n 4^{-n} = \frac{1/4}{1-1/4} = \frac{1}{3} < \infty$, you have $y \in \ell^2$ with $\|y\| = \frac{1}{\sqrt{3}}$. For $x \in \ell^2$, by the Cauchy–Schwarz inequality in ℓ^2 :

$$\sum_{n=1}^{\infty} \left| \frac{x_n}{2^n} \right| \leq \left(\sum_n x_n^2 \right)^{1/2} \left(\sum_n 4^{-n} \right)^{1/2} = \frac{\|x\|}{\sqrt{3}} < \infty,$$

so the series converges absolutely and φ is well defined, with $|\varphi(x)| \leq \frac{1}{\sqrt{3}} \|x\|$. Linearity is immediate from the linearity of convergent series. The bound above says that φ is *bounded*, and *for linear functionals bounded is the same as continuous*. $\square$

(b) By construction, $\varphi(x) = \sum_n x_n 2^{-n} = \langle x, y \rangle$ with $y = (2^{-n})_n$: *this is the representative, and it is unique by the Riesz theorem* (if there were two, $y - y'$ would be orthogonal to all of ℓ^2 , in particular to itself, so $y = y'$). The theorem also gives the norm:

$$\|\varphi\|_{(\ell^2)^*} = \|y\|_{\ell^2} = \left(\sum_{n \geq 1} 4^{-n} \right)^{1/2} = \frac{1}{\sqrt{3}}.$$

Verification of the equality of norms without citing the theorem (in case it is asked): “ $\leq$ ” is the bound of (a); for “ $\geq$ ” evaluate at the extremal $x = y/\|y\|$: $\varphi(x) = \langle y, y \rangle / \|y\| = \|y\|$. $\square \quad \square$

Exercise B.5 (Fréchet derivative and optimality). Let $H = L^2[0, 1]$, let $g \in H$ be fixed, and let $J: H \rightarrow \mathbb{R}$ be given by

$$J(u) = \int_0^1 (u(t) - g(t))^2 dt = \|u - g\|^2.$$

(a) Compute the Fréchet derivative $J'(u)$ by checking the definition.

(b) Use the first order necessary condition to find the global minimizer of J and justify that it is indeed a minimum.

Solution. (a) Expand, using the bilinearity of the inner product:

$$J(u+h) - J(u) = \|(u-g) + h\|^2 - \|u-g\|^2 = 2\langle u-g, h \rangle + \|h\|^2.$$

Candidate: the term that is linear in h is $Ah := 2\langle u-g, h \rangle$, which is linear and bounded by Cauchy–Schwarz ($|Ah| \leq 2\|u-g\| \|h\|$), so $A \in H^*$. *Remainder:*

$$\frac{|J(u+h) - J(u) - Ah|}{\|h\|} = \frac{\|h\|^2}{\|h\|} = \|h\| \xrightarrow{h \rightarrow 0} 0.$$

By definition, J is Fréchet differentiable at every u with $J'(u)h = 2\langle u-g, h \rangle$; the gradient (through Riesz) is $\nabla J(u) = 2(u-g)$. $\square$

(b) J is defined on all of H (every point is interior). *Necessary condition:* if u^* is a local minimum, $J'(u^*) = 0$, that is $\langle u^* - g, h \rangle = 0$ for every $h \in H$. Taking $h = u^* - g$: $\|u^* - g\|^2 = 0$, so $u^* = g$ (in L^2). *Sufficiency:* for every u , $J(u) = \|u - g\|^2 \geq 0 = J(g)$, so g is a *global* minimum. (Alternative: J is convex because it is the composition of the squared norm with a translation, and for differentiable convex functions the condition $J'(u^*) = 0$ is sufficient for a global minimum.) $\square$

Exercise B.6 (Semicontinuity and the Berge theorem). (a) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = 1$ if $x \geq 0$ and $f(x) = 0$ if $x < 0$. Study whether f is usc and/or lsc at $x_0 = 0$.

(b) For $p \geq 0$ consider the problem

$$v(p) = \max_{x \in [0,1]} \left(px - \frac{x^2}{2} \right).$$

Check the hypotheses of the Berge maximum theorem, compute $x^*(p)$ and $v(p)$ explicitly, and verify the conclusions of the theorem.

Solution. (a) Compute the one sided limits: $\lim_{x \rightarrow 0^-} f(x) = 0$ and $\lim_{x \rightarrow 0^+} f(x) = 1$, so that $\limsup_{x \rightarrow 0} f(x) = 1$ and $\liminf_{x \rightarrow 0} f(x) = 0$. Since $f(0) = 1$:

$$\begin{aligned} \limsup_{x \rightarrow 0} f(x) = 1 \leq f(0) = 1 &\Rightarrow f \text{ is usc at } 0; \\ \liminf_{x \rightarrow 0} f(x) = 0 < 1 = f(0) &\Rightarrow f \text{ is not lsc at } 0. \end{aligned}$$

(Memory rule: at a jump, the function is usc at the point if “it keeps the value from above”.) $\square$

(b) *Berge hypotheses:* the objective function $f(x, p) = px - \frac{x^2}{2}$ is a polynomial in (x, p) , hence *continuous* on $\mathbb{R} \times \mathbb{R}$. The constraint correspondence is constant, $\Gamma(p) = [0, 1]$: it is *nonempty valued*, *compact valued* ($[0, 1]$ is closed and bounded in $\mathbb{R}$: Heine–Borel) and *continuous* (every constant correspondence is uhc and lhc trivially: for uhc take the sequence itself, for lhc the constant sequence $y_n = y$). *All the hypotheses of the Berge maximum theorem hold.*

Explicit computation: for fixed p , $f(\cdot, p)$ is strictly concave in x (second derivative $-1 < 0$) with interior critical point $x = p$. Since the maximum on $[0, 1]$ of a concave function is the

critical point when it is feasible and the nearest corner otherwise:

$$x^*(p) = \min\{p, 1\} = \begin{cases} p, & 0 \leq p \leq 1, \\ 1, & p > 1, \end{cases} \quad v(p) = \begin{cases} \frac{p^2}{2}, & 0 \leq p \leq 1, \\ p - \frac{1}{2}, & p > 1. \end{cases}$$

Checking the conclusions: v is continuous at $p = 1$ ($\frac{1}{2} = 1 - \frac{1}{2}$), as Berge predicts. The maximizer is unique for each p (strict concavity), and the function $p \mapsto x^*(p) = \min\{p, 1\}$ is continuous: *a usc and single valued argmax is a continuous function*, which is exactly the improvement you get when you add strict concavity. $\square$

In economic terms: $x^*(p)$ is a supply function with capacity saturation at $x = 1$; Berge guarantees that the optimal profit $v(p)$ does not jump even when the constraint binds. $\square$

B.5 Overall strategy for the final exam day [P2]

Exam strategy

- **One line mental map:** normed $\rightarrow$ (complete) Banach $\rightarrow$ (with inner product) Hilbert $\rightarrow$ (dual) Riesz $\rightarrow$ (optimize) Fréchet and Berge. If you identify which box the problem lives in, you already know which theorems are available.
- **Every equality of norms is two inequalities:** in operator norms and in $\|\varphi\| = \|y\|$ of Riesz, always organize the solution into “ $\leq$ ” (general bound) and “ $\geq$ ” (extremal). The grader looks for that structure.
- **Check hypotheses out loud:** “ M is finite dimensional, hence closed”; “ ℓ^2 is Hilbert”; “ $[0, 1]$ is compact by Heine–Borel”; “ φ is bounded by Cauchy–Schwarz”. Each of these sentences is worth points and keeps you from using theorems outside their scope.
- **Integral computations without mistakes:** in projections and derivatives of functionals, half of the problem is $\int_0^1 t^k dt = \frac{1}{k+1}$ done calmly. Check the orthogonality of the residual: it is a free check of your computations.
- **Minimal counterexamples from memory:** unbounded derivative on $(C^1, \|\cdot\|_\infty)$; $C[0, 1]$ with $\|\cdot\|_1$ not complete; $\|\cdot\|_\infty$ fails the parallelogram law; $f(x, y) = x^3y/(x^6 + y^2)$ Gâteaux without Fréchet; a step function usc but not lsc.
- **Close each problem with a stated conclusion:** “therefore $\|T\| = 1$ ”, “so $u^* = g$ is a global minimum”. Open endings lose easy points.

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